Algebra I

Lecture Notes & Transcription

Course: **Algebra I**

Lecturer: **Prof. Rahul Pandharipande**

Department of Mathematics – ETH Zürich

Fall Semester 2026

Notice & Transcription Disclaimer

AI-Generated Notes: This document was automatically generated using AI/LLM systems from the lecture recordings and board content.

Unofficial Document: This is **not** official course material of ETH Zürich or the lecturers, and has **not** been manually corrected or proofread by the staff (unreviewed draft).

Disclaimer: Artificial intelligence can make errors (e.g., in notation, indices, or logical derivations). Use at your own risk — without warranty of any kind regarding accuracy, completeness, or relevance to exams.

Contents

Prerequisites

Background from the First-Year Courses . . . 7

0.1 Notation, Sets and Maps . . . 7

0.2 Equivalence Relations and Partial Orders . . . 8

0.3 The Integers . . . 9

0.4 Groups, Fields and the Complex Numbers . . . 10

0.5 Linear Algebra . . . 11

0.6 Analysis . . . 12

Week 1

Wednesday, 16 Sep — Gaussian Integers, Ring Axioms, Ring Homomorphisms, and Ideals . . . 14

1.1 Gaussian Integers and Primes . . . 14

1.1.1 Units and Primes in $\mathbb{Z}$ and $\mathbb{Z}[i]$ . . . 14

1.1.2 When Does a Rational Prime Remain Prime in $\mathbb{Z}[i]$? . . . 17

1.1.3 Complex Conjugation and the Sum of Two Squares . . . 17

1.1.4 Fermat's Theorem on Sums of Two Squares . . . 19

1.2 The Axioms and Basic Properties of Rings . . . 21

1.2.1 Definition of a Ring . . . 21

1.2.2 Elementary Property: $0 \cdot a = 0$ . . . 22

1.2.3 The Zero Ring $(1 = 0)$ . . . 23

1.2.4 Additive Inverses and the Element $-a$ . . . 24

1.2.5 Integer Multiples and the Canonical Map $\mathbb{Z} \to R$ . . . 25

1.3 Ring Homomorphisms . . . 27

1.3.1 Standing Convention: Commutativity and Units . . . 27

1.3.2 Definition of a Ring Homomorphism . . . 28

1.3.3 Preservation of the Additive Identity: $f(0) = 0$ . . . 29

1.3.4 Why Identity Preservation Must Be Axiomatized . . . 30

1.3.5 Examples of Ring Homomorphisms: The Canonical Map $\mathbb{Z} \to R$ . . . 32

1.3.6 Inclusion Homomorphisms and Subrings . . . 33

1.3.7 The Ring of Residue Classes $\mathbb{Z}/n\mathbb{Z}$ and the Projection Homomorphism . . . 34

1.4 The Kernel of a Ring Homomorphism and Ideals . . . 35
1.4.1 Definition of the Kernel . . . 35
1.4.2 Properties of the Kernel . . . 37
1.4.3 Ideals in a Ring . . . 38
1.4.4 The Kernel as an Ideal . . . 40
1.4.5 Ideals in a Field . . . 41
1.5 Outlook: Zorn's Lemma and Vector Space Bases . . . 43
1.5.1 Existence of Bases in Infinite-Dimensional Vector Spaces . . . 43

Week 2

Tuesday, 22 Sep — Homomorphisms, Substructures, Ideals, and Category Theory . . . 46
2.1 Administrative Remarks and Course Logistics . . . 46
2.2 Review of Algebraic Structures . . . 47
2.2.1 Groups and Uniqueness of Identity and Inverses . . . 47
2.2.2 Rings and Fields . . . 49
2.3 Group Homomorphisms and Isomorphisms . . . 51
2.3.1 Definition of a Group Homomorphism . . . 51
2.3.2 Preservation of the Neutral Element . . . 52
2.3.3 Group Isomorphisms and Characterization . . . 53
2.3.4 Examples: Isomorphisms of the Real Numbers . . . 54
2.4 Kernels and Subgroups . . . 59
2.4.1 The Kernel of a Group Homomorphism . . . 59
2.4.2 Properties of the Kernel . . . 60
2.4.3 Definition of a Subgroup . . . 62
2.4.4 Kernel and Image as Subgroups . . . 62
2.4.5 Pedagogical Aside: Mathematical Reading and Textbooks . . . 63
2.5 Ring Homomorphisms, Ideals, and Subrings . . . 64
2.5.1 Definition of a Ring Homomorphism . . . 64
2.5.2 Ring Isomorphisms and Symmetry Breaking . . . 65
2.5.3 Ideals in Rings and the Kernel . . . 66
2.5.4 Definition of a Subring . . . 67
2.5.5 Verification of the Subring Property for the Image . . . 68
2.6 Examples of Subrings and Ideals . . . 69
2.6.1 Even Integers in the Ring of Integers . . . 69
2.6.2 Intermediate Subrings between $\mathbb{Z}$ and $\mathbb{Q}$ . . . 71
2.6.3 Polynomial Rings: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$ . . . 72
2.6.4 Subrings of the Integers $\mathbb{Z}$ . . . 74

2.7 Ideals in the Ring of Integers . . . 75
2.7.1 Classification of Ideals in $\mathbb{Z}$ . . . 75
2.7.2 Historical Context: Ideal Numbers and Hilbert's Epitaph . . . 76
2.7.3 Proof: Every Ideal of $\mathbb{Z}$ is Principal . . . 77
Proof of Theorem 2.35 . . . 78
2.7.4 Trivial Ideals and Comparison with Fields . . . 81
2.8 Ideals in Rings of Smooth Functions . . . 82
2.8.1 The Ring $C^\infty(\mathbb{R})$ and Compactly Supported Functions . . . 82
2.8.2 The Ideal of Compactly Supported Smooth Functions . . . 83
2.8.3 A Strictly Larger Ideal: Functions Vanishing on a Right Half-Line . . . 85
2.8.4 The Restriction Homomorphism on Function Spaces . . . 86
2.9 Introduction to Category Theory . . . 87
2.9.1 Definition of a Category . . . 87
2.9.2 The Category of Groups . . . 88
2.9.3 Categories of Rings and Vector Spaces . . . 89
2.9.4 Interlude: Non-Surjectivity of the Restriction Map . . . 89
2.9.5 Axioms of a Category: Homs, Composition, and Identity . . . 91
2.9.6 Categorical Definition of Isomorphism . . . 93
2.9.7 Definition of a Functor . . . 96
2.9.8 Examples of Functors: Forgetful Functor from Rings to Groups . . . 97
2.9.9 The Double Dual Functor on Vector Spaces . . . 98
Wednesday, 23 Sep — Categories, Integral Domains, and the Group of Units . . . 102
2.10 Course Logistics and Announcements . . . 102
2.11 Review of Categories: Data and Axioms . . . 103
2.11.1 Categorical Isomorphisms . . . 105
2.11.2 Monomorphisms and Epimorphisms . . . 106
2.12 Integral Domains . . . 109
2.12.1 Zero Divisors and the Definition of Integral Domains . . . 109
2.12.2 Student Discussion: Monomorphisms in Linear Algebra . . . 111
2.12.3 Characterization of $\mathbb{Z}/m\mathbb{Z}$ as a Domain . . . 112
Proof of Proposition 2.63 . . . 113
2.12.4 Boundary Cases and Principle of Lecturing . . . 115
2.13 Divisibility and Units in Commutative Rings . . . 116
2.13.1 Divisibility in General Rings . . . 116
2.13.2 Units and Multiplicative Inverses . . . 117
2.13.3 Linguistic Conventions: The Units, The Unit, and A Unit . . . 118
2.13.4 The Group of Units $\mathcal{U}_R$ . . . 119
2.14 Units in Residue Class Rings $\mathbb{Z}/m\mathbb{Z}$ . . . 120
2.14.1 Characterization of Units via Coprimality . . . 120
2.14.2 The Units of $\mathbb{Z}/7\mathbb{Z}$ and Cyclic Generator Calculations . . . 122
2.14.3 Isomorphism with the Additive Cyclic Group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$ . . . 125

Week 3

Tuesday, 29 Sep — Groups of Units, Fields of Fractions, and Polynomial Rings 129

3.1 Review: The Group of Units of a Ring . . . 129
3.1.1 Units in the Integers and the Gaussian Integers . . . 130
3.1.2 The Group of Units of $\mathbb{Z}/m\mathbb{Z}$ . . . 133
3.1.2.1 Example: Units of $\mathbb{Z}/10\mathbb{Z}$ . . . 133
3.2 Euler's Totient Function . . . 135
3.2.1 Evaluating $\phi(p)$ and $\phi(p^2)$ for a Prime p . . . 135
3.3 Fields of Fractions . . . 137
3.3.1 The Universal Property of $\mathbb{Q}$ . . . 138
3.3.2 Construction and Well-Definedness of the Field Extension . . . 139
3.4 The Field of Fractions of an Integral Domain . . . 146
3.4.1 Construction of the Set of Formal Fractions . . . 147
3.4.2 Digression: Equivalence Relations and Partitions . . . 147
3.4.3 The Equivalence Relation on Formal Fractions . . . 149
3.5 The Field of Fractions and Its Canonical Embedding . . . 154
3.5.1 Construction of the Field of Fractions . . . 154
3.5.2 The Canonical Embedding of R into F . . . 154
3.6 Field Operations on the Field of Fractions . . . 156
3.6.1 Definition of Zero, One, and Addition on F . . . 156
3.6.2 Well-Definedness of Addition . . . 157
3.6.3 Definition and Well-Definedness of Multiplication on F . . . 158
3.6.4 Compatibility of the Operations with the Canonical Embedding . . . 159
3.6.5 Summary of the Field of Fractions Construction . . . 161
3.7 Examples of Fields of Fractions . . . 163
3.7.1 Fields of Fractions of Intermediate Rings and Fields . . . 163
3.8 Polynomial Rings . . . 166
3.8.1 Informal Definition of Polynomials as Formal Expressions . . . 167
3.8.2 Examples and Constant Polynomials . . . 167
3.8.3 Formal Rigorous Definition of $R[x]$ as Sequences . . . 168
3.8.4 The Indeterminate and the Ring Operations on $R[x]$ . . . 169
3.8.5 Addition in $R[x]$ . . . 171
3.8.6 Multiplication in $R[x]$ and the Cauchy Convolution . . . 171
3.8.7 Conclusion of the Lecture . . . 173

Wednesday, 30 Sep — Field of Fractions, Polynomial Rings, and Formal Power Series . . . 175

3.9 Field of Fractions and Ring Constructions . . . 175
3.9.1 Field of Fractions of an Integral Domain . . . 175

3.10 Polynomial Rings . . . 178
3.10.1 Definition and Structure of $R[x]$ . . . 178
3.10.2 The Universal Property of Polynomial Rings . . . 179
Proof of Theorem 3.38 . . . 181
3.10.3 Units in the Polynomial Ring over an Integral Domain . . . 184
3.11 Formal Power Series Rings . . . 189
3.11.1 Definition and Ring Operations in $R[[x]]$ . . . 189
3.11.2 Units in the Formal Power Series Ring . . . 194
Proof of Theorem 3.42 . . . 196

Week 4

Tuesday, 06 Oct — Polynomial Rings, Laurent Series, Quotient Rings, and the First Isomorphism Theorem . . . 200
4.1 Review: Polynomial Rings and Power Series . . . 200
4.1.1 The Binomial Formula in Commutative Rings . . . 202
4.1.2 Characteristic p and the Frobenius Identity . . . 203
4.2 Integral Domains, Fields of Fractions, and Laurent Series . . . 204
4.2.1 Polynomial Rings and Power Series Rings as Integral Domains . . . 204
4.2.2 Fields of Fractions: Rational Functions and Laurent Series . . . 206
4.2.3 Formal Laurent Series with Finitely Many Negative Powers . . . 207
4.3 Ideals and Ring Homomorphisms . . . 210
4.3.1 Definition and Axioms of an Ideal . . . 210
4.3.2 The Linear Algebra Analogy: Kernels, Images, and Isomorphisms . . . 211
4.3.3 Ring Homomorphisms vs. Linear Transformations . . . 212
4.3.4 Injectivity and the Zero Ideal . . . 213
Proof of Proposition 4.16 . . . 214
4.4 The Construction of Quotient Rings . . . 216
4.4.1 Step 1: The Underlying Set and Equivalence Relation Modulo an Ideal . . . 217
4.4.2 Naming Conventions for Equivalence Classes (Cosets) . . . 219
4.4.3 Step 2 & 3: Distinguished Elements and the Well-Definedness of Addition . . . 220
4.4.4 Step 4: Multiplication and the Crucial Role of the Ideal Absorption Property . . . 223
4.4.5 Recap: The Crucial Role of the Ideal Axioms . . . 224
4.4.6 Completion of the Quotient Ring Structure . . . 225
4.5 The First Isomorphism Theorem for Rings . . . 227
Proof of the First Isomorphism Theorem . . . 228
4.5.1 Step 1: Construction and Well-Definedness of the Induced Map $\bar{f}$ . . . 228
4.5.2 Step 2: Surjectivity of $\bar{f}$ onto $\mathrm{Im}(f)$ . . . 232
4.5.3 Step 3: Injectivity and Trivial Kernel of $\bar{f}$ . . . 233
4.5.4 Recapitulation and the Anatomy of the Theorem . . . 235

4.6 Ideals Generated by Elements . . . 236
4.6.1 Quotients of Univariate Polynomial Rings . . . 238
4.6.2 The Quotient $\mathbb{Q}[x]/(x^2+1)$ and the Gaussian Rationals . . . 240
4.6.3 Construction of the Evaluation Homomorphism at i . . . 241
4.6.4 Image of the Evaluation Homomorphism and the First Isomorphism Theorem . . . 243
4.6.5 Computing the Kernel via Polynomial Long Division . . . 244

Wednesday, 7 Oct — Quotient Rings, the Correspondence Theorem for Ideals, and Maximal Ideals . . . 249

4.7 Review: Ideals and the Quotient Ring Construction . . . 249
4.7.1 The Canonical Quotient Map . . . 250
4.7.2 Review: The First Isomorphism Theorem for Rings . . . 253
4.8 Pullbacks of Ideals under Ring Homomorphisms . . . 254
4.9 The Correspondence Theorem for Ideals . . . 257
4.9.1 Sets of Ideals and Notation . . . 257
4.9.2 Statement of the Lattice Correspondence . . . 258
Proof of the Correspondence Theorem for Ideals . . . 259
4.9.2.1 Well-Definedness of the Correspondence Maps . . . 259
4.9.2.2 Bijectivity of the Correspondence . . . 263
4.10 Maximal Ideals . . . 267

Notation and Facts Assumed by the Lectures

Background from the First-Year Courses

About This Chapter

This chapter is not part of the lectures. It collects what the lectures take for granted from the first-year courses (Linear Algebra, Analysis, and the Foundations course *"Grundstrukturen"*), together with the notation of these notes. The lectures point here wherever they use one of these facts.

- Results are stated as **facts**, without proof: they are standard first-year material. Bézout's identity and Euclid's lemma get a two-line argument, because the lectures build on them.
- The first lecture of the course (Tuesday, 15 September) was not recorded. According to the later lectures, it introduced groups, rings and fields and discussed factorization in $\mathbb{Z}$; these notes start with the second lecture. The lecturer restates groups and fields in Week 2 and rings in Week 1.

Notation, Sets and Maps

Notation 0.1 (*Number Sets and Inclusions*)**:**

- $\mathbb{N} = \{0, 1, 2, \dots\}$, also written $\mathbb{N}_0$ to stress that 0 is included, and $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. Subscripts restrict a set: $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$, $\mathbb{Z}_{\geq 2}$, $\mathbb{R}_{>0}$.
- For a field K, $K^* = K \setminus \{0\}$. The group of units of a ring R (Week 2, Proposition 2.70) appears in these notes as $\mathcal{U}_R$, $U(R)$, $\mathrm{Unit}(R)$, $R^\times$ and, from Week 3 on, R^* (Proposition 3.3). For a field the two meanings of K^* agree, since its units are exactly its non-zero elements.
- $A \subset B$ means that every element of A lies in B; equality is allowed, as in the lecturer's usage. A proper inclusion is written $A \subsetneq B$.
- $|X|$ (or $\#X$) is the number of elements of a finite set X; for a finite group it is called the *"order"*.
- $X \times Y = \{(x, y) \mid x \in X,\ y \in Y\}$ is the *"Cartesian product"*. A binary operation on X is a map $X \times X \to X$; the set of pairs used for the field of fractions (Definition 3.12) is a subset of $R \times R$.
- Y^X is the set of all maps $X \to Y$. In particular $R^{\mathbb{N}_0}$ is the set of all sequences $(a_0, a_1, a_2, \dots)$ with entries in R; polynomials and power series are defined as such sequences (Definition 3.34, Definition 3.40).

Definition 0.2 (*Maps*)**:** A map $f\colon X \to Y$ is *"injective"* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$, *"surjective"* if every $y \in Y$ is of the form $f(x)$, and *"bijective"* if it is both. The composition of f

with $g\colon Y \to Z$ is $g \circ f\colon X \to Z$, $x \mapsto g(f(x))$; the identity of X is id_X (also written 1_X); the restriction of f to $U \subset X$ is $f|_U\colon U \to Y$.

Fact 0.3 (*Inverse Maps*): A map $f\colon X \to Y$ is bijective if and only if there is a map $g\colon Y \to X$ with $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. This g is unique; it is the *"inverse map"* f^{-1}.

Fact 0.4 (*Cardinality*): Two sets have the same *"cardinality"* if there is a bijection between them. A set is *"countable"* if it is finite or has the cardinality of $\mathbb{N}$: $\mathbb{Z}$ and $\mathbb{Q}$ are countable, while $\mathbb{R}$ is *"uncountable"* (Cantor's diagonal argument). The sets $\mathbb{R}_{>0}$ and $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ have the same cardinality as $\mathbb{R}$; for $\mathbb{R}_{>0}$, a bijection is exp (Fact 0.37). A bijection says nothing about algebraic structure: the groups $(\mathbb{R}^*, \cdot)$ and $(\mathbb{R}_{>0}, \cdot)$ have the same cardinality but are not isomorphic (Week 2, Proposition 2.15).

Definition 0.5 (*Image and Preimage of a Subset*): Let $f\colon X \to Y$ be a map. For $A \subset X$, the *"image"* of A is $f(A) = \{f(a) \mid a \in A\} \subset Y$; the image of f is $\mathrm{im}(f) = f(X)$. For $B \subset Y$, the *"preimage"* (or pullback) of B is

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\} \subset X.$$

Important Remark 0.6 (*Two Meanings of f^{-1}*): The preimage $f^{-1}(B)$ is defined for **every** map f, bijective or not; it is a set, possibly empty. Only when f is bijective does f^{-1} also denote the inverse map (Fact 0.3), and then the two readings agree: the preimage of B is the image of B under the inverse map. The kernel of a homomorphism is a preimage, $\ker(f) = f^{-1}(\{0\})$.

Fact 0.7 (*Images and Preimages*): Let $f\colon X \to Y$ be a map, $A, A' \subset X$ and $B, B' \subset Y$.

- **(i)** Preimages respect inclusions, intersections and unions: $B \subset B' \implies f^{-1}(B) \subset f^{-1}(B')$, $f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$, $f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$. Images respect inclusions and unions, but in general not intersections.
- **(ii)** $f(f^{-1}(B)) \subset B$, with equality for every B if and only if f is surjective.
- **(iii)** $A \subset f^{-1}(f(A))$, with equality for every A if and only if f is injective.

The Correspondence Theorem (Proposition 4.52, Week 4) uses (ii) for the surjective quotient map, and replaces injectivity in (iii) by the hypothesis that the ideal contains the kernel.

Equivalence Relations and Partial Orders

Definition 0.8 (*Equivalence Relation*): An *"equivalence relation"* on a set T is a relation $\equiv$ on T that is

- **(i) reflexive:** $x \equiv x$ for all $x \in T$;
- **(ii) symmetric:** $x \equiv y \implies y \equiv x$;
- **(iii) transitive:** $x \equiv y$ and $y \equiv z \implies x \equiv z$.

Definition 0.9 (*Equivalence Classes and Quotient Set*): The *"equivalence class"* of $x \in T$ is $[x] = \{y \in T \mid y \equiv x\}$, and the *"quotient set"* $T/\!\equiv$ (written $T_{\equiv}$ on the lecturer's board) is the set of all equivalence classes. Two classes are either equal or disjoint, $[x] = [y]$ holds exactly when $x \equiv y$, and the classes cover T: they form a *"partition"* of T.

Maps Defined on Classes: A rule that defines a value for a class through one of its elements (a *"representative"*) defines a map on $T/\!\equiv$ only if equivalent representatives give the same value; the rule is then called *"well-defined"*. The lectures carry out this check for the operations of $\mathbb{Z}/n\mathbb{Z}$ (Definition 1.24), for the extension $g\colon \mathbb{Q} \to F$ (Claim 3.9) and for the field of fractions (Proposition 3.22).

Definition 0.10 (*Partial Orders and Maximal Elements*): A *"partial order"* on a set P is a relation $\leq$ that is reflexive, transitive and *"antisymmetric"* ($x \leq y$ and $y \leq x$ imply $x = y$). Two elements need not be comparable; the standard example is a set of subsets, ordered by inclusion. A *"chain"* is a subset in which any two elements are comparable (a totally ordered subset). An *"upper bound"* of $A \subset P$ is an element $b \in P$ with $a \leq b$ for all $a \in A$. An element $m \in P$ is *"maximal"* if nothing in P lies strictly above it: $m \leq x$ implies $x = m$. A maximal element need not be a largest element, and there can be several. The lectures use these notions for Zorn's lemma (Week 1, Remark 1.34) and for maximal ideals, which are the maximal elements of the set of proper ideals ordered by inclusion (Week 4, Definition 4.54); $\mathbb{Z}$ has infinitely many of them, the ideals $p\mathbb{Z}$ for primes p.

The Integers

Fact 0.11 (*Well-Ordering Principle*): Every non-empty subset of $\mathbb{N}$ has a least element. (This is equivalent to the principle of induction, Fact 0.12.) The lectures use it to find the generator of an ideal of $\mathbb{Z}$ (Theorem 2.35).

Fact 0.12 (*Induction*): Let $P(n)$ be a statement for each $n \in \mathbb{N}$. If $P(0)$ holds, and $P(n)$ implies $P(n+1)$ for every n, then $P(n)$ holds for all $n \in \mathbb{N}$. The same holds with any starting value $n_0 \in \mathbb{Z}$ in place of 0.

Fact 0.13 (*Binomial Theorem*): For $n \in \mathbb{N}$ and $0 \leq a \leq n$, the *"binomial coefficient"* $\binom{n}{a} = \frac{n!}{a!\,(n-a)!}$ is a positive integer, and $\binom{n}{a} = \binom{n-1}{a-1} + \binom{n-1}{a}$ for $1 \leq a \leq n-1$ (Pascal's rule). For all $x, y \in \mathbb{C}$,

$$(x+y)^n = \sum_{a=0}^{n} \binom{n}{a} x^a y^{n-a},$$

by induction on n using Pascal's rule. The proof uses only commutativity and distributivity, so the formula holds in every commutative ring (Theorem 4.2, Week 4).

Fact 0.14 (*Division with Remainder*): For $y \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$ there are unique integers q, r with

$$y = qn + r \quad \text{and} \quad 0 \leq r < n.$$

Definition 0.15 (*Divisibility, Greatest Common Divisor, Primes*): For $a, b \in \mathbb{Z}$, a *"divides"* b, written $a \mid b$, if $b = ac$ for some $c \in \mathbb{Z}$. For a, b not both 0, $\gcd(a,b)$ is the largest integer dividing both; the lecturer also writes (a,b). The integers a, b are *"coprime"* (relatively prime) if $\gcd(a,b) = 1$. A *"prime"* is an integer $p \geq 2$ whose only positive divisors are 1 and p.

Fact 0.16 (*Bézout's Identity*): For $a, b \in \mathbb{Z}$, not both 0, there are $x, y \in \mathbb{Z}$ with

$$ax + by = \gcd(a,b).$$

In particular, a and b are coprime if and only if $ax + by = 1$ for some $x, y \in \mathbb{Z}$.

Why It Holds: The set $I = \{ax + by \mid x, y \in \mathbb{Z}\}$ is an ideal of $\mathbb{Z}$ (Definition 1.30), so $I = d\mathbb{Z}$ with $d > 0$ by Theorem 2.35 (Week 2). Since $a, b \in I$, d divides a and b; and every common divisor of a and b divides every element of I, in particular d. Hence $d = \gcd(a,b)$. Conversely, if $ax + by = 1$, every common divisor of a and b divides 1.

Fact 0.17 (*Euclid's Lemma*): Let p be a prime. If $p \mid ab$, then $p \mid a$ or $p \mid b$.

Why It Holds: If $p \nmid a$, then $\gcd(a,p) = 1$, since the only positive divisors of p are 1 and p. By Bézout, $ax + py = 1$ for some $x, y \in \mathbb{Z}$, so $b = ab\,x + p\,by$, and p divides both terms on the right.

Example 0.18 (*The Euclidean Algorithm*): The coefficients in Bézout's identity can be computed. Since $\gcd(y,n) = \gcd(n,r)$ whenever $y = qn + r$, repeated division with remainder (Fact 0.14) shrinks the pair until the remainder is 0; the last non-zero remainder is the gcd. For $a = 252$, $b = 198$:

$$252 = 1 \cdot 198 + 54, \qquad 198 = 3 \cdot 54 + 36,$$
$$54 = 1 \cdot 36 + 18, \qquad 36 = 2 \cdot 18 + 0,$$

so $\gcd(252, 198) = 18$. Solving the equations backwards for the remainders,

$$18 = 54 - 36 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198 = 4 \cdot (252 - 198) - 198 = 4 \cdot 252 - 5 \cdot 198.$$

If $\gcd(a,m) = 1$, the same back-substitution gives $ax + my = 1$, so $\bar{x}$ is the inverse of $\bar{a}$ in $\mathbb{Z}/m\mathbb{Z}$ (Theorem 2.71).

Fact 0.19 (*Unique Factorization in $\mathbb{Z}$*): Every integer $n \geq 2$ is a product of primes, and this product is unique up to the order of the factors.

Definition 0.20 (*Congruence Modulo n*): Let $n \in \mathbb{Z}_{>0}$. Integers a, b are *"congruent modulo n"*, written $a \equiv b \pmod{n}$, if $n \mid a - b$.

Fact 0.21 (*Residue Classes*): Congruence modulo n is an equivalence relation on $\mathbb{Z}$, and it is compatible with the operations: $a \equiv a'$ and $b \equiv b'$ imply $a + b \equiv a' + b'$ and $ab \equiv a'b' \pmod{n}$. Its classes are the *"residue classes"* $\bar{a} = a + n\mathbb{Z}$; by division with remainder there are exactly n of them, $\bar{0}, \bar{1}, \ldots, \overline{n-1}$. The lecturer recalls the resulting ring $\mathbb{Z}/n\mathbb{Z}$ in Week 1 (Definition 1.24).

Groups, Fields and the Complex Numbers

Definition 0.22 (*Group*): A *"group"* $(G, \cdot, e)$ is a set G with an associative operation $\cdot$, a neutral element e ($ea = ae = a$ for all a), and an inverse for every element ($ab = ba = e$ for some b). It is *"abelian"* if $ab = ba$ for all a, b; an abelian group is often written additively as $(G, +, 0)$, with inverses $-a$. Examples: $(\mathbb{Z}, +, 0)$ and $(\mathbb{R}^*, \cdot, 1)$. The lecturer restates the definition in Week 2 (Definition 2.1).

Definition 0.23 (*Group Homomorphism*): A map $f\colon G \to H$ between groups is a *"homomorphism"* if $f(ab) = f(a)f(b)$ for all $a, b \in G$, and an *"isomorphism"* if it is moreover bijective (Week 2, Definition 2.10).

Definition 0.24 (*Order of an Element, Cyclic Groups*): Let G be a group with neutral element e and $a \in G$. The *"order"* of a is the least $n \geq 1$ with $a^n = e$ (or ∞ if there is none); in additive notation the condition reads $na = 0$. The group G is *"cyclic"*, with *"generator"* a, if every element of G is a power a^k, $k \in \mathbb{Z}$. A finite group G is cyclic with generator a exactly when a has order $|G|$, and then $G = \{e, a, a^2, \ldots, a^{|G|-1}\}$. Example: $(\mathbb{Z}/n\mathbb{Z}, +, \bar{0})$ is cyclic with generator $\bar{1}$. An isomorphism maps elements of order n to elements of order n; the lectures use this in Week 2 (Proposition 2.15, Proposition 2.73).

Definition 0.25 (*Field*): A *"field"* $(K, +, \cdot, 0, 1)$ is a set with two operations such that $(K, +, 0)$ is an abelian group, $(K \setminus \{0\}, \cdot, 1)$ is an abelian group, and $a(b + c) = ab + ac$ for all a, b, c. In particular $0 \neq 1$. Examples are $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Z}/p\mathbb{Z}$ for a prime p (in the lectures: Proposition 2.63 and Theorem 2.71). A *"subfield"* of a field is a subring (Week 2, Definition 2.26) that is itself a field, such as $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ or the Gaussian rationals $\mathbb{Q}(i) \subset \mathbb{C}$ (Week 3, Example 3.28). The lecturer restates fields in Week 2 (Definition 2.9).

Fact 0.26 (*Complex Conjugation and Modulus*): For $z = a + bi \in \mathbb{C}$ ($a, b \in \mathbb{R}$, $i^2 = -1$), the *"complex conjugate"* is $\bar{z} = a - bi$ and the *"modulus"* is $|z| = \sqrt{a^2 + b^2}$. For all $z, w \in \mathbb{C}$:

$$\overline{z + w} = \bar{z} + \bar{w}, \qquad \overline{zw} = \bar{z}\,\bar{w}, \qquad z\bar{z} = |z|^2, \qquad |zw| = |z|\,|w|,$$

and $\bar{z} = z$ exactly when $z \in \mathbb{R}$. So conjugation respects sums and products, and under multiplication the lengths of complex numbers multiply (and their angles add).

Linear Algebra

Definition 0.27 (*Linear Map, Kernel and Image*): Let V, W be vector spaces over a field K. A map $T\colon V \to W$ is *"linear"* if $T(u+v) = T(u) + T(v)$ and $T(\lambda v) = \lambda T(v)$ for all $u, v \in V$ and $\lambda \in K$. Its *"kernel"* is $\ker(T) = \{v \in V \mid T(v) = 0\}$ and its *"image"* is $\operatorname{im}(T) = T(V)$. The linear maps $V \to W$ form a vector space $\operatorname{Hom}_K(V, W)$ under pointwise operations; the lectures reuse the notation Hom for the morphisms of any category (Definition 2.44).

Fact 0.28 (*Kernel, Image and Rank–Nullity*): For a linear map $T\colon V \to W$, $\ker(T) \subset V$ and $\operatorname{im}(T) \subset W$ are subspaces, and T is injective if and only if $\ker(T) = \{0\}$. If V is finite-dimensional,

$$\dim V = \dim \ker(T) + \dim \operatorname{im}(T).$$

Fact 0.29 (*Span, Linear Independence and Bases*): A family of vectors of V *"spans"* V if every vector of V is a finite linear combination of them, and it is *"linearly independent"* if the only finite linear combination of them that equals 0 is the one with all coefficients 0; a *"basis"* is a linearly independent family that spans V. Every vector space spanned by finitely many vectors has a finite basis, and all its bases have the same number of elements, its *"dimension"*. That every vector space has a basis is proved with Zorn's lemma, which the course takes up itself (Week 1, Question 1.33).

Fact 0.30 (*Matrices*): The $n \times n$ matrices with entries in a field K form a ring $M_n(K)$ under matrix addition and multiplication, with the identity matrix as its 1; for $n \geq 2$ it is not commutative. Its units are the invertible matrices, those with non-zero determinant, and they form the group $\mathrm{GL}_n(K)$. Linear maps $K^n \to K^n$ correspond to matrices and composition to the matrix product, so a linear map between vector spaces of the same finite dimension is an isomorphism exactly when its matrix with respect to bases is invertible. The lectures meet matrix rings only as examples of non-commutative rings (Week 1) and of isomorphisms in $\mathbf{Vect}_K$ (Week 2, Definition 2.50).

Fact 0.31 (*Division with Remainder for Polynomials*): Let K be a field, $f, g \in K[x]$ polynomials with coefficients in K, and $g \neq 0$. There are unique $q, r \in K[x]$ with

$$f = qg + r \quad \text{and} \quad r = 0 \text{ or } \deg(r) < \deg(g).$$

This is long division of polynomials: as long as $\deg(f) \geq \deg(g)$, subtracting $\frac{a}{b}x^{\deg f - \deg g}\, g$, where a and b are the leading coefficients of f and g, lowers the degree of f; dividing by b is where K has to be a field. It is the polynomial analogue of Fact 0.14; the lectures use it for $g = x^2 + 1$ (Week 4, Lemma 4.41).

Definition 0.32 (*Quotient Vector Space*): Let $U \subset V$ be a subspace. The relation $v \equiv w \iff v - w \in U$ is an equivalence relation on V, with classes the cosets $v + U$. The *"quotient space"* V/U is the set of these cosets with

$$(v + U) + (w + U) = (v + w) + U, \qquad \lambda(v + U) = \lambda v + U;$$

these operations are well-defined, V/U is a vector space, and the projection $\pi\colon V \to V/U$, $v \mapsto v + U$, is linear and surjective with $\ker(\pi) = U$. If V is finite-dimensional, $\dim V/U = \dim V - \dim U$.

Fact 0.33 (*First Isomorphism Theorem for Linear Maps*): For a linear map $T\colon V \to W$, the map $\bar{T}\colon V/\ker(T) \to \operatorname{im}(T)$, $v + \ker(T) \mapsto T(v)$, is a well-defined isomorphism of vector spaces. With $\dim V/\ker(T) = \dim V - \dim \ker(T)$ this gives rank–nullity again (Fact 0.28).

The Model for Quotient Rings: The quotient ring R/I of Week 4 follows this pattern step by step, with the ideal I in the role of the subspace U: congruence modulo I (Definition 4.17), the set of classes (Definition 4.19), well-defined operations (Proposition 4.22, Proposition 4.24), and the First Isomorphism Theorem $R/\ker(f) \cong \operatorname{im}(f)$ (Theorem 4.27). The new ingredient is multiplication: a subspace only has to absorb scalars, an ideal has to absorb all of R, and that is exactly what makes the product of classes well-defined.

Definition 0.34 (*Dual Space and Dual Map*): The *"dual space"* of a vector space V over K is $V^* = \operatorname{Hom}_K(V, K)$, the vector space of linear maps $V \to K$. A linear map $T\colon V \to W$ has the *"dual map"* $T^*\colon W^* \to V^*$, $T^*(\varphi) = \varphi \circ T$; it satisfies $(S \circ T)^* = T^* \circ S^*$ and $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

Fact 0.35 (*The Double Dual*): For every vector space V, the map $\mathrm{ev}\colon V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is linear and injective. If V is finite-dimensional, then $\dim V^* = \dim V$ and ev is an isomorphism. (Used in Week 2, Example 2.53.)

Analysis

Fact 0.36 (*The Real Numbers*): $\mathbb{R}$ is a complete ordered field: every non-empty subset that is bounded above has a supremum, and every Cauchy sequence converges. It can be constructed from $\mathbb{Q}$ by Cauchy sequences or by Dedekind cuts (Week 3, Wednesday).

Fact 0.37 (*Exponential Function and Logarithm*): The exponential function $\exp\colon \mathbb{R} \to \mathbb{R}_{>0}$ is a continuous, strictly increasing bijection with $\exp(0) = 1$ and $\exp(x+y) = \exp(x)\exp(y)$; its inverse is the natural logarithm $\ln\colon \mathbb{R}_{>0} \to \mathbb{R}$ (Week 2, Proposition 2.16).

Definition 0.38 (*Open Sets and Closure in $\mathbb{R}$*): A subset $U \subset \mathbb{R}$ is *"open"* if around each of its points it contains a whole interval $(x-\varepsilon, x+\varepsilon)$, $\varepsilon > 0$; examples are open intervals and $\mathbb{R}_{>0}$. A subset is *"closed"* if its complement is open. The *"closure"* $\overline{A}$ of $A \subset \mathbb{R}$ is the smallest closed set containing A: the set of all limits of convergent sequences in A. The lectures use these notions for the support of a function and for restricting functions to an open set (Week 2, Definition 2.39, Definition 2.42).

Fact 0.39 (*Compact Sets and Smooth Functions*): A subset of $\mathbb{R}$ is *"compact"* if and only if it is closed and bounded (Heine–Borel); finite unions of compact sets, and closed subsets of a compact set, are compact. $C^\infty(\mathbb{R})$ denotes the infinitely differentiable functions $\mathbb{R} \to \mathbb{R}$; sums and products of such functions are again infinitely differentiable (Week 2, Definition 2.38).

Fact 0.40 (*Limits and Continuity*): A function $f\colon U \to \mathbb{R}$ on an open set $U \subset \mathbb{R}$ is *"continuous"* at $a \in U$ if $\lim_{x \to a} f(x) = f(a)$, and continuous if this holds at every point of U. Sums, products and compositions of continuous functions are continuous, and every differentiable function is continuous, so the functions in $C^\infty(\mathbb{R})$ are continuous. If $g\colon \mathbb{R} \to \mathbb{R}$ is continuous, the one-sided limit $\lim_{x \to 0^+} g(x)$ exists and equals $g(0)$. So a function on $\mathbb{R}_{>0}$ with no limit at 0^+, such as $x \mapsto \sin(1/x)$, is not the restriction of a continuous function on $\mathbb{R}$ (Week 2, Proposition 2.48).

Fact 0.41 (*Smooth Functions Vanishing on a Half-Line*): The function

$$h(x) = \begin{cases} e^{-1/x} & x > 0, \\ 0 & x \le 0 \end{cases}$$

lies in $C^\infty(\mathbb{R})$: all its derivatives tend to 0 as $x \to 0^+$, because $e^{-1/x}$ goes to 0 faster than any power of x. So h vanishes exactly on $(-\infty, 0]$; the function $x \mapsto h(-x)$ vanishes exactly on $[0, \infty)$, and $h(x-a)\,h(b-x)$ is a smooth function that is non-zero exactly on (a, b) (a *"bump*

function"). Such functions show that the ideals of $C^\infty(\mathbb{R})$ in Week 2 are not zero and differ from each other (Proposition 2.40, Proposition 2.41).

Fact 0.42 (*Geometric Series and Cauchy Product*): For $z \in \mathbb{C}$ with $|z| < 1$,

$$\sum_{k=0}^{\infty} z^k = \frac{1}{1-z}.$$

If $\sum_n a_n$ and $\sum_n b_n$ converge absolutely, then so does their *"Cauchy product"* $\sum_n c_n$ with $c_n = \sum_{i=0}^{n} a_i b_{n-i}$, and $\sum_n c_n = \left(\sum_n a_n\right)\left(\sum_n b_n\right)$. The lectures use the same coefficient formula to multiply polynomials and formal power series, where no convergence is involved (Definition 3.36, Definition 3.41).

Wednesday 16 September 2026

Gaussian Integers, Ring Axioms, Ring Homomorphisms, and Ideals

Course Content Overview:
Gaussian Integers, Ring Axioms, Homomorphisms & Ideals

In this lecture, we bridge arithmetic and abstract algebra by exploring the following core themes:

- **Gaussian Integers and Primes:** Comparing factorization and units in $\mathbb{Z}$ and $\mathbb{Z}[i]$, and establishing how the primality of an ordinary integer prime in $\mathbb{Z}[i]$ connects directly to Fermat's two-square theorem (an odd prime p is a sum of two squares $a^2 + b^2$ if and only if $p \equiv 1 \pmod 4$).
- **Axioms and Elementary Properties of Rings:** Formalizing the ring axioms $(R, +, \cdot, 0, 1)$, deriving fundamental arithmetic identities such as $0 \cdot a = 0$ and $(-1) \cdot a = -a$, analyzing the degenerate zero ring ($1 = 0$), and defining scalar multiples by integers.
- **Ring Homomorphisms:** Defining structure-preserving maps between rings, explaining why identity preservation ($f(1) = 1$) must be explicitly required while $f(0) = 0$ is automatic, and exploring canonical examples including standard subring inclusions, modular reduction $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, and the initial evaluation map $\mathbb{Z} \to R$.
- **Ideals and Field Structure:** Introducing the kernel $\ker(f)$, defining ideals $I \subset R$ via additive closure and multiplicative absorption, proving that every kernel is an ideal, and classifying the ideals of a field (establishing that fields possess only $\{0\}$ and F).
- **Outlook:** Connecting the existence of bases in infinite-dimensional vector spaces to Zorn's lemma and the Axiom of Choice.

Gaussian Integers and Primes

Units and Primes in $\mathbb{Z}$ and $\mathbb{Z}[i]$

Spoken Transcript [00:00:00 - 00:00:53]

Okay, so we'll try to start. Maybe you could sit down.

On the practical aspects, there was some progress: the website is more or less correct

now, I think, so you can look for the information on the website.

The second issue is that people asked, and now the lectures will be recorded. You can see yourself — or maybe you can only see me, but anyway — the lectures will be recorded, so you can find them later on some website somewhere.

At the beginning, I said that the website is more or less in reasonable form; in particular, the assignment for September 25th is there. Some of the technical parts we're still working on, so we'll get back to you on that, but...

Administrative Opening

The lecturer makes brief administrative announcements regarding the course website, the availability of the first homework assignment due on September 25th, and lecture video recordings.

Spoken Transcript [00:00:53 - 00:02:48]

If you spent last night thinking about the Gaussian integers *[writes $\mathbb{Z}[i]$ on the blackboard]*, then... well, maybe for the people who didn't spend the whole night thinking about the Gaussian integers, I'll say a few things here.

As I said in the previous lecture, we discussed $\mathbb{Z}$ and its factorization properties. And I said you can discuss the same thing for the Gaussian integers. It will be part of the theory of the course, but it's nice to think about the example.

The first thing is: what are... for $\mathbb{Z}$ *[writes $\mathbb{Z}$]*, there were the units, that's ± 1 *[writes $\{\pm 1\}$]*. And those confused our multiplication always, because you could always put in as many -1's and $+1$'s as you want.

So, what's the analogue here? What are the units, what are the invertible elements in here? Well, of course, there's still ± 1, but there's also $\pm i$ *[writes $\{\pm 1, \pm i\}$]*. These are also invertible, because $i^2 = -1$, and then you can multiply it by 1 (i.e., $(-i) \cdot i = 1$).

Is that clear that these four objects play in some sense the role of the two there? Is that clear?

So, these are the units. It means if you write down any multiplication, like any unique factorization into primes, before you had to worry about ± 1; now you have to worry about more, okay?

In particular, the notion of prime here was that p is prime *[writes p prime]* if the factors are ± 1 and $\pm p$ *[writes $\{\pm 1, \pm p\}$]*. That's on the $\mathbb{Z}$ side. Here, if we have some element — maybe I call it q is prime *[writes q prime]* — then we can consider factorizations here; but now there's a lot more: there's four choices, there could be any one of these. ± 1 and $\pm i$ are factors, and also $\pm q$ and $\pm iq$ *[writes $\{\pm 1, \pm q, \pm i, \pm iq\}$ in two rows]*. These are all factors. That's the analogous statement.

This is all some kind of nonsense. It's just saying that, you know, this is more confusion with these extra i's.

Units and Primes in the Gaussian Integers

Definition 1.1 (*Gaussian Integers*): The ring of *"Gaussian integers"*, denoted by $\mathbb{Z}[i]$, is the subring of the complex numbers $\mathbb{C}$ given by:

$$\mathbb{Z}[i] := \{a + bi \in \mathbb{C} \mid a, b \in \mathbb{Z}\}, \quad \text{where } i^2 = -1.$$

Proposition 1.2 (*Units in $\mathbb{Z}$ and $\mathbb{Z}[i]$*): An element u in a ring is called a *"unit"* if it possesses a multiplicative inverse in that ring.

- **Units in $\mathbb{Z}$:** The group of units in $\mathbb{Z}$ consists of exactly two elements:
$$\mathbb{Z}^\times = \{1, -1\} = \{\pm 1\}.$$
- **Units in $\mathbb{Z}[i]$:** The group of units in $\mathbb{Z}[i]$ consists of exactly four elements:
$$\mathbb{Z}[i]^\times = \{1, -1, i, -i\} = \{\pm 1, \pm i\}.$$

Definition 1.3 (*Prime Elements and Divisors*): In an integral domain (Definition 2.62, Week 2), a non-zero element that is not a unit is called prime (irreducible) if its only divisors are units or units multiplied by the element itself:

- **Primes in $\mathbb{Z}$:** An integer $p \in \mathbb{Z} \setminus \{0, \pm 1\}$ is prime if its only factors are:
$$\pm 1 \quad \text{and} \quad \pm p.$$
- **Primes in $\mathbb{Z}[i]$:** An element $q \in \mathbb{Z}[i] \setminus \{0, \pm 1, \pm i\}$ is prime if its only factors are the four units and their associated multiples:
$$\pm 1, \quad \pm i, \quad \pm q, \quad \pm iq.$$

Algebraic Intuition: The Concept of Associates: In a unique factorization domain (such as $\mathbb{Z}$, by Fact 0.19, or $\mathbb{Z}[i]$, as the course shows later), factorizations into irreducible elements are unique up to multiplication by invertible elements (*"units"*) and order. Two elements $x, y \in R$ are called *"associates"* if $x = u \cdot y$ for some unit $u \in R^\times$.

- **Associates in $\mathbb{Z}$:** The only units are ± 1, meaning every prime p has two associates: p and $-p$.
- **Associates in $\mathbb{Z}[i]$:** Since $i \cdot (-i) = 1$ (i.e., $(-i) \cdot i = 1$), the imaginary units i and $-i$ are also units. Consequently, every Gaussian integer q has four associates:
$$q, \quad -q, \quad iq, \quad -iq.$$

Verification of Invertibility via the Algebraic Norm: The elements $\pm 1, \pm i$ are clearly invertible in $\mathbb{Z}[i]$ since $(\pm 1)^2 = 1$ and $i \cdot (-i) = -i^2 = 1$.

To confirm that these are the **only** units in $\mathbb{Z}[i]$, we use the multiplicative field norm $N \colon \mathbb{Z}[i] \to \mathbb{Z}_{\geq 0}$ defined by:

$$N(a + bi) := (a + bi)\overline{(a + bi)} = a^2 + b^2.$$

If $\alpha \in \mathbb{Z}[i]$ is a unit, there exists $\beta \in \mathbb{Z}[i]$ such that $\alpha\beta = 1$. Taking the norm on both sides gives:

$$N(\alpha)N(\beta) = N(\alpha\beta) = N(1) = 1.$$

Since $N(\alpha), N(\beta) \in \mathbb{Z}_{\geq 0}$, we must have $N(\alpha) = a^2 + b^2 = 1$. The only integer solutions to $a^2 + b^2 = 1$ are $(a, b) \in \{(\pm 1, 0), (0, \pm 1)\}$, corresponding precisely to ± 1 and $\pm i$. The lecturer writes out this argument in Week 3, there with the complex modulus $|a + bi| = \sqrt{N(a + bi)}$ (Proposition 3.2).

When Does a Rational Prime Remain Prime in $\mathbb{Z}[i]$?

Spoken Transcript [00:02:48 - 00:04:08]

So, let's do something that's not nonsense. A math question you can ask, which is a very nice question, is: if I take a prime, an actual prime in the integers *[writes $p \in \mathbb{Z}$]*, and I view that prime, that p, as an element of the Gaussian integers *[writes and view that p as an element of $\mathbb{Z}[i]$]*, which it is — because remember, the Gaussian integers are the set of integers plus i times an integer *[writes $\mathbb{Z}[i] = \{n + im \mid n, m \in \mathbb{Z}\}$]*... So, the integers naturally sit inside the Gaussian integers *[writes $\mathbb{Z} \subset \mathbb{Z}[i]$]* by just an integer going here. Is that clear?

If I take a prime, an honest prime in the integers — so this would be, you know, like $2, 3, 5, 7, 11, \ldots$ *[writes $p = 2, 3, 5, 7, 11, \ldots$]* and so on — take one of them, and I consider the Gaussian integers: does it stay prime? Or maybe say: When does p stay prime? *[writes When does p stay prime?]*

Meaning: it's born prime in the integers; when does it stay prime when I consider it as a Gaussian integer? That's a real question. And we had an example in the class last time.

Embedding of $\mathbb{Z}$ and the Prime Divisibility Question

The ring of integers embeds naturally into the Gaussian integers via the inclusion map:

$$\mathbb{Z} \hookrightarrow \mathbb{Z}[i], \quad n \mapsto n + 0i.$$

Question 1.4 (*When Does an Integer Prime Stay Prime in $\mathbb{Z}[i]$?*): Let $p \in \mathbb{Z}$ be an integer prime ($p \in \{2, 3, 5, 7, 11, 13, \ldots\}$). When viewed as an element of $\mathbb{Z}[i]$:

When does p remain prime (irreducible) in $\mathbb{Z}[i]$?

Why Primes Can Factor in Larger Rings: An element that has no non-trivial divisors in $\mathbb{Z}$ might acquire non-trivial divisors in $\mathbb{Z}[i]$ because $\mathbb{Z}[i]$ contains new elements with non-zero imaginary parts. For example, 5 cannot be factored in $\mathbb{Z}$, but in $\mathbb{Z}[i]$ it factors as $5 = (2 + i)(2 - i)$, where neither factor is a unit.

To investigate whether p factors non-trivially in $\mathbb{Z}[i]$, suppose there exists a non-unit divisor:

$$(a + bi) \mid p \quad \text{in } \mathbb{Z}[i] \quad (a, b \in \mathbb{Z}). \tag{1.1}$$

Complex Conjugation and the Sum of Two Squares

Spoken Transcript [00:04:08 - 00:05:18]

But now we try to think about it mathematically. We have p, and we have to consider whether it's divisible in the Gaussian integers. So, we think, okay, maybe it's divisible. How do I do that? Well, maybe that means $a + bi$ divides p *[writes $(a + bi) \mid p$]* in the Gaussian integers. Let's try to think what that means. Now, if $a + bi$ divides p, that means $a + bi$ times something equals p — that's the definition of divisibility. If that

happened, I could just complex conjugate the whole thing. So, if this happens — you have to think about this — then $a - bi$ also divides p *[writes ⇓ and below it $(a - bi) \mid p$]*.

Do you believe that statement? This has to do with complex conjugation, and it depends on what your feeling is for the complex numbers. If you don't understand the statement, you should think about it, because what does this mean? This means p is equal to something something something times $a + bi$ *[writes $p = \ldots (a + bi)$]*. That's what divisibility means. And I just take the complex conjugate of everything. But the complex conjugate of p is p, and all the something something something gets complex con— I don't care, but at the end, this one becomes that *[points to $a - bi$ and writes $p = \ldots (a - bi)$]*. So, that means that that also happens, right?

Complex Conjugation of Divisors

Proposition 1.5 (*Conjugate Divisor Property*): Let $p \in \mathbb{Z}$. If $(a + bi) \mid p$ in $\mathbb{Z}[i]$, then:

$$(a - bi) \mid p \quad \text{in } \mathbb{Z}[i]. \tag{1.2}$$

Proof

By definition of divisibility in $\mathbb{Z}[i]$, $(a + bi) \mid p$ means there exists an element $z \in \mathbb{Z}[i]$ such that:

$$p = z \cdot (a + bi).$$

Applying the complex conjugation automorphism (Fact 0.26) to both sides yields:

$$\overline{p} = \overline{z \cdot (a + bi)} = \overline{z} \cdot \overline{(a + bi)}.$$

Since $p \in \mathbb{Z}$ is real, $\overline{p} = p$. Furthermore, $\overline{a + bi} = a - bi$, and $\overline{z} \in \mathbb{Z}[i]$. Thus,

$$p = \overline{z} \cdot (a - bi),$$

which proves that $(a - bi) \mid p$ in $\mathbb{Z}[i]$. Q.E.D.

Spoken Transcript [00:05:18 - 00:06:08]

Then you might think — and this is the kind of thing we will make precise in this class — that actually then both of them divide *[draws a curved arrow and writes $(a + bi)(a - bi) \mid p$]*. That already requires some statement about unique factorization. And this is now a nice integer: $a^2 + b^2$ divides p *[writes ⇓ and $a^2 + b^2 \mid p$]*. If it's an integer, then p has to be of the form $a^2 + b^2$ *[writes $p = a^2 + b^2$]*. So, it has to be a prime that's a sum of two squares.

This argument will show that if you want p to stay prime... so, if p factors, then it's a sum of two squares. And this now connects to this world of trying to express primes as a sum of two squares, and there's facts about this.

Reduction to the Sum of Two Squares

Multiplying the two conjugate factors together yields an ordinary integer:

$$(a + bi)(a - bi) = a^2 + b^2 \in \mathbb{Z}.$$

Using the field norm $N(x+yi) = x^2+y^2 = (x+yi)(x-yi)$, which is multiplicative, the equation $p = (a+bi)z$ implies:

$$N(p) = p^2 = N(a+bi) \cdot N(z) = (a^2+b^2) \cdot N(z).$$

Since p is a prime in $\mathbb{Z}$, the integer a^2+b^2 dividing p^2 is 1, p or p^2. But $a^2+b^2 = 1$ would make $a+bi$ a unit (as $(a+bi)(a-bi) = 1$), and $a^2+b^2 = p^2$ would force $N(z) = 1$, making z a unit and $a+bi = pz^{-1}$ an associate of p. So if $a+bi$ is neither a unit nor an associate of p, then:

$$a^2+b^2 = p. \tag{1.3}$$

Rigorous Derivation via the Multiplicative Norm: The blackboard derivation outlines the necessity of p being a sum of two squares. Rigorously, this follows from the multiplicative norm:

(1) **Norm Equation:** If $p = z \cdot (a+bi)$ with $z = c+di$, taking norms yields:
$$p^2 = N(p) = N(z)N(a+bi) = (c^2+d^2)(a^2+b^2).$$
(2) **Non-Trivial Factors:** Because $a+bi$ is not a unit, $N(a+bi) = a^2+b^2 > 1$. Because z is not a unit, $N(z) = c^2+d^2 > 1$.
(3) **Prime Factorization in $\mathbb{Z}$:** Since p is prime in $\mathbb{Z}$, the only divisors of p^2 in $\mathbb{Z}_{>0}$ are $1, p, p^2$. Thus, we must have:
$$a^2+b^2 = p \quad \text{and} \quad c^2+d^2 = p.$$

Connection Between Gaussian Primes and Sums of Squares: This algebraic deduction establishes a profound bridge:

- If an integer prime p factors non-trivially in $\mathbb{Z}[i]$ as $p = (a+bi)(a-bi)$, then p must be expressible as the sum of two integer squares: $p = a^2+b^2$.
- By contrapositive, if p cannot be expressed as the sum of two squares, it cannot factor non-trivially into non-unit Gaussian integers, and hence p must remain prime in $\mathbb{Z}[i]$.
- Conversely, $p = a^2+b^2$ gives the factorization $p = (a+bi)(a-bi)$ with $N(a \pm bi) = p > 1$, so neither factor is a unit. Hence, p stays prime in $\mathbb{Z}[i]$ **if and only if** p is not a sum of two squares.

Intermediate Steps Added by Transcriber

The lecturer argues that both $a+bi$ and $a-bi$ divide p, so their product a^2+b^2 divides p, and remarks that this step *"already requires some statement about unique factorization"* (Fact 0.19, or Euclid's lemma, Fact 0.17). The norm computation above reaches (1.3) without that statement: it only uses $N(zw) = N(z)\,N(w)$ and the primality of p in $\mathbb{Z}$.

Fermat's Theorem on Sums of Two Squares

Spoken Transcript [00:06:08 - 00:08:12]

The simple fact is that if p is congruent to $3 \pmod 4$ *[writes If $p \equiv 3 \pmod 4$]*, this is impossible *[writes this is impossible]*. You can never write a prime that is $3 \pmod 4$ as

a sum of two squares. That's easy to prove; I think it's part of the exercise, I'm not sure.

But the one that's tricky is: if p is congruent to $1 \pmod 4$ *[writes If $p \equiv 1 \pmod 4$]*, then you can. You can solve *[writes then you can solve]* $p = a^2 + b^2$ *[writes $p = a^2 + b^2$]* in the integers. This is a theorem that's due to Fermat *[writes (Fermat)]*. And this is one of the problems on the problem set.

So, the answer to this question finally is that if you have a prime, and you want to know whether it stays prime in the Gaussian integers, the whole thing that matters is whether p is equal to 1 or $3 \pmod 4$. It's kind of an interesting question.

And as I said, we will return to these matters in the class, but it already shows us that the very naive idea of thinking about how things factor leads to interesting mathematics, and in this case it goes in a number-theoretic direction. That's this question about whether a prime can be expressed as a sum of two squares — a really basic question.

We had an example yesterday, I remind you; the answer yesterday had to do with $5 = 2^2 + 1^2$ *[writes $5 = 2^2 + 1^2$]*. So, 5 is a prime in the integers, 5 is $1 \pmod 4$, and you can express it as a sum of two squares. The thing that's tricky on the board here is this statement: that if p is equal to $1 \pmod 4$, then you can solve this equation, and that is the exercise on the homework, on the first homework set. There's a way to go through that — a very clever way to go through that, not the way that Fermat used.

Fermat's Two-Square Theorem and Gaussian Primes

Theorem 1.6 (*Fermat's Two-Square Theorem*): Let p be an odd integer prime ($p > 2$).

(i) If $p \equiv 3 \pmod 4$, then p cannot be written as the sum of two squares ($p \neq a^2 + b^2$). Consequently, p remains prime in $\mathbb{Z}[i]$.

(ii) If $p \equiv 1 \pmod 4$, then p can be written uniquely (up to sign and order) as the sum of two squares:

$$p = a^2 + b^2 = (a + bi)(a - bi) \quad (a, b \in \mathbb{Z}).$$

Consequently, p factors in $\mathbb{Z}[i]$ and does not remain prime.

Example 1.7 (*Decomposition of 5 vs. Primality of 3 in $\mathbb{Z}[i]$*):

- **The prime 5:** Since $5 \equiv 1 \pmod 4$, we have:
$$5 = 2^2 + 1^2 = (2 + i)(2 - i).$$
Thus, 5 is composite in $\mathbb{Z}[i]$.
- **The prime 3:** Since $3 \equiv 3 \pmod 4$, and squares modulo 4 can only be 0 or 1 ($0^2 \equiv 0$, $1^2 \equiv 1$, $2^2 \equiv 0$, $3^2 \equiv 1$), the sum of two squares can only be $0, 1$, or $2 \pmod 4$. Hence, $a^2 + b^2 \equiv 3 \pmod 4$ has no integer solutions, and 3 remains prime in $\mathbb{Z}[i]$.

Classification of Integer Primes in $\mathbb{Z}[i]$: Every integer prime $p \in \mathbb{Z}$ falls into one of three categories in $\mathbb{Z}[i]$:

- **Ramified Prime ($p = 2$):** $2 = 1^2 + 1^2 = (1 + i)(1 - i) = -i(1 + i)^2$.
- **Split Primes ($p \equiv 1 \pmod 4$):** $p = a^2 + b^2 = (a + bi)(a - bi)$ splits into two distinct non-associate Gaussian primes (e.g., $5, 13, 17, 29$).

- **Inert Primes ($p \equiv 3 \pmod 4$):** p cannot be written as $a^2 + b^2$, so p remains prime in $\mathbb{Z}[i]$ (e.g., $3, 7, 11, 19, 23$).

Beyond the Lecture: The Prime 2

The even prime is not covered by Theorem 1.6: $2 = 1^2 + 1^2 = (1+i)(1-i)$, and since $1 - i = -i(1+i)$, in fact $2 = -i\,(1+i)^2$, so 2 does not stay prime in $\mathbb{Z}[i]$.

Board Transition

The lecturer erases the center and right chalkboards with a sponge to prepare for reviewing the abstract definition and axioms of a ring.

The Axioms and Basic Properties of Rings

Definition of a Ring

Spoken Transcript [00:08:15 - 00:09:45]

All right. So, then the main direction of the lecture last time was about rings, which we revisit now.

So, we go back to rings and the foundational material. I had defined a ring: R with multiplication, 0, and 1 *[writes $(R, +, \cdot, 0, 1)$]*. This is a ring *[writes is a ring]*. And the axioms, if we rewrite them now differently: there's the axioms for $+$, and this axiom for $+$, we can say just that $(R, +, 0)$ is an abelian group *[writes $+ \implies (R, +, 0)$ abelian group]*. This is just the same axioms written differently; those are just the group axioms. Is that clear?

Then there's the axiom for multiplication *[writes $\cdot \implies$]*, and these were — now I don't have a more efficient way to say this — it's associative *[writes associative]*, and then there's the axiom of the unit *[writes $1 \cdot a = a \cdot 1 = 1$, a slip for $= a$]*. And then there's the axioms together *[writes $+, \cdot \implies$]*, and these are two times the distributive law *[writes $\times 2$ Dist]*, which I don't write, okay?

This is just the definition which we've seen and you've seen before. And the only thing I've done now is abbreviated the $+$ as saying that if I look at the additive structure, it's an abelian group, because that's what the axioms were.

Axiomatic Definition of a Ring

Definition 1.8 (*Ring*): A *"ring"* is a tuple $(R, +, \cdot, 0, 1)$ consisting of a set R, two binary operations $+$ (addition) and $\cdot$ (multiplication), and two distinguished elements $0, 1 \in R$ satisfying the following axioms:

(1) Additive Abelian Group Axioms: $(R, +, 0)$ is an abelian group:

- **Associativity:** $\forall a, b, c \in R$: $(a + b) + c = a + (b + c)$.

- **Identity:** $\forall a \in R\colon a + 0 = 0 + a = a$.
- **Inverse:** $\forall a \in R\ \exists(-a) \in R\colon a + (-a) = (-a) + a = 0$.
- **Commutativity:** $\forall a, b \in R\colon a + b = b + a$.

(2) Multiplicative Monoid Axioms: Multiplication is associative and possesses a unit element 1:

- **Associativity:** $\forall a, b, c \in R\colon (a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- **Identity (Unit):** $\forall a \in R\colon 1 \cdot a = a \cdot 1 = a$.

(3) Distributivity Axioms: Multiplication distributes over addition from both the left and the right:

- **Left Distributivity:** $\forall a, b, c \in R\colon a \cdot (b + c) = (a \cdot b) + (a \cdot c)$.
- **Right Distributivity:** $\forall a, b, c \in R\colon (a + b) \cdot c = (a \cdot c) + (b \cdot c)$.

Structural Synthesis of the Ring Axioms: Summarizing the additive structure compactly as an abelian group $(R, +, 0)$ packages four standard axioms (associativity, commutativity, neutral element, and inverses) into a single standard algebraic concept. The distributive laws act as the essential bridge coupling the additive abelian group structure with the multiplicative monoid structure.

Two words that are easy to mix up: **abelian** refers to the addition ($a + b = b + a$ is part of the axioms), whereas **commutative ring** refers to the multiplication. The definition above does not require $a \cdot b = b \cdot a$; the standing convention of this course adds it (Remark 1.17). Nor does it require $1 \neq 0$: the case $1 = 0$ is allowed and treated in Proposition 1.10.

Elementary Property: $0 \cdot a = 0$

Spoken Transcript [00:09:45 - 00:11:32]

So, then there's a bunch of things to say at the beginning: some basic properties *[writes Basic Properties]*. And these are all in the books, but in these axioms, there are lots of things that you're familiar with from the integers and arithmetic that are not in the axioms. Like, one question is: what happens if I take $0 \times a$ *[writes $0 \cdot a$]*? Because 0 is an object that lives on the additive side, and I'm multiplying it, so it's on the multiplicative side. You can wonder what this is, and you can imagine what the answer is going to be.

To prove it — if we're going to formally prove it — you, of course, have to use the distributive law. The distributive law is the only thing that relates the addition to the multiplication. So, we're going to try to prove this, and to start, we know we have to use the distributive law somewhere. One way to start is we can write this as $(0 + 0) \times a$ *[writes $= (0 + 0) \cdot a$]*. We have not used the distributive law yet; this uses the fact that $0 + 0 = 0$, which is the neutral part of the group law. And now we use the distributive law *[writes $= (0 \cdot a) + (0 \cdot a)$]*: $(0 \cdot a) + (0 \cdot a)$.

Okay. And now we're happy, because we have this equation, and then we go back to using the additive properties, which is the left cancellation — the additive inverse, okay? So, now we can cancel *[cancels $0 \cdot a$ on both sides with chalk]*, and we happily see that $0 = 0 \times a$ *[writes $0 = 0 \cdot a$ and draws a chalk box around it]*, which is, of course, the basic law of arithmetic. It's not an axiom, but it follows immediately from the axioms,

which is why it's not an axiom. You could actually just put it as an axiom, it wouldn't hurt anybody, but... Okay.

Derivation of Multiplication by Zero

Proposition 1.9 (*Multiplication by Zero*): In any ring R, for every element $a \in R$:

$$0 \cdot a = 0 \quad \text{and} \quad a \cdot 0 = 0. \tag{1.4}$$

Proof

Because 0 is the additive identity of $(R, +, 0)$, we have $0 + 0 = 0$. Substituting this into $0 \cdot a$:

$$\begin{aligned} 0 \cdot a &= (0 + 0) \cdot a \\ &= (0 \cdot a) + (0 \cdot a) \qquad \text{(by right distributivity).} \end{aligned} \tag{1.5}$$

Because $(R, +, 0)$ is an abelian group, the element $0 \cdot a \in R$ possesses a unique additive inverse $-(0 \cdot a) \in R$. Adding $-(0 \cdot a)$ to both sides of (1.5):

$$\begin{aligned} -(0 \cdot a) + (0 \cdot a) &= -(0 \cdot a) + \big((0 \cdot a) + (0 \cdot a)\big) \\ 0 &= \big(-(0 \cdot a) + (0 \cdot a)\big) + (0 \cdot a) \qquad \text{(by associativity)} \\ 0 &= 0 + (0 \cdot a) \\ 0 &= 0 \cdot a. \end{aligned}$$

By the exact same argument using left distributivity, $a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$, which upon adding $-(a \cdot 0)$ yields $a \cdot 0 = 0$. Q.E.D.

Why Distributivity is Indispensable: The zero element 0 is defined exclusively through addition ($a + 0 = a$). To deduce its behavior under multiplication ($0 \cdot a = 0$), one must invoke the distributive law, as it provides the only axiomatic connection linking addition to multiplication.

The Zero Ring ($1 = 0$)

Spoken Transcript [00:11:32 - 00:12:43]

In particular, if $1 = 0$ *[writes If $1 = 0$, then]* — because I said for a ring we allow this extremely unpleasant boundary case where 1 is equal to 0, the degenerate case... So, if that happens — if we have a ring where $1 = 0$ — then we have a, for any a, is equal to $a \times 1$ *[writes $a = a \cdot 1$]*, which is the multiplication axiom. And then, if we use $1 = 0$, this is equal to $a \times 0$ *[writes $= a \cdot 0$]* using $1 = 0$, and then this is 0 *[writes $= 0$]*.

So, it's a rather degenerate case: if you have a ring where $1 = 0$, then all elements are equal to 0. It's not a super exciting ring; there's only one element. If I say this formally: if you have $0 = 1$, this implies your ring as a set consists of just one element *[writes If $0 = 1 \implies R = \{0\}$]*. Anyway, it's a degenerate case, it's unpleasant to think about, but we carry it around because it fits well in certain statements of theorems.

The Degenerate Zero Ring

Proposition 1.10 (*Characterization of the Zero Ring*): Let R be a ring. If $1 = 0$ in R, then R is the trivial ring containing only a single element:

$$R = \{0\}.$$

Proof

Assume $1 = 0$. Let $a \in R$ be an arbitrary element. Using the unit axiom ($a = a \cdot 1$), the hypothesis $1 = 0$, and Proposition 1.9:

$$a = a \cdot 1 = a \cdot 0 = 0.$$

Because every element $a \in R$ must equal 0, the set R contains only the zero element:

$$R = \{0\}.$$

Q.E.D.

Remark 1.11 (*The Zero Ring in Abstract Algebra*): The trivial ring $R = \{0\}$ is often called the *"zero ring"*. While fields strictly require $1 \neq 0$, the general definition of a ring intentionally admits $1 = 0$ so that the category of rings has a terminal object (the zero ring: every ring has exactly one homomorphism to it) and quotient rings R/I remain rings even when $I = R$. Definitions that must exclude it say so explicitly, as for integral domains (Definition 2.62, Week 2).

Additive Inverses and the Element $-a$

Spoken Transcript [00:12:43 - 00:15:05]

Other things that are kind of convention: if you take -1... 1 is in the ring because it's part of the data *[writes $1 \in R$]*. Because of the additive structure, the group axioms say that there exists an additive inverse. So, by convention, -1 is by definition the additive inverse of 1 *[writes -1 by def the add. inv. of 1]*, and you can prove that it's unique. Maybe I leave that as an exercise. Given the ring, there's 1, and there's the additive inverse. Additive inverses are unique, you can prove that. So, that's -1.

And then $-a$, for any element, is defined to be $(-1) \times a$ *[writes $-a := (-1) \cdot a$]*; this is not a surprising definition.

And then the little lemma — maybe it doesn't even get to be called a lemma — is that $-a$ is the additive inverse of a *[writes Lemma: $-a$ is the additive inverse of a.]*. That's very convenient; it would be really unpleasant if that weren't the case.

The proof of this is: we should start with $0 = (1 + (-1)) \cdot a$ *[writes Proof: $0 = (1 + (-1)) \cdot a$]*... that's 0. And we've already proven that that's that (i.e., $0 \cdot a = 0$, Proposition 1.9), okay? I've used two steps here: here on this side, this thing is 0, and we've already proved $0 \times a = 0$, so... okay? And then I use the distributive law and distribute this side as a, and on this side as plus $(-1) \cdot a$ *[writes $= a + (-1) \cdot a$]*, and that exhibits this $(-1) \times a$ as the additive inverse of a.

So, there's a number of things — I think we spent enough time on this — but the way the axioms work, they work so that it fits with your intuition about how multiplication works.

Are you happy with this? You can also look in the book; there'll be a couple of pages of playing around with these. It's not deep, but it's good to be familiar with it.

Additive Inverses and Multiplication by -1

Definition 1.12 (*The Element -1 and the Notation $-a$*)**:** Let R be a ring.

(i) We denote by $-1 \in R$ the unique additive inverse (Proposition 2.7, Week 2) of the multiplicative identity $1 \in R$, so that:

$$1 + (-1) = 0.$$

(ii) For any element $a \in R$, we define:

$$-a := (-1) \cdot a. \tag{1.6}$$

Lemma 1.13 (*$-a$ is the Additive Inverse of a*)**:** For any element $a \in R$, the element $-a = (-1) \cdot a$ is the additive inverse of a, that is:

$$a + (-a) = 0.$$

Proof

Using Proposition 1.9 ($0 \cdot a = 0$), Definition 1.12 ($1 + (-1) = 0$), and distributivity:

$$\begin{aligned} 0 &= 0 \cdot a \\ &= \big(1 + (-1)\big) \cdot a \\ &= (1 \cdot a) + \big((-1) \cdot a\big) && \text{(by distributivity)} \\ &= a + (-a) && \text{(since } 1 \cdot a = a \text{ and } (-1) \cdot a = -a\text{).} \end{aligned}$$

Since $a + \big((-1) \cdot a\big) = 0$, the element $(-1) \cdot a$ satisfies the defining property of the additive inverse of a. By uniqueness of inverses in $(R, +, 0)$, this confirms that $-a = (-1) \cdot a$ is indeed the unique additive inverse of a. Q.E.D.

Integer Multiples and the Canonical Map $\mathbb{Z} \to R$

Spoken Transcript [00:15:05 - 00:17:10]

Another convention about a ring is: I said that if I have a as an element of the ring *[writes $a \in R$]*, $-a$ *[writes $-a \in R$]* we just write as $-a$, and by definition $-a$ is just $(-1) \times a$, and it's a canonical element of the ring.

There's something else: we can also look at $a + a$ *[writes $a + a$]*, and this is defined as $2a$ *[writes $= 2a$]*. You can take $a + a + a$ *[writes $a + a + a$]* n times *[draws an underbrace labeled n]*; this is what this term na is defined to be *[writes $= na$]*.

So, in any ring, if I have an element a in a ring *[writes $a \in R$]*, I can take 17 times that element *[writes $17a \in R$]*. That is an element of the ring, and I just mean you add a to itself 17 times. It's well-defined, just using the ring structure. You have no objections?

And I can also do -17 times *[writes $-17a \in R$]*, because I just, you know, either add $-a$ to itself 17 times, or I add a to itself 17 times and take the minus, and they're the same. All right?

In particular, there's a map from $\mathbb{Z}$ to the ring *[writes $\mathbb{Z} \xrightarrow{\phi} R$]*. And this map requires no choices. This map sends 0 to 0 *[writes $0 \mapsto 0$]*, sends 1 to 1 *[writes $1 \mapsto 1$]*, sends 2 to $1+1$ *[writes $2 \mapsto 1+1$]*, sends 3 to $1+1+1$ ...whatever *[writes $3 \mapsto 1+1+1$]*, okay? And it sends -1 to the additive inverse *[writes $-1 \mapsto -1$]*.

So, if you give me a ring, I know how to map $\mathbb{Z}$ to that ring. There's nothing going on with this map: I just map 1 to 1, and then everything else I just add up.

This is not just any map. It turns out this is what's called a ring homomorphism. So, we have to define this.

Scalar Multiplication by Integers and the Canonical Map $\mathbb{Z} \to R$

Definition 1.14 (*Multiples of Ring Elements by Integers*): Let R be a ring and $a \in R$. For any integer $n \in \mathbb{Z}$, the element $n \cdot a \in R$ is defined inductively:

$$n \cdot a := \begin{cases} 0_R & \text{if } n = 0, \\ \underbrace{a + a + \cdots + a}_{n \text{ times}} & \text{if } n > 0, \\ \underbrace{(-a) + (-a) + \cdots + (-a)}_{|n| \text{ times}} & \text{if } n < 0. \end{cases}$$

In particular, for any specific integer such as $n = 17$ or $n = -17$:

$$17a \in R \quad \text{and} \quad -17a = -(17a) \in R.$$

Remark 1.15 (*Two Ways to Form $-17 \cdot a$*): The equality $n \cdot (-a) = -(n \cdot a)$ holds for $n > 0$: adding $-a$ to itself n times gives the same element as adding a to itself n times and taking the additive inverse, because $(n \cdot a) + (n \cdot (-a)) = n \cdot (a + (-a)) = 0$ by commutativity and associativity of $+$. This is what the lecturer means by *"they're the same"*.

A Harmless Clash of Notation: In $n \cdot a$ the factor n is an integer, not an element of R, so this dot is not the ring multiplication. The two readings never conflict: $0 \cdot a = 0_R$ holds for the integer 0 by definition and for the ring element 0 by Proposition 1.9, and in general $n \cdot a = (n \cdot 1_R) \cdot a$ by the distributive law.

Definition 1.16 (*Canonical Map from $\mathbb{Z}$ into a Ring*): For every ring R, there exists a canonical map $\phi \colon \mathbb{Z} \to R$ defined by evaluating integer multiples on the multiplicative unit 1_R:

$$\phi(n) := n \cdot 1_R.$$

Explicitly, the map sends:

$$\begin{aligned} 0 &\mapsto 0_R, \\ 1 &\mapsto 1_R, \\ 2 &\mapsto 1_R + 1_R, \\ 3 &\mapsto 1_R + 1_R + 1_R, \\ -1 &\mapsto -1_R, \quad \dots \end{aligned}$$

Universal Initial Property of the Integers $\mathbb{Z}$: This map $\phi \colon \mathbb{Z} \to R$ requires absolutely no choices: it is entirely determined by the ring structure of R. As we will

see next (Example 1.22), this map is not merely a set-theoretic function, but a ring homomorphism. Going beyond the lecture: it is even the unique ring homomorphism from $\mathbb{Z}$ to R (any such g has $g(1) = 1_R$, and additivity forces $g(n) = n \cdot 1_R$), establishing that $\mathbb{Z}$ is the *"initial object"* in the category of rings with unit (categories: Definition 2.49, Week 2).

Ring Homomorphisms

Standing Convention: Commutativity and Units

Spoken Transcript [00:17:10 - 00:18:44]

Let R_1 and R_2 be two rings *[writes let R_1 and R_2 be two rings]*.

Now, maybe I'll say a little bit more about these rings. For this class, as I said, if I consider totally abstract rings — like totally arbitrary rings here *[points to ring axioms on the upper left board]* — I have not assumed the multiplication is commutative. For a lot of things we don't, and sometimes we do, but when I say *"rings"* in this class from now on, I will mean a ring that has commutative multiplication *[writes Ring: commutative mult.]* and always has a unit: 1 *[writes unit 1]*.

If you look in the literature about what the definition of a ring is, of course a ring does not require the multiplication to be commutative, and that's reasonable. If you get off the plane in Tokyo and you meet a mathematician and you say, *"I'm working on a ring,"* that person will not know whether it's commutative or not; you have to tell them. And also, you can define rings without units. So, the fancy words for this would be a **commutative ring with unit**. That's just telling the person that that's the version of ring you're signing up for. You can study anything you want, but in this class, we will be considering for the most part commutative rings with unit. I just wanted to say that.

Convention: Commutative Rings with Unit

Remark 1.17 (*Standing Convention for Rings in this Course*): In the general mathematical literature, rings are not necessarily required to have commutative multiplication, nor are they always required to possess a multiplicative identity 1 (sometimes called *"rngs"* or pseudo-rings).

Standing Convention: In this course, unless explicitly stated otherwise, the term *"ring"* will always designate a **commutative ring with unit**: a ring with unit 1 in the sense of Definition 1.8 such that $a \cdot b = b \cdot a$ for all $a, b \in R$. The zero ring ($1 = 0$, Proposition 1.10) is still allowed.

Terminology Across the Literature: Rings vs. Commutative Rings

Mathematical literature reflects varying conventions:

- **Commutative Algebra & Algebraic Geometry (Atiyah–Macdonald, Eisenbud):** A *"ring"* implicitly means a commutative ring with multiplicative identity 1.

- **General Algebra (Bourbaki, Lang, Artin):** A *"ring"* possesses a multiplicative identity 1, but multiplication is not required to be commutative.
- **Functional Analysis & Operator Algebras:** Rings often lack a multiplicative identity 1 (sometimes called *"rngs"* or pseudo-rings).

In this course, Prof. Pandharipande adheres to the commutative algebra convention: unless explicitly stated otherwise, all rings are commutative and unital. The zero ring, where $1 = 0$, is allowed (Proposition 1.10).

Beyond the Lecture: What Changes Without Commutativity

Typical non-commutative rings are matrix rings $M_n(K)$ for $n \geq 2$ (Fact 0.30) and endomorphism rings. Because this course assumes commutativity, the following distinctions never come up in the later lectures:

- **Ideals (Definition 1.30):** left ideals ($r \cdot x \in I$), right ideals ($x \cdot r \in I$) and two-sided ideals ($r \cdot x \cdot s \in I$) coincide in a commutative ring. In general they differ, and kernels of ring homomorphisms are always two-sided ideals.
- **Divisibility (Definition 2.65, Week 2):** in general one distinguishes left divisibility ($b = a \cdot c$) from right divisibility ($b = c' \cdot a$).
- **Units (Definition 2.66, Week 2):** in general a unit must have a two-sided inverse ($uw = wu = 1$). The units still form a group, with $(u_1u_2)^{-1} = u_2^{-1}u_1^{-1}$, but it can be non-abelian, e.g., $U(M_n(K)) = \mathrm{GL}_n(K)$ for $n \geq 2$.
- **Integral domains (Definition 2.62, Week 2):** the term presupposes commutativity. A non-commutative ring without zero divisors is just called a **domain**, e.g., the real quaternions $\mathbb{H}$.

Definition of a Ring Homomorphism

Spoken Transcript [00:18:44 - 00:20:03]

Okay, so, the definition of a homomorphism: a homomorphism *[writes A homomorphism]* f from R_1 to R_2 *[writes $f : R_1 \to R_2$]* — maybe I put here: a ring homomorphism *[writes Ring above homomorphism]* — is defined to satisfy some properties.

The first is that $f(a+b) = f(a) + f(b)$ *[writes (i) $f(a+b) = f(a) + f(b)$]*. This is saying that this map respects the linear structure, the additive structure of the two rings. So, this should remind you from linear algebra class: there's the idea of linear homomorphisms of vector spaces, and it has this linear condition — that's this linear condition.

The second condition is that it respects the multiplication, so you can imagine what's going to be written here *[writes (ii) $f(ab) = f(a)f(b)$]*, okay? And the third condition is that it respects the identity: $f(1) = 1$ *[writes (iii) $f(1) = 1$]*.

Those are the three conditions. That's the definition of a homomorphism of rings.

Definition of a Ring Homomorphism

Definition 1.18 (*Ring Homomorphism*): Let R_1 and R_2 be rings. A mapping $f: R_1 \to R_2$ is called a *"ring homomorphism"* if it satisfies the following three conditions for all $a, b \in R_1$:

(i) Additive Compatibility:

$$f(a+b) = f(a) + f(b). \tag{1.7}$$

(ii) Multiplicative Compatibility:

$$f(a \cdot b) = f(a) \cdot f(b). \tag{1.8}$$

(iii) Identity Preservation:

$$f(1_{R_1}) = 1_{R_2}. \tag{1.9}$$

Comparison to Linear Algebra and Group Theory: Condition **(i)** asserts that f is a group homomorphism between the additive groups $(R_1, +)$ and $(R_2, +)$, mirroring the linearity condition $T(u+v) = T(u) + T(v)$ from linear algebra. Condition **(ii)** ensures compatibility with the multiplicative monoid structure. Condition **(iii)** explicitly guarantees that the multiplicative unit of the domain maps to the multiplicative unit of the codomain.

Preservation of the Additive Identity: $f(0) = 0$

Spoken Transcript [00:20:03 - 00:22:25]

There are some puzzling aspects about this definition. You could say, *"Well, why don't I say it respects the zero?"* That's a very reasonable question. So, you could say: what about $f(0)$ *[writes What about $f(0)$?]*?

Well, we can write $f(0+0)$ *[writes $f(0+0)$]*. I want to use property **(i)**: $f(0+0) = f(0) + f(0)$ *[writes $= f(0) + f(0)$]*, right? That's just applying condition **(i)** of the homomorphism *[writes Condition (i) $\Longrightarrow$ in front of it]*. But now, $0+0$ is 0, we know that, because 0 is the neutral element for addition. So, I can rewrite this as $f(0) = f(0) + f(0)$ *[writes $f(0) = f(0) + f(0)$]*, okay?

And then I can use left cancellation by taking the inverse. I've said that several times, and I hope you know what I mean by that: if I take such an equation... Let me just do it once formally, because I said it several times. I have one $f(0)$ here and two $f(0)$'s here, and I want to get rid of the $f(0)$ on both sides. So, what I do — and this is what I mean by left cancellation — is that there exists some $b \in R_2$ *[writes $\exists b \in R_2$]* such that $b + f(0) = 0$ *[writes $, b + f(0) = 0$]*. That's the group property on R_2.

And then I just add b to both sides of this. So, I get $b + f(0) = b + f(0) + f(0)$ *[writes $b + f(0) = b + f(0) + f(0)$]*. I use this cancellation, I use the additive inverse of $f(0)$ on both sides, and then, since that equals 0, this is 0 *[writes 0 under $b + f(0)$]*, that's 0 *[writes $= 0 + f(0)$ under the right side]*, and finally I conclude that $f(0) = 0$ *[writes $\boxed{f(0) = 0}$]*.

So, it is true that $f(0) = 0$ (i.e., $f(0_{R_1}) = 0_{R_2}$), but I don't have to put it here. It's just traditional not to. You could put it here, it wouldn't make any difference.

Automatic Preservation of the Additive Identity

Proposition 1.19 (*Preservation of Zero*): Let $f\colon R_1 \to R_2$ be a ring homomorphism. Then f automatically preserves the additive identity:

$$f(0_{R_1}) = 0_{R_2}. \tag{1.10}$$

Proof

Because 0_{R_1} is the additive identity of R_1, we have $0_{R_1} + 0_{R_1} = 0_{R_1}$. Applying property **(i)** of Definition 1.18:

$$f(0_{R_1}) = f(0_{R_1} + 0_{R_1}) = f(0_{R_1}) + f(0_{R_1}). \tag{1.11}$$

Since $(R_2, +, 0_{R_2})$ is an abelian group, the element $f(0_{R_1}) \in R_2$ possesses an additive inverse $b \in R_2$ such that $b + f(0_{R_1}) = 0_{R_2}$. Adding b to both sides of (1.11):

$$\begin{aligned} b + f(0_{R_1}) &= b + \big(f(0_{R_1}) + f(0_{R_1})\big) \\ 0_{R_2} &= \big(b + f(0_{R_1})\big) + f(0_{R_1}) \qquad \text{(by associativity in } R_2\text{)} \\ 0_{R_2} &= 0_{R_2} + f(0_{R_1}) \\ 0_{R_2} &= f(0_{R_1}). \end{aligned}$$

Thus, $f(0_{R_1}) = 0_{R_2}$. Q.E.D.

Why Identity Preservation Must Be Axiomatized

Spoken Transcript [00:22:25 - 00:24:16]

Then maybe the question is: *"Well, then why do I put $f(1) = 1$?"* That's a reasonable question. They're treated not symmetrically.

And the answer to that... *[erases the cancellation calculation for $f(0)$ on the board]*

Yeah, it has to do with the fact that I cannot... I mean, maybe this is what you're hinting at, but I don't have this cancellation property in multiplication. I have this cancellation — the existence of an additive inverse — that is an axiom of the ring, so I can use it. But there is not an existence of a multiplicative inverse, so I cannot just follow these steps. So, I have to assume it by hand.

This is about the definitions, and as I said, the strange thing about mathematics is that when it's taught, the definitions are first, but somehow when it's understood, the definitions are last. So, it's somehow a constant difficulty. *[erases more board space and begins writing]*

So, the point is: we don't have *[writes We don't have]* multiplicative inverses *[writes mult. inverses]* — we don't always have multiplicative inverses — in R_1 *[writes in R]*, and so, we can't *[writes so we can't]* use this argument *[writes use that argument and an arrow pointing to it]* — the one that's in the corner there (i.e., the cancellation argument for $f(0) = 0$, Proposition 1.19) *[points at the proof of $f(0) = 0$ in the corner of the board]*.

So, to fix that, we just say at the beginning: $f(1) = 1$. We don't leave it to chance. You could hope, maybe, that even though we couldn't use this argument, maybe you could find some different argument. You could hope that, but it's not true.

Spoken Transcript [00:24:16 - 00:25:08]

For example, I could take a bad example *[writes A bad example]*: I could take R and map it to... sorry, I could take $R_1 \to R_2$ *[writes $R_1 \to R_2$]* and map it so everything goes to zero. Say they're both healthy rings — in fact, they could both be $\mathbb{Z}$ *[writes $\mathbb{Z}$ above R_1 and $\mathbb{Z}$ above R_2]*. I could map every element to zero.

If you map every element to zero, you certainly are not mapping 1 to 1. But you won't get a contradiction with every element being zero; all the other properties will be satisfied.

So, it's not a good idea to think about this, but the fact of the matter is: we need to just assume it. *[crosses out the bad example]* Okay.

The Asymmetry Between Addition and Multiplication

Board State Note:

"We don't have mult. inverses in R, so we can't use that argument."

Proposition 1.20 (*Multiplicative Idempotence of $f(1)$*): Let $f: R_1 \to R_2$ satisfy conditions **(i)** and **(ii)** of a ring homomorphism. Because $1_{R_1} \cdot 1_{R_1} = 1_{R_1}$, multiplicativity implies:

$$f(1_{R_1}) = f(1_{R_1} \cdot 1_{R_1}) = f(1_{R_1}) \cdot f(1_{R_1}) = f(1_{R_1})^2. \tag{1.12}$$

Thus, $f(1_{R_1})$ is an *"idempotent"* element in R_2. However, because elements in a general ring do not necessarily possess multiplicative inverses, we cannot cancel $f(1_{R_1})$ to deduce that $f(1_{R_1}) = 1_{R_2}$.

Why $f(1) = 1$ Does Not Follow from $f(ab) = f(a)f(b)$: One might wonder why $f(1) = 1$ is required as an explicit axiom in Definition 1.18, whereas $f(0) = 0$ is a provable theorem.

- **Additive Case:** In the additive group $(R_2, +)$, every element has an additive inverse. Hence, from $f(0) = f(0) + f(0)$, we can subtract $f(0)$ to obtain $f(0) = 0$.
- **Multiplicative Case:** Using multiplicativity, we know that $f(1) = f(1 \cdot 1) = f(1) \cdot f(1)$. However, in a general ring R_2, $f(1)$ does not necessarily possess a multiplicative inverse! While $x^2 = x$ in a field (or in a division ring, where every non-zero element is invertible but multiplication need not commute) implies $x \in \{0, 1\}$, in a general ring there can be non-trivial idempotents ($e^2 = e$ with $e \neq 0, 1$), or $f(1)$ could simply equal 0.

No cleverer argument can help either: the zero map in Example 1.21 satisfies the first two conditions but not $f(1) = 1$.

Counterexample: The Non-Unital Zero Map

Example 1.21 (*The Trivial Zero Map Fails Condition **(iii)*):** Let $R_1 = R_2 = \mathbb{Z}$ (or any non-zero rings). Consider the zero map:

$$f: R_1 \to R_2, \quad f(a) := 0 \quad \forall a \in R_1.$$

This map satisfies conditions **(i)** and **(ii)**:

- **Addition:** $f(a+b) = 0 = 0+0 = f(a)+f(b)$,
- **Multiplication:** $f(a \cdot b) = 0 = 0 \cdot 0 = f(a) \cdot f(b)$.

However, $f(1_{R_1}) = 0 \neq 1_{R_2}$. Without demanding condition **(iii)** ($f(1) = 1$), the zero map would be considered a ring homomorphism between non-zero rings.

Examples of Ring Homomorphisms: The Canonical Map $\mathbb{Z} \to R$

Spoken Transcript [00:25:08 - 00:27:20]

Then there's lots of examples *[writes Examples]* of these ring homomorphisms, and it's the main way we compare rings — by homomorphisms.

For example, the thing I said here (i.e., the canonical map of Definition 1.16): $\mathbb{Z}$ goes to R *[writes* $\mathbb{Z} \xrightarrow{\phi} R$*]*, this basic map ϕ. This is a ring homomorphism *[writes This is a ring hom.]*. Not only does $\mathbb{Z}$ map to R by just seeing where 1 goes, it's actually a homomorphism. And to check it's a homomorphism, you have to do something.

I'm not going to do this for you, you have to do it, because it's some kind of internal tautology about addition... Where does 2 go? We said 2 goes to $1+1$ *[writes* $2 \mapsto 1+1$*]*. Where does 3 go? 3 goes to $1+1+1$ *[writes* $3 \mapsto 1+1+1$*]*. So, then we'd have to check that if I add these two, 5 goes to five 1's, which looks pretty good.

And then you'd also have to check: where does 6 go? 6 is 2×3, and this goes to $(1+1) \times (1+1+1)$ *[writes* $2 \cdot 3 \mapsto (1+1)(1+1+1)$*]*. And now you'd use the distributive law a lot of times. If you use the distributive law a lot of times, you'll get this to be $1+1+1+1+1+1$ *[writes* $= (1+1+1+1+1+1)$*]*, which is the definition of where 6 goes. Okay?

So, if you do this, you'll see that there's essentially nothing going on except understanding that the distributive law is making this a homomorphism.

This kind of mathematics that's on this kind of foundational, definitional level, you pretty much have to do for yourself to understand, you know, because it has a certain kind of logical... And I hope this is clear. But there is some content — there's a very small amount of content: it's about how this definition of 2 as $1+1$ interacts with the distributive law. I think most people find this not so controversial, but if you want to understand it, you have to go through the mental effort of checking the things yourself. On the board, I could convince you of anything, probably. Okay.

The Canonical Homomorphism $\mathbb{Z} \to R$

Example 1.22 (*Canonical Homomorphism from the Integers*)**:** For any ring $(R, +, \cdot, 0_R, 1_R)$, the canonical evaluation map:

$$\phi\colon \mathbb{Z} \to R, \quad \phi(n) := n \cdot 1_R$$

is a ring homomorphism.

Verification of Homomorphism Axioms:

- **Preservation of Unit:** $\phi(1) = 1 \cdot 1_R = 1_R$.

- **Additive Compatibility (e.g., $2+3=5$):**
$$\begin{aligned}\phi(2)+\phi(3) &= (1_R+1_R)+(1_R+1_R+1_R)\\ &= 1_R+1_R+1_R+1_R+1_R = \phi(5) = \phi(2+3).\end{aligned}$$
- **Multiplicative Compatibility (e.g., $2\cdot 3=6$):** Using repeated applications of the distributive laws:
$$\begin{aligned}\phi(2)\cdot\phi(3) &= (1_R+1_R)(1_R+1_R+1_R)\\ &= 1_R(1_R+1_R+1_R)+1_R(1_R+1_R+1_R)\\ &= (1_R+1_R+1_R)+(1_R+1_R+1_R)\\ &= 1_R+1_R+1_R+1_R+1_R+1_R = \phi(6) = \phi(2\cdot 3).\end{aligned}$$

The General Check the Lecturer Leaves to You: The two computations above are instances of two identities, valid for all $m,n\in\mathbb{Z}$:

- **Additivity:** $(m+n)\cdot 1_R = m\cdot 1_R + n\cdot 1_R$. For $m,n\geq 0$ both sides are a sum of $m+n$ copies of 1_R (associativity of $+$); the cases with negative m or n follow by adding inverses on both sides (Definition 1.14, Remark 1.15).
- **Multiplicativity:** $(m\cdot 1_R)(n\cdot 1_R) = (mn)\cdot 1_R$. Expanding the left side with the distributive laws gives $|mn|$ products $1_R\cdot 1_R = 1_R$, taken with the sign of mn, exactly as in the computation for $2\cdot 3=6$.

Together with $\phi(1)=1_R$, this proves that ϕ is a ring homomorphism.

Universal Initial Property of $\mathbb{Z}$: In category theory, this property establishes that $\mathbb{Z}$ is the *"initial object"* in the category of rings (Example 2.46, Week 2): for every ring R, there exists a **unique** ring homomorphism from $\mathbb{Z}$ to R.

Inclusion Homomorphisms and Subrings

Spoken Transcript [00:27:20 - 00:28:24]

That's a very simple example of a homomorphism, but in fact there's lots of... all the ones we've seen so far: if you take $\mathbb{Z}$ inside $\mathbb{Q}$ *[writes $\mathbb{Z}\subset\mathbb{Q}$]*, if you view this as a map — just the embedding map, if you view an integer as a rational number — then this gives me a homomorphism. And that's a homomorphism inside $\mathbb{R}$ *[writes $\subset\mathbb{R}$]*, and this is a homomorphism inside $\mathbb{C}$ *[writes $\subset\mathbb{C}$]*. So, these inclusions are all homomorphisms. And I can also include this inside the ring of polynomials *[writes $\cap$ under each ring and below them $\mathbb{Z}[x]\subset\mathbb{Q}[x]\subset\mathbb{R}[x]\subset\mathbb{C}[x]$]* — polynomials in one variable.

These are many of the rings that we will study, but if you start at the bottom, $\mathbb{Z}$, they just include, and all these inclusions are homomorphisms. Because they all respect... you know, whenever I include, I'm not changing the definition of addition or multiplication for the smaller object, and 1 is always going to 1.

So, this is an example. Let's try some different kinds of examples... *[draws a vertical line dividing the board]*

Inclusion Homomorphisms of Standard Rings

Example 1.23 (*Canonical Inclusions*): The canonical inclusion embeddings of the standard number systems:

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R} \hookrightarrow \mathbb{C} \tag{1.13}$$

and their single-variable polynomial rings:

$$\mathbb{Z}[x] \hookrightarrow \mathbb{Q}[x] \hookrightarrow \mathbb{R}[x] \hookrightarrow \mathbb{C}[x] \tag{1.14}$$

are all injective ring homomorphisms. For any subring $S \subseteq R$ (defined formally in Week 2, Definition 2.26), the inclusion map $\iota\colon S \hookrightarrow R$ defined by $\iota(s) = s$ is a ring homomorphism because S inherits the operations and the unit $1_S = 1_R$ directly from R.

The Ring of Residue Classes $\mathbb{Z}/n\mathbb{Z}$ and the Projection Homomorphism

Spoken Transcript [00:28:24 - 00:30:48]

One example is that you've studied this ring $\mathbb{Z}/n\mathbb{Z}$ *[writes $\mathbb{Z}/n\mathbb{Z}$]*... Well, if we haven't, let me at least say what it is, because this also plays a role in the class. So, this ring here, this set here... *[writes $(\mathbb{Z}/n\mathbb{Z}, +, \cdot, \bar{0}, \bar{1})$]*. This is a standard name; probably in other courses it's also given other names. But the words are "$\mathbb{Z}$ mod $n\mathbb{Z}$", and this is the set of residue classes mod n *[writes $= \{$set of residue classes mod $n\}$]*. If you want to be explicit, this is equal to the set $\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}$ *[writes $= \{\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}\}$]*, because n is zero. Is this notation reasonable notation? It's kind of semi-standard mathematical notation. There's exactly n elements of this; it's just the residue classes mod n.

And you can add residue classes, you can multiply them, there's the zero, and there's the one. So, if I take this whole data together, this exactly is a commutative ring with unit *[writes Ring!]*. So, it's a ring for us. It's a ring of a slightly different nature than the other ones because it has finitely many elements. It's an interesting ring to think about. In some sense, one of the simplest and most confusing ones is this one *[writes $\mathbb{Z}/2\mathbb{Z}$]*: this is a ring with two elements *[writes $= \{\bar{0}, \bar{1}\}$]*, zero and one. That's kind of a fun ring, but it's a full-fledged ring. You know, a ring is a ring no matter how small.

And we get here: $\mathbb{Z}$ goes to $\mathbb{Z}/n\mathbb{Z}$ *[writes $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$]*; this is also a homomorphism *[writes f over the arrow]*. It's a ring homomorphism where I take $f(a)$ — for any integer a — to $\bar{a}$ *[writes $f(a) = \bar{a}$]*. Okay? So, that's another example.

Residue Class Rings and the Reduction Modulo n

Definition 1.24 (*Ring of Residue Classes Modulo n*): For any integer $n \geq 1$, the ring of "*residue classes modulo n*" (denoted $\mathbb{Z}/n\mathbb{Z}$ or $\mathbb{Z}_n$) is the set:

$$\mathbb{Z}/n\mathbb{Z} := \{\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}\}, \tag{1.15}$$

equipped with modular addition and multiplication:

$$\bar{a} + \bar{b} := \overline{a+b}, \quad \bar{a} \cdot \bar{b} := \overline{a \cdot b},$$

with additive identity $\bar{0}$ and multiplicative identity $\bar{1}$. Here $\bar{a}$ denotes the residue class of $a \in \mathbb{Z}$, and $\bar{a} = \bar{b}$ if and only if $n \mid (a-b)$ (congruence modulo n, Definition 0.20

and Fact 0.21).

Why the Operations Are Well-Defined: The formulas use representatives a, b of the classes, so they must not depend on that choice. If $\bar{a} = \bar{a}'$ and $\bar{b} = \bar{b}'$, say $a' = a + kn$ and $b' = b + \ell n$, then

$$a' + b' = (a + b) + (k + \ell)n, \qquad a'b' = ab + (a\ell + kb + k\ell n)n,$$

so $\overline{a' + b'} = \overline{a + b}$ and $\overline{a'b'} = \overline{ab}$. The ring axioms are then inherited from $\mathbb{Z}$, and so is commutativity. For $n = 1$ one gets the zero ring $\{\bar{0}\}$.

Example 1.25 (*The Two-Element Field $\mathbb{Z}/2\mathbb{Z}$*): For $n = 2$, $\mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$ is the smallest non-trivial commutative ring with unit ($1 \neq 0$). The addition and multiplication tables are given by:

$+$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$

$\cdot$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{0}$	$\bar{1}$

This is a non-trivial commutative ring with unit despite containing only two elements.

Proposition 1.26 (*Canonical Projection Homomorphism*): The reduction map modulo n (written f on the board):

$$\pi\colon \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}, \quad a \mapsto \bar{a} := a \bmod n \tag{1.16}$$

is a surjective ring homomorphism.

Proof

For any $a, b \in \mathbb{Z}$:

- **(i) Additivity:** $\pi(a + b) = \overline{a + b} = \bar{a} + \bar{b} = \pi(a) + \pi(b)$.
- **(ii) Multiplicativity:** $\pi(a \cdot b) = \overline{a \cdot b} = \bar{a} \cdot \bar{b} = \pi(a) \cdot \pi(b)$.
- **(iii) Unit Preservation:** $\pi(1) = \bar{1}$.

Surjectivity holds because every residue class $\bar{k} \in \mathbb{Z}/n\mathbb{Z}$ is the image of the integer $k \in \mathbb{Z}$. Q.E.D.

The Kernel of a Ring Homomorphism and Ideals

Definition of the Kernel

Spoken Transcript [00:30:49 - 00:32:06]

There are lots of interesting examples; I'm not going to get to them all today. But I want to say that *[pushes up the lower board]* for homomorphisms, like for vector spaces, whenever you have one, you can consider the kernel and the image. *[erases the board that came down: the definition of a ring homomorphism and the proof of $f(0) = 0$]* And it turns out, in a maybe surprising way, that for ring theory, the notion of the kernel turns out to be much more important than the notion of the image.

In linear algebra, if you have a homomorphism between two vector spaces — did

you use the word *"homomorphism"*? You could also say *"linear transformation"*; both those words are used — if you have a linear transformation between two vector spaces, there's the notion of its kernel (those are the elements that go to zero) and the notion of its image. And they're both subvector spaces. In linear algebra, these things are kind of on similar footing. But in ring theory, when you have a homomorphism, there's again the kernel and the image. So, we will start with the kernel first; probably next time we'll say a little bit about the image.

So, this is an interesting object. Here we have: let...

Spoken Transcript [00:32:06 - 00:32:56]

Let f from R_1 to R_2 *[writes Let $f: R_1 \to R_2$]* be a homomorphism of rings *[writes be a hom. of rings]*. Then we define the kernel of f: $\ker(f)$ *[writes $\ker(f) =$]* — it's those elements in the first ring, because the kernel lives in the first ring *[writes $\{a \in R_1 \mid$]*, such that $f(a)$ is equal to zero *[writes $f(a) = 0\}$]*. Okay? I hope this definition is clear: it's everybody that goes to zero.

In particular, you could say: what is the kernel of this? *[writes $\ker(f) \subset \mathbb{Z} \xrightarrow{f} \mathbb{Z}/n\mathbb{Z}$]* What is the kernel here?

Student Answer

Multiples of n.

Spoken Transcript [00:32:57 - 00:33:17]

Yeah. The kernel here is exactly all multiples of n, and this is often denoted $n\mathbb{Z}$ *[writes $n\mathbb{Z}$ under $\ker(f)$]*. This is equal to the set of multiples of n *[writes $\subset$ "mults of n"]*. That's the kernel.

Definition of the Kernel of a Ring Homomorphism

Definition 1.27 (*Kernel of a Ring Homomorphism*): Let R_1, R_2 be rings and let $f: R_1 \to R_2$ be a ring homomorphism. The *"kernel"* of f, denoted by $\ker(f)$, is the pre-image of the zero element 0_{R_2}:

$$\ker(f) := \{a \in R_1 \mid f(a) = 0_{R_2}\} \subseteq R_1. \tag{1.17}$$

Example 1.28 (*Kernel of the Modular Reduction Homomorphism*): For the canonical projection $\pi: \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ given by $\pi(a) = \bar{a}$, an integer a maps to $\bar{0}$ if and only if a is a multiple of n. Hence:

$$\ker(\pi) = n\mathbb{Z} := \{k \cdot n \mid k \in \mathbb{Z}\}. \tag{1.18}$$

The Kernel as a Group Kernel: Since f respects addition and $f(0_{R_1}) = 0_{R_2}$ (Proposition 1.19), $\ker(f)$ is the kernel of the group homomorphism $f: (R_1, +, 0) \to (R_2, +, 0)$ in the sense of Week 2 (Definition 2.17). The multiplication is what makes the ring kernel special, see Proposition 1.29.

***Insight:* Kernels versus Images in Ring Theory**

In linear algebra, the kernel and image of a linear map $T: V \to W$ enjoy a symmetric status: both are vector subspaces, connected by the rank-nullity theorem $\dim V = \dim \ker(T) + \dim \operatorname{im}(T)$ (Fact 0.28).

In ring theory, however, their algebraic roles diverge profoundly:

- The image $\operatorname{im}(f) \subseteq R_2$ is merely a **subring** of the codomain.
- The kernel $\ker(f) \subseteq R_1$ is not merely a subring, but an **ideal** of the domain: it is closed under addition and absorbs multiplication from the entire ring ($r \cdot a \in \ker(f)$ for all $r \in R_1$ and $a \in \ker(f)$).

This absorption property makes the kernel the precise algebraic structure needed to construct quotient rings $R_1/\ker(f)$ and establish the First Isomorphism Theorem for rings ($R_1/\ker(f) \cong \operatorname{im}(f)$).

Properties of the Kernel

Spoken Transcript [00:33:17 - 00:34:49]

So, the kernel has very specific properties *[writes Properties of the ker(f)]*. The first property is that it's closed under addition: if we have a and b in the kernel of f *[writes bullet: $a, b \in \ker(f)$]*, this implies $a+b$ is in the kernel *[writes $\implies (a+b) \in \ker(f)$]*. This is the linear algebra property. All of these things are easy to prove. To check whether $a+b$ is in the kernel, I have to calculate $f(a+b)$ *[writes $f(a+b) =$]*. Using the definition of homomorphism, this is $f(a)+f(b)$ *[writes $= f(a)+f(b)$]*. Then, using the fact that a and b are both in the kernel, well, this one is zero *[writes 0 under $f(a)$]* and that one is zero *[writes 0 under $f(b)$]*, so the whole thing is zero *[writes $= 0$]* — and so, it's in the kernel, right? I should have said even earlier: we already know that zero is in the kernel *[writes bullet: $0 \in \ker(f)$]*.

If a is in the kernel *[writes bullet: If $a \in \ker(f)$]* and r is any element of the ring *[writes , $r \in R$]*, then $r \times a$ is in the kernel *[writes $\implies ra \in \ker(f)$]*.

Spoken Transcript [00:34:49 - 00:36:11]

That's a wider property. These are the linear properties — we'll say more about them in a moment — but this property is about the multiplication. It says that if I take an element of the kernel, I can then go pick any element of the ring and multiply it, and it'll be in the kernel. So, it's a more interesting property. You can see that here: if you have an element of the kernel, it's n times something. If I multiply it by any integer, it's still n times something.

This is not hard to prove either. We have to calculate $f(r \times a)$ *[writes $f(ra) =$]*, and then, of course, we use the multiplicative property *[writes $= f(r)f(a)$]*. And now, this is the point: in the multiplicative property, in order to deduce that this product is zero, we only need one of the factors to be zero. For the addition, we needed both, but for the multiplication, we only need to know that one of the factors is zero, because that'll kill the product. And that's already there because a is in the kernel. So, this is zero *[writes 0*

under $f(a)$], and so, this is zero *[writes $= 0$]*, okay?

And so, in particular, if a is in the kernel, $-a$ is in the kernel *[writes bullet: $a \in \ker(f) \implies -a \in \ker(f)$]*. That's the additive inverse, because for $-a$, r is equal to -1. So, those are all properties enjoyed by the kernel...

Algebraic Properties of the Kernel

Proposition 1.29 (*Axiomatic Properties of the Kernel*): Let $f: R_1 \to R_2$ be a ring homomorphism. The kernel $\ker(f) \subseteq R_1$ satisfies the following properties:

(1) Containment of Zero: $0_{R_1} \in \ker(f)$.

(2) Closure under Addition: $\forall a, b \in \ker(f) \implies a + b \in \ker(f)$.

(3) Closure under Additive Inverses: $\forall a \in \ker(f) \implies -a \in \ker(f)$.

(4) Multiplicative Absorption (Ideal Property): $\forall a \in \ker(f), \forall r \in R_1 \implies r \cdot a \in \ker(f)$.

Proof

We verify each property directly using the axioms of a ring homomorphism:

- **Property (1):** By Proposition 1.19, $f(0_{R_1}) = 0_{R_2}$, so $0_{R_1} \in \ker(f)$.
- **Property (2):** If $a, b \in \ker(f)$, then $f(a) = 0$ and $f(b) = 0$. By additivity:
$$f(a+b) = f(a) + f(b) = 0 + 0 = 0 \implies a + b \in \ker(f).$$
- **Property (4):** If $a \in \ker(f)$ and $r \in R_1$, then $f(a) = 0_{R_2}$. By multiplicativity and Proposition 1.9:
$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0_{R_2} = 0_{R_2} \implies r \cdot a \in \ker(f).$$
- **Property (3):** Setting $r = -1_{R_1}$ in Property (4) and using Lemma 1.13:
$$-a = (-1_{R_1}) \cdot a \in \ker(f).$$

Q.E.D.

Why Multiplicative Absorption Works: Additive vs. Multiplicative Absorption: In the additive closure proof (Property (2)), both $f(a)$ and $f(b)$ had to be zero: knowing only $f(a) = 0$ would leave the sum $f(a) + f(b) = f(b)$ arbitrary. In contrast, for Property (4), multiplication by zero in a ring is absorbing: as soon as one factor in a product is zero, the entire product vanishes ($f(r) \cdot 0_{R_2} = 0_{R_2}$), regardless of whether $f(r)$ is zero or not. Thus, multiplying an element of $\ker(f)$ by **any** element $r \in R_1$ from the ambient ring keeps the result inside $\ker(f)$. This strong absorption property distinguishes ring kernels from mere subrings (Definition 2.26, Week 2).

Ideals in a Ring

Spoken Transcript [00:36:11 - 00:38:07]

Those properties can be said in a more concise way, which is a more natural way to say it, and it comes with a definition: the definition of an ideal. *[pushes up the board and*

erases the lower board] This is the definition of an ideal I in R *[writes Def: Ideal $I \subset R$]*.

The properties are: 0 must be in I *[writes $\bullet\ 0 \in I$]*. And the second way to say things is that if I take I, and I take the $+$ from R, and I take the 0, this is an abelian group *[writes $\bullet\ (I,+,0)$ abelian group]*. Remember, whenever I write R, I always mean R with the two operations and this data *[writes $= (R,+,\cdot,0,1)$ after $I \subset R$]*. So, to be an ideal, you have to have zero, and then when I consider this subset with $+$ and 0, it should be an abelian group. This means it's closed under addition, and it's closed under additive inverses, okay?

And the second thing is that for every a in I and for every r in R, ra remains in I *[writes $\bullet\ \forall a \in I, \forall r \in R \implies ra \in I$]*. Okay? That's the ideal condition.

This is an extremely important definition for the class. It tells you various subsets of R are ideals, and they have this linear property, which is kind of like linear algebra, but then they also have this multiplicative property, which is not like linear algebra. And the role of an ideal will play a little bit like the role of a subspace in linear algebra.

Definition of an Ideal

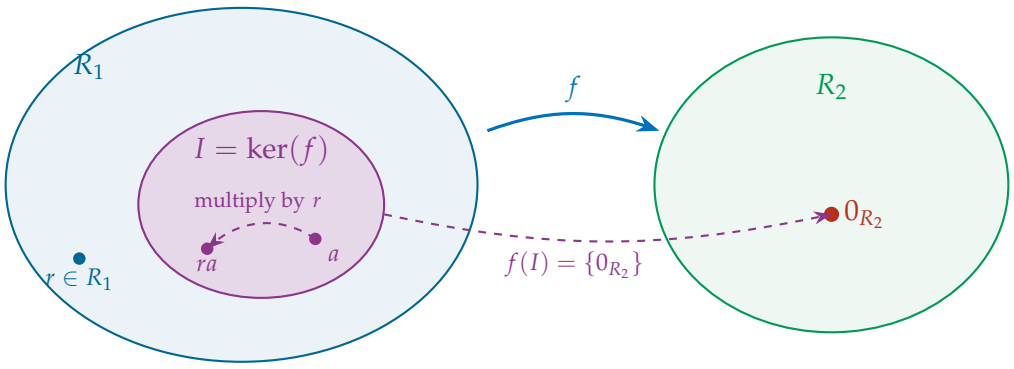
Definition 1.30 (*Ideal*): Let $R = (R,+,\cdot,0,1)$ be a commutative ring with unit. A subset $I \subseteq R$ is called an *"ideal"* of R if it satisfies:

(1) Additive Subgroup: $(I,+,0)$ is an abelian subgroup of $(R,+,0)$:

- $0 \in I$,
- $\forall a,b \in I \implies a+b \in I$,
- $\forall a \in I \implies -a \in I$.

(2) Multiplicative Absorption (Ideal Condition):

$$\forall a \in I,\ \forall r \in R \implies r \cdot a \in I. \tag{1.19}$$



Ideals as the Ring-Theoretic Analogue of Normal Subgroups and Subspaces: In linear algebra, kernels of linear maps are vector subspaces ($T(u+v) = T(u)+T(v)$ and $T(cv) = cT(v)$). A vector subspace $W \subseteq V$ is closed under vector addition ($w_1 + w_2 \in W$) and scalar multiplication by field scalars ($\lambda \cdot w \in W$ for $\lambda \in k$).

In group theory, kernels of group homomorphisms are normal subgroups ($gNg^{-1} \subseteq N$).

In ring theory, kernels of ring homomorphisms are *"ideals"*: they are closed under addition ($a+b \in I$) and absorb multiplication by **any** element from the entire ambient ring R ($r \cdot a \in I$ for all $r \in R$).

Notation: Ideals in the Lecture and in Textbooks

Many textbooks write $I \trianglelefteq R$ (or $I \triangleleft R$) for "I *is an ideal of* R". The lecture does not use this symbol: on the board, the lecturer writes $I \subset R$ and says in words that I is an ideal ("*Def: Ideal* $I \subset R$"). These notes follow the board. His $\subset$ allows equality, like $\subseteq$; he explains this convention when he defines maximal ideals (Definition 4.54, Week 4).

The Kernel as an Ideal

Spoken Transcript [00:38:07 - 00:39:13]

The proposition we've proven with this is:

Proposition *[writes Prop:]*: The kernel of a homomorphism $f\colon R_1 \to R_2$ *[writes The ker of a hom* $f\colon R_1 \to R_2$*]* is an ideal *[writes is an ideal]* in R_1 *[writes in* R_1*]*.

That's what we've proven on the board up there (Proposition 1.29): with the kernel, you always get an ideal. And rings can have... we will do some examples next time, I'm running out of time here, but for $\mathbb{Z}$, if I take inside here $n\mathbb{Z}$ *[writes* $\mathbb{Z} \supset n\mathbb{Z} = I$*]*, that's an ideal (it is the kernel of π, Example 1.28). Rings can have lots of ideals.

What about a field? I'll just say one more thing about this. Is that clear?

We have lots of examples of rings and lots of examples of ideals, but suppose F is a field *[writes* F *is field]*, $(F, +, \cdot, 0, 1)$ *[writes* $(F, +, \cdot, 0, 1)$*]*. What are its ideals? *[writes What are the ideals?]*

The Kernel is an Ideal

Proposition 1.31 (*Kernel as an Ideal*)**:** Let $f\colon R_1 \to R_2$ be a ring homomorphism. Then $\ker(f)$ is an ideal of R_1:

$$\ker(f) \subset R_1 \text{ is an ideal.}$$

Proof

This is the synthesis of Proposition 1.29:

(1) $(\ker(f), +, 0)$ is an abelian subgroup of $(R_1, +, 0)$ because $0 \in \ker(f)$, $a + b \in \ker(f)$ for all $a, b \in \ker(f)$, and $-a \in \ker(f)$ for all $a \in \ker(f)$.

(2) Multiplicative absorption holds because for all $a \in \ker(f)$ and all $r \in R_1$:

$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0 = 0 \implies r \cdot a \in \ker(f).$$

Hence, $\ker(f)$ satisfies all axioms of an ideal. Q.E.D.

Ideals in a Field

Spoken Transcript [00:39:13 - 00:40:05]

What are its ideals? What are the ideals of a field? Because a field is a ring, so it has the notion of ideals. Every abstract concept of a ring, we can apply to a field, because a field is a particular type of ring: it's a ring where we have multiplicative inverses for non-zero elements. So, what could be an ideal here? What are the ideals?

I'll start with one simple thing: if I just take $I = \{0\}$ *[writes $I = \{0\}$]*. Then zero is in I. It is an abelian subgroup just by itself, because $0 + 0 = 0$, and everything... you know, zero is sufficient unto itself *[laughter]*, one might say. And it has this multiplicative property, because if I multiply by anybody, it stays zero. So, zero is always an ideal.

So, what else? Yeah? *[points at student]*

Student Answer

The whole field?

Spoken Transcript [00:40:07 - 00:40:21]

The whole field is always an ideal: ideal equals the whole field *[writes $I = F$]*. And, of course, there can be no doubt about that, because if you take the whole thing, it's an abelian group and you're not going to get out of it by multiplying.

Can there be any more? Why not?

Student Answer

Because as soon as you have another element, then you have the inverse...

Spoken Transcript [00:40:26 - 00:40:28]

Multiplicative inverse.

Student Answer, continued

...multiplicative inverse, so you have one...

Spoken Transcript [00:40:30 - 00:40:37]

One, and then you have everything.

Okay, so, this argument is correct. So, the claim is: for a field, these are the only ideals *[writes "only ideals!" next to $I = \{0\}$ and $I = F$]*.

Spoken Transcript [00:40:37 - 00:42:20]

Which is an important thing to always keep in mind. And the reason is the argument he said: suppose that I have an ideal, and suppose this ideal is not this silly zero ideal *[writes $I \neq (0)$]*. That means there exists somebody in the ideal that's not zero *[writes $\exists\, a \in I,\, a \neq 0$]*. That somebody, because it's a field, has a multiplicative inverse *[writes $\exists\, a^{-1}$ so that]*. There exists an a^{-1}, so that $a^{-1} \cdot a$ is also in the ideal *[writes $a^{-1}a \in I$]*, because you're allowed to take anybody and multiply — this is anybody, and this is somebody in the ideal. So, if we start with a in the ideal, I can multiply by anybody; in particular, I can multiply by its inverse. But then this is one *[writes $1 \in I$ above $a^{-1}a$, joined by $=$]*!

Once I have one in the ideal, then I can easily get anybody in the ideal, because I can just take any $r \times 1$. Okay?

It's an elementary argument, and it's very good to understand. So, fields have very few ideals — extremely few. But rings can have quite a number of ideals. As I said here, this is the simplest example to see: the ideals in the integers come from multiples of any integer. And ideals in... yeah, we will see some examples of it. But was this argument clear? Sometimes I do these things very quickly, it's just, you know...

The standard thing to say here is that you can view fields as sellouts, because they've lost all their ideals. *[laughter]*

Classification of Ideals in a Field

Theorem 1.32 (*Ideals in a Field*): Let F be a field. The only ideals of F are the trivial zero ideal $\{0\}$ and the unit ideal F:

$$I \subset F \text{ an ideal} \implies I = \{0\} \quad \text{or} \quad I = F. \tag{1.20}$$

Proof

Let $I \subseteq F$ be an ideal of F. Clearly, $I = \{0\}$ and $I = F$ are ideals.

Now, suppose $I \neq \{0\}$. Then there exists a non-zero element:

$$a \in I \quad \text{with } a \neq 0.$$

Since F is a field, every non-zero element has a multiplicative inverse $a^{-1} \in F$. By the multiplicative absorption property of ideals (Definition 1.30), choosing $r = a^{-1} \in F$:

$$a^{-1} \cdot a \in I \implies 1_F \in I.$$

Once the unit 1_F lies in I, applying absorption once more with any arbitrary element $r \in F$:

$$r = r \cdot 1_F \in I \quad \forall r \in F.$$

Therefore, $F \subseteq I$, which forces $I = F$. Q.E.D.

Implications for Field Homomorphisms: A direct consequence of this theorem is that every ring homomorphism $f\colon F \to R$ from a field F to a non-zero ring R is automatically **injective**:

$$\ker(f) \subset F \text{ an ideal} \implies \ker(f) = \{0\} \quad (\text{since } f(1_F) = 1_R \neq 0_R \implies \ker(f) \neq F).$$

Because $\ker(f) = \{0\}$, field homomorphisms into non-zero rings are always embeddings ($f(a) = f(b)$ gives $f(a-b) = 0$, so $a = b$). This is why the extension $g\colon \mathbb{Q} \to F$ in Week 3 is automatically injective (Theorem 3.7).

Insight: Fields as "Sellouts"

The lecturer's humorous pedagogical quip — that *"fields are sellouts because they have lost all their ideals"* — highlights an essential contrast between field theory and ring theory:

- In a field F, the abundance of units ($F^\times = F \setminus \{0\}$) destroys all non-trivial ideals, as any non-zero element $a \in I$ immediately pulls the identity into the ideal: $1 = a^{-1} \cdot a \in I$ (with $r = a^{-1} \in F$ and $a \in I$), forcing $I = F$.
- In general rings (such as $\mathbb{Z}$ or polynomial rings), the scarcity of units allows rich, complex lattices of proper non-zero ideals to flourish (e.g., the ideals $n\mathbb{Z} \subset \mathbb{Z}$, which by Theorem 2.35 are all the ideals of $\mathbb{Z}$), which form the foundational building blocks for quotient rings R/I and algebraic geometry.

Outlook: Zorn's Lemma and Vector Space Bases

Existence of Bases in Infinite-Dimensional Vector Spaces

Spoken Transcript [00:42:20 - 00:43:29]

For this week, I did some material that was billed to the second week, and I didn't cover Zorn, so I'll say two words about Zorn and then I'll say a little bit more next week.

The first question is about linear algebra, which is this statement which *[erases the lower left board and prepares it]* probably you've seen in the first year, which is a strange statement, but it's about... So, first of all, I will spend some time also next week on Zorn's lemma, so you can read about that in the book *[writes Zorn's Lemma]*.

And it's related to the statement:

Let V be a vector space over a field k *[writes Let V be a vector space over a field k]*.

Does V have a basis *[writes Does V have a basis?]*?

Zorn's Lemma and the Existence of Vector Space Bases

Question 1.33 (*Existence of a Basis for Vector Spaces*): Let V be a vector space over a field k. Does V always have a basis?

Finite-Dimensional versus Infinite-Dimensional Vector Spaces:

- **Finite-Dimensional Spaces:** In standard first-year linear algebra, if V is spanned by finitely many vectors, an explicit basis is easily constructed by extracting a maximal linearly independent subset.
- **Infinite-Dimensional Spaces:** When V is infinite-dimensional (e.g., function spaces such as $C(\mathbb{R}, \mathbb{R})$ or $\mathbb{R}^{\mathbb{N}}$), constructing an explicit (Hamel) basis by hand is impossible. Proving that **every** vector space possesses a basis requires non-

constructive set-theoretic axioms—specifically, *"Zorn's Lemma"*, which is equivalent to the Axiom of Choice (AC).

Remark 1.34 (*Infinite-Dimensional Vector Spaces and the Axiom of Choice*): For finite-dimensional vector spaces, the existence of a basis is constructive via Gaussian elimination or the Steinitz exchange lemma (Fact 0.29). For arbitrary, infinite-dimensional vector spaces (such as function spaces $C(\mathbb{R},\mathbb{R})$ or sequence spaces $\mathbb{R}^{\mathbb{N}}$), the assertion that **every vector space possesses a basis** (known as a *"Hamel basis"*) cannot be proven constructively in Zermelo–Fraenkel set theory (ZF). It is logically equivalent to the *"Axiom of Choice"* (AC) and is standardly established by applying *"Zorn's Lemma"* to the poset of linearly independent subsets ordered by inclusion.

Beyond the Lecture: How Zorn's Lemma Gives a Basis

Zorn's lemma says: if P is a non-empty partially ordered set (Definition 0.10) in which every chain (totally ordered subset) has an upper bound in P, then P has a maximal element. For a vector space V over a field k:

(1) Let P be the set of linearly independent subsets of V, ordered by inclusion. It is non-empty, since $\emptyset \in P$.

(2) The union of a chain in P is again linearly independent: a linear relation involves only finitely many vectors, and these all lie in one member of the chain. So every chain has an upper bound in P.

(3) Zorn's lemma gives a maximal element $B \in P$. If some $v \in V$ were not in the span of B, then $B \cup \{v\}$ would be linearly independent and strictly larger than B. Hence B spans V, and B is a basis.

Spoken Transcript [00:43:29 - 00:44:01]

Did you consider this question in your first year?

Of course, if it's finite-dimensional, of course it has a basis. And this issue is about whether, if you have some massively infinite-dimensional field (i.e., vector space)... has a basis. And the traditional way to do this is an application of Zorn's lemma, which is related to the axiom of choice in logic — not my favorite direction to think about, but we will say a little bit about it. But...

Okay, and I think that's it. All right, so... *[The lecturer puts down the chalk, unclips the microphone, students applaud, and the lecture concludes.]*

Summary of Key Concepts

This lecture established core bridges connecting concrete arithmetic with axiomatic ring theory:

- **Gaussian Integers $\mathbb{Z}[i]$ and Primes:** The ring $\mathbb{Z}[i] = \{a+bi \mid a,b \in \mathbb{Z}\}$ possesses four units $\{\pm 1, \pm i\}$. An ordinary prime $p \in \mathbb{Z}$ factors in $\mathbb{Z}[i]$ if and only if $p = a^2+b^2$, which occurs precisely when $p = 2$ or $p \equiv 1 \pmod 4$ (Fermat's two-square theorem). Integer primes $p \equiv 3 \pmod 4$ remain prime in $\mathbb{Z}[i]$.

- **Rings and Elementary Arithmetic:** A ring $(R, +, \cdot, 0, 1)$ consists of an additive abelian group $(R, +, 0)$ and a multiplicative monoid $(R, \cdot, 1)$ coupled by the distributive laws. Distributivity yields the universal identities $0 \cdot a = 0$ and $(-1) \cdot a = -a$. When $1 = 0$, the ring collapses to the degenerate zero ring $R = \{0\}$.
- **Canonical Evaluation Map $\mathbb{Z} \to R$:** Defining integer multiples $n \cdot a$ yields the unique canonical ring homomorphism $\phi\colon \mathbb{Z} \to R$ with $\phi(n) = n \cdot 1_R$, establishing $\mathbb{Z}$ as the initial object in the category of unital rings.
- **Ring Homomorphisms:** A ring homomorphism preserves addition, multiplication, and the multiplicative identity ($f(1) = 1$). While $f(0) = 0$ is automatic via additive group cancellation, $f(1) = 1$ must be explicitly axiomatized because elements in arbitrary rings generally lack multiplicative inverses.
- **Ideals and Kernels:** An ideal $I \subset R$ is an additive subgroup that multiplicatively absorbs the entire ambient ring ($r \cdot a \in I$ for all $a \in I, r \in R$). The kernel $\ker(f) = f^{-1}(\{0\})$ of every ring homomorphism is an ideal.
- **Ideals in Fields:** Because every non-zero element in a field F is invertible, any non-zero ideal contains 1_F and equals F. Thus, fields have only the trivial ideals $\{0\}$ and F, ensuring that every homomorphism from a field into a non-zero ring is injective.
- **Outlook on Zorn's Lemma:** The fundamental problem of whether every infinite-dimensional vector space possesses a basis relies on non-constructive set theory via Zorn's Lemma.

Tuesday 22 September 2026

Homomorphisms, Substructures, Ideals, and Category Theory

Course Content Overview

This lecture explores the foundational algebraic structures of modern mathematics—groups, rings, and fields—and the structure-preserving mappings between them, concluding with an introduction to category theory:

- **Groups, Rings, and Fields:** Review of the defining axioms; intrinsic uniqueness of group identities and inverses; the axiomatic hierarchy connecting rings and fields to additive abelian groups.
- **Homomorphisms and Isomorphisms:** Group homomorphisms and isomorphisms; characterization via two-sided inverse homomorphisms; classification of additive and multiplicative groups of real numbers via the exponential map.
- **Kernels, Images, and Substructures:** Structural properties of kernels and images; formal definition of subgroups ($H \leq G$); ring homomorphisms and subrings ($S \subseteq R$).
- **Ideals and Symmetry Breaking:** The breakdown of structural symmetry in rings, where kernels are ideals ($I \subset R$) and images are subrings; classification of ideals in $\mathbb{Z}$ ($I = n\mathbb{Z}$) via the Euclidean division algorithm; ideals in rings of smooth functions $C^\infty(\mathbb{R})$.
- **Foundations of Category Theory:** Formal definition of categories (objects, morphisms, composition, unit laws); standard categories (**Grp**, **Ring**, $\mathbf{Vect}_{\mathbb{C}}$); categorical formulation of isomorphisms; covariant functors, including the forgetful functor and the double dual endofunctor on vector spaces.

Administrative Remarks and Course Logistics

Spoken Transcript [00:00:00 - 00:01:48]

Okay, so, we start, and there has been some progress on the website. There should be instructions there. First of all, you should be able to register for the problem sessions now. The second thing is that there should be instructions on the website about how to upload your homework. And the third thing is about the quiz, and I think there's a link there, too. We'll see if it works on Friday. The rule for the quiz will be that you can take it anytime from 6 in the morning on Friday to midnight. And if this is not enough

freedom, we can also discuss it. It's not so...

Another course hack is that if you look at this Metaphor page: in 2018 *[writes "Metaphor 2018" on the board]* it was the last time I taught this course. That Metaphor page is there. This course is going to follow roughly the same path. In fact, the topics and homework — I mean, there's maybe some slight switching order of lectures — but you can find all the information there. In particular, all the problem sets that Lika will eventually upload, you can already look at them now if you want to get ahead. And also, there are all the answers to all the problem sets; you can also look at them if you want. But in the modern world, it's easy to find answers to these questions, so, I don't see there's any reason to hide them from you. You can find everything there.

The other thing is that if you want to ask me a question, I'm happy to talk about math with anybody. You have to find me. Normally I have an office, but I was kicked out because of this *"Fenstersanierung"*. And it's going to last forever. So, I have a secret office on this floor. If you find me, you can talk to me. You'd have to find it — it's actually pretty hard to find. But if you do, you can knock on the door. Sometimes I'm busy, sometimes I'm not, but I'm happy to talk to anyone. But you have to find the office yourself.

On the Board

"Metaphor 2018"

Review of Algebraic Structures

Groups and Uniqueness of Identity and Inverses

Spoken Transcript [00:01:48 - 00:03:20]

Okay, so, last week we discussed kind of carefully these three objects which are the centerpiece for this course: G is a group. *[writes $(G, \cdot, e)$ on the board]* And last week I used a star for the product. No one does that. That was purely for didactic purposes, and probably pointless didactic purposes. The standard for a group is dot, but as I said, you can name this anything you want.

So, this is the standard data of the group. And the group — I won't repeat everything, but it has the three axioms: *[draws a brace next to $(G, \cdot, e)$]* associativity, *[writes "ass."]* existence of the neutral element, *[writes "exist. neutral"]* and existence of inverses. *[writes "exist. of inverses"]* This is written now in slightly more abstract language than the explicit equations.

And there are many things I didn't say, because they are all in some sense part of the definition. Like, I didn't say explicitly... You could ask yourself: this neutral element, could there be some other element that also has the neutrality property? *[writes "Could there be another $\hat{e} \in G$?"]* Could there be... These are things you should think about yourself. I can't teach you everything; it's not possible.

Could there be another $\hat{e} \in G$ that has the neutrality property? And the answer is

no, *[writes "No!"]* because then you could look at $e\hat{e}$. *[writes $e\hat{e} = \hat{e}$ and e below]* If I use the fact that e is a neutral element, this equals $\hat{e}$; and if $\hat{e}$ is a pretend neutral element, then this equals e, okay? Is that clear?

Group Axioms and Uniqueness of the Neutral Element

Definition 2.1 (*Group*): A *"group"* is a triple $(G, \cdot, e)$ consisting of a set G, a binary operation $\cdot\colon G \times G \to G$, and a distinguished element $e \in G$ satisfying the following three axioms:

(i) Associativity (ass.): $\forall a, b, c \in G\colon (a \cdot b) \cdot c = a \cdot (b \cdot c)$.

(ii) Existence of Neutral Element (exist. neutral): $\forall a \in G\colon a \cdot e = e \cdot a = a$.

(iii) Existence of Inverses (exist. of inverses): $\forall a \in G\ \exists b \in G\colon a \cdot b = b \cdot a = e$.

Question 2.2: Could there be another neutral element $\hat{e} \in G$?

Answer 2.3: No!

Proposition 2.4 (*Uniqueness of the Neutral Element*): Let $(G, \cdot, e)$ be a group. The neutral element $e \in G$ is unique.

Proof

Suppose $\hat{e} \in G$ also satisfies the neutrality property:

$$a \cdot \hat{e} = \hat{e} \cdot a = a \quad \forall a \in G.$$

Evaluating the product $e \cdot \hat{e}$:

$$e \cdot \hat{e} = \hat{e} \qquad \text{(since } e \text{ is a neutral element)},$$
$$e \cdot \hat{e} = e \qquad \text{(since } \hat{e} \text{ is a neutral element)}.$$

Equating both expressions yields:

$$\hat{e} = e \cdot \hat{e} = e.$$

Hence, the neutral element in any group is strictly unique. Q.E.D.

Spoken Transcript [00:03:20 - 00:05:05]

And then other questions you could ask: *[writes "If $x \in G$, $\exists y\colon xy = e$"]* if x is an element of the group, then this existence of inverses says that there exists a y such that xy equals the neutral element. You could say: Could there be another y, $\hat{y}$, where $x\hat{y} = e$? *[writes "Could there be another $\hat{y}\colon x\hat{y} = e$?"]* And the answer is no again. *[writes "No!"]*

And how do you prove this? Well, you look at this equation and you multiply both sides by the first one: $y(x\hat{y}) = y$. And then this thing is equal, by the original property... The neutral element property is on both sides, so, this thing is equal to e, so, this is equal to $\hat{y}$, *[writes $\hat{y} = (yx)\hat{y} = y$ and circles yx]* okay? Is that clear?

All of this is buried, in some sense, in the definitions. You could define a group by saying that there exists a neutral element and it's unique, and there exist inverses and they're unique. You don't have to, but you could do that. That's making the definition bigger.

You can also try to make the definition smaller. There's some kind of strange direction where people think that, well, there's three axioms, but we'd like a group to only have two axioms. And some guy in the '60s found some crazy way to write it with just one axiom, but it's just logically equivalent. You can go look that up if you want. I don't remember; it's some complicated axioms, you can imagine.

Uniqueness of Inverse Elements

Question 2.5: Let $x \in G$ and let $y \in G$ satisfy $xy = e$. Could there be another $\hat{y} \in G$ such that $x\hat{y} = e$?

Answer 2.6: No!

Proposition 2.7 (*Uniqueness of Inverses*): Let $(G, \cdot, e)$ be a group, $x \in G$, and $y \in G$ an inverse of x ($xy = yx = e$). If $\hat{y} \in G$ satisfies $x\hat{y} = e$, then $\hat{y} = y$. In particular, the inverse of x is unique.

Proof

Assume there exists $\hat{y} \in G$ such that $x\hat{y} = e$. Since $yx = e$ (circled on the board), associativity gives:

$$\hat{y} = (yx)\hat{y} = y(x\hat{y}) = y \cdot e = y.$$

Thus, for every element $x \in G$, its inverse is uniquely determined; it is denoted x^{-1}.

Q.E.D.

Redundancy in Axiomatic Definitions: Axiom sets are often chosen for pedagogical simplicity rather than strict minimal independence:

- **Uniqueness is Inherent:** Although we only postulated the **existence** of identities and inverses, uniqueness is an algebraic consequence of associativity.
- **Axiom Minimization:** It is possible to formalize groups using fewer axioms (or even a single equational axiom, as found by Higman and Neumann), but the standard three axioms remain the universal textbook standard.

Rings and Fields

Spoken Transcript [00:05:05 - 00:06:54]

Okay, so, that's the group, and the other structure we had was this ring (Recall: Definition 1.8), *[writes $(R, +, \cdot, 0, 1)$ on the board]* which I won't repeat all over again. But the way to think about this is that if I look at just the additive part, this is an abelian group, *[writes $(R, +, 0)$ abelian group]* and then I have some axioms for multiplication, *[writes "axioms for $\cdot$"]* and then two times the distributive law. *[writes "2× Distr"]* Those were the axioms for the ring, and we did some examples.

Certainly, one part of this class is really internalizing these axioms. And there's some kind of idea about mathematics which is some kind of asymptotic understanding — which is not just that you understand it, but the really asymptotic understanding that you hope for, which you never really achieve, is that not only do you understand it, but

you cannot imagine the state of not understanding it. That's the thing you should really aim for. And you can only do that asymptotically. You can never really achieve that, but...

This is just at the axiomatic level, and then there was the notion of a field. *[writes "Field"]* The group is by itself an algebraic object; the field is a particular case of a ring, *[writes "Field is a particular case of a Ring"]* meaning a field is a ring with some additional structure, and the additional structure is that non-zero elements have multiplicative inverses. So, you can think about the whole world of rings: some of them are fields. But rings and groups are separate, although they are related by the fact that the additive part of the ring is an abelian group.

Axiomatic Organization of Rings and Fields

Recap: The ring axioms were introduced in Week 1 (Definition 1.8), together with the standing convention that rings are commutative (Remark 1.17). The lecturer repeats them in the compact form below.

Definition 2.8 (*Ring with Identity*): A *"ring"* is a 5-tuple $(R, +, \cdot, 0, 1)$ such that:

(i) Additive Abelian Group: $(R, +, 0)$ is an abelian group (associative, commutative, with additive identity 0 and additive inverses $-a$).

(ii) Multiplicative Monoid: Multiplication $\cdot$ is associative and possesses a multiplicative identity $1 \in R$:

$$(a \cdot b) \cdot c = a \cdot (b \cdot c), \quad 1 \cdot a = a \cdot 1 = a \quad \forall a, b, c \in R.$$

(iii) Distributive Laws ($2\times$ Distr): Multiplication distributes over addition from both sides:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c),$$
$$(a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

In this course, the multiplication is moreover commutative (Remark 1.17). The case $1 = 0$ is allowed; it gives the zero ring (Proposition 1.10).

Definition 2.9 (*Field*): A *"field"* is a commutative ring $(R, +, \cdot, 0, 1)$ with $0 \neq 1$ in which every non-zero element possesses a multiplicative inverse:

$$\forall a \in R \setminus \{0\} \; \exists a^{-1} \in R \colon a \cdot a^{-1} = 1.$$

Equivalently, a field is a ring whose non-zero elements form an abelian group under multiplication:

$$(R \setminus \{0\}, \cdot, 1) \text{ is an abelian group.}$$

***Insight:* Asymptotic Understanding in Mathematics**

The lecturer highlights a profound pedagogical concept: true mathematical mastery is not merely recognizing or memorizing formal axioms, but reaching an *"asymptotic understanding"* where a conceptual framework becomes so deeply internalized that one can no longer imagine the mental state of not understanding it.

***Insight:* The Hierarchy of Algebraic Structures**

Algebra organizes mathematical systems by layering structural axioms:

- **Groups** $(G, \cdot, e)$**:** A single operation with associativity, identity, and inverses.
- **Rings** $(R, +, \cdot, 0, 1)$**:** Two operations where $(R, +, 0)$ is an abelian group, multiplication is associative with 1 (and, in this course, commutative), and the operations are tied together via the distributive laws.
- **Fields** $(F, +, \cdot, 0, 1)$**:** Commutative rings where $(F \setminus \{0\}, \cdot, 1)$ is also an abelian group.

Board Transition

The lecturer erases the far-left chalkboard to transition from algebraic structures to structure-preserving maps between them.

Group Homomorphisms and Isomorphisms

Definition of a Group Homomorphism

Spoken Transcript [00:07:18 - 00:09:02]

Okay, so, those were the structures. And in algebra, after you have structures, there's the notion of the maps that respect the structures, which are the homomorphisms. And you've seen this before, and we discussed it a little bit last week, but I want to put them on equal footing here.

If you have two groups, *[writes $G \to \hat{G}$]* then we want to consider maps that respect the group structure. A map f is called a group homomorphism *[writes $f\colon G \to \hat{G}$, "f is a group homomorphism"]* if... That's the definition of this term: *"group homomorphism"*. If $f(a \cdot b) = f(a) \cdot f(b)$ *[writes $f(a \cdot b) = f(a) \cdot f(b)$]* for all $a, b \in G$. *[writes $\forall a, b \in G$]*

Last week, I did something complicated where G was $(G, \star)$ and $\hat{G}$ was $(\hat{G}, \hat{\star})$ to be completely clear about the products being different. I'm never going to do that again; no one ever does that. So, here, when you read this, this multiplication is the operation in the first one, and that's the operation in the second one. Is that clear? They're both just called dot.

And in fact, many people just don't write it at all. It would be kind of normal to look in a book and there'd just be nothing, because that's the standard way you deal with algebra. You don't write x times x — that's a bad example — you don't write x times y, you just write xy.

Group Homomorphisms

Definition 2.10 (*Group Homomorphism*): Let $(G, \cdot)$ and $(\hat{G}, \cdot)$ be groups. A map $f\colon G \to \hat{G}$ is called a *"group homomorphism"* if it respects the group operations:

$$f(a \cdot b) = f(a) \cdot f(b) \quad \forall a, b \in G. \tag{2.1}$$

In standard algebraic juxtaposition without explicit multiplication symbols:

$$f(ab) = f(a)f(b) \quad \forall a, b \in G.$$

Operational Interpretation of the Definition: In (2.1), the operation on the left-hand side $(a \cdot b)$ is the group multiplication in the domain G, whereas the operation on the right-hand side $(f(a) \cdot f(b))$ is the group multiplication in the codomain $\hat{G}$.

Preservation of the Neutral Element

Spoken Transcript [00:09:02 - 00:10:40]

Okay, so, that's the definition of a group homomorphism. It respects the structure. It says you can multiply here and then go over by f, or you can go over separately and multiply. That's what respecting the structure means.

Then you could say: What about the neutral element? Why am I not saying anything about the neutral element? I could. I could add that $f(e)$ is equal to e *[writes $f(e) = \hat{e}$]* — or if I want to be clear, $\hat{e}$, the e on that side. I could add that, but I don't have to. Because if I put e here, if I just use this property, I put $f(e \cdot e)$... *[writes $f(e \cdot e) = f(e) \cdot f(e)$]* There's only one thing you can do always.

So, I just stick e in here. Do you agree? And then I use the neutral property on this side to get $f(e) = f(e) \cdot f(e)$. *[writes $f(e) = f(e) \cdot f(e)$]* And then I can use the cancellation property of the group: $f(e)$ has some inverse. There exists some $y \in \hat{G}$ such that $y \cdot f(e) = \hat{e}$, *[writes $y \in \hat{G}$: $yf(e) = \hat{e}$]* right? This is the cancellation property. We've used it many times, and at some point I will stop explaining it, but anyway, here... We multiply by y on both sides, and we get $y \cdot f(e)$... *[writes $yf(e) = y(f(e)f(e))$]*

So, then this is $\hat{e}$... Then we use the... Sorry, what are we doing here? $\hat{e} \cdot f(e)$, *[writes $\hat{e} = \hat{e}f(e)$]* and this is $f(e)$. *[writes $\hat{e} = f(e)$]* So, $f(e) = \hat{e}$, okay? So, I don't have to, but you could put it in there.

Automatic Preservation of Identity

Proposition 2.11 (*Identity Preservation*): Let $f: G \to \hat{G}$ be a group homomorphism, where $e \in G$ and $\hat{e} \in \hat{G}$ denote the respective neutral elements. Then:

$$f(e) = \hat{e}.$$

Proof

Applying the homomorphism property to $e = e \cdot e \in G$:

$$\begin{aligned} & f(e \cdot e) = f(e) \cdot f(e) \\ \implies\ & f(e) = f(e) \cdot f(e). \end{aligned} \tag{2.2}$$

Since $f(e) \in \hat{G}$, it possesses an inverse $y \in \hat{G}$ such that $y \cdot f(e) = \hat{e}$. Multiplying both sides of (2.2) on the left by y:

$$\begin{aligned} y \cdot f(e) &= y \cdot \big(f(e) \cdot f(e)\big) \\ \hat{e} &= \big(y \cdot f(e)\big) \cdot f(e) && \text{(associativity in } \hat{G}\text{)} \\ \hat{e} &= \hat{e} \cdot f(e) \\ \hat{e} &= f(e). \end{aligned}$$

Thus, $f(e) = \hat{e}$. **Q.E.D.**

Group Isomorphisms and Characterization

Spoken Transcript [00:10:40 - 00:11:45]

So, that's for a group. The group is the algebraic structure, and if you have two groups, you compare them with a homomorphism: it's a map that respects the structure. And you could say that the next thing you see — this is following the pattern in analysis, and in linear algebra also: there you have two vector spaces, and the map is a linear transformation. That's the notion of the map that respects the vector space structure on both sides.

And then there are some standard things to say: we say f is a group isomorphism *[writes "f is a group isomorphism"]* if f is a group homomorphism *[writes "if f is a group hom."]* and f is bijective. *[writes "and f is bijective"]* Okay? This should not surprise you. You've seen these terms in linear algebra, right? Which language did you learn linear algebra in?

Student Answer

English.

Spoken Transcript [00:11:49 - 00:12:43]

English, okay. Well, that's no problem then.

And there's another way to say this: f is a group isomorphism *[writes "f is group isom."]* if and only if *[writes "iff"]* there exists an inverse — there exists a group homomorphism in the other direction. There exists a group homomorphism *[writes $\exists g\colon \hat{G} \to G$ is hom.]* such that if I do f first and then g, I get the identity on the first group; and if I do g first and then f, I get the identity on the second group. *[writes "such that $g \circ f = 1_G$, $f \circ g = 1_{\hat{G}}$"]*

These are equivalent, okay? So, if you have the bottom, it's certainly bijective, because you have the forward and backward maps. And if you have the top, since it's bijective, there exists an inverse function, and you have to show that that's a homomorphism, which I leave to you to show. I hope this is clear.

Group Isomorphisms

Definition 2.12 (*Group Isomorphism*): A group homomorphism $f: G \to \hat{G}$ is called a *"group isomorphism"* if f is bijective. Two groups G and $\hat{G}$ are called *"isomorphic"* (written $G \cong \hat{G}$) if there exists a group isomorphism between them.

Proposition 2.13 (*Alternative Characterization of Isomorphism*): A map $f: G \to \hat{G}$ is a group isomorphism if and only if there exists a group homomorphism $g: \hat{G} \to G$ such that:

$$g \circ f = 1_G \quad \text{and} \quad f \circ g = 1_{\hat{G}}, \tag{2.3}$$

where $1_G = \mathrm{id}_G$ denotes the identity map of G. (For maps of sets, this is the characterization of bijections in Fact 0.3; the new point is that g is again a homomorphism.)

Proof

Direction ($\Leftarrow$): If homomorphisms f and g satisfy (2.3), then g serves as a set-theoretic two-sided inverse of f. Hence, f is a bijection, and since it is given as a homomorphism, it is an isomorphism.

Direction ($\Rightarrow$): Suppose f is a bijective group homomorphism. As a bijection, there exists a unique set-theoretic inverse map $g = f^{-1}: \hat{G} \to G$ satisfying (2.3). It remains to show that g is a homomorphism. Let $\hat{a}, \hat{b} \in \hat{G}$, and let $a = g(\hat{a})$, $b = g(\hat{b}) \in G$. Then $f(a) = \hat{a}$ and $f(b) = \hat{b}$. Since f is a homomorphism:

$$f(ab) = f(a)f(b) = \hat{a}\hat{b}.$$

Applying $g = f^{-1}$ to both sides:

$$g(\hat{a}\hat{b}) = ab = g(\hat{a})g(\hat{b}).$$

Thus, g is a group homomorphism. Q.E.D.

Examples: Isomorphisms of the Real Numbers

Spoken Transcript [00:12:43 - 00:14:43]

All right, so, maybe an example.

If I take the real numbers, one group is the real numbers with addition and zero: *[writes $(\mathbb{R}, +, 0)$]*. This is a rather basic group: real numbers, zero, and you can add them.

Another group, which you maybe don't think about as much, is that you could take the non-zero real numbers, multiplication, and one: *[writes $(\mathbb{R}^*, \cdot, 1)$]*. Here, I just remind you — this is pretty standard — when you have a field and you put a star, *[writes note: $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$]* this usually, but not always, means you take the field and throw out zero. Is that clear? So, this is the non-zero real numbers with multiplication and one. That is also a group. Are you happy with that? Because you can invert non-zero numbers.

And there's another one here: I can take the non-zero ones, but there's kind of two ways to be non-zero: you can be positive, or you can be negative. So, I could take the positive ones: *[writes $(\mathbb{R}^*_{>0}, \cdot, 1)$]*. That's still a group. Is that clear? If I try to take the

negative ones, I kind of get in trouble, because if I multiply negative ones, I get positive, and one is not there. So, anyway, there are these three groups.

Which of these are isomorphic, or which ones are not isomorphic? Do you know this? Well, maybe I will write the question: Which are isomorphic? *[writes "Which are isomorphic?"]* Which means: For which ones can I find an isomorphism?

Three Real Number Groups

Consider the following three groups based on subsets of $\mathbb{R}$:

(1) The additive group of real numbers:

$$(\mathbb{R}, +, 0).$$

(2) The multiplicative group of non-zero real numbers:

$$(\mathbb{R}^*, \cdot, 1), \quad \text{where } \mathbb{R}^* := \mathbb{R} \setminus \{0\}.$$

(3) The multiplicative group of strictly positive real numbers:

$$(\mathbb{R}^*_{>0}, \cdot, 1), \quad \text{where } \mathbb{R}^*_{>0} := \{x \in \mathbb{R} \mid x > 0\}.$$

Question 2.14: Which of these three groups are isomorphic to each other?

Board Transition

The lecturer pulls down and wipes clean a section of the lower chalkboard to systematically investigate potential isomorphisms between these groups.

Spoken Transcript [00:15:26 - 00:16:21]

Let's start with this — this is maybe the more confusing one. If I take the non-zero real numbers, multiplication, and 1, *[writes $(\mathbb{R}^*, \cdot, 1)$]* and I take the positive ones, multiplication, and 1, *[writes $(\mathbb{R}^*_{>0}, \cdot, 1)$]* could these be isomorphic? *[writes "Could these be isom.?"]*

It's kind of strange, because if you draw the picture, one of them is kind of both sides, and the other one is just one side. *[sketches real lines for $\mathbb{R}^*$ and $\mathbb{R}^*_{>0}$]* So, it seems kind of unlikely, although there are still bijections that do this. So, the question is: Is there a bijection which respects... You know, there are set-theoretic bijections, because both sets are just uncountable sets; they have the same cardinality.

What do you think? Yeah?

Student Answer

No, because we run into a problem for -1, because, um, if we had if $f(-1)$...

Spoken Transcript [00:16:31 - 00:16:33]

Which way is f going?

Student Answer, continued

So, f goes from, uh, $\mathbb{R}^*$ to $\mathbb{R}^*_{>0}$.

Spoken Transcript [00:16:36 - 00:16:40]

Yeah. *[writes $f\colon \mathbb{R}^* \to \mathbb{R}^*_{>0}$]*

Student Answer, continued

And then f of minus $f(1)$...

Spoken Transcript [00:16:42 - 00:16:45]

Has to be 1, that's the... *[writes $f(1) = 1$]*

Student Answer, continued

Yeah, that's 1. But that's also $f(-1)$...

Spoken Transcript [00:16:47 - 00:16:50]

Yeah, so, $f(-1)$ equals what? That's a good... *[writes $f(-1) = ?$]*

Student Answer, continued

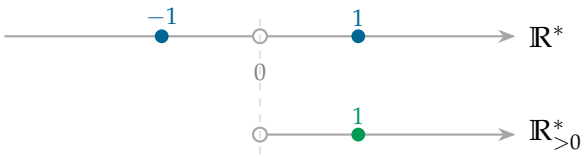
$f(-1) \cdot f(-1)$ will be 1, so the only positive number such that, uh, its times itself is 1 would be 1. So, $f(-1)$ has to be 1. So, $f(1) = f(-1)$, and then it's not bijective.

Spoken Transcript [00:17:10 - 00:18:00]

Yeah, this is all correct. The simplest way to say it is that $f(-1)$ has to be a square root of 1, *[writes $f(-1) \cdot f(-1) = f((-1) \cdot (-1)) = f(1) = 1$]* by this law. And there's only one square root of 1 in the positive numbers, and that's already taken by 1. So, it violates bijectivity.

Do you understand this statement? This was maybe not said as clearly as possible, but the point is that 1 has two elements in here — there are two elements which, when you multiply them by themselves, you get 1: $+1$ and -1. And those two elements have to go to two different elements here, because it's a bijection. If there were a bijection, they would have to go to two different elements. But here, there's only one element whose square is 1. So, it's impossible that there is a bijection, okay? This is a solid argument; this is the standard argument.

Non-Isomorphism of $(\mathbb{R}^*, \cdot, 1)$ and $(\mathbb{R}^*_{>0}, \cdot, 1)$



Proposition 2.15: The groups $(\mathbb{R}^*, \cdot, 1)$ and $(\mathbb{R}^*_{>0}, \cdot, 1)$ are not isomorphic:

$$(\mathbb{R}^*, \cdot, 1) \ncong (\mathbb{R}^*_{>0}, \cdot, 1).$$

Proof

Suppose for contradiction there exists an isomorphism $f\colon \mathbb{R}^* \to \mathbb{R}^*_{>0}$. By Proposition 2.11, any homomorphism must map the identity element to the identity element:

$$f(1) = 1.$$

Now consider the element $-1 \in \mathbb{R}^*$. Since $(-1)\cdot(-1) = 1$, the homomorphism property implies:

$$f(-1)^2 = f(-1)\cdot f(-1) = f\big((-1)\cdot(-1)\big) = f(1) = 1.$$

Thus, $f(-1)$ must be a solution to the equation $y^2 = 1$ in the codomain $\mathbb{R}^*_{>0}$. However, in $\mathbb{R}^*_{>0}$, all elements are strictly positive, and the only positive real number satisfying $y^2 = 1$ is $y = 1$:

$$f(-1) = 1.$$

This implies $f(-1) = f(1) = 1$. Since $-1 \neq 1$, f cannot be injective, contradicting the assumption that f is an isomorphism. Q.E.D.

Algebraic Invariants: Elements of Order 2: The proof uses an *"algebraic invariant"*, a property preserved by every isomorphism. In $(\mathbb{R}^*, \cdot, 1)$, the element -1 has order 2 ($x \neq e$ with $x^2 = e$, Definition 0.24), while $(\mathbb{R}^*_{>0}, \cdot, 1)$ has no element of order 2. An isomorphism maps elements of order 2 to elements of order 2, so the two groups are not isomorphic, although as sets they have the same cardinality (Fact 0.4).

Spoken Transcript [00:18:00 - 00:18:23]

Now, what about this one? *[writes $(\mathbb{R}, +, 0)$ and $(\mathbb{R}^*_{>0}, \cdot, 1)$]* $(\mathbb{R}, +, 0)$, $(\mathbb{R}^*_{>0}, \cdot, 1)$. If I make any mistakes on the board, you have to tell me, all right?

So, what about this? Could these be isomorphic? Yeah?

Student Answer

I think it could. We can take, like, f from the first one, from $\mathbb{R}, +$ to $\mathbb{R}^*_{>0}$ by the function e to the power of x.

Spoken Transcript [00:18:32 - 00:19:27]

Yeah, yeah. This is like the most famous function in mathematics, and this is its purpose, actually. This is by exp: *[writes exp and arrow]* $\exp(x)$. So, $f(x) = \exp(x)$. *[writes $f(x) = \exp(x)$]* You take a number, and you get a positive number. If you take 0, you get 1.

And the main property of exp is that it intertwines the addition here with the multiplication there. So, it's actually the most famous function in mathematics, basically. And this is its role: it gives you an isomorphism — a group isomorphism between the additive structure and the multiplicative structure.

So, that's a kind of weird thing. It's saying that, from the point of view of the group, the group doesn't see the difference between the additive structure and the multiplicative structure of the reals.

Isomorphism via the Exponential Map

Proposition 2.16: The additive group of real numbers is isomorphic to the multiplicative group of positive real numbers:

$$(\mathbb{R}, +, 0) \cong (\mathbb{R}^*_{>0}, \cdot, 1).$$

Proof

Define the map $f\colon \mathbb{R} \to \mathbb{R}^*_{>0}$ by the natural exponential function:

$$f(x) := \exp(x) = e^x.$$

We verify that f is a group isomorphism:

(i) Homomorphism Property: For any $x, y \in \mathbb{R}$, by the fundamental exponent law (Fact 0.37):

$$f(x+y) = \exp(x+y) = \exp(x) \cdot \exp(y) = f(x) \cdot f(y).$$

Thus, f converts addition in $(\mathbb{R}, +)$ into multiplication in $(\mathbb{R}^*_{>0}, \cdot)$.

(ii) Identity Preservation:

$$f(0) = \exp(0) = 1.$$

(iii) Bijectivity: The exponential function $\exp\colon \mathbb{R} \to \mathbb{R}^*_{>0}$ is strictly monotonically increasing and continuous, with image $(0, \infty) = \mathbb{R}^*_{>0}$. It possesses a two-sided inverse given by the natural logarithm:

$$g\colon \mathbb{R}^*_{>0} \to \mathbb{R}, \quad g(y) := \ln(y),$$

satisfying $g(f(x)) = \ln(e^x) = x$ and $f(g(y)) = e^{\ln(y)} = y$.

Therefore, $f = \exp$ is a bijective group homomorphism, establishing $(\mathbb{R}, +, 0) \cong (\mathbb{R}^*_{>0}, \cdot, 1)$. Q.E.D.

***Insight:* The Fundamental Role of the Exponential Function**

From an algebraic perspective, the exponential function $x \mapsto e^x$ is not merely an analytic power series or a solution to $y' = y$; its primary algebraic purpose is to establish an exact

equivalence (*"group isomorphism"*) between the continuous additive world $(\mathbb{R}, +)$ and the continuous positive multiplicative world $(\mathbb{R}^*_{>0}, \cdot)$.

Kernels and Subgroups

The Kernel of a Group Homomorphism

Spoken Transcript [00:19:27 - 00:20:55]

Okay, so, maybe I'll say two more things about it.

There's a notion of a subgroup. We did this for the ring (Recall: Definition 1.27, Proposition 1.29); I'm going to repeat it all for the ring in one second. If you have this map, $f\colon G \to \hat{G}$ — and I'm not going to keep saying *"group homomorphism"*; the standard is to say *"homomorphism"*. It is a homomorphism, which is abbreviated hom, *[writes "is a hom"]* and you're supposed to understand, since I'm talking about groups, that it's a group homomorphism. All right?

If f is a group homomorphism, then in the spirit of linear algebra — which is the algebra you're supposed to be most familiar with — if you have a linear transformation between two vector spaces, the thing you immediately study is the kernel and the image of that linear transformation. We can do that also with groups: we can consider the kernel and the image. *[writes "Then we can consider* $\ker$, Im*"]* Let's start with the kernel. What properties does this kernel enjoy?

The kernel of a group homomorphism is defined to be those elements in G such that when I apply f, I get the neutral element on the second side: *[writes* $\ker(f) := \{x \in G \mid f(x) = \hat{e}\}$*]*. In the linear algebra language, that's things that went to zero. Now it's the elements that go to the neutral element. Is that clear?

Definition of the Kernel

Definition 2.17 (*Kernel*): Let $f\colon G \to \hat{G}$ be a group homomorphism, with $\hat{e} \in \hat{G}$ the neutral element of $\hat{G}$. The *"kernel"* of f, denoted $\ker(f)$, is defined as the preimage of the neutral element:

$$\ker(f) := \{x \in G \mid f(x) = \hat{e}\}. \tag{2.4}$$

For a ring homomorphism, viewed as a homomorphism of the additive groups, this is the kernel of Week 1 (Definition 1.27).

Analogy with Linear Algebra: In linear algebra, the kernel of a linear map $T\colon V \to W$ consists of all vectors mapped to the additive identity $\mathbf{0}_W$: $\ker(T) = \{v \in V \mid T(v) = \mathbf{0}_W\}$. In group theory, $\ker(f)$ similarly collects all elements mapped to the neutral element $\hat{e}$ of the target group.

Properties of the Kernel

Spoken Transcript [00:20:55 - 00:22:06]

Let's try to look at properties of this. Well, the neutral element itself is in here: *[writes $e \in \ker(f)$]*. That's one property, because it's one of the properties of the homomorphism: the homomorphism takes the neutral element here to the neutral element there. So, that's there, right?

The second property: suppose a and b are in the kernel of f. *[writes "suppose $a, b \in \ker(f)$, then"]* What can we say? We can consider what happens to $f(ab)$. There's only one thing we can do here. These proofs at this level are kind of strong proofs, meaning that there's only one thing you can do at every stage, and you just have to do it.

This is $f(a)f(b)$. *[writes $f(ab) = f(a)f(b)$]* And now, the assumption is that they're both in the kernel, so, these are both $\hat{e}$, *[writes $= \hat{e} \cdot \hat{e}$]* and then we use the identity property: that's $\hat{e}$. *[writes $= \hat{e}$]* What have we learned here? That says that if a and b are in the kernel, we have proven that $a \cdot b$ is in the kernel, *[writes $ab \in \ker(f)$]* okay?

Kernel Properties: Identity and Closure Under Products

Property 1 (Identity): The neutral element $e \in G$ always belongs to the kernel:

$$e \in \ker(f) \quad (\text{since } f(e) = \hat{e} \text{ by Proposition 2.11}). \tag{2.5}$$

Property 2 (Closure under Multiplication): If $a, b \in \ker(f)$, then their product ab is also in the kernel:

$$\begin{aligned} f(ab) &= f(a) \cdot f(b) && \text{(homomorphism property)} \\ &= \hat{e} \cdot \hat{e} && \text{(since } a, b \in \ker(f)\text{)} \\ &= \hat{e}. \end{aligned} \tag{2.6}$$

Thus, $ab \in \ker(f)$.

Board Transition

The lecturer pushes the middle boards upwards and cleans the lower chalkboard to address the final property of the kernel.

Spoken Transcript [00:22:06 - 00:23:03]

So, we're looking at properties of the kernel, and we have discovered that the neutral element is in the kernel, and the kernel is closed under the group law, the product.

We might be curious now whether the inverses are there also. That's the third property of the kernel.

Spoken Transcript [00:23:03 - 00:25:05]

The third property of the kernel is that if x is in the kernel of f, *[writes "if $x \in$*

$\ker(f)$ "] then the inverse... I should make this comment that very often the inverse for the group law is denoted x inverse: *[writes "inverse for the group law is denoted x^{-1}"]*. That's just the name of it. The axioms say there exists an inverse; this is very often the name that's given to it.

This kind of presupposes that the product you're calling dot. If the product that you're calling is plus, then you usually use the minus sign here. But this is just notation. You could call it anything you want — you could call it Fred or an elephant, anything. We're calling it this, okay?

Then the inverse is also in the kernel: *[writes "then $x^{-1} \in \ker(f)$"]*.

And how are we going to prove this? I'm not going to go through this proof, but you have to start out like this: *[writes $f(x \cdot x^{-1})$]* $f(x \cdot x^{-1})$. That's the one thing you know. On the one hand, this is $f(e)$, and that's $\hat{e}$. *[writes $= f(e) = \hat{e}$]* On the other hand, this is $f(x) \cdot f(x^{-1})$, *[writes $= f(x) \cdot f(x^{-1})$]* and x is assumed to be in the kernel, so, this is equal to... Actually, I'm finishing it now, unfortunately. *[writes $= \hat{e} \cdot f(x^{-1}) = f(x^{-1})$]*

So, we get that the neutral element is equal to the neutral element times that, so, that must be the neutral element, okay? This implies that $x^{-1} \in \ker(f)$. *[writes $\hat{e} = f(x^{-1}) \implies x^{-1} \in \ker(f)$]* Is that clear? We have the neutral element, we do all this chain of things, and we get that the neutral element is the neutral element times this. We could go one more if we want to be really explicit: then we have the full chain, the neutral element is equal to that.

So, the kernel enjoys these three properties: it contains the neutral element of G, the source group; it's closed under the group product; and it's closed under inverses.

Consolidated Properties of the Kernel

Proposition 2.18 (*Properties of the Kernel*): Let $f\colon G \to \hat{G}$ be a group homomorphism. The kernel $\ker(f) \subseteq G$ satisfies the following three fundamental structural properties:

(1) Contains the Identity: $e \in \ker(f)$.
(2) Closure under Products: $\forall a, b \in \ker(f) \implies ab \in \ker(f)$.
(3) Closure under Inverses: $\forall x \in \ker(f) \implies x^{-1} \in \ker(f)$.

Proof of Property (3)

Let $x \in \ker(f)$, so that $f(x) = \hat{e}$. Consider the evaluation of $f(x \cdot x^{-1})$:

$$f(x \cdot x^{-1}) = f(e) = \hat{e} \qquad \text{(by Proposition 2.11)}.$$

On the other hand, by the homomorphism property:

$$\begin{aligned} f(x \cdot x^{-1}) &= f(x) \cdot f(x^{-1}) \\ &= \hat{e} \cdot f(x^{-1}) \qquad \text{(since } x \in \ker(f)\text{)} \\ &= f(x^{-1}). \end{aligned}$$

Equating both expressions yields:

$$f(x^{-1}) = \hat{e} \implies x^{-1} \in \ker(f).$$

This completes the proof of all three properties. Q.E.D.

Definition of a Subgroup

Spoken Transcript [00:25:18 - 00:26:29]

This is the definition of a subgroup. We will come back to this in a second.

Definition: *[writes "Def:"]* If H is a subset of G, *[writes $H \subset G$]* then H is a subgroup *[writes "then H is a subgroup"]* if zero is in H *[writes "if $0 \in H$"]* — as we've shown — H is closed under the product, *[writes "H is closed under the product"]* and H is closed under inverses, *[writes "and H is closed under inverse"]* meaning if an element is in H, then its inverse is also in H, okay?

In particular, this implies that... That's the definition of a subgroup. This implies that if I take H, and I take the group law — which makes sense on H because H is closed under it — and I take the zero, which also makes sense because zero is in H, then this is a group: *[writes $(H, \cdot, 0)$ is group]*. Which is a very nice thing: a subgroup is a group. It would be terrible if it weren't, right?

That's the definition of a subgroup.

Definition of a Subgroup

Definition 2.19 (*Subgroup*): Let $(G, \cdot, e)$ be a group. A subset $H \subseteq G$ is called a *"subgroup"* of G (denoted $H \leq G$) if it satisfies the following three conditions:

1. **Contains the Identity:** $e \in H$ (often denoted $0 \in H$ in additive notation).
2. **Closure under Multiplication:** $\forall a, b \in H \implies a \cdot b \in H$.
3. **Closure under Inverses:** $\forall x \in H \implies x^{-1} \in H$.

In particular, the restriction of the group law to H endows $(H, \cdot, e)$ with the structure of a group in its own right.

Inherited Axioms: Associativity holds automatically for elements in H because it holds for all elements in the ambient group G. Therefore, checking identity and closures under products and inverses is necessary and sufficient for H to form a group.

Kernel and Image as Subgroups

Spoken Transcript [00:26:29 - 00:28:05]

And now we've proven a little lemma here — proposition, or however you want to call it: The kernel of f is a subgroup of G: *[writes "Prop The $\ker(f)$ is subgroup $\leq G$"]*. Is that clear?

That's the investigation of the kernel and the notion of a subgroup, but there's also the image. And so, proposition — and I leave this for you, you can prove it: The image of f is a subgroup of $\hat{G}$, *[writes "Prop The $\mathrm{Im}(f)$ is subgroup $\leq \hat{G}$"]* okay?

How do you prove this? You just check that the image contains the neutral element. It has to, because the neutral element from the first group goes to the neutral element of the second group, so, the image has to contain the neutral element. And then you have

to check that if you have two elements in the image, when you apply the group law for $\hat{G}$, they stay in the image; and the same for inverses. The proofs are all just the same: you just follow your nose. There's only one thing you can do, and you can do it.

The Kernel and Image are Subgroups

Proposition 2.20 (*Kernel as a Subgroup*): Let $f: G \to \hat{G}$ be a group homomorphism. Then the kernel of f is a subgroup of the domain:

$$\ker(f) \leq G.$$

Proposition 2.21 (*Image as a Subgroup*): Let $f: G \to \hat{G}$ be a group homomorphism. Then the image of f, defined as

$$\mathrm{Im}(f) := f(G) = \{f(x) \in \hat{G} \mid x \in G\},$$

is a subgroup of the codomain:

$$\mathrm{Im}(f) \leq \hat{G}.$$

Proof of Proposition 2.21

We verify the three subgroup axioms for $\mathrm{Im}(f) \subseteq \hat{G}$:

(1) Identity: By Proposition 2.11, $f(e) = \hat{e}$. Since $e \in G$, $\hat{e} \in \mathrm{Im}(f)$.

(2) Closure under Multiplication: Let $\hat{a}, \hat{b} \in \mathrm{Im}(f)$. Then there exist $a, b \in G$ such that $f(a) = \hat{a}$ and $f(b) = \hat{b}$. Since G is a group, $ab \in G$, and by the homomorphism property:

$$\hat{a} \cdot \hat{b} = f(a) \cdot f(b) = f(ab) \in \mathrm{Im}(f).$$

(3) Closure under Inverses: Let $\hat{x} \in \mathrm{Im}(f)$ with $\hat{x} = f(x)$ for some $x \in G$. Since $x^{-1} \in G$:

$$\hat{x} \cdot f(x^{-1}) = f(x) \cdot f(x^{-1}) = f(x \cdot x^{-1}) = f(e) = \hat{e}.$$

By uniqueness of inverses in $\hat{G}$, $(\hat{x})^{-1} = f(x^{-1}) \in \mathrm{Im}(f)$.

Thus, $\mathrm{Im}(f) \leq \hat{G}$. Q.E.D.

Pedagogical Aside: Mathematical Reading and Textbooks

Spoken Transcript [00:28:05 - 00:29:38]

This is a nice situation for groups: the understanding of the group is given by these very simple axioms. There are lots of examples — in fact, part of this class is the examples for groups, which will come later. But anyway, there's the notion of a homomorphism, which respects the group structure; and whenever you have one, there's the kernel and the image.

This one I leave to you. You can look in the books, too, but it's not worth looking in the books. Because if you read a book... Actually, I'm very much for reading books; this is maybe the wrong way to order the discussion. You should read the books — I take it back! But the problem with reading such a simple thing in a book is that you have to learn what notation the author is using and all that. It's much simpler just to write the

proof of this yourself.

Actually, I'm a big fan of books. That's one way to understand this kind of classical history of mathematics: read a lot of math books. And I know that somehow in the modern world this is less popular, but in my generation we read a lot of math books.

And on the topic of books: there's the book for the course, which was picked a while ago — not by me even — and it has the advantage that it's freely available on the ETH website, so, you can just click it. Then there are some other references I gave, which are, I think, a little bit better books: they're more advanced in perspective, they're written better. But I think they're not freely available on the webpage, so, it wasn't chosen for this class. But it's a good idea to read books; you'll learn something.

Textbook Study and Literature Recommendations

The lecturer reflects on the pedagogical value of reading mathematical literature: while textbooks are invaluable for developing deep perspective, elementary verifications (such as checking subgroup axioms for the image) are best proven independently by the student to avoid the cognitive overhead of unfamiliar author-specific notations.

Ring Homomorphisms, Ideals, and Subrings

Definition of a Ring Homomorphism

Spoken Transcript [00:29:39 - 00:31:07]

Okay, so, for the ring, *[writes* $(R, +, \cdot, 0, 1)$*]* we had also defined what a homomorphism is (Recall: Definition 1.18) — a ring homomorphism. If I have two rings, *[writes* $R \to \hat{R}$*]* then we had defined that f is a... This is from last week: a ring hom, *[writes "is a ring hom"]* which, as I said, I will just say *"hom"*, and you're supposed to understand they're rings. And the axioms that a ring hom has to satisfy, if you remember: the first one is this one: $f(a+b) = f(a) + f(b)$ *[writes* $f(a+b) = f(a) + f(b)$*]*.

The interpretation of this is: the additive part of this is a group, and the additive part of that is a group, and the ring hom should be a group hom for those parts. Is that clear? In words, additively, f is a group hom for the plus part, okay? *[writes "additively* f *is group hom for* $+$*"]*

Of course, we also have that it respects multiplication: $f(ab) = f(a)f(b)$, *[writes* $f(ab) = f(a)f(b)$*]* and then we also have that the identity has to be respected: $f(1) = 1$ *[writes* $f(1) = 1$*]*.

Those are the notions of a ring homomorphism from last time. I just write it here.

Definition of a Ring Homomorphism

Recap: Ring homomorphisms were defined in Week 1 (Definition 1.18), where it was also shown that $f(0) = 0$ follows from additivity (Proposition 1.19) while $f(1) = 1$ must be required (Example 1.21). The lecturer repeats the definition and now phrases it

with the group notions of this lecture.

Definition 2.22 (*Ring Homomorphism*): Let $(R, +, \cdot, 0, 1)$ and $(\hat{R}, +, \cdot, \hat{0}, \hat{1})$ be rings. A map $f\colon R \to \hat{R}$ is called a *"ring homomorphism"* if it satisfies:

- **(i) Additivity:** $f(a+b) = f(a) + f(b)$ for all $a, b \in R$ (i.e., $f\colon (R, +, 0) \to (\hat{R}, +, \hat{0})$ is a group homomorphism of additive groups).
- **(ii) Multiplicativity:** $f(a \cdot b) = f(a) \cdot f(b)$ for all $a, b \in R$.
- **(iii) Identity Preservation:** $f(1) = \hat{1}$.

Why Must $f(1) = \hat{1}$ Be Explicitly Required?: For group homomorphisms, identity preservation $f(e) = \hat{e}$ is automatic from cancellation in a group (Proposition 2.11). In a ring, multiplicative inverses are generally not available; hence $f(1) \cdot f(1) = f(1)$ does not imply $f(1) = \hat{1}$ (for instance, the zero map $f(x) = \hat{0}$ preserves addition and multiplication, but sends $1 \mapsto \hat{0}$). Therefore, identity preservation $f(1) = \hat{1}$ must be postulated as an explicit axiom for rings with identity.

Ring Isomorphisms and Symmetry Breaking

Spoken Transcript [00:31:07 - 00:32:57]

That's the notion of a ring homomorphism. And again, we can think about ring isomorphisms: a ring isom is a bijective ring hom. *[writes "Ring isom is a bijective ring hom."]* I won't write it, but the same thing is true: another way to think about it is that if there's an isomorphism between two rings, that means that the inverse is a ring homomorphism in the other direction, just as I said with groups.

Then we can do the parallel study: if $f\colon R \to \hat{R}$ is a ring homomorphism, *[writes $f\colon R \to \hat{R}$ is ring hom.]* there are two things I can study as before: the kernel of f and the image of f. *[writes $\ker(f), \mathrm{Im}(f)$]*

We studied the kernel last week (Recall: Definition 1.27), if you remember. And now the matter is different. For groups, there was the notion of a subgroup, and it happened that both the kernel and the image were subgroups: the former was a subgroup of the source group, and the latter was a subgroup of the target group.

But for rings, the symmetry is broken. So, one has to concentrate a bit here. And by the way, that's also true for vector spaces: for vector spaces, you have a linear map between two vector spaces, the kernel is a subspace of the first one, and the image is a subspace of the second one. From the point of view of the group, that's a very familiar structure. For rings, something different happens, so, it requires some concentration.

Ring Isomorphisms and Symmetry Breaking

Definition 2.23 (*Ring Isomorphism*): A ring homomorphism $f\colon R \to \hat{R}$ is called a *"ring isomorphism"* if f is bijective. Equivalently, f is an isomorphism if and only if there exists a ring homomorphism $g\colon \hat{R} \to R$ such that $g \circ f = \mathrm{id}_R$ and $f \circ g = \mathrm{id}_{\hat{R}}$ (as for groups, Proposition 2.13).

Symmetry Breaking: Groups and Vector Spaces vs. Rings: In groups and vector spaces, kernel and image share the exact same algebraic classification:

- **Groups:** $\ker(f) \leq G$ (subgroup) and $\mathrm{Im}(f) \leq \hat{G}$ (subgroup).
- **Vector Spaces:** $\ker(T) \leq V$ (subspace) and $\mathrm{Im}(T) \leq W$ (subspace).
- **Rings (Symmetry Broken!):** The image $\mathrm{Im}(f)$ is a **subring** of $\hat{R}$, but the kernel $\ker(f)$ is **not** a subring (unless $\hat{R} = \{0\}$, since $1 \notin \ker(f)$ because $f(1) = \hat{1} \neq \hat{0}$). Instead, the kernel is an **ideal**!

Ideals in Rings and the Kernel

Spoken Transcript [00:32:57 - 00:35:06]

And for the kernel of f, we had proven last week — I think this is on the videotape even, in the second lecture last week — that the kernel of f is an ideal (Recall: Definition 1.30, Proposition 1.31). *[writes "for the $\ker(f)$, we had proven last week that $\ker(f)$ is an ideal"]* We proved that.

An ideal — and this is an extremely important concept — *[writes "Ideal $I \subset R$ satisfies"]* an ideal I of a ring satisfies the following. We defined it last time; I'll just repeat it. It satisfies one condition on the additive structure: if I take $(I,+,0)$ inside $(R,+,0)$, this is a subgroup: *[writes $(I,+,0) \subset (R,+,0)$ subgroup]*. That's the additive structure.

We wrote the axioms differently last time, but we interpret it now. For an ideal, there are kind of two pieces: one is how it works additively, and it must be a subgroup. That is not a surprise, because we already said that f on the additive structures is a group homomorphism, and we already know that the kernel from the point of view of the group homomorphism is an additive subgroup, so, it should be an additive subgroup. This is logically the same as what I explained last week.

And the second condition has to do with the multiplicative part. You can see that it's going to be different now, because we have these constraints here. The kernel satisfies this property: if I take any element of the ring and any element of the ideal, then $r \cdot x$ is in the ideal: *[writes $\forall r \in R, x \in I \implies rx \in I$]*.

These are the two conditions that define an ideal. And the proposition last week was: the kernel is an ideal. All right? That's just review.

Definition of an Ideal

Recap: Ideals were defined in Week 1 (Definition 1.30), and the kernel of a ring homomorphism was shown to be one (Proposition 1.31). The lecturer restates both; with the group notions of this lecture, the additive half of the proof becomes a one-liner.

Definition 2.24 (*Ideal of a Ring*): Let $(R,+,\cdot,0,1)$ be a commutative ring. A subset $I \subseteq R$ is called an "*ideal*" if it satisfies:

(1) Additive Subgroup: $(I,+,0) \leq (R,+,0)$, meaning:

$$0 \in I, \quad \forall x,y \in I \implies x+y \in I, \quad \text{and} \quad \forall x \in I \implies -x \in I.$$

(2) Absorption of Multiplication:

$$\forall r \in R,\ \forall x \in I \implies r \cdot x \in I. \tag{2.7}$$

Proposition 2.25 (*The Kernel of a Ring Homomorphism is an Ideal*)**:** Let $f\colon R \to \hat{R}$ be a ring homomorphism. Then:

$$\ker(f) := \{x \in R \mid f(x) = \hat{0}\}$$

is an ideal of R.

Proof

Since f is a group homomorphism between the additive groups $(R, +, 0)$ and $(\hat{R}, +, \hat{0})$, by Proposition 2.20, $(\ker(f), +, 0)$ is an additive subgroup.

To verify multiplicative absorption, let $r \in R$ and $x \in \ker(f)$ (so $f(x) = \hat{0}$). By the multiplicativity of f:

$$f(r \cdot x) = f(r) \cdot f(x) = f(r) \cdot \hat{0} = \hat{0}.$$

Hence, $r \cdot x \in \ker(f)$. Thus, $\ker(f)$ is an ideal of R. Q.E.D.

Spoken Transcript [00:35:06 - 00:36:04]

What about the image? Is the image an ideal? What is the image? The answer is: that's not true.

The ideal is the notion for the kernel. The image is something else: the image is a subring.

Board Transition

The lecturer pushes the middle boards upwards and cleans the lower chalkboard to formalize the definition of a subring.

Definition of a Subring

Spoken Transcript [00:36:04 - 00:38:17]

So, what is a subring?

Let R be a ring: *[writes "Let R be a ring $(R, +, \cdot, 0, 1)$"]*. Whenever I say *"let R be a ring"*, that means I have in mind all the data. In this class, when I say *"let R be a ring"*, it means I have in mind the whole data — you can ask me about it, or maybe we share the idea of which ring it is in common — but when you say *"let R be a ring"*, it means the whole thing.

A subring S of R satisfies the following properties — this is the definition: *[writes "A subring $S \subset R$ satisfies:"]*. And it shouldn't be surprising to you. The first property is that 0 and 1 should be in the subring: *[writes "(i) $0, 1 \in S$"]*. Just like in a group, you should start with the neutral elements.

The second is that S is closed under the two operations: *[writes "(ii) S is closed under*

$+, \cdot$ *"]*. That means whenever I have elements of the subring, I can either add them or multiply them, and they stay in the subring, okay?

And the third is that $(S, +, \cdot, 0, 1)$ is a ring: *[writes "(iii) $(S, +, \cdot, 0, 1)$ is a ring"]*. Now it makes sense to ask this thing to be a ring.

You can say it in different ways, but that's what a subring is. A subring inside here has 0 and 1, it's closed under the operations, and together it's a ring. This means it has additive inverses and things like that, okay?

And so, the proposition here is: *[writes "Prop:"]* if I have a ring homomorphism $f\colon R \to \hat{R}$, *[writes $f\colon R \to \hat{R}$ is ring hom.]* then the image of f in the second one is a subring: *[writes $\mathrm{Im}(f) \subset \hat{R}$ is a subring]*.

As I said, the kernel is an ideal, and the image is a subring.

Definition of a Subring and Image Property

Definition 2.26 (*Subring*): Let $(R, +, \cdot, 0, 1)$ be a ring. A subset $S \subset R$ is called a *"subring"* of R if:

- **(i) Neutral Elements:** $0 \in S$ and $1 \in S$.
- **(ii) Closure under Operations:** For all $x, y \in S$, $x + y \in S$ and $x \cdot y \in S$.
- **(iii) Ring Structure:** $(S, +, \cdot, 0, 1)$ forms a ring with the operations restricted from R (in particular, S contains additive inverses: $\forall x \in S \implies -x \in S$).

Proposition 2.27 (*Image as a Subring*): Let $f\colon R \to \hat{R}$ be a ring homomorphism. Then the image $\mathrm{Im}(f) \subset \hat{R}$ is a subring of $\hat{R}$.

Verification of the Subring Property for the Image

Spoken Transcript [00:38:20 - 00:39:53]

Formally, we should prove this. But this is again a strong proof: you get on this road and there are just no turns, you just have to go on it. So, I'm not really going to give you this proof; you should think about it. But you have to check... *[writes "Check:"]*

Let's just start — I'm definitely not going to finish it. You just have to check that the image satisfies these properties. Does the image have 0? The answer is yes, because on the additive structure it's a homomorphism of groups, and 0 goes to 0. So, the image has 0; that's correct: *[writes $\mathrm{Im}(f) \ni \hat{0}$ (grp hom)]*.

Does the image have 1? *[writes $\mathrm{Im}(f) \ni \hat{1}$]* Well, that's part of the definition of a homomorphism, so, it has 1. *[writes ✓]*

That's kind of a start. Then you have to check whether it's closed under plus and dot, and you would just check that by writing out the equations if you have two elements here. I leave this for you. And whether this is a ring: then you have to check whether it has additive inverses, but that's also clear. So, I leave this as an exercise: *[writes "left as an exercise"]*.

Formally, you just have to take this, start checking everything, and use the definitions. That's the only thing you could do.

Verification of the Subring Axioms for the Image

Proof of Proposition 2.27

We systematically verify conditions (i)–(iii) of Definition 2.26 for $\mathrm{Im}(f) \subset \hat{R}$:

(i) Neutral Elements:

- Since $f\colon (R,+,0) \to (\hat{R},+,\hat{0})$ is a group homomorphism, $f(0) = \hat{0}$, so $\hat{0} \in \mathrm{Im}(f)$.
- By condition (iii) of Definition 1.18, $f(1) = \hat{1}$, so $\hat{1} \in \mathrm{Im}(f)$.

(ii) Closure under Addition and Multiplication: Let $\hat{x}, \hat{y} \in \mathrm{Im}(f)$ with $\hat{x} = f(x)$ and $\hat{y} = f(y)$ for some $x, y \in R$. Then:

$$\hat{x} + \hat{y} = f(x) + f(y) = f(x+y) \in \mathrm{Im}(f),$$
$$\hat{x} \cdot \hat{y} = f(x) \cdot f(y) = f(x \cdot y) \in \mathrm{Im}(f).$$

(iii) Additive Inverses: For $\hat{x} = f(x) \in \mathrm{Im}(f)$, we have $-\hat{x} = -f(x) = f(-x) \in \mathrm{Im}(f)$; here $f(-x) = -f(x)$ because group homomorphisms preserve inverses (shown in the proof of Proposition 2.21). Associativity, commutativity and distributivity hold identically as they are inherited from $\hat{R}$.

Therefore, $\mathrm{Im}(f)$ is a subring of $\hat{R}$. Q.E.D.

Insight: **Broken Symmetry: Subrings vs. Ideals**

In group theory and linear algebra, substructures are uniform: the kernel and image of a homomorphism are both subgroups (or vector subspaces). In ring theory, this symmetry is broken:

- The **kernel** $\ker(f)$ is an **ideal** (it absorbs multiplication from the entire ring, but does not contain 1 unless $\hat{R} = \{0\}$).
- The **image** $\mathrm{Im}(f)$ is a **subring** (it contains the identity 1, but in general does not absorb multiplication from outside the image).

Examples of Subrings and Ideals

Even Integers in the Ring of Integers

Spoken Transcript [00:39:53 - 00:40:57]

It's good to have these ideas extremely clear in your head: you have the group, you have the ring, there's a notion of group hom, there's a notion of ring hom, and there's kernel and image in both cases. That's four different cases to keep in your head. For the group, the kernel and image are subgroups. For the ring, something different happens: the image is a subring, but the kernel is an ideal.

Now we should do some examples. Let's try here: *[writes $2\mathbb{Z} \subset \mathbb{Z}$]*.

$2\mathbb{Z}$ inside $\mathbb{Z}$. $\mathbb{Z}$ is a ring: *[writes "$\mathbb{Z}$ ring"]*. I can take the even integers: *[writes "$2\mathbb{Z}$ even integers"]*. Even integers — that's kind of standard notation for the even integers.

What is this inside that? Is it an ideal? Is it a subring? What's the answer? Yeah?

Student Answer

I think it is an ideal.

Spoken Transcript [00:40:59 - 00:41:32]

It's an ideal, so, claim: this is an ideal: *[writes "claim: $2\mathbb{Z}$ ideal"]*. Why is it an ideal? We have to check that additively it's a subgroup, and that's just saying that if you add two even numbers, you get an even number; zero is an even number; and if you have an even number, the negative of it is an even number. So, you're totally solid on the subgroup.

Then you have to check that if you take any other element of the integers and multiply by an even number, you stay even. That looks pretty good. The answer is yes: *[writes "YES"]*.

What about a subring? Is it a subring? Yeah?

Student Answer

No, because we don't have the 1.

Spoken Transcript [00:41:37 - 00:41:44]

Yeah, so, it's not a subring. Ideal: yes; subring: no: *[writes "subring no"]*.

The Even Integers: Ideal vs. Subring

Example 2.28 (*The Even Integers $2\mathbb{Z} \subset \mathbb{Z}$*): Consider the set of even integers $2\mathbb{Z} := \{2k \mid k \in \mathbb{Z}\}$ as a subset of the ring $(\mathbb{Z}, +, \cdot, 0, 1)$.

- **$2\mathbb{Z}$ is an ideal of $\mathbb{Z}$ (YES):**
  - **(a)** Additively, $2\mathbb{Z} \leq \mathbb{Z}$ since $0 \in 2\mathbb{Z}$, the sum of two evens is even ($2k + 2\ell = 2(k + \ell)$), and the negative of an even is even ($-(2k) = 2(-k)$).
  - **(b)** Multiplicatively, $2\mathbb{Z}$ absorbs products from all of $\mathbb{Z}$: for any $r \in \mathbb{Z}$ and $2k \in 2\mathbb{Z}$, we have $r \cdot (2k) = 2(rk) \in 2\mathbb{Z}$.
- **$2\mathbb{Z}$ is not a subring of $\mathbb{Z}$ (NO):** By Definition 2.26, a subring must contain the multiplicative identity of the ambient ring ($1 \in \mathbb{Z}$). Since 1 is odd, $1 \notin 2\mathbb{Z}$.

Intermediate Subrings between $\mathbb{Z}$ and $\mathbb{Q}$

Spoken Transcript [00:41:44 - 00:42:37]

Let's try something else: *[writes $\mathbb{Z} \subset \mathbb{Q}$]*.

This is maybe a trickier question. Certainly $\mathbb{Z}$ is a subring of $\mathbb{Q}$: *[writes $\mathbb{Z} \subset S \subset \mathbb{Q}$]*. I hope that's not controversial. You accept that $\mathbb{Z}$ is a subring of $\mathbb{Q}$?

Can you find a subring that's in between $\mathbb{Z}$ and $\mathbb{Q}$? Does there exist a subring in between? *[writes "Does there exist a subring in between?"]*

That means that these are strictly not equal: *[writes $\mathbb{Z} \subsetneq S \subsetneq \mathbb{Q}$]*.

How about that? Can you do that? You have three minutes.

Student Proposal

Look at pairs of integers, and the denominator is just powers of two.

Spoken Transcript [00:42:43 - 00:44:07]

Okay, that's good. He is saying that we let S be the set of rational numbers of this form: *[writes $S = \{\frac{a}{2^n} \mid a \in \mathbb{Z}, n \geq 0\}$]* where a is an integer and n is also an integer. Is this exactly what you're saying?

That's a proposal. One thing that's good about this proposal is that this set S is certainly more than $\mathbb{Z}$ because it has $3/2$ in it, and it's certainly less than $\mathbb{Q}$ because it doesn't have $2/3$ in it. So, that part is solid.

Then the claim is that you have to check that this satisfies all these properties. What are the properties? Having 0 and 1 — basically the answer is yes, it's going to satisfy them, because you need to know how to add fractions, which I don't teach you. I've never taught anyone how to add fractions; I'm not going to teach you! But when you think about the rule for adding fractions, you keep 2^n in the bottom; that's the main thing here. When you add fractions, the denominator doesn't give you things that you didn't put in. And so, the answer is yes, this is true: *[writes "YES"]*.

That's an example, and that's a very important one. Of course, 2 is not important here: you could have put 17 here, and it would be fine: *[writes 17^n]*. That's another example. And in fact, being prime is not that important either.

Dyadic Rationals as an Intermediate Subring

Example 2.29 (*Dyadic Rationals $\mathbb{Z}[1/2]$*): There exists a strict intermediate subring $\mathbb{Z} \subsetneq S \subsetneq \mathbb{Q}$. Consider the set of dyadic rationals:

$$S := \left\{ \frac{a}{2^n} \;\middle|\; a \in \mathbb{Z},\ n \in \mathbb{Z}_{\geq 0} \right\} = \mathbb{Z}\left[\frac{1}{2}\right]. \tag{2.8}$$

- **Proper Inclusions:**
  - For $n = 0$, $\frac{a}{2^0} = a \in S$, so $\mathbb{Z} \subseteq S$. Moreover, $\frac{1}{2} = \frac{1}{2^1} \in S \setminus \mathbb{Z}$, so $\mathbb{Z} \subsetneq S$.
  - Clearly $S \subset \mathbb{Q}$. Moreover, $\frac{1}{3} \in \mathbb{Q} \setminus S$ (since $3 \nmid 2^n$ for any $n \geq 0$), so $S \subsetneq \mathbb{Q}$.

- **Subring Axioms:**
  - **(i)** $0 = \frac{0}{2^0} \in S$ and $1 = \frac{1}{2^0} \in S$.
  - **(ii)** Closed under addition: For $\frac{a}{2^n}, \frac{b}{2^m} \in S$ with $n \leq m$:
$$\frac{a}{2^n} + \frac{b}{2^m} = \frac{a \cdot 2^{m-n} + b}{2^m} \in S.$$
  - **(iii)** Closed under multiplication:
$$\frac{a}{2^n} \cdot \frac{b}{2^m} = \frac{a \cdot b}{2^{n+m}} \in S.$$
  - **(iv)** Closed under additive inverses: $-\left(\frac{a}{2^n}\right) = \frac{-a}{2^n} \in S$.

Generalization: Ring of Fractions / Localization: As observed by the lecturer, the prime 2 can be replaced by any integer $k \geq 2$ (e.g., 17):
$$\mathbb{Z}\left[\frac{1}{k}\right] := \left\{ \frac{a}{k^n} \;\middle|\; a \in \mathbb{Z},\ n \geq 0 \right\}$$
forms a valid intermediate subring between $\mathbb{Z}$ and $\mathbb{Q}$. In abstract algebra, this construction represents the simplest instance of the *"localization"* of a commutative ring at a multiplicative set, here $\{1, k, k^2, \dots\}$: the elements of that set are made invertible. The case $k = 3$ returns in Week 3 (Example 3.29).

Polynomial Rings: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$

Spoken Transcript [00:44:07 - 00:45:04]

Another example — I should have maybe not... Do we get a bell? I don't remember. Do I get to go to the bell, or do I have to stop?

Let's try one more example before the bell, hopefully. If I look at polynomials here, these are polynomials in the variable x — that's a ring: *[writes $\mathbb{Z}[x]$.]* Inside it are polynomials in the variable x^2: *[writes $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$.]*

Every polynomial in the variable x^2 is a polynomial in x, so, this is a subset. I hope that's clear. Is this an ideal, or a subring, or both, or neither, or what?

You ran out of time. Okay, we take a break; you can think about it.

Polynomial Ring Example Question

Question 2.30: Let $\mathbb{Z}[x]$ denote the ring of polynomials with integer coefficients in the indeterminate x. Consider the subset of polynomials in x^2:
$$\mathbb{Z}[x^2] := \left\{ a_0 + a_1 x^2 + a_2 x^4 + \cdots + a_m x^{2m} \;\middle|\; m \geq 0,\ a_i \in \mathbb{Z} \right\} \subset \mathbb{Z}[x].$$
Is $\mathbb{Z}[x^2]$ an ideal, a subring, both, or neither in $\mathbb{Z}[x]$?

Mid-Lecture Break

The lecturer announces a short pause for the students to consider whether $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$ is an ideal or a subring. The lecture resumes at timestamp 00:45:05.

Spoken Transcript [00:45:05 - 00:45:20]

Okay, so, maybe we start. Or is it a minute? I don't know whether we get a bell; I don't remember, but anyway. What's the answer to this question?

Student Answer

I think it's a subring, but not an ideal.

Spoken Transcript [00:45:24 - 00:45:44]

Yeah, so, subring: yes; ideal: no: *[writes "Subring YES", "Ideal no"]*.

For the subring, again, you just have to check that everything's there, and if you look at it, well... Oh, we do get a bell! *[bell rings in background]*

Why is it not an ideal? Yeah?

Student Answer

Take x^2, and multiply by x.

Spoken Transcript [00:45:48 - 00:46:21]

x^2 is in this, *[writes $x^2 \in \mathbb{Z}[x^2]$]* and if it were an ideal, then I should be able to take any r in the ring and multiply it by this x^2. So, if I take any r in the ring and multiply by x^2, it should stay in here if it were an ideal: *[writes $r \in \mathbb{Z}[x] \implies r \cdot x^2 \in \mathbb{Z}[x^2]$]*.

I could take r to be x, *[writes $r = x$]* and then this would be odd and wouldn't be in there, okay? *[writes $x \cdot x^2 = x^3 \notin \mathbb{Z}[x^2]$]* You can check for yourself.

These are not supposed to be hard questions; it's just about whether you have absorbed the definitions. Which, as I said, is not an easy task.

Resolution: Subring vs. Ideal for $\mathbb{Z}[x^2]$

Example 2.31 (*Polynomials in x^2: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$*):

- **$\mathbb{Z}[x^2]$ is a subring of $\mathbb{Z}[x]$ (YES):**
  - **(a)** Contains $0 = 0 \cdot x^0$ and $1 = 1 \cdot x^0$.
  - **(b)** Closed under addition: The sum of two polynomials with only even powers of x again has only even powers of x.
  - **(c)** Closed under multiplication: The product of two even powers is an even power: $x^{2k} \cdot x^{2\ell} = x^{2(k+\ell)}$.
  - **(d)** Closed under additive inverses: Negating coefficients preserves even powers.
- **$\mathbb{Z}[x^2]$ is not an ideal of $\mathbb{Z}[x]$ (NO):** It fails the absorption property (2.7): for

$x^2 \in \mathbb{Z}[x^2]$, an ideal would need $r \in \mathbb{Z}[x] \implies r \cdot x^2 \in \mathbb{Z}[x^2]$. Taking $r = x$:

$$x \cdot x^2 = x^3 \notin \mathbb{Z}[x^2],$$

because x^3 has an odd degree.

Subrings of the Integers $\mathbb{Z}$

Spoken Transcript [00:46:22 - 00:46:52]

Okay, so, let's go to the integers: *[writes $\mathbb{Z}$]*. What are the subrings? *[writes "What are the subrings?"]*

We've seen that the rational numbers have lots of interesting subrings, because we found some: there are the integers, and you could invert 2, invert 3, 17. But what are the subrings of $\mathbb{Z}$? Yeah?

Student Answer

Just $\mathbb{Z}$.

Spoken Transcript [00:46:53 - 00:47:40]

Yeah. There are no real interesting subrings: only $\mathbb{Z}$ in $\mathbb{Z}$. *[writes "Only $\mathbb{Z} \subset \mathbb{Z}$"]* That's the only subring.

The whole ring is always a subring, just by definition. And this is the only one. Why is that? Because the subring would have to contain 1. Once it contains 1, it contains every positive number. But the subring also is a subgroup, so, it has to contain the negative ones, too.

So, there's nothing you can do.

Subrings of $\mathbb{Z}$

Proposition 2.32 (*Uniqueness of the Subring of $\mathbb{Z}$*): The only subring of $(\mathbb{Z}, +, \cdot, 0, 1)$ is $\mathbb{Z}$ itself.

Proof

Let $S \subseteq \mathbb{Z}$ be a subring.

(1) By condition (i) of Definition 2.26, $1 \in S$.

(2) Since S is closed under addition, by induction $n = \underbrace{1+1+\cdots+1}_{n \text{ times}} \in S$ for all positive integers $n \in \mathbb{N}_{>0}$.

(3) Since $0 \in S$ and S contains additive inverses (condition (iii)), $-n \in S$ for all $n \in \mathbb{N}_{>0}$.

Thus, S contains all integers: $S = \mathbb{Z}$. Q.E.D.

Contrast: Ideals vs. Subrings in $\mathbb{Z}$: This result highlights the sharp contrast between subrings and ideals in the ring of integers $\mathbb{Z}$:

- **Subrings of $\mathbb{Z}$:** There is only **one** subring, namely $\mathbb{Z}$ itself (because a subring must contain 1, which generates all of $\mathbb{Z}$ additively).
- **Ideals of $\mathbb{Z}$:** There are **infinitely many** distinct ideals of $\mathbb{Z}$, each of the form $n\mathbb{Z}$ for $n \in \mathbb{N}_{\geq 0}$ (because an ideal does not need to contain 1; see Theorem 2.35 below).

Ideals in the Ring of Integers

Classification of Ideals in $\mathbb{Z}$

Spoken Transcript [00:47:40 - 00:47:46]

So, what are the ideals? *[writes "What are the ideals of $\mathbb{Z}$?"]* What would you guess the answer to this question is? Yeah?

Student Answer

The set of multiples...

Spoken Transcript [00:47:48 - 00:49:02]

Yeah, so, you pick some n — *[writes $n \in \mathbb{Z}$]* you pick some integer — and you look at the multiples of n: *[writes $n\mathbb{Z} \subset \mathbb{Z}$]*.

The first comment is: this is certainly an ideal. *[points at $n\mathbb{Z} \subset \mathbb{Z}$]* That's the standard notation: n times $\mathbb{Z}$ are the multiples. If $n = 2$, it's the even numbers; when $n = 3$, it's the numbers divisible by 3, and so on. So, various comments on this: *[writes "Comment 1. Certainly an ideal"]*

1. This is certainly an ideal. Just because of the way we checked carefully that 2 — the even ones — were an ideal; it's the same argument, okay? I hope you understand that.

And the second is that it's not sensitive to whether you put $+n$ or $-n$ here, because inside $\mathbb{Z}$ is -1, so, if you put $+7$ or -7, it's the same set. Is that also clear? *[writes • $\pm n$]*

So, this doesn't matter. Typically, when you write $n\mathbb{Z}$, *[writes $n \geq 0$]* you should take $n \geq 0$. You don't have to, but that's the convention. In the literature, you'll never see minus... It's just a weird thing to do. I guess you could do it, but it's just a weird thing to do; there's no reason.

Ideals Generated by Multiples of Integers

Recap: The set $n\mathbb{Z}$ first appeared in Week 1 as the kernel of the reduction map $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ (Example 1.28).

Definition 2.33 (*Principal Ideal Generated by an Integer*): For any integer $n \in \mathbb{Z}$, the set of all multiples of n is denoted:

$$n\mathbb{Z} := \{n \cdot k \mid k \in \mathbb{Z}\} \subset \mathbb{Z}. \tag{2.9}$$

Remark 2.34 (*Properties and Conventional Normalization of $n\mathbb{Z}$*):

- **$n\mathbb{Z}$ is an ideal:** For any $n \in \mathbb{Z}$, $n\mathbb{Z}$ is an additive subgroup of $\mathbb{Z}$ and absorbs multiplication: for any $r \in \mathbb{Z}$ and $nk \in n\mathbb{Z}$, $r \cdot (nk) = n(rk) \in n\mathbb{Z}$.
- **Sign Invariance:** Since $-1 \in \mathbb{Z}$ is an invertible unit, $n\mathbb{Z} = (-n)\mathbb{Z}$ for every $n \in \mathbb{Z}$.
- **Non-negative Normalization:** By universal algebraic convention, we normalize generators to $n \geq 0$.

Historical Context: Ideal Numbers and Hilbert's Epitaph

Spoken Transcript [00:49:02 - 00:49:48]

We're going to prove these are all the ideals, but this is an important example because of the word *"ideal"*: Where does it come from? The idea is that it's an idealized number, an idealized integer. For the integers, the ideals are the integers; but when you think about other rings, the ideals are some kind of idea of a number. Maybe this is a slightly old-fashioned point of view, but...

In some sense, the origin of this word *"ideal"* here is *"ideal number"*. This terminology goes back to about 1900, and there were lots of people around, but certainly one person who played a role was the great David Hilbert, who was a mathematician in Göttingen. A piece of math culture: Does anyone know what the epitaph on the grave of David Hilbert is?

Spoken Transcript [00:49:48 - 00:49:52]

Not one person? Yeah?

Student Question

What is an epitaph?

Spoken Transcript [00:49:52 - 00:50:20]

Ah, epitaph! Very good. You have a gravestone, *[draws a gravestone shape on the board with name and motto lines]* and what's written here: there's the person's name, and then there's like a little motto or slogan. That's the epitaph. What's the word for that in German?

This is a real piece of math culture: What's the epitaph on David Hilbert's grave? These are the kinds of things you learn to be a mathematician, you know; it's part of being in the culture. Yeah, do you know it?

Student Answer

Wir müssen wissen, wir werden wissen.

Spoken Transcript [00:50:20 - 00:50:43]

Yeah! *"Wir müssen wissen, wir werden wissen."* Two lines. It's in Göttingen. *[erases the gravestone drawing]*

I have to teach you some culture, too, you know! Okay, but that's the origin of this n. So, we have to prove something.

Mathematical Culture: David Hilbert's Gravestone



Insight:* Historical Origin of Ideals and Kummer's *"Ideal Numbers"

The terminology *"ideal"* originates from Ernst Kummer's concept of **ideal numbers** (developed in the 1840s to restore unique prime factorization in cyclotomic integer rings while attempting to prove Fermat's Last Theorem), later formalized by Richard Dedekind and David Hilbert. For the integers $\mathbb{Z}$, every ideal is simply the set of multiples of an actual integer $n\mathbb{Z}$. In more general rings where unique factorization fails, ideals serve as the generalized, *"idealized"* numbers that recover unique prime decomposition.

Proof: Every Ideal of $\mathbb{Z}$ is Principal

Classification Theorem for Ideals in $\mathbb{Z}$

Theorem 2.35 (*Ideals of $\mathbb{Z}$ are Principal*): Every ideal $I \subset \mathbb{Z}$ is of the form:

$$I = n\mathbb{Z}$$

for a unique integer $n \geq 0$.

Proof of Theorem 2.35

Spoken Transcript [00:50:43 - 00:52:00]

[writes "Proof:"] What do we have to prove? Let I be an ideal in $\mathbb{Z}$: *[writes "Let $I \subset \mathbb{Z}$ be an ideal."]*.

I want to prove it's this. What am I going to do here? Maybe you prove it.

I don't like proofs that much, you know. Actually, about math, what I like is truth: proof is good, but truth is better. And this stuff is true, so, you know... But anyway, we should prove it; it's a math course.

If you want to prove it, you have to think: well, okay, Case 1... *[writes "Case (i) $I = (0) \implies 0\mathbb{Z}$"]*

Case 1 is that I could be just the zero set. That's fine. And this means it's 0 times $\mathbb{Z}$. It's kind of a degenerate case, but that's possible. And that's an ideal: zero is an ideal.

Case 2 is that I has some non-zero element. *[writes "Case (2) I has some non-zero element"]* It has to have zero, because I is an additive subgroup. If it has only zero, you're in this case. If you're not in this case, you have to have some non-zero element. Let's call that element x: *[writes $x \in I, x \neq 0, x \in \mathbb{Z}$]*.

Case Analysis: Zero vs. Non-Zero Ideal

Case (i) ($I = (0)$): If I contains only 0, then $I = (0) \implies I = 0\mathbb{Z}$, which corresponds to $n = 0$.

Case (ii) (I has some non-zero element): There exists $x \in I$ with $x \neq 0$.

Spoken Transcript [00:52:00 - 00:53:05]

Now, x is an integer. It has a non-zero element, so, x is an integer; then we can multiply by ± 1 and get a positive integer: *[writes $\pm 1 \cdot x \implies$ get $x \in \mathbb{Z}_{>0}$]*. One of these is a positive integer.

So, if it has a non-zero element, then I has a positive integer: *[writes "I has positive integer"]*. I hope that's clear.

Existence of Positive Elements in I

Since $(I, +, 0)$ is an additive subgroup of $\mathbb{Z}$, for any $x \in I$, its additive inverse $-x = (-1) \cdot x$ is also in I. Since $x \neq 0$, exactly one of x or $-x$ is strictly positive:

$$x \in I \setminus \{0\} \implies |x| \in I \cap \mathbb{Z}_{>0}.$$

Therefore, the subset of positive integers in I is non-empty:

$$I \cap \mathbb{Z}_{>0} \neq \emptyset.$$

Spoken Transcript [00:53:05 - 00:53:19]

And if it has a positive integer, what am I going to say now? This is where you pretend to be like an LLM: What am I going to say?

Yeah?

Student Answer

It has a minimum.

Spoken Transcript [00:53:19 - 00:53:46]

Yeah, it has a least positive integer. That's the next line: *[moves to the lower left board, writes] [writes "I has a least positive integer"]*. As I said, for basically every property of the integers, at some point you have to use this principle. Once it has a least positive integer, let's call that one n. So, n is equal to the least positive integer of I: *[writes n = least pos. int. of I]*.

Application of the Well-Ordering Principle

By the *"Well-Ordering Principle"* of $\mathbb{N}$ (every non-empty subset of positive integers possesses a least element, Fact 0.11), the non-empty set $I \cap \mathbb{Z}_{>0}$ contains a unique minimum element:

$$n := \min(I \cap \mathbb{Z}_{>0}). \tag{2.10}$$

Spoken Transcript [00:53:46 - 00:54:50]

Now what we have to show is that I is actually $n\mathbb{Z}$. That's what we want to show, all right? *[writes "We have now to show $I = n\mathbb{Z}$"]*

Suppose not: *[writes "Suppose not"]*. Then there's some element $y \in I$ that's not a multiple. Well, maybe I don't want to suppose not — you can do it two different ways. You look at y: y is some element of I: *[writes $y \in I$]*.

There's only one thing you can do now, which is the division algorithm. The division algorithm says y is equal to some multiple of n plus a remainder term, right? *[writes $y = qn + r$ by division]* That's long division.

And the important part of this remainder term is that $0 \leq r < n$: *[writes $0 \leq r < n$]*.

Division Algorithm Setup

We show $I = n\mathbb{Z}$:

- **Inclusion $n\mathbb{Z} \subseteq I$:** Since $n \in I$ and I is an ideal, by multiplicative absorption $\forall k \in \mathbb{Z} \implies kn \in I$, so $n\mathbb{Z} \subseteq I$.
- **Inclusion $I \subseteq n\mathbb{Z}$:** Let $y \in I$ be arbitrary. By the *"Euclidean Division Algorithm"*

(Fact 0.14), there exist unique quotient and remainder integers $q, r \in \mathbb{Z}$ such that:

$$y = q \cdot n + r \quad \text{with} \quad 0 \leq r < n. \tag{2.11}$$

Spoken Transcript [00:54:50 - 00:56:10]

Consider this element $y \in I$. Suppose it's not in $n\mathbb{Z}$: that means that it's not divisible, so, there is a remainder term. But this remainder term is equal to $y - qn$, and so, the remainder term is also in I. Because y is in I, and n is in I, so, qn is in I; and since I is an additive subgroup, this is also in I: *[writes $r = y - qn \in I$]*.

It's not here, so, we can't have zero, because otherwise it would be divisible, right? So, this r is strictly between zero and n. That's the logic: we say *"suppose not"*, and we use the division algorithm. It can't be zero, because otherwise it would divide, so, long division says r is between zero and n. But then r is a lesser positive integer in I, which is a contradiction: *[writes "but r is a less pos. int. in I (contradiction)"]*.

I don't know what the symbol for contradiction is. Anyway. Okay, that's the proof.

Spoken Transcript [00:56:10 - 00:56:31]

We used what you have to use for integers: you use this fact that non-empty subsets of positive integers have a least element. Once you have that least element, you know what you're trying to prove, and then you use the long division algorithm and the remainder. This is what you always do with these kinds of elementary properties of integers.

That proves this claim. *[draws a separating line on the board]*

Consolidated Proof: Ideals of $\mathbb{Z}$

Complete Derivation of Theorem 2.35

Let $I \subset \mathbb{Z}$ be an ideal.

Case 1: $I = \{0\}$**:** Then $I = 0\mathbb{Z}$, and $n = 0$ is unique.

Case 2: $I \neq \{0\}$**:** Choose any non-zero $x \in I$. Since I is an additive subgroup, $-x = (-1) \cdot x \in I$. Thus, $|x| \in I \cap \mathbb{Z}_{>0} \neq \emptyset$.

By the Well-Ordering Principle of $\mathbb{N}$, the non-empty set of positive elements in I has a least element:

$$n := \min(I \cap \mathbb{Z}_{>0}) > 0.$$

We show that $I = n\mathbb{Z}$:

- Since $n \in I$ and I is an ideal, $k \cdot n \in I$ for all $k \in \mathbb{Z}$, whence $n\mathbb{Z} \subseteq I$.
- Conversely, let $y \in I$ be arbitrary. Applying the Euclidean division theorem to divide y by n:

  $$y = q \cdot n + r \quad \text{with} \quad 0 \leq r < n.$$

  Rearranging for the remainder yields:

  $$r = y - q \cdot n.$$

Since $y \in I$ and $n \in I$ (which implies $q \cdot n \in I$), closure under subtraction implies $r \in I$.

If $r > 0$, then $r \in I \cap \mathbb{Z}_{>0}$ is a strictly positive element of I satisfying $r < n$, directly contradicting the minimality of $n = \min(I \cap \mathbb{Z}_{>0})$.

Therefore, we must have $r = 0$, which gives $y = q \cdot n \in n\mathbb{Z}$.

Hence, $I = n\mathbb{Z}$. Uniqueness follows because distinct non-negative integers generate distinct sets of positive multiples ($n = \min(n\mathbb{Z} \cap \mathbb{Z}_{>0})$ for $n > 0$).

Q.E.D.

Q.E.D.

Insight: Principal Ideal Domains (PIDs)

An integral domain (Definition 2.62) in which every ideal is generated by a single element (i.e., $I = aR$) is called a *"Principal Ideal Domain"* (*"PID"*). Theorem 2.35 proves that the ring of integers $\mathbb{Z}$ is a PID.

Trivial Ideals and Comparison with Fields

Spoken Transcript [00:56:31 - 00:57:05]

That's the most basic property of ideals in the integers. For any ring R, we've seen before that we have: zero is an ideal, and the whole ring is an ideal: *[writes $R \to (0)$ is ideal, R is ideal]*.

We haven't discussed fields at all today, but we discussed last time that if you have a field, these are the only two ideals (Recall: Theorem 1.32). But the integers have a lot of ideals: all these $n\mathbb{Z}$'s are ideals. These are all ideals.

Trivial Ideals in Rings and Fields

Recap: That a field has only these two ideals was shown in Week 1 (Theorem 1.32).

Proposition 2.36 (*Trivial Ideals*): For any ring $(R, +, \cdot, 0, 1)$, the zero ideal $\{0\} = (0)$ and the entire ring $R = (1)$ are always ideals (called *"trivial ideals"*). Here $(a) = aR = \{ra \mid r \in R\}$ denotes the ideal generated by a, defined in Week 4 (Definition 4.32).

Proposition 2.37 (*Ideals in a Field*): If F is a field, its only ideals are (0) and F.

Proof

If F is a field and $I \subseteq F$ is an ideal containing a non-zero element $x \in I \setminus \{0\}$, then x possesses a multiplicative inverse $x^{-1} \in F$. By the ideal absorption property:

$$1 = x^{-1} \cdot x \in I.$$

Once $1 \in I$, for any $r \in F$ we have $r = r \cdot 1 \in I$, so $I = F$. Q.E.D.

Beyond the Lecture: PID vs. Field Structure

The ring of integers $\mathbb{Z}$ is a Principal Ideal Domain, but not a field, because it contains infinitely many non-trivial proper ideals $n\mathbb{Z}$ for $n \geq 2$. In contrast, in a field F, every non-zero element is invertible, instantly forcing any non-zero ideal to absorb 1 and expand to the entire ring.

Ideals in Rings of Smooth Functions

The Ring $C^\infty(\mathbb{R})$ and Compactly Supported Functions

Spoken Transcript [00:57:05 - 00:58:27]

Let's try another example. As I said, this course has several purposes, but one of them is to give you some way of looking at objects you know in mathematics and thinking about them in terms of algebra. An object you certainly know in mathematics, if I take the real numbers — I think there's a whole class on real numbers — is that you can take C^∞ functions: *[writes $\mathbb{R} \to C^\infty(\mathbb{R})$]*.

Are you happy with this? You don't have to take C^∞; you could take 17 times differentiable functions, but we'll just go to infinity. These are infinitely differentiable functions: *[writes "$C^\infty(\mathbb{R})$ ∞ diff. functions"]*.

Are you happy with this terminology? Is it what you use? It's standard. An element here is some function $f\colon \mathbb{R} \to \mathbb{R}$, which I can differentiate as many times as I want, so, it has a nice graph. I can graph it like this: *[draws a wave curve]*. There it is.

And this beautiful object is a ring, *[writes $(C^\infty(\mathbb{R}), +, \cdot, 0, 1)$]* because why? You can add functions, you can multiply functions, zero is just the constant function zero, and one is the constant function one. Is that clear? It's a ring: *[writes "Ring"]*.

Yeah?

Student Question

What is written after $C^\infty(\mathbb{R})$?

Spoken Transcript [00:58:27 - 00:58:45]

Here? Which board?

Here? Infinitely differentiable functions.

Yeah, sorry, good. My handwriting's not good: functions. And it's not because I have this wrist injury — it's not because of that! I just have bad handwriting.

Board Cleaning and Aside on Wrist Injury

The lecturer erases the lower left blackboard while humorously assuring students that his messy handwriting is innate and unrelated to his wrist brace.

Spoken Transcript [00:58:45 - 00:59:33]

And I don't even know how I got this wrist injury. It's not like I'm an MMA fighter or anything like that. It just started hurting at one point.

I thought it was going to be a career-ending injury, but it turns out that if I wear this thing, it's fine.

The Ring of Smooth Functions

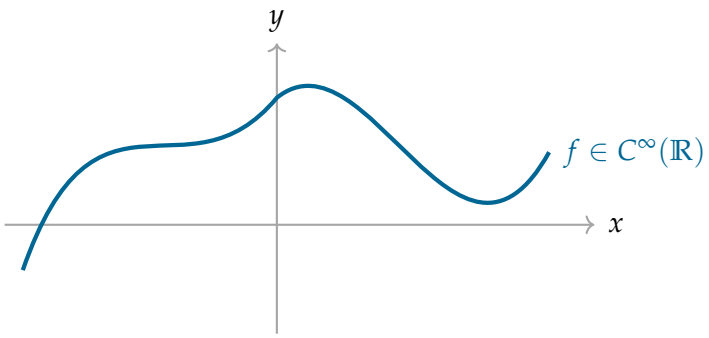
Definition 2.38 (*Ring of Smooth Functions $C^\infty(\mathbb{R})$*): The set of infinitely differentiable real-valued functions on $\mathbb{R}$,

$$C^\infty(\mathbb{R}) := \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ is smooth}\},$$

forms a commutative ring $(C^\infty(\mathbb{R}), +, \cdot, 0, 1)$ under pointwise operations:

$$(f+g)(x) := f(x) + g(x),$$
$$(f \cdot g)(x) := f(x) \cdot g(x),$$

with additive identity $0(x) \equiv 0$ and multiplicative identity $1(x) \equiv 1$. Sums and products of smooth functions are smooth (Fact 0.39), so the operations stay inside $C^\infty(\mathbb{R})$.



The Ideal of Compactly Supported Smooth Functions

Spoken Transcript [00:59:33 - 01:01:30]

There are a lot of things to say about this ring. This class is not primarily about such rings of differentiable functions — that's some other class — but I can show you something interesting in it.

There's an interesting ideal inside $C^\infty(\mathbb{R})$: *[writes $I \subset C^\infty(\mathbb{R})$]* functions with compact support: *[writes "functions with compact support"]*.

First, you have to understand what this is as a set: it's the set of functions with compact support. A general function can do whatever it wants, totally unconstrained. But a function with compact support means that... Are you happy with this analysis terminology: *"compact support"*? It means that its entire life happens somewhere compact. It has to be zero here, and then suddenly it comes to life, and then it goes to zero. It's like these hat functions that look like that, okay? *[draws a bump function graph on the board]* That's a function with compact support.

Is this an ideal? You have to ask: Who's in here? To be an ideal, it has to be the case that zero is there; and very fortunately, zero has compact support. What does not have compact support is the constant one function; that does not have compact support. So, whatever it is, it's surely not a subring, because it does not have the constant one function.

Then you have to check that if you add compactly supported things, they have compact support; that's true. And the interesting thing is: if I take a function of compact support and multiply it by any function, it still has compact support. Because once this function is zero, it just cannot be brought back to life, no matter how much you multiply. That's an example.

Compactly Supported Smooth Functions as an Ideal

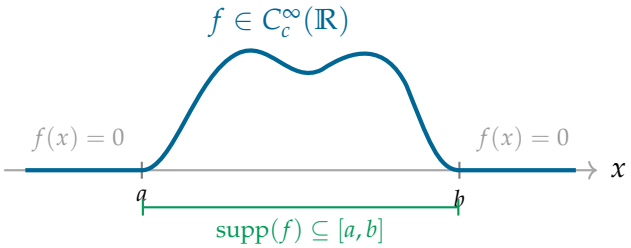
Definition 2.39 (*Compactly Supported Smooth Functions*): The *"support"* of a function $f\colon \mathbb{R} \to \mathbb{R}$ is the topological closure (Definition 0.38) of the set where f does not vanish:

$$\operatorname{supp}(f) := \overline{\{x \in \mathbb{R} \mid f(x) \neq 0\}}.$$

The set of smooth functions with compact support is denoted:

$$C_c^\infty(\mathbb{R}) := \{f \in C^\infty(\mathbb{R}) \mid \operatorname{supp}(f) \text{ is compact in } \mathbb{R}\}.$$

By Heine–Borel, $\operatorname{supp}(f)$ is compact if and only if it is bounded (Fact 0.39).



Proposition 2.40 ($C_c^\infty(\mathbb{R})$ *is an Ideal, but not a Subring*): The set of compactly supported smooth functions $I := C_c^\infty(\mathbb{R}) \subset C^\infty(\mathbb{R})$ is an ideal of $C^\infty(\mathbb{R})$, but is not a subring.

Proof

We check the ideal properties:

(1) Additive Subgroup: The zero function has $\operatorname{supp}(0) = \emptyset$, which is compact, so $0 \in C_c^\infty(\mathbb{R})$. For $f, g \in C_c^\infty(\mathbb{R})$, $\operatorname{supp}(f+g) \subseteq \operatorname{supp}(f) \cup \operatorname{supp}(g)$. The union of two compact sets is compact, so $f+g \in C_c^\infty(\mathbb{R})$ and $-f \in C_c^\infty(\mathbb{R})$.

(2) Multiplicative Absorption: Let $g \in C^\infty(\mathbb{R})$ be an **arbitrary** smooth function, and let $f \in C_c^\infty(\mathbb{R})$. Since $(g \cdot f)(x) = 0$ whenever $f(x) = 0$, we have:

$$\operatorname{supp}(g \cdot f) \subseteq \operatorname{supp}(f).$$

As a closed subset of the compact set $\operatorname{supp}(f)$, $\operatorname{supp}(g \cdot f)$ is compact. Thus, $g \cdot f \in C_c^\infty(\mathbb{R})$.

However, the multiplicative identity $1(x) \equiv 1$ has support $\operatorname{supp}(1) = \mathbb{R}$, which is not compact. Hence $1 \notin C_c^\infty(\mathbb{R})$, so $C_c^\infty(\mathbb{R})$ is not a subring. Q.E.D.

A Strictly Larger Ideal: Functions Vanishing on a Right Half-Line

Spoken Transcript [01:01:30 - 01:03:45]

You can ask other questions in analysis about this. I'm not going to talk about that so much, but this is a kind of interesting ideal.

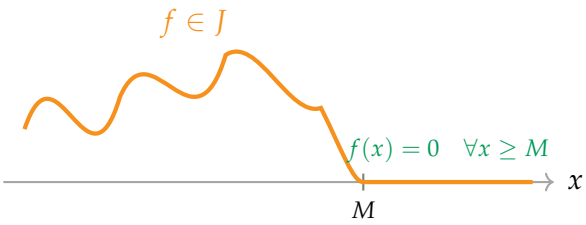
Maybe I'll ask you a question: Is there some ideal J that's bigger? *[writes $I \subsetneq J \neq C^\infty(\mathbb{R})$]* Can you find an ideal that's bigger than this, and not the whole ring, of course? Because that's always an ideal that's bigger.

One of the strange things about the way math is taught is that we take all these subjects and divide them. The traditional vision of classical education in mathematics is that there's an algebra direction, an analysis direction, and a geometry direction. But actually, it doesn't really make sense to separate them all: it's all just one way of looking at numbers. I don't know why I said that, but can you think of a way to make this ideal bigger?

We used the fact that it's zero on both sides here, but that's not necessary, right? Can't we just say: functions that eventually go to zero on the right? Those are more, and it satisfies the same thing. So, here you can say J — I'll write it colloquially: functions that become zero for all sufficiently large x: *[writes $J = \{f \text{ that become } 0 \text{ for all suff. large } x\}$]*.

Those are functions that look like... They do whatever they want, and then eventually they become zero: *[draws a second graph: function wiggles on the left then becomes identically zero on the right]*. These functions are certainly inside there, but there are more of them! As these drawings perfectly prove, there are more of these functions. And these things are also an ideal, because if you multiply by something that's zero here, it's going to stay zero. You might wonder: Are there bigger and bigger ideals? These are kind of interesting questions we're not going to address now, unfortunately.

Construction of a Strictly Larger Proper Ideal



Proposition 2.41 (*A Strictly Larger Proper Ideal*)**:** Let J be the set of smooth functions that vanish identically on some right ray $[M, \infty)$:

$$J := \{f \in C^\infty(\mathbb{R}) \mid \exists M \in \mathbb{R} \text{ such that } f(x) = 0 \text{ for all } x \geq M\}. \tag{2.12}$$

Then J is an ideal of $C^\infty(\mathbb{R})$, and with $I = C_c^\infty(\mathbb{R})$:

$$I \subsetneq J \neq C^\infty(\mathbb{R}).$$

Proof

- **J is an ideal:** If $f(x) = 0$ for $x \geq M_1$ and $g(x) = 0$ for $x \geq M_2$, then $(f+g)(x) = 0$ for $x \geq \max(M_1, M_2)$; also $0 \in J$ and $-f = (-1) \cdot f \in J$. Furthermore, for any

$h \in C^\infty(\mathbb{R})$, $(h \cdot f)(x) = h(x) \cdot 0 = 0$ for all $x \geq M_1$.

- **Proper inclusions:** If $f \in C_c^\infty(\mathbb{R})$, its support is bounded, so it vanishes for large positive x, whence $C_c^\infty(\mathbb{R}) \subseteq J$. A smooth function that is zero for $x \geq 0$ but non-zero on all of $(-\infty, 0)$, such as $x \mapsto e^{1/x}$ for $x < 0$ and 0 for $x \geq 0$ (Fact 0.41), belongs to J but not to $C_c^\infty(\mathbb{R})$, so $C_c^\infty(\mathbb{R}) \subsetneq J$. Finally, the constant function $1 \notin J$, so $J \subsetneq C^\infty(\mathbb{R})$.

Q.E.D.

The Restriction Homomorphism on Function Spaces

Spoken Transcript [01:03:45 - 01:05:45]

Maybe I'll say one more thing, just for fun. If I look at C^∞ functions on $\mathbb{R}$, *[writes $C^\infty(\mathbb{R})$]* there's a ring homomorphism to C^∞ functions on $\mathbb{R}_{>0}$: *[writes $\to C^\infty(\mathbb{R}_{>0})$]*.

This is a good example of a homomorphism, and this is called restriction; maybe I'll call it Res: *[writes "Res" over the arrow]*.

If I have a C^∞ function here, it's a function from all of $\mathbb{R}$ to $\mathbb{R}$: *[writes $f : \mathbb{R} \to \mathbb{R}$]*. And if I have one, I can restrict it: $f|_{\mathbb{R}_{>0}} : \mathbb{R}_{>0} \to \mathbb{R}$: *[writes $f_{\text{Res}} : \mathbb{R}_{>0} \to \mathbb{R}$]*.

Is that clear? I can restrict a function. I have a function here; this is a ring. Every element here I can just restrict, and that gives me a map. First, it's a map on sets, and then you have to show it respects zero — of course, the zero function restricts to the zero function; and if I restrict a function that is constant one, it's the function here that is constant one. Also, addition and multiplication behave well under restriction. So, this is a legitimate ring homomorphism: *[writes "Ring hom"]*.

For function spaces, restriction is a very useful way of making ring homomorphisms.

Okay, now your analysis question: Is this surjective? That's properly a question in your analysis class. You can think about that. If you understood all the examples in your analysis class, you should be able to tell me the answer.

Spoken Transcript [01:05:45 - 01:06:13]

It's just been a long time since I taught analysis — like 25 years, I think.

[begins cleaning the center and left chalkboards] Are you prepared to tell me?

Okay, you can think about it.

Board Transition: Preparing for Foundations

The lecturer cleans the center and left chalkboards thoroughly, transitioning from rings of functions to foundational category theory.

The Restriction Homomorphism

Definition 2.42 (*Restriction Homomorphism*): Let $U \subseteq \mathbb{R}$ be an open subset (Definition 0.38; for example, $U = \mathbb{R}_{>0}$). The *"restriction map"*

$$C^\infty(\mathbb{R}) \xrightarrow{\text{Res}} C^\infty(U), \qquad (f\colon \mathbb{R} \to \mathbb{R}) \longmapsto (f_{\text{Res}} := f|_U\colon U \to \mathbb{R}),$$

is a ring homomorphism:

(i) $\text{Res}(f+g) = (f+g)|_U = f|_U + g|_U = \text{Res}(f) + \text{Res}(g)$.
(ii) $\text{Res}(f \cdot g) = (f \cdot g)|_U = f|_U \cdot g|_U = \text{Res}(f) \cdot \text{Res}(g)$.
(iii) $\text{Res}(1_{\mathbb{R}}) = 1_U$.

Question 2.43: Is the restriction homomorphism $\text{Res}\colon C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R}_{>0})$ surjective? (The lecturer leaves this to the students; the answer follows in Proposition 2.48.)

Introduction to Category Theory

Definition of a Category

Spoken Transcript [01:06:13 - 01:07:09]

All right, so, there's a topic that's in some of the reading. There are two foundational topics which I've avoided discussing, but I think I'll discuss now, and it has to do with category theory and Zorn's lemma.

You had this foundations, this Grundstrukturen class. Did you discuss categories there?

You can say no.

Student Answer

No.

Spoken Transcript [01:07:09 - 01:07:57]

Okay, so, category. We discuss categories: *[writes "Category"]*.

A category consists of *[writes "Category consists of"]* objects *[writes "Objects"]* and homs: *[writes "and Homs"]*.

Homs are sometimes called morphisms, *[writes* $\begin{cases} \textit{morphisms} \\ \textit{arrows} \end{cases}$ *]* so, the language differs. Different authors take a different choice of language: sometimes homs are called morphisms; sometimes they're called arrows. The first thing to know about a category: it has objects and it has homs.

More specifically, given two objects in the category — *[writes "Given A, B objects in Category,"]* given A and B objects in the category — there is a set of homs, and this set has a name: $\text{Hom}(A, B)$: *[writes* $\text{Hom}(A, B)$*]*.

Definition of a Category

Definition 2.44 (*Category*): A *"category"* $\mathcal{C}$ consists of the following data:

(1) A collection of *"objects"*, denoted $\mathrm{Obj}(\mathcal{C})$.

(2) For every pair of objects $A, B \in \mathrm{Obj}(\mathcal{C})$, a set of *"morphisms"* (also called *"homs"* or *"arrows"*), denoted:

$$\mathrm{Hom}_{\mathcal{C}}(A,B) \quad (\text{or } \mathrm{Mor}(A,B)).$$

Terminology Across Literature: In modern mathematical literature:

- **Objects:** The structural spaces (e.g., sets, groups, rings, vector spaces, topological spaces).
- **Homs / Morphisms / Arrows:** The structure-preserving mappings between objects ($f\colon A \to B$).

The Category of Groups

Spoken Transcript [01:07:57 - 01:09:08]

Of course, there are some conditions this has to satisfy, but before I give you the conditions, I will give you the model: for example, the category of groups: *[writes "category of Groups"]*.

The category of groups is a category. This category has objects, *[writes "Objects = grp"]* and the objects are groups: *[writes $(G,\cdot,e)$]*.

An object is a group; that means it's a group with an operation and a neutral element. Those are the objects. When you think about the category of groups, somehow you think about this world where all these groups are: they're all over the place, lots of different groups; every possible group is in here. That's in the category.

And what are the arrows going to be? The homs: if I have two particular groups, *[writes $\mathrm{Hom}(G,\hat{G})$]* then this is going to be the set of group homomorphisms *[writes "= set of group homomorphisms"]* from G to $\hat{G}$: *[writes "from G to $\hat{G}$"]*.

The Category of Groups Grp

Example 2.45 (*The Category of Groups*): In the category of groups, standardly denoted **Grp**:

- **Objects:** All groups $(G,\cdot,e)$.
- **Morphisms:** For any two groups $G, \hat{G}$, the set of morphisms from G to $\hat{G}$ is the set of all group homomorphisms:

$$\mathrm{Hom}_{\mathbf{Grp}}(G,\hat{G}) := \{f\colon G \to \hat{G} \mid f \text{ is a group homomorphism}\}. \tag{2.13}$$

Categories of Rings and Vector Spaces

Spoken Transcript [01:09:10 - 01:10:24]

Okay, that's the model. Is that clear?

The idea of a category is that we want to think about groups and their relationships abstractly. We can think about the category of groups, we think of all groups, and if I pick two groups in that world of all groups, I get the set of homs, okay?

As I said, there are going to be some conditions. But another example is the category of rings: *[writes "Category of Rings"]*. That's also fine.

There, the objects are rings, as we've defined in this class, meaning the objects are rings with the full data: *[writes $(R, +, \cdot, 0, 1)$]*.

And the homs, which will not surprise you, are ring homomorphisms: *[writes "are ring homs"]*.

From last year, you could consider the category of vector spaces over whatever field you want — say, vector spaces over the complex numbers: *[writes "Category of Vector Spaces/$\mathbb{C}$"]*. In the category of vector spaces over the complex numbers, the objects are vector spaces over the complex numbers, and the homs are linear maps: *[writes "Homs are lin. maps"]*.

Those are all examples of categories.

Categories of Rings and Vector Spaces

Example 2.46 (*The Category of Rings* Ring): In the category of rings, standardly denoted **Ring**:

- **Objects:** Rings with identity $(R, +, \cdot, 0, 1)$.
- **Morphisms:** Ring homomorphisms $f : R \to \hat{R}$ (satisfying additivity, multiplicativity, and $f(1) = \hat{1}$).

Example 2.47 (*The Category of Complex Vector Spaces* $\mathbf{Vect}_{\mathbb{C}}$): In the category of complex vector spaces, denoted $\mathbf{Vect}_{\mathbb{C}}$:

- **Objects:** Vector spaces V over the field $\mathbb{C}$.
- **Morphisms:** $\mathbb{C}$-linear transformations $T : V \to W$.

Interlude: Non-Surjectivity of the Restriction Map

Spoken Transcript [01:10:24 - 01:10:54]

Then, of course, I have to give you the conditions. To be a category, if I want to tell you what a category is, I have to tell you what the objects are; given two objects, I have to tell you what the homs are; and the resulting structure has to have some properties.

All of these will have that property.

Spoken Transcript [01:10:54 - 01:11:01]

[points at the restriction map equation on the right board] Did you figure this out?

Yeah?

Student Answer

I think $\sin(1/x)$...

Spoken Transcript [01:11:01 - 01:11:29]

Yeah, so, the answer is no, it's not surjective.

I'll draw the picture. A picture is worth a thousand words!

The picture is: you have an element here that's not there. This is zero, and this is this element — that one: *[draws a sketch of $\sin(1/x)$ oscillating wildly as $x \to 0^+$]*.

You should recognize that from your analysis class, I would think. Maybe not drawn with such enthusiasm!

You know this function, right?

Resolution: The Restriction Map is Not Surjective

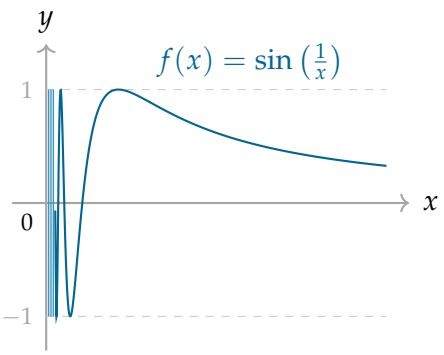
Proposition 2.48 (*Non-Surjectivity of Smooth Restriction*): The restriction homomorphism $\mathrm{Res}\colon C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R}_{>0})$ is not surjective.

Proof

Consider the function $f\colon \mathbb{R}_{>0} \to \mathbb{R}$ defined by:

$$f(x) := \sin\left(\frac{1}{x}\right).$$

By standard calculus rules, f is infinitely differentiable on $(0,\infty)$, so $f \in C^\infty(\mathbb{R}_{>0})$. However, as $x \to 0^+$, $f(x)$ oscillates infinitely often between -1 and 1, so $\lim_{x\to 0^+} f(x)$ does not exist. Consequently, f cannot even be extended continuously to $x = 0$ (Fact 0.40), let alone smoothly to all of $\mathbb{R}$. Thus, $f \notin \mathrm{Im}(\mathrm{Res})$. Q.E.D.



Spoken Transcript [01:11:29 - 01:12:07]

[erases the graph of $\sin(1/x)$ and thoroughly cleans the lower blackboard to prepare for the category axioms]

Axioms of a Category: Homs, Composition, and Identity

Spoken Transcript [01:12:07 - 01:14:06]

Okay, so, the properties that the homs have to satisfy for a category are very simple properties.

The first property, as I said, is that $\mathrm{Hom}(A,B)$ is a set: *[writes "• $\mathrm{Hom}(A,B)$ is a set"]*. That's almost a ridiculous property, and we're not going to think about it too much, but it's...

The second is that there is composition: *[writes "• Composition"]*.

Composition means that if I have three objects, A, B, and C, *[writes $A \quad B \quad C$]* then here I have the set of homs from A to B, *[writes $\mathrm{Hom}(A,B)$]* and I also have homs from B to C: *[writes $\times\,\mathrm{Hom}(B,C)$]*. This composition means that if I'm giving you the category, I have to tell you the composition rule. Whenever I have this diagram: A, B, and C, there has to be a rule, which is called composition, that tells you how, if you have a hom from A to B and a hom from B to C, this gives me a hom from A to C: *[writes $\to\,\mathrm{Hom}(A,C)$]*.

That is part of the data of the category. If I'm telling you a category, it's up to me to tell you how to compose homs. And in all examples, you just compose them, okay?

So, what's the content? For groups, of course, if I have a hom from G to $\hat{G}$ and another one from $\hat{G}$ to G'', you can compose them as functions — you can always compose functions — but then you have to show that the composition is a group hom, all right? Which I leave to you to check.

That's the content of this: to say that this category of groups is a category, there are groups, there are homs, and there's a natural composition of homs; and it's very important that when I have a group hom here and a group hom here, when I compose them, I get a group hom. Okay?

Axioms of a Category: Hom-Sets and Composition

Axiom 1 (Hom-Sets): For any two objects $A, B \in \mathrm{Obj}(\mathcal{C})$, the collection $\mathrm{Hom}_{\mathcal{C}}(A,B)$ is a well-defined set.

Axiom 2 (Composition Law): For any three objects $A, B, C \in \mathrm{Obj}(\mathcal{C})$, there is a composition mapping:

$$\begin{aligned} \circ\colon \mathrm{Hom}_{\mathcal{C}}(A,B) \times \mathrm{Hom}_{\mathcal{C}}(B,C) &\longrightarrow \mathrm{Hom}_{\mathcal{C}}(A,C) \\ (f,g) &\longmapsto g \circ f. \end{aligned}$$

Spoken Transcript [01:14:06 - 01:16:30]

And there's one more: inside $\mathrm{Hom}(A, A)$ — *[writes* $\bullet \mathrm{Hom}(A, A) \ni \mathrm{Id}_A$*]* which is legal to do; I said you pick two objects of the category, and they could be the same object — there is a distinguished element called the identity. That's part of the data, okay?

There's one more condition, which is how the identity relates to composition, which is not going to be a surprise to you.

If we go back to this example of groups: objects are groups, homs are group homs, composition is composition of functions — and we're lucky that composition of functions takes group homs to group homs — and there is an identity: if you take the group and you do nothing, that is definitely a group hom. So, that's here.

And then there's the final property: if you take the identity on A — we take here A, A, and B *[writes* $A \quad A \quad B$*]* — then I can take the identity here, *[writes* $\mathrm{Id}_A \in \mathrm{Hom}(A, A)$*]* and I can take any element f here, *[writes* $f \in \mathrm{Hom}(A, B)$*]* and I can compose it, because composition is given data. I do the identity first and then I do f: $f \circ \mathrm{Id}_A$. *[writes* $f \circ \mathrm{Id}_A$*]* This has to be an element of $\mathrm{Hom}(A, B)$. *[writes* $\mathrm{Hom}(A, B)$*]* And what do you think the axiom is going to say?

What do you think the axiom is going to say here? If I do the identity first, and then I do f, what do you think this should be equal to?

Yeah: f! *[writes* $= f$*]* That's the identity axiom. This is an axiom of the category.

And also the other way: you can also have A, B, and B. *[writes* $A \quad B \quad B$*]* If I put f here and I put Id_B, *[writes* $A \xrightarrow{f} B \xrightarrow{\mathrm{Id}_B} B$*]* then if you do f first and then Id_B, this should just be f, okay? *[writes* $\mathrm{Id}_B \circ f = f$*]* That's actually the full category stuff.

Axioms of a Category: Identity and Unit Laws

Axiom 3 (Identity Morphisms): For every object $A \in \mathrm{Obj}(\mathcal{C})$, there exists a distinguished morphism, called the *"identity morphism"*:

$$\mathrm{Id}_A \in \mathrm{Hom}_{\mathcal{C}}(A, A).$$

Axiom 4 (Identity Laws / Unit Axioms): For every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$:

$$f \circ \mathrm{Id}_A = f \in \mathrm{Hom}_{\mathcal{C}}(A, B), \tag{2.14}$$

$$\mathrm{Id}_B \circ f = f \in \mathrm{Hom}_{\mathcal{C}}(A, B). \tag{2.15}$$

Spoken Transcript [01:16:30 - 01:17:26]

The data of this category is supposed to capture what happens when we think about these algebraic objects.

Of course, it can be applied to many other settings. But if you think about groups, what is the abstract structure of the group? We have objects, we have homs, the homs compose, there is an identity hom, and they satisfy these identity properties.

It's the same for rings, and for vector spaces.

And then there's a whole development which most people find kind of dull, but anyway...

Is there a question about what the definition of a category is?

I'll just say it again: a category has objects, it has homs, the homs have to be endowed with a composition rule, there is an identity element, and we have this property. Okay?

Consolidated Definition of a Category

Definition 2.49 (*Formal Category*): A *"category"* $\mathcal{C}$ consists of:

(1) A collection of objects $\mathrm{Obj}(\mathcal{C})$.

(2) A set of morphisms $\mathrm{Hom}_{\mathcal{C}}(A,B)$ for every pair $A,B \in \mathrm{Obj}(\mathcal{C})$.

(3) A composition operation:

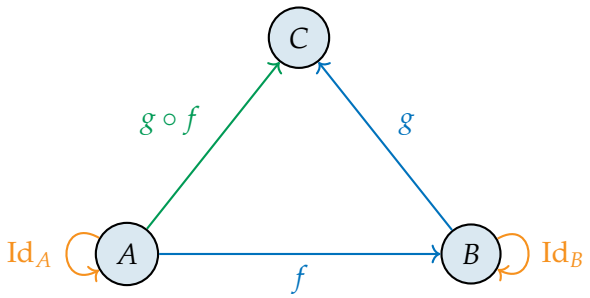
$$\circ\colon \mathrm{Hom}_{\mathcal{C}}(A,B) \times \mathrm{Hom}_{\mathcal{C}}(B,C) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(A,C), \quad (f,g) \longmapsto g \circ f,$$

satisfying **associativity**:

$$h \circ (g \circ f) = (h \circ g) \circ f \quad \forall f \in \mathrm{Hom}(A,B),\ g \in \mathrm{Hom}(B,C),\ h \in \mathrm{Hom}(C,D).$$

(4) For each object $A \in \mathrm{Obj}(\mathcal{C})$, an identity morphism $\mathrm{Id}_A \in \mathrm{Hom}_{\mathcal{C}}(A,A)$ satisfying the **unit laws**:

$$f \circ \mathrm{Id}_A = f \quad \text{and} \quad \mathrm{Id}_B \circ f = f \quad \forall f \in \mathrm{Hom}_{\mathcal{C}}(A,B).$$



Associativity in Categories: Although the lecturer did not explicitly write associativity on the board, associativity of composition, $(h \circ g) \circ f = h \circ (g \circ f)$, is an indispensable standard axiom of any category, guaranteeing that diagrammatic paths compose unambiguously.

Categorical Definition of Isomorphism

Spoken Transcript [01:17:26 - 01:19:18]

I just assert that all these things — groups, rings, fields, vector spaces... *[writes "Groups, Rings, Fields, Vector spaces..."]* and many other structures — give you categories.

If you have a category, built into the category is a notion of isomorphic objects.

Suppose we have a category: Cat is a category: *[writes "Suppose"] [writes "Cat is a category"]*. I want to define the notion of isomorphism purely in categorical terms: *[writes "I want to define the notion of isomorphism"] [writes "purely in categorical terms."]*.

There is a very straightforward way to do this. That's why I say just "Cat *is a cate-*

gory" — that means you don't know anything about it. All you know is that there are some objects, they have homs, and they satisfy these principles.

Suppose I have a category, and I take two objects. I want to know: Is there a notion of isomorphism in that category? Which is to say: Is there a purely arrow-theoretic notion of isomorphism? Category-theoretic.

Sorry... *[adjusts microphone wire]* Oh, here we go.

And the answer is yes, and I will explain this.

Spoken Transcript [01:19:18 - 01:20:08]

In the reading section, there are a few pages on the definition of these categories — the reading for this week.

Are the readings for this week already on the website? No? Maybe they are. In any case, as I said, if you ever want to know what's on the website, you can go to 2018; it's all there. You don't have to wait.

Course Literature and Reading Assignments

The lecturer reminds students that reading assignments detailing category theory are provided in the syllabus materials, which are also archived on the 2018 course website.

Spoken Transcript [01:20:08 - 01:21:17]

I was in the middle of trying to explain the categorical definition of isomorphism.

I have this category Cat, *[writes* Cat*]* and I have two objects: A and B are in the objects of my Cat: *[writes* $A, B \in \mathrm{Obj}(\mathrm{Cat})$*]*. Definition: *[writes "Def:"]* if I have $f \in \mathrm{Hom}(A, B)$ — that means if I have an arrow here: *[writes* $f \in \mathrm{Hom}(A, B)$*, draws* $A \xrightarrow{f} B$*]* — then f is an isomorphism — this is a definition, not a theorem — *[writes "then* f *is an isomorphism"]* if... *[writes "if"]*

Now you're supposed to complete this for me: What would you do here?

What do you normally need for an isomorphism? You need a map the other way. If there exists a g, which is an arrow in the other direction, *[writes* $\exists g \in \mathrm{Hom}(B, A)$*]* — that means there exists a g going in the other direction: *[draws* $B \xrightarrow{g} A$*]* — such that what?

Yeah?

Student Answer

The two compositions are the identity.

Spoken Transcript [01:21:21 - 01:21:46]

Yeah, the two compositions give the identity. There exists a g in the other direction

such that *[writes "such that"]* if I do g first, then f — then you have to concentrate: this is the identity on B: *[writes $f \circ g = \mathrm{Id}_B$]* — and if I do f first, then g, this is the identity on A: *[writes $g \circ f = \mathrm{Id}_A$]*.

That's the definition! That's a purely category-theoretic definition of isomorphism.

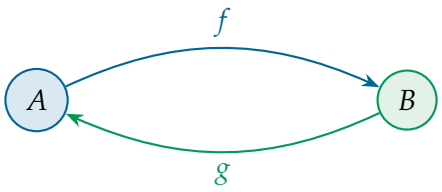
Is that confusing or clear?

Categorical Definition of Isomorphism

Definition 2.50 (*Isomorphism in a Category*): Let $\mathcal{C}$ be a category, and let $A, B \in \mathrm{Obj}(\mathcal{C})$. A morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ (denoted $f\colon A \to B$) is called an *"isomorphism"* if there exists a morphism $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$ (denoted $g\colon B \to A$) such that:

$$f \circ g = \mathrm{Id}_B \quad \text{and} \quad g \circ f = \mathrm{Id}_A\,. \tag{2.16}$$

Two objects A and B are called *"isomorphic"* (written $A \cong B$) if there exists an isomorphism between them.



Arrow-Theoretic vs. Element-Theoretic Isomorphism: In standard concrete algebra, an isomorphism is typically introduced as a **bijective homomorphism**. In abstract category theory, where objects may not have *"elements"* and morphisms are not necessarily *"functions"*, bijectivity cannot even be formulated. Instead, category theory characterizes isomorphisms purely through arrows via the existence of a two-sided inverse morphism g. In concrete categories this recovers the familiar notions:

- In **Grp**, categorical isomorphisms are precisely bijective group homomorphisms (Proposition 2.13).
- In **Ring**, categorical isomorphisms are bijective ring homomorphisms (Definition 2.23).
- In $\mathbf{Vect}_K$, categorical isomorphisms are bijective linear transformations (invertible matrices in finite dimension, Fact 0.30).

***Insight:* The Power of Category-Theoretic Abstraction**

Category theory unifies mathematics by defining concepts purely in terms of external interactions (*"morphisms"*) between systems rather than internal set elements. The definition of an isomorphism $f \circ g = \mathrm{Id}_B$ and $g \circ f = \mathrm{Id}_A$ is universally identical across group theory, ring theory, linear algebra, topology, and algebraic geometry.

Spoken Transcript [01:21:46 - 01:22:27]

In particular, once I tell you the categories of groups or rings, I tell you the homs, the identity, and this property, that already tells you what the notion of isomorphism is. I told you anyway what the notion of isomorphism is, but in some sense I didn't have to: it was just this categorical notion of isomorphism.

And there's one more thing that I hope I'll finish before this ends, because this is kind of boring and dry, this category stuff. But it's kind of an organizational way to think about these structures.

There's nothing deep happening here, really nothing. It's just language.

Definition of a Functor

Spoken Transcript [01:22:27 - 01:23:55]

There's the notion of a functor: *[writes "Functor"]*.

This is maybe cooler: *[writes "Functor"]*.

A functor is kind of a notion of a map from one category to another. A category is a thing, and you might have two categories, and you want to compare them — just like we have groups, and we might want to compare groups. Functors, in some sense, compare two different categories *[writes "Functor Compare categories"]* — or it could be the same category: compare categories.

The idea of a functor is: if I have some category here, and I have another category, $\widehat{\mathrm{Cat}}$, *[writes* $\mathrm{Cat} \xrightarrow{F} \widehat{\mathrm{Cat}}$*]* then the functor F is written as a map from one category to the other.

And what's the data of the functor? *[writes "Data:"]*

If I'm giving you the functor, then it's on me to provide all this data. The data is that for every object A in the first category, *[writes "for* A *object in* Cat*,"]* I have to give you an object in the second category: $F(A) \in \widehat{\mathrm{Cat}}$: *[writes* $F(A) \in \widehat{\mathrm{Cat}}$*]*.

That's part of the data of the functor. This is like, you know, if I give you a function, it's my responsibility to tell you where every point goes; that's totally on me if I'm giving you the function. Here, if I'm giving you a functor, I have to tell you, whenever I have an object on one side, how to get the object on the other side. But that's not enough.

Spoken Transcript [01:23:55 - 01:25:22]

[cleans a section of the right chalkboard] I have to tell you whenever I have a hom for Cat, the first one: *[writes* $\mathrm{Hom}(A, B)$ *for* Cat*]*. If I have A and I have an arrow here — *[writes* $A \xrightarrow{f} B$*]* that's on the Cat side — I've already told you what these objects are, *[writes* $F(A) \quad F(B)$*]* but I also have to tell you how this hom goes to a hom between these on the other side: *[writes* $F(A) \xrightarrow{F(f)} F(B)$*]*.

That's, again, on me to do that. So, it's a lot of things I have to tell you.

And it can't just be anything: it has to respect all of the structure, which means this

rule must take the identity to the identity, and it must respect composition — which I don't write, but it's kind of clear what it is: *[writes "must respect identity, composition"]*.

You can look that up, and I will give you an example. Let me just say it: it must respect identity and composition. You can look in the book, or you can imagine it yourself: there's only one thing you can do here.

That's the notion of a functor.

Definition of a Functor

Definition 2.51 (*Covariant Functor*): Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A "*functor*" (or "*covariant functor*") $F\colon \mathcal{C} \to \mathcal{D}$ consists of the following data:

(1) Object Assignment: To every object $A \in \mathrm{Obj}(\mathcal{C})$, an object $F(A) \in \mathrm{Obj}(\mathcal{D})$.

(2) Morphism Assignment: To every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A,B)$, a morphism:

$$F(f) \in \mathrm{Hom}_{\mathcal{D}}\big(F(A),F(B)\big),$$

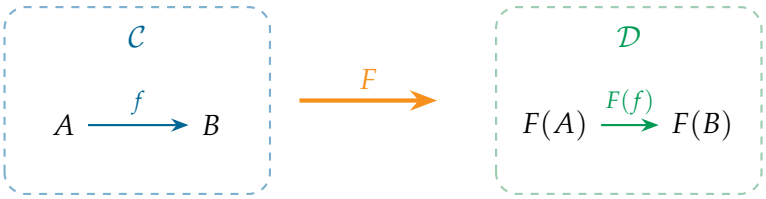
satisfying the following structural compatibility axioms:

(i) Preservation of Identity: For every object $A \in \mathrm{Obj}(\mathcal{C})$:

$$F(\mathrm{Id}_A) = \mathrm{Id}_{F(A)}.$$

(ii) Preservation of Composition: For all morphisms $f \in \mathrm{Hom}_{\mathcal{C}}(A,B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B,C)$:

$$F(g \circ f) = F(g) \circ F(f).$$



Examples of Functors: Forgetful Functor from Rings to Groups

Spoken Transcript [01:25:22 - 01:27:44]

And what's an example of a functor? *[erases the middle chalkboard]*

There's one that's been staring us in the face the whole time: if I take the category of rings, there's a way to get to the category of groups. This is going to be totally underwhelming, but it's just taking the additive structure of the ring. That's a rule to go from rings to groups, and that is a functor. I'll write it.

You can criticize this whole discussion, this lecture, as being basically nonsensical, and that's basically the correct point of view, I think. But it's kind of useful eventually. Maybe I should say: it's not an incorrect point of view; let's put it like that.

The first example of a functor — this whole direction is not really the point of this class — I take the category of rings: *[writes Ring]*. Sometimes people take terminology like Ring to denote the category of rings, *[underlines Ring]* and Grp to denote the category of groups: *[writes Group]*. There's a functor F: *[writes Ring $\xrightarrow{F}$ Group]*. This

functor is defined by: if I take a ring, which is all of this data, $(R,+,\cdot,0,1)$, *[writes $F(R,+,\cdot,0,1)$]* I need to make a group out of it. And so, I could just do this: $(R,+,0)$: *[writes $=(R,+,0)$]*. Totally underwhelming, but anyway, that's definitely a group.

This functor associates the ring to a group, so, it gives an object here to an object there. And then you have to check that it does all the things that you want. I have to also tell you: if I have a hom on the ring side, how do I get a hom on the group side? Well, the hom on the ring side is just a way to get from elements from one ring to the other; it also gives you an additive homomorphism here.

Again, almost nothing is happening; it's just reinterpretation. But that's an example, kind of a dumb example.

The Forgetful / Additive Functor from Rings to Groups

Example 2.52 (*The Additive Functor F: Ring $\to$ Grp*): There is a canonical functor:

$$F\colon \mathbf{Ring} \longrightarrow \mathbf{Grp}$$

defined by forgetting the multiplicative structure:

- **On Objects:** For every ring $(R,+,\cdot,0,1) \in \mathrm{Obj}(\mathbf{Ring})$:
$$F(R,+,\cdot,0,1) := (R,+,0) \in \mathrm{Obj}(\mathbf{Grp}).$$
- **On Morphisms:** For any ring homomorphism $f\colon R \to \hat{R}$, $F(f)$ is the same underlying mapping regarded as an additive group homomorphism:
$$F(f) := f\colon (R,+,0) \longrightarrow (\hat{R},+,\hat{0}).$$

Since ring homomorphisms satisfy $f(a+b) = f(a)+f(b)$ and $f(0) = \hat{0}$, $F(f)$ is a group homomorphism. Furthermore, $F(\mathrm{Id}_R) = \mathrm{Id}_{(R,+)}$ and $F(g\circ f) = F(g)\circ F(f)$, verifying all functor axioms.

Forgetful Functors in Algebra: This construction is the standard prototype of a *"forgetful functor"*: it strips away (forgets) a layer of algebraic structure (the multiplication $\cdot$ and unit 1), leaving only the underlying additive group $(R,+,0)$.

The Double Dual Functor on Vector Spaces

Spoken Transcript [01:27:44 - 01:30:03]

A more interesting one you know — this is actually a much more interesting example that you learned, but probably...

If I take vector spaces... I can do anything I want, but let's just take complex vector spaces, because I like the complex numbers: *[writes "VectorSpace/$\mathbb{C}$"]*. Is that clear? This is the category of complex vector spaces: the objects are complex vector spaces, and the homs are linear maps. You should be totally happy to think about this category; in some sense, your whole linear algebra class was about this category.

There's a very nice functor here to itself: *[writes "VectorSpace/$\mathbb{C}$ $\xrightarrow{F}$ VectorSpace/$\mathbb{C}$"]*. There's no reason you can't have a functor from the category to itself.

This functor takes V: *[writes V]*. What can you do with V? One thing you can do is

take its dual: *[writes V^*]*. If you take its dual, it reverses the arrows. Technically, what I've defined is a covariant functor, which means that the composition goes forward. The dual switches the arrows; it's confusing. But something that doesn't switch the arrows is the double dual: *[writes $(V^*)^*$]*.

That's a really beautiful thing you can do: take a vector space to its double dual. To every vector space, you can associate its double dual. And the claim is that this is a functor: *[writes "is a functor"]*.

It takes an object here and canonically associates an object here, and also the maps. In linear algebra, if you have a map between vector spaces, $T\colon V \to W$, *[writes $V \xrightarrow{f} W$]* that gives you a map the other way: $T^*\colon W^* \to V^*$: *[writes $W^* \xrightarrow{f^*} V^*$]*. You get a map this way — the dual — and then if you do it again, you get a map in the correct direction: $(V^*)^* \xrightarrow{(T^*)^*} (W^*)^*$: *[writes $(V^*)^* \xrightarrow{f^{**}} (W^*)^*$]* V double dual to W double dual.

This is to say that this functor takes arrows to the T double dual arrow: *[writes $F(f) = f^{**}$]*. You can think about it.

Student Question

Is the dimension of vector spaces also only finite or can it also be infinite?

Spoken Transcript [01:30:08 - 01:30:50]

Well, I didn't say anything.

You could, if you want, say you want the category of finite-dimensional vector spaces. That's a nice category inside, but the way I said it, it's anything.

For finite-dimensional vector spaces, it happens that the double dual is isomorphic. I'm not saying any of that; I'm just saying there's a motion giving a vector space another one, and that's the double dual, and it's functorial in this sense.

You've seen this double dual business, right? Usually, in linear algebra, you spend all your time dualing and double dualing.

Okay, good, that's it. *[students applaud]*

The Double Dual Endofunctor on Vector Spaces

Example 2.53 (*The Double Dual Functor*): Let $\mathbf{Vect}_{\mathbb{C}}$ denote the category of complex vector spaces and $\mathbb{C}$-linear maps. The double dual construction defines an *"endofunctor"* (a functor from a category to itself):

$$(-)^{**}\colon \mathbf{Vect}_{\mathbb{C}} \longrightarrow \mathbf{Vect}_{\mathbb{C}}$$

defined by:

- **On Objects:** For every vector space $V \in \mathrm{Obj}(\mathbf{Vect}_{\mathbb{C}})$:
$$V \longmapsto V^{**} := (V^*)^*, \quad \text{where } V^* := \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C}).$$
- **On Morphisms:** For any linear map $f\colon V \to W$, the dual map $f^*\colon W^* \to V^*$

(Definition 0.34) reverses direction:

$$f^*(\varphi) := \varphi \circ f \quad \forall \varphi \in W^*.$$

Applying the dual operation a second time reverses the arrow direction once more, restoring covariance:

$$f^{**}\colon V^{**} \longrightarrow W^{**}, \quad (f^{**}(\psi))(\varphi) := \psi(f^*(\varphi)) = \psi(\varphi \circ f) \quad \forall \psi \in V^{**}.$$

$$V \xrightarrow{\quad f \quad} W$$

$$V^* \xleftarrow{\quad f^* \quad} W^*$$

$$V^{**} \xrightarrow{\quad f^{**} = (f^*)^* \quad} W^{**}$$

Covariance, Contravariance, and Canonical Isomorphism:

- **Single Dual is Contravariant:** The mapping $V \mapsto V^*$ and $f \mapsto f^*$ reverses arrow directions ($f\colon V \to W \implies f^*\colon W^* \to V^*$), making the dual functor a *"contravariant functor"*.
- **Functoriality:** $(\mathrm{Id}_V)^{**} = (\mathrm{Id}_{V^*})^* = \mathrm{Id}_{V^{**}}$, and for $U \xrightarrow{g} V \xrightarrow{f} W$ the dual satisfies $(f \circ g)^* = g^* \circ f^*$, so
$$(f \circ g)^{**} = (g^* \circ f^*)^* = f^{**} \circ g^{**}.$$
- **Double Dual is Covariant:** Dualizing twice reverses the arrows twice, so $(-)^{**}$ preserves morphism directions ($f\colon V \to W \implies f^{**}\colon V^{**} \to W^{**}$), forming a valid covariant functor.
- **Finite vs. Infinite Dimensions:** For finite-dimensional vector spaces, the evaluation mapping $\mathrm{ev}\colon V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is a canonical isomorphism ($V \cong V^{**}$, Fact 0.35). For infinite-dimensional spaces, ev remains an injective linear map, but is generally not surjective. The functorial construction $V \mapsto V^{**}$, however, is universally defined across all vector spaces regardless of dimension.

Summary of Key Concepts

This lecture established the following central algebraic and categorical foundations:

- **Uniqueness of Identity and Inverses:** In any group $(G, \cdot, e)$, associativity inherently guarantees that both the identity element and inverse elements are strictly unique.
- **Homomorphisms and Isomorphisms:** Group homomorphisms preserve products and automatically map identities to identities ($f(e) = \hat{e}$). Two groups are isomorphic if and only if there exists a two-sided inverse homomorphism. Notably, the exponential function establishes an isomorphism between the additive reals $(\mathbb{R}, +, 0)$ and the strictly positive multiplicative reals $(\mathbb{R}^*_{>0}, \cdot, 1)$.
- **Subgroups and Symmetry Breaking in Rings:** For group homomorphisms, both the kernel $\ker(f) \leq G$ and image $\mathrm{Im}(f) \leq \hat{G}$ are subgroups. In ring theory,

this symmetry breaks: the image $\mathrm{Im}(f) \subset \hat{R}$ is a subring, whereas the kernel $\ker(f) \subset R$ is an ideal absorbing ring multiplication.

- **Ideals of $\mathbb{Z}$ and $C^\infty(\mathbb{R})$:** Every ideal of $\mathbb{Z}$ is principal ($I = n\mathbb{Z}$ with $n \geq 0$), proven via the Well-Ordering Principle and Euclidean division. In the ring of smooth functions $C^\infty(\mathbb{R})$, compactly supported functions $C_c^\infty(\mathbb{R})$ form an ideal that is strictly contained within larger proper ideals, such as functions vanishing on a right half-line $[M, \infty)$.
- **Categories and Functors:** A category $\mathcal{C}$ formalizes collections of objects and morphism sets with associative composition and unit laws. Isomorphisms are characterized purely arrow-theoretically by two-sided inverses. Functors formalize structure-preserving maps between categories, exemplified by the forgetful functor $\mathbf{Ring} \to \mathbf{Grp}$ and the covariant double dual endofunctor $(-)^{**}\colon \mathbf{Vect}_{\mathbb{C}} \to \mathbf{Vect}_{\mathbb{C}}$.

Wednesday 23 September 2026

Categories, Integral Domains, and the Group of Units

Course Content Overview

This lecture begins with administrative guidance regarding enrollment, problem set submissions, and exam scope before reviewing the foundational abstractions of category theory and advancing into commutative ring theory. The main topics covered are:

- **Category Theory Review:** Foundational data (objects, hom-sets, composition laws, identity morphisms), category axioms (associativity and identity laws), and categorical formulations of isomorphisms, monomorphisms (left-cancellation / injectivity), and epimorphisms (right-cancellation / surjectivity).
- **Integral Domains:** Definition of zero divisors and integral domains (with explicit treatment of the non-triviality condition $R \neq \{0\}$), followed by a complete proof that $\mathbb{Z}/m\mathbb{Z}$ is an integral domain if and only if m is prime, along with edge-case considerations for $m \leq 1$.
- **Divisibility and Units in Commutative Rings:** Generalization of divisibility to arbitrary commutative rings, definitions of units and multiplicative inverses, linguistic conventions (*"the units"*, *"the unit"*, and *"a unit"*), and the proof that the units form an abelian group $(\mathcal{U}_R, \cdot, 1)$.
- **Units in Modular Arithmetic:** The units of $\mathbb{Z}/m\mathbb{Z}$ are the classes $\bar{x}$ with $\gcd(x, m) = 1$; explicit computation of $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$, and the construction of an explicit group isomorphism with the additive cyclic group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$ generated by powers of $a = \bar{3} \pmod 7$.

Course Logistics and Announcements

Spoken Transcript [00:00:00 - 00:01:26]

Okay, so, we start. I wanted to make this comment: There were several questions about the course. The first one is, if you want to watch the lectures, you have to be enrolled. So, if you're not enrolled, it will stop you. And there's not much I can do about that, or if there is, I don't know what to do. But the idea is you should enroll in the course if you want to watch the lectures.

Other people had some difficulty uploading the problem sets. Maybe you have to be enrolled in the course for that, too. I don't know, but many people were successful; so, if you have difficulty, find someone who was successful, because I don't know how

to solve the problem in another way.

A question that's more about the actual course was that every week there are these suggested pages to read in the book. And the question is: How do they relate to the course? That's a good question.

And the answer is that, roughly speaking, I do that so you can find the parallel material on which I'm lecturing in the book. And I don't always follow exactly the order there, I don't follow exactly the notation, I don't follow exactly the text of the definitions. But I do something that's logically equivalent, so it can be helpful. As far as matters like the exam, I will only test you on material that has appeared in the lecture. Okay? So, this is my best statement about this so far. Okay.

Administrative Policies

The lecturer outlines three organizational guidelines:

- **Lecture Access:** Enrolled status is required to stream or access recorded lecture videos.
- **Problem Sets:** Submission portals are restricted to enrolled students; technical submission difficulties should be resolved peer-to-peer or via course administrators.
- **Course Literature vs. Exam Scope:** Weekly reading assignments provide parallel textbook treatment, but examinations are strictly based on definitions, theorems, and proofs presented during the live lectures.

Review of Categories: Data and Axioms

Spoken Transcript [00:01:26 - 00:02:53]

So, the last topic yesterday was categories, and I wanted to say a few things.

When I had this discussion about categories yesterday, it was mixed in with all the examples. So, I want to just very quickly give the pure abstract definition and two small points. But this course is not about category theory. It's a topic here because it's natural to introduce when you talk about these algebraic objects; so, anyway, the course is not about category theory.

A category *[writes "Category"]* has data, two types of data *[writes "Cat* $\Rightarrow$*"]*: It has objects *[writes "Obj* $\ni A$*"]*, as we said, and it has homs *[writes "Homs* $\mathrm{Hom}(A, B)$*"]*. So, maybe A can be an object; and for homs, if you pick any two objects, you get a hom-space *[writes* $\mathrm{Hom}(A, B)$ *with a brace]*, sometimes called morphisms.

And the properties, the abstract properties of this hom-space — I said them, but... — is that it's a set *[writes "•* Set*"]*. That's a technical property; that's to avoid some contradictions in foundations, which I'm not going to discuss now. The second is that there's a composition law *[writes "•* $Comp\ Law$*"]*. And the third is that there are identity elements — existence of identity, so *[writes "•* $1_A \in \mathrm{Hom}(A, B)$*", a slip for* $\mathrm{Hom}(A, A)$*]*...

Okay, that's the data. Is there any question about this data?

Constituents of a Category: Data

Recap: Categories were introduced the day before, mixed in with examples (Definition 2.44, Definition 2.49). The lecturer now repeats the abstract definition on its own.

Definition 2.54 (*Category Data*): A "*category*" $\mathcal{C}$ consists of the following mathematical data:

- **Objects:** A class of objects, denoted $\mathrm{Obj}(\mathcal{C})$. We write $A \in \mathrm{Obj}(\mathcal{C})$ (or simply $A \in \mathcal{C}$) to denote that A is an object of $\mathcal{C}$.
- **Morphism Sets (Homs):** For any pair of objects $A, B \in \mathrm{Obj}(\mathcal{C})$, a set of morphisms from A to B, denoted:
$$\mathrm{Hom}(A,B) \quad \text{or} \quad \mathrm{Hom}_{\mathcal{C}}(A,B).$$
An element $f \in \mathrm{Hom}(A,B)$ is represented diagrammatically as an arrow $f\colon A \to B$.
- **Composition Law:** For any three objects $A, B, C \in \mathrm{Obj}(\mathcal{C})$, a binary composition mapping:
$$\begin{aligned} \circ\colon \mathrm{Hom}(B,C) \times \mathrm{Hom}(A,B) &\to \mathrm{Hom}(A,C), \\ (g,f) &\mapsto g \circ f. \end{aligned}$$
- **Identities:** For each object $A \in \mathrm{Obj}(\mathcal{C})$, a designated morphism called the identity morphism on A:
$$\mathrm{Id}_A \in \mathrm{Hom}(A,A) \quad (\text{or } 1_A \in \mathrm{Hom}(A,A)).$$

Foundational Remark on Hom-Sets: The requirement that $\mathrm{Hom}(A,B)$ forms a legitimate set (rather than a proper class, a collection too large to be a set, like the collection of all sets) is a standard set-theoretic convention defining a **locally small category**. This prevents Russell-style self-referential paradoxes when indexing collections of arrows.

Spoken Transcript [00:02:53 - 00:03:45]

And then there are the axioms. The axioms... *[writes "Axioms"]* The first one is that the composition law is associative *[writes "– Comp Law Associative"]*, which means, you know, if you have three things, you can compose them in the two different ways. I hope I don't have to say that more.

And then the second is the axioms about the identity, which is roughly: If I do this, $f \circ \mathrm{Id}_A$, this is f *[writes "$f \cdot \mathrm{Id}_A = f$"]*, and the other way around also *[writes "$\mathrm{Id}_B \cdot g = g$"]*.

So, the identity doesn't change the... We just discussed this last time, so, I wanted to say.

Associativity and Identity Axioms

Definition 2.55 (*Category Axioms*): The data of a category $\mathcal{C}$ must satisfy the following two fundamental axioms:

(1) Associativity of Composition: For all objects $A, B, C, D \in \mathrm{Obj}(\mathcal{C})$ and all mor-

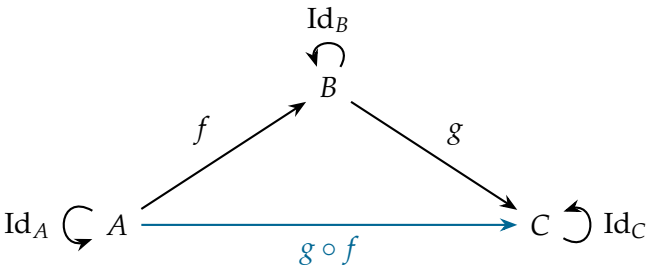
phisms $f \in \mathrm{Hom}(A,B)$, $g \in \mathrm{Hom}(B,C)$, and $h \in \mathrm{Hom}(C,D)$, composition is associative:

$$(h \circ g) \circ f = h \circ (g \circ f). \tag{2.17}$$

(2) Identity Axioms (Neutrality): For all morphisms $f, g \in \mathrm{Hom}(A,B)$, composition with the respective identity morphisms changes nothing (on the board, with a dot for the composition: $f \cdot \mathrm{Id}_A = f$ and $\mathrm{Id}_B \cdot g = g$):

$$f \circ \mathrm{Id}_A = f, \tag{2.18}$$

$$\mathrm{Id}_B \circ g = g. \tag{2.19}$$



Categorical Isomorphisms

Spoken Transcript [00:03:45 - 00:05:37]

So, that's the abstract... There's really not that much here. We discussed this last time, and we had lots of examples.

And so, you could say: Well, why do we want to do this? And as I said, many concepts that we talk about when we talk about groups, rings, fields, vector spaces, they're categorical concepts, meaning that they only depend on this structure. So, it simplifies some kind of thought to think about them all together.

I already told you about isomorphism. *[writes "• Isomorphism"]*

The categorical isomorphism is that if I have this arrow from A to B *[writes "$A \xrightarrow{f} B$"]* — this picture means $f \in \mathrm{Hom}(A,B)$ *[writes "$f \in \mathrm{Hom}(A,B)$"]*. In category theory, you can say this, or you can say that; you mean the same thing.

This f is an isomorphism *[writes "f is an isom."]* — that's a definition — if and only if there exists g the other way *[writes "$\iff \exists A \xleftarrow{g} B$"]*, such that *[writes "such that"]* the two compositions are identities.

And if you want to say which identity, you have to be careful. So, g starts with B, so you do g first, then f back — this should be the identity in B *[writes "$f \circ g = \mathrm{Id}_B$"]*. And this is the identity in A *[writes "$g \circ f = \mathrm{Id}_A$"]*.

So, that's the abstract definition of isomorphism. And in all of the categories we discussed, that coincides with the notion of isomorphism in the category that you already know, like, for example, isomorphism of vector spaces.

Categorical Isomorphism

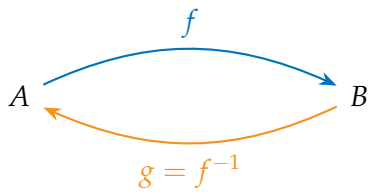
Recap: The categorical isomorphism was defined the day before (Definition 2.50); the lecturer restates it.

Definition 2.56 (*Isomorphism*): Let $\mathcal{C}$ be a category and let $f \in \mathrm{Hom}(A, B)$ (written $f\colon A \to B$). The morphism f is an *"isomorphism"* if and only if there exists a morphism $g \in \mathrm{Hom}(B, A)$ (written $g\colon B \to A$) such that:

$$f \circ g = \mathrm{Id}_B, \tag{2.20}$$

$$g \circ f = \mathrm{Id}_A\,. \tag{2.21}$$

If such a morphism g exists, it is uniquely determined, called the *"inverse"* of f, and denoted $f^{-1} := g$. Two objects A and B are said to be *"isomorphic"* (written $A \cong B$) if there exists an isomorphism between them.



$f \circ g = \mathrm{Id}_B$ and $g \circ f = \mathrm{Id}_A$

Consistency across Concrete Categories: In concrete categories where objects are sets endowed with structure and morphisms are structure-preserving maps (homomorphisms):

- In the category **Set** (objects: all sets; morphisms: all maps between them), isomorphisms are precisely bijective functions (Fact 0.3).
- In **Grp**, **Ring**, and $\mathbf{Vect}_K$, isomorphisms are bijective homomorphisms (for groups: Proposition 2.13).

The power of the categorical definition is that it never inspects elements $x \in A$; it characterizes invertibility purely through diagrammatic composition.

Monomorphisms and Epimorphisms

Spoken Transcript [00:05:37 - 00:07:22]

There are two other notions, and that's the last thing I want to say. And this is the notion of monomorphism. *[writes "• Monomorphism"]*

So, f from A to B is a monomorphism *[writes "$A \xrightarrow{f} B$ is a monomorphism"]* — that's a long word, that's a definition — if *[writes "if"]* for all g_1 arrows... this is another object X, and another one... *[writes "$\forall X \xrightarrow{g_1} A, X \xrightarrow{g_2} A$"]* We take any two, then the composition *[writes "then $f \circ g_1 = f \circ g_2$"]* implies $g_1 = g_2$ *[writes "$\implies g_1 = g_2$"]*.

I hope I wrote that clearly here, okay?

It's a little bit complicated to say. So, I'm defining: If I have an arrow in the category, it may or may not be a monomorphism. I'm defining a monomorphism.

And the condition of being a monomorphism means that whenever I take an object X and map it two ways to A, the composition with f being equal implies they were

equal to start with. Of course, if they're equal to start with, the composition is equal, but it's the other implication. That's the definition of a monomorphism.

You know, you have to think about this and unwind it. But once you have this definition, you can imagine: What is a monomorphism in all of those categories you know? And what do you think it is?

Spoken Transcript [00:07:22 - 00:08:27]

So, I leave this for you to think about, but roughly speaking, a monomorphism is injective. *[writes "(injective)" and circles it on the blackboard]*

Just imagine they're sets. If this is injective, if you take any two maps, you can compare them after you do f. So, you have to think about it.

But this is the definition. Once you take this definition, you can say: Okay, I can take the category of sets and functions, and what is a monomorphism there? And the answer is injective. That's the *"mono"* in monomorphism.

This is a categorical formulation of being injective.

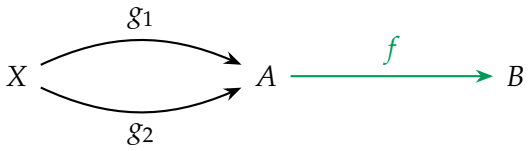
I don't know if you find this appealing or not, but that's what this discussion is about. Because the nice thing of this formulation is that it applies to all examples of categories. Every single one we discussed, and all the other ones you can think about, there's a notion of monomorphism.

And you can imagine this one has a sibling, or a partner — I don't know their relationship — which is... Whenever you talk about injective, you can also talk about surjective, and the categorical term there is epimorphism.

Left-Cancellable Morphisms: Monomorphisms

Definition 2.57 (*Monomorphism*): Let $\mathcal{C}$ be a category. A morphism $f \in \mathrm{Hom}(A, B)$ (or $f\colon A \to B$) is called a *"monomorphism"* (often abbreviated as a **monic morphism** or **mono**) if it is left-cancellable: for every object $X \in \mathrm{Obj}(\mathcal{C})$ and all pairs of parallel morphisms $g_1, g_2 \in \mathrm{Hom}(X, A)$,

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2. \tag{2.22}$$



$f \circ g_1 = f \circ g_2 \implies g_1 = g_2$ (Categorical Injective)

Why Monomorphisms Characterize Injectivity: In the category of sets (**Set**), suppose $f\colon A \to B$ is a monomorphism. To test whether f is injective on points $a_1, a_2 \in A$, let $X = \{*\}$ be a one-element set and define two maps $g_1, g_2\colon \{*\} \to A$ by $g_1(*) = a_1$ and $g_2(*) = a_2$. Then $(f \circ g_1)(*) = f(a_1)$ and $(f \circ g_2)(*) = f(a_2)$. Thus:

$$f(a_1) = f(a_2) \implies f \circ g_1 = f \circ g_2 \implies g_1 = g_2 \implies a_1 = a_2.$$

Hence, left-cancellation with respect to arbitrary test objects X precisely recovers

element-wise injectivity without mentioning elements. Conversely, an injective f is a monomorphism: if $f \circ g_1 = f \circ g_2$, then $f(g_1(x)) = f(g_2(x))$ for every $x \in X$, so $g_1(x) = g_2(x)$ by injectivity, that is, $g_1 = g_2$.

Spoken Transcript [00:08:27 - 00:09:20]

So, you can define epimorphism. *[writes "epimorphism"]* If you have f from A to B, it is an epimorphism *[writes "$A \xrightarrow{f} B$ is an epimorphism"]* if *[writes "if"]* for all...

And this is now about this kind of feeling of mathematics: How do you think I'm going to define the opposite notion of injective?

Like here we had injective by X mapping to A. So, what do you think I should do here?

You should guess. Guessing is like one of the most important tools of a mathematician.

Yeah?

Dualizing the Arrows

Well, you do the same, except we only switch like with the arrows...

Spoken Transcript [00:09:25 - 00:10:37]

This is a long answer. You should just say... Here we map X to A, so the opposite of that is mapping from B. It's like the other side.

So, for every $B \xrightarrow{g_1} X$, $B \xrightarrow{g_2} X$ *[writes "$\forall B \xrightarrow{g_1} X, B \xrightarrow{g_2} X$"]*, then...

This is the kind of opposite of that, by just flipping everything on the other side. So, for every one of these, then we can consider f. In this one, it was the g's first, then the f. Now it's the f first, then the g's: $g_1 \circ f = g_2 \circ f$ implies $g_1 = g_2$ *[writes "then $g_1 \circ f = g_2 \circ f \implies g_1 = g_2$"]*.

So, that's the categorical notion of epimorphism, and it's somehow just dual to the monomorphism. If you like this world, it almost makes epimorphism and monomorphism the same definition up to this flipping. And thinking about things categorically is good for such understanding.

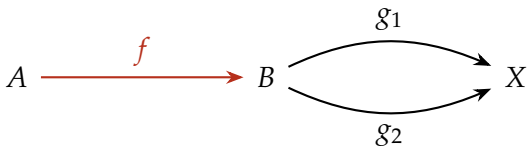
If you look at this definition, and you look at it for the examples we know, this is surjectivity.

Right-Cancellable Morphisms: Epimorphisms

Definition 2.58 (*Epimorphism*): Let $\mathcal{C}$ be a category. A morphism $f \in \mathrm{Hom}(A, B)$ (or $f\colon A \to B$) is called an *"epimorphism"* (often abbreviated as an **epic morphism** or **epi**) if it is right-cancellable: for every object $X \in \mathrm{Obj}(\mathcal{C})$ and all pairs of parallel morphisms

$g_1, g_2 \in \mathrm{Hom}(B, X)$,

$$g_1 \circ f = g_2 \circ f \implies g_1 = g_2. \tag{2.23}$$



$g_1 \circ f = g_2 \circ f \implies g_1 = g_2$ (Categorical Surjective)

Epimorphisms Need Not Be Surjective

The lecturer says that for the examples we know, epimorphism means surjectivity. This holds in **Set**, **Grp** and $\mathbf{Vect}_K$, but not in **Ring**: the inclusion $\iota\colon \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism without being surjective. Indeed, if $g_1, g_2\colon \mathbb{Q} \to R$ are ring homomorphisms with $g_1 \circ \iota = g_2 \circ \iota$, then for $b \neq 0$ the element $g_k(b)$ is invertible with inverse $g_k(\frac{1}{b})$, so $g_1(\frac{a}{b}) = g_1(a)\, g_1(b)^{-1} = g_2(a)\, g_2(b)^{-1} = g_2(\frac{a}{b})$. For monomorphisms the slogan is safe: in **Set**, **Grp**, **Ring** and $\mathbf{Vect}_K$ they are exactly the injective homomorphisms.

***Insight:* Categorical Duality: Reversing Arrows**

Monomorphisms and epimorphisms represent our first concrete encounter with *"categorical duality"*. If we construct the opposite category $\mathcal{C}^{\mathrm{op}}$ by reversing all morphism arrows ($f^{\mathrm{op}}\colon B \to A$ for every $f\colon A \to B$), then:

$$f \text{ is a monomorphism in } \mathcal{C} \iff f^{\mathrm{op}} \text{ is an epimorphism in } \mathcal{C}^{\mathrm{op}}.$$

Every theorem or statement about monomorphisms immediately yields an identical, dual statement about epimorphisms simply by reversing the order of all composition arrows.

Spoken Transcript [00:10:37 - 00:11:03]

Okay, so, that's the end of this discussion about categories. If you like it, you can go read more. As I said, this course is not a course in categories. There's a bunch of other things that I would say here, but I'm not going to, because... yeah.

Maybe I will at the end if I have time.

Integral Domains

Zero Divisors and the Definition of Integral Domains

Spoken Transcript [00:11:03 - 00:12:35]

So, let us return to rings now. *[writes "Rings $(R, +, \cdot, 0, 1)$"]*

This was kind of a foundational side, so, we have this ring $(R, +, \cdot, 0, 1)$ (Recall: commutative with unit, Remark 1.17).

And the rings here, there's kind of a different flavor of rings. There's the rings that are like $\mathbb{Z}$ *[writes "$\mathbb{Z}$"]*, where when you have things that are non-zero and you multiply them, they stay non-zero. That's one of the principles of $\mathbb{Z}$.

But it's not true for rings like $\mathbb{Z}/6\mathbb{Z}$, this ring *[writes "$\mathbb{Z}/6\mathbb{Z}$"]*. I can take two elements here, like the residue class of 2 times the residue class of 3, neither of which is zero. So, this is not zero *[writes "$\bar{2} \neq \bar{0}, \bar{3} \neq \bar{0}$ in $\mathbb{Z}/6\mathbb{Z}$"]*, and that is not zero.

So, I take two elements that are legitimately not zero, but I can multiply them; and when I multiply these residue classes, this is six, and that is zero *[writes "$\bar{2} \cdot \bar{3} = \bar{6} = \bar{0}$"]*.

That's a different type of multiplication. It's a multiplication where when you multiply things, you get zero, which is unexpected from the point of view of the integers. And this is also related to the fact that there's no hope of having a multiplicative inverse if you're mapping things to zero.

So, there are these two differences, and there's a definition for this.

Zero Divisors in Modular Arithmetic

Let $(R, +, \cdot, 0, 1)$ be a commutative ring (Remark 1.17).

Example 2.59 (*Multiplication in $\mathbb{Z}/6\mathbb{Z}$*): In the ring of residue classes modulo 6 (Definition 1.24), denoted $\mathbb{Z}/6\mathbb{Z}$ (or $\mathbb{Z}_6$):

$$\bar{2} \neq \bar{0},$$
$$\bar{3} \neq \bar{0}.$$

However, their product in the ring is:

$$\bar{2} \cdot \bar{3} = \overline{2 \cdot 3} = \bar{6} = \bar{0} \in \mathbb{Z}/6\mathbb{Z}.$$

Zero Divisors vs. Invertibility: Non-zero elements whose product is zero are called *"zero divisors"*. If an element $a \in R$ is a zero divisor (with $a \cdot b = 0$ for some $b \neq 0$), it can never possess a multiplicative inverse a^{-1}:

$$b = 1 \cdot b = (a^{-1} \cdot a) \cdot b = a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0 = 0,$$

which contradicts $b \neq 0$. Hence, rings containing zero divisors fail to exhibit regular arithmetic cancellation.

Spoken Transcript [00:12:35 - 00:14:26]

A ring R is an integral domain *[writes "Def. A ring R is an integral domain"]* — so it's a definition of this term, integral domain. And some authors would just say it's a domain. Also, if you speak about rings, very often mathematicians will say it's a domain.

If... whenever I have a and b, neither of which is zero — so, maybe I'll just be really explicit: whenever I have $a \neq 0$ and $b \neq 0$ *[writes "if $\forall a \neq 0, b \neq 0$"]* — this implies the product is not equal to zero *[writes "$\implies ab \neq 0$"]*.

So, that's the definition. That's a really basic definition in ring theory, and it separates this type of ring from that type of ring. Is that clear?

Fields, for example — for example, fields are always domains *[writes "For example: Fields are always domains."]* because you have the multiplicative inverse, if you think

about it, okay?

So, see, I did that already there: I called it a domain, not an integral domain, because it takes a lot of syllables to say integral domain. Formally, this is defined as an integral domain; colloquially, almost everyone says domain.

Definition of an Integral Domain (Initial Board State)

Definition 2.60 (*Integral Domain / Domain*)**:** A commutative ring R is called an "*integral domain*" (or simply a "*domain*") if for all $a, b \in R$:

$$a \neq 0 \quad \text{and} \quad b \neq 0 \implies a \cdot b \neq 0. \tag{2.24}$$

Equivalently, taking the contrapositive:

$$a \cdot b = 0 \implies a = 0 \quad \text{or} \quad b = 0.$$

Example 2.61 (*Fields are Domains*)**:** Every field F is an integral domain: if $a \neq 0$ and $a \cdot b = 0$, then multiplying by the inverse $a^{-1} \in F$ immediately yields:

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

Student Discussion: Monomorphisms in Linear Algebra

Spoken Transcript [00:14:26 - 00:14:43]

These terms, monomorphism and epimorphism, have you ever heard these terms before? In which context?

Context: Linear Algebra

Linear algebra.

Spoken Transcript [00:14:45 - 00:14:50]

Ah, okay. And they were just used synonymously with injection and surjection?

Terminology Shift

They were introduced, but then we went back to using injective and surjective.

Spoken Transcript [00:14:54 - 00:14:57]

Okay, yeah.

Homomorphisms Context

Still they were like injective and surjective homomorphisms.

Spoken Transcript [00:15:02 - 00:15:33]

Yes, exactly. Yes, injective and surjective homomorphisms.

That's precisely what this categorical thing says, right? Because in the category of groups, the arrows are homomorphisms; and when a homomorphism is a monomorphism, it's an injective homomorphism.

In my experience with undergraduates, some really like the idea of categories, and they really like to think in that language, and others like the examples. So, it can be... yeah, either way is okay.

Characterization of $\mathbb{Z}/m\mathbb{Z}$ as a Domain

Spoken Transcript [00:15:33 - 00:16:03]

All right, so, here we come to the first proposition, which is something my guess is you are already familiar with, but we do it again.

We take this ring, $(\mathbb{Z}/m\mathbb{Z}, +, \cdot, \bar{0}, \bar{1})$ *[writes "Prop. $(\mathbb{Z}/m\mathbb{Z}, +, \cdot, \bar{0}, \bar{1})$ $m \in \mathbb{Z}_{>0}$"]*, and the claim is that this $\mathbb{Z}/m\mathbb{Z}$ is a domain if and only if... And what am I going to write here?

Condition for Domain

m is prime.

Spoken Transcript [00:16:05 - 00:17:09]

Yeah, m is prime. m is prime. *[writes "$\mathbb{Z}/m\mathbb{Z}$ is a domain $\iff$ m is prime"]*

Oh, yeah, I should say one thing about domains which is important, which I didn't say, and it's kind of annoying. Part of the domain... I should put that in because it's... *[pulls down the middle blackboard to modify the earlier definition]*

It's a super annoying thing to talk about, but I talk about any ring here, so my definition has to deal with the zero ring — the one that, you know, is unpleasant to think about. And you could think about: What would this definition be for the zero ring? Because for all a and b, there are no a's and b's that are not equal to zero. So, formally this statement, in the formal deductive logical sense, is true for a zero ring.

So, it is part of the definition: a ring R which is not the zero ring is a domain *[adds $\neq 0$ under the R of the definition on the board]*. Zero rings are not allowed to be domains, by fiat, in the definition. *[pushes the blackboard back up]*

Correction: Exclusion of the Zero Ring

The lecturer corrects Definition 2.60 by adding condition (i):

Definition 2.62 (*Integral Domain (Formal Standard Definition)*): A commutative ring R is an *"integral domain"* (or *"domain"*) if:

(i) $R \neq 0$ (that is, $1_R \neq 0_R$: the zero ring is excluded; cf. Proposition 1.10).
(ii) For all $a, b \in R$:

$$a \neq 0 \quad \text{and} \quad b \neq 0 \implies a \cdot b \neq 0.$$

Why the Zero Ring Satisfies the Conditional Vacuously: In the zero ring $R = \{0\}$, the only element is 0. Therefore, there exist no non-zero elements $a, b \in R$. The premise $(a \neq 0 \wedge b \neq 0)$ is vacuously false for all elements, making the logical implication $(a \neq 0 \wedge b \neq 0 \implies ab \neq 0)$ vacuously true! To avoid having the degenerate zero ring classified as a domain, mathematicians add $R \neq 0$ explicitly by fiat.

Proposition: When Residue Class Rings are Domains

Proposition 2.63: Let $m \in \mathbb{Z}_{>0}$ be a positive integer. The residue class ring $(\mathbb{Z}/m\mathbb{Z}, +, \cdot, \bar{0}, \bar{1})$ is an integral domain if and only if m is a prime number:

$$\mathbb{Z}/m\mathbb{Z} \text{ is an integral domain} \iff m \text{ is prime.} \tag{2.25}$$

Proof of Proposition 2.63

Spoken Transcript [00:17:09 - 00:19:24]

Okay, $\mathbb{Z}/m\mathbb{Z}$ is a domain if and only if m is prime. So, let's try to prove this.

Proof. *[writes "Proof"]*

This is like one of the issues I said: this issue about the zero ring is a little bit problematic everywhere in the statements. Because for any statement you write, *"let R be a ring"*, you have to think — if you want that statement to be literally true in the mathematical sense from the definitions in this class — what that statement could mean for the zero ring and whether it's true. And, as I said, I don't view it as a deep or really important thing to do, but in terms of bookkeeping and saying things correctly, you have to say it.

Okay, so, let's try to prove this: it is a domain if and only if m is prime.

(1): Suppose m is not prime. *[writes "(1) Suppose m is not prime"]*

We have m here as some positive integer. If m is not prime, then this means that m admits a factorization into ab *[writes "$m = ab$"]*, where a is strictly greater than 1, strictly less than m *[writes "$1 < a < m$"]*, and b is strictly like this: $1 < b < m$ *[writes "$1 < b < m$"]*. That's what it means: If it's not prime, I can factor it; and if I can factor it, I break it in two, into smaller pieces.

And then if I look at this equation as a residue class, this equals that, which is equal to zero in $\mathbb{Z}/m\mathbb{Z}$ *[writes "$\bar{a} \cdot \bar{b} = \bar{m} = \bar{0} \in \mathbb{Z}/m\mathbb{Z}$"]*. And this shows that $\bar{a}$ is not zero *[writes "$\bar{a} \neq \bar{0}$"]*, and this shows that $\bar{b}$ is not zero *[writes "$\bar{b} \neq \bar{0}$"]*.

Okay? So, this says that if m is not prime, there's no way for it to be a domain, because it factors, and those factors precisely give the violation of that property (i.e., condition **(ii)** of Definition 2.62). All right?

Proof of Proposition: Direction 1 (Non-Prime Case)

Direction 1: m Not Prime $\implies$ $\mathbb{Z}/m\mathbb{Z}$ Not a Domain.

Suppose m is not prime. If $m = 1$, then $\mathbb{Z}/1\mathbb{Z} = \{\bar{0}\}$ is the zero ring, which is not a domain by condition (i) of Definition 2.62 (see also Remark 2.64). Otherwise $m > 1$ is composite and admits a proper non-trivial factorization in $\mathbb{Z}$:

$$\begin{aligned} m &= a \cdot b, \\ 1 &< a < m, \\ 1 &< b < m. \end{aligned} \tag{2.26}$$

Passing to the residue classes in the quotient ring $\mathbb{Z}/m\mathbb{Z}$:

$$\bar{a} \cdot \bar{b} = \overline{a \cdot b} = \bar{m} = \bar{0} \in \mathbb{Z}/m\mathbb{Z}. \tag{2.27}$$

However, because $1 < a < m$ and $1 < b < m$, neither a nor b is divisible by m. Therefore:

$$\bar{a} \neq \bar{0} \quad \text{and} \quad \bar{b} \neq \bar{0}. \tag{2.28}$$

The existence of non-zero elements $\bar{a}, \bar{b} \in \mathbb{Z}/m\mathbb{Z}$ satisfying $\bar{a} \cdot \bar{b} = \bar{0}$ demonstrates that $\mathbb{Z}/m\mathbb{Z}$ possesses zero divisors, so $\mathbb{Z}/m\mathbb{Z}$ is not an integral domain.

Spoken Transcript [00:19:24 - 00:21:50]

So, then the other thing is: Suppose it is prime.

Suppose $m = p$ is prime. *[writes "Suppose $m = p$ is prime"]* Then let a and b be integers *[writes "Then let $a, b \in \mathbb{Z}$"]* such that $\bar{a} \cdot \bar{b} = \bar{0}$ in $\mathbb{Z}/p\mathbb{Z}$ *[writes "such that $\bar{a} \cdot \bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$"]*.

All right? What are we going to do here? If I translate this equation on residue classes to integers, this means that $a \cdot b$ is equal to some integer $d \cdot p$ *[writes "$ab = dp$"]*.

I am using this notation and arithmetic in $\mathbb{Z}/m\mathbb{Z}$ freely here, because I assume you've seen this before. Is this true? If it's not, you should tell me. All right. If you haven't seen it, you should study it.

So, if I take these two residue classes and they multiply to zero, that exactly means that the product is divisible by p, and that is that difference. Is that clear?

And then this means that p divides ab, because ab is d times p *[writes "$p \mid ab$"]*. And then there's the principle of Euclid we discussed in the first week that says: either $p \mid a$ or $p \mid b$ *[writes "$\implies$ Euclid either $p \mid a$ or $p \mid b$"]*. It wasn't such a long time ago; it was just last week, right?

And if this is true, then that means $\bar{a} = \bar{0}$ *[writes "$\implies \bar{a} = \bar{0}$"]*; and if this is true, that means $\bar{b} = \bar{0}$ *[writes "or $\bar{b} = \bar{0}$"]*.

So, we've proven that if the product is zero in $\mathbb{Z}/p\mathbb{Z}$, the only way that happens is one of them is zero. And that's logically equivalent to this: If they were both not zero, then the product is not zero. Or the logical equivalent, which is the contrapositive, is that if the product is zero, then one of the factors must be zero, which we've established. So, this proposition is proven.

Proof of Proposition: Direction 2 (Prime Case via Euclid's Lemma)

Direction 2: $m = p$ Prime $\implies \mathbb{Z}/p\mathbb{Z}$ Is a Domain.

Suppose $m = p$ is a prime number. Let $\bar{a}, \bar{b} \in \mathbb{Z}/p\mathbb{Z}$ and assume that their product vanishes:

$$\bar{a} \cdot \bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}. \tag{2.29}$$

By definition of modular multiplication, this implies $\overline{a \cdot b} = \bar{0}$, meaning that $a \cdot b$ is congruent to $0 \pmod{p}$. Hence, there exists some integer $d \in \mathbb{Z}$ such that:

$$a \cdot b = d \cdot p \implies p \mid (a \cdot b). \tag{2.30}$$

By Euclid's Lemma (which holds whenever p is prime, Fact 0.17):

$$p \mid (a \cdot b) \implies p \mid a \quad \text{or} \quad p \mid b. \tag{2.31}$$

Translating each case back into modular equivalence in $\mathbb{Z}/p\mathbb{Z}$:

- If $p \mid a$, then $\bar{a} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$.
- If $p \mid b$, then $\bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$.

Thus, $\bar{a} \cdot \bar{b} = \bar{0}$ implies $\bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$. This establishes that $\mathbb{Z}/p\mathbb{Z}$ has no zero divisors. Because $p \geq 2$, $\mathbb{Z}/p\mathbb{Z}$ contains at least two distinct elements ($\bar{0} \neq \bar{1}$), so $\mathbb{Z}/p\mathbb{Z}$ is not the zero ring. Therefore, $\mathbb{Z}/p\mathbb{Z}$ is an integral domain.

Q.E.D.

Boundary Cases and Principle of Lecturing

Spoken Transcript [00:21:50 - 00:23:18]

You could say again in this proposition: When someone writes this, I here put m to be a positive number, and now it matches carefully. But these boundary cases... There's some aspect of this foundational part where you have to be careful yourself about the boundary cases.

So, if I put $m = 1$, then I get the zero ring; but 1 is not considered a prime number, so it matches. You could say: What happens if I put zero? It's not here. We don't think about the primality of zero, so zero is left out of this. But you could still say: If m is zero, this is $\mathbb{Z}$, and it is a domain. Okay?

All right.

Sometimes being too clear is actually more confusing in lectures. There's this kind of principle of lecturing where you lie about all the little details, and the lectures come out cleaner if you do that. Nah. And it's not clear didactically that that isn't a better way to do it. *[erases the middle and right blackboards]*

There's a certain kind of mathematician that takes pleasure in these edge cases, and you can tell that I'm not one of them.

Pedagogical Analysis of Boundary Cases

Remark 2.64 (*Boundary Cases: $m = 1$ and $m = 0$*): The proposition specifies $m \in$

$\mathbb{Z}_{>0}$. Let us examine the boundary cases:

- **Case $m = 1$:** The ideal is $1\mathbb{Z} = \mathbb{Z}$. The quotient ring is the zero ring:
$$\mathbb{Z}/1\mathbb{Z} \cong \{0\},$$
where $\bar{1} = \bar{0}$. Since the definition of an integral domain explicitly requires $R \neq \{0\}$, $\mathbb{Z}/1\mathbb{Z}$ is **not** an integral domain. This perfectly matches the number-theoretic convention that 1 is not a prime number.
- **Case $m = 0$:** The zero ideal is $0\mathbb{Z} = \{0\}$. The quotient ring is canonically isomorphic to the integers:
$$\mathbb{Z}/0\mathbb{Z} \cong \mathbb{Z}.$$
The integers $\mathbb{Z}$ are the prototypical integral domain. In modern commutative algebra, $\{0\} \subset \mathbb{Z}$ is a prime ideal precisely because $\mathbb{Z}/\{0\} \cong \mathbb{Z}$ is an integral domain!

Divisibility and Units in Commutative Rings

Divisibility in General Rings

Spoken Transcript [00:23:18 - 00:24:29]

Okay, so, something more fun is we get to talk about units, and this is actually pretty fun.

This has to do with every ring; so, let's say here, we take a ring R *[writes "R ring"]*. It's our ring.

And this is now related to last week; in fact, it was the point of the discussion last week. There we discussed the ring $\mathbb{Z}$, but now we can take any ring.

So, the definition here is that if we have elements a and b in the ring *[writes "Def. if $a, b \in R$"]*, then we say — this is the definition of the term — we say a divides b *[writes "we say a divides b"]*. What this means is there exists some element of the ring c *[writes "$\exists c \in R$"]* such that $a \cdot c = b$ *[writes "$ac = b$"]*.

Okay? That's just the actual definition of division, and it's the same definition as with the integers. So, this is a notion of division for any ring.

Divisibility Relations

Let R be a ring (commutative, as always in this course).

Definition 2.65 (*Divisibility in a Ring*): Let $a, b \in R$. We say that "*a divides b*", denoted:
$$a \mid b, \tag{2.32}$$
if there exists an element $c \in R$ such that:
$$a \cdot c = b. \tag{2.33}$$
In this case, we also say that b is a "*multiple*" of a, or that a is a "*divisor*" of b.

Connection with Principal Ideals: In ideal-theoretic language, $a \mid b$ in a com-

mutative ring R is equivalent to the statement that the principal ideal generated by b is contained in the principal ideal generated by a (principal ideals $(a) = aR$: Definition 4.32, Week 4):

$$a \mid b \iff b \in aR \iff (b) \subset (a).$$

Units and Multiplicative Inverses

Spoken Transcript [00:24:29 - 00:26:22]

Now, $u \in R$ is a unit *[writes "$u \in R$ is a unit"]* — that's the definition of a unit. The definition is: if u divides 1 *[writes "if $u \mid 1$"]*.

Oh, I should say that also: *"a divides b"*, that's how you say it in English, and the symbol for that is $a \mid b$ *[writes "$(a \mid b)$"]*.

So, here this is now a definition: An element is a unit if u divides 1. And that is equivalent to: There exists a w such that *[writes "$\iff \exists w$ such that "]* $u \cdot w = 1$ *[writes "$uw = 1$"]*. I never have to worry about the order here, because I said that for the rings for this course, unless I tell you explicitly otherwise, we're always in commutative rings. Okay?

So, this is the definition, this is just the equivalence unwinding division; and if I want to unwind this using English — so, this is the definition of division *[writes "Def of div"]* — this is just English, which we say: if and only if u has a multiplicative inverse *[writes "$\iff u$ has mult. inverse in R"]*. Multiplicative inverse in R.

This is a really fundamental definition: a unit in a ring.

Are you happy with this? That's the definition, this is just translation, and that's what you say: u has a multiplicative inverse, w is the multiplicative inverse. Is there any problem with this?

Invertible Elements

Definition 2.66 (*Unit in a Ring*): Let R be a commutative ring with unity 1. An element $u \in R$ is called a *"unit"* if u divides 1:

$$u \mid 1. \tag{2.34}$$

By the definition of divisibility (Definition 2.65), this is equivalent to:

$$\exists w \in R \quad \text{such that} \quad u \cdot w = 1. \tag{2.35}$$

In words, u is a unit if and only if u has a multiplicative inverse in R. The element w is uniquely determined and denoted $u^{-1} := w$.

Equivalence of Characterizations on the Blackboard: The blackboard outlines three equivalent perspectives on units:

(1) **Divisibility Perspective:** $u \mid 1$ (the element divides the multiplicative identity).
(2) **Algebraic Perspective (Unwinding Division):** $\exists w \in R$ such that $u \cdot w = 1$.
(3) **Verbal Perspective (English):** u possesses a multiplicative inverse in R.

Since all rings considered here are commutative, left-inverses and right-inverses coincide.

Linguistic Conventions: The Units, The Unit, and A Unit

Spoken Transcript [00:26:22 - 00:28:16]

Actually, people asked me some questions about language, and, you know, English is a language I know, which means that I actually speak it without thinking. It was pointed out to me that the use of *"unit"* is subtle; and I was doing it already in the first week, and if you were sensitive to language, you'd have seen.

So, for $\mathbb{Z}$ here, what are the units for $\mathbb{Z}$? *[writes "What are units in $\mathbb{Z}$ ="]* In $\mathbb{Z}$, and it comes as no surprise, this is $+1$ and -1 *[writes "$\{1, -1\}$"]*. Those are the only invertible elements in it.

So, the inertia of language: *"The units"* *[writes "The units:"]* — if I say *"the units"* here, that means all of them: $\{1, -1\}$ *[writes "$\{1, -1\}$"]*. The units.

If I say *"the unit"* *[writes "The unit:"]* — this is standard use of mathematical English — if I say *"the unit"*, that's just 1 *[writes "1"]*. That's the one. If I have a ring and say, *"let's consider the unit"*, that means 1.

You could say *"a unit"* *[writes "A unit:"]*. If I say, *"let's consider a ring, and let's consider a unit"*, that could be either one *[writes "could be either"]*. We don't know; it could be either. A unit: this could be either.

So, that's actually standard usage. It's a little confusing, but once you get used to it, you know, it's just language. Is it clear? It was pointed out to me that I was doing that, and anyway, that's right. But if someone comes to you with a ring and says *"consider the unit"*, that person means 1.

Units of the Integers and Linguistic Distinctions

Example 2.67 (*Units of the Ring of Integers $\mathbb{Z}$*): In the ring of integers $\mathbb{Z}$, the only elements possessing multiplicative inverses are 1 and -1:

$$\text{The units of } \mathbb{Z} = \{1, -1\}. \tag{2.36}$$

Remark 2.68 (*Linguistic Distinction in Mathematical English*): In mathematical English, the word *"unit"* carries three distinct grammatical usages depending on the article:

- ***"The units"* (Plural):** Refers to the entire set of all invertible elements in the ring R, denoted $\mathcal{U}_R$ or $R^\times$. For $\mathbb{Z}$, the units are $\{1, -1\}$.
- ***"The unit"* (Singular, Definite Article):** Refers specifically to the multiplicative identity element $1_R \in R$.
- ***"A unit"* (Singular, Indefinite Article):** Refers to an arbitrary invertible element $u \in R$ (i.e., any element $u \in \mathcal{U}_R$). For $\mathbb{Z}$, *"a unit"* could be either 1 or -1.

The Group of Units $\mathcal{U}_R$

Spoken Transcript [00:28:16 - 00:29:52]

We discussed the units of the Gaussian integers (Recall: $\pm 1, \pm i$, Week 1, Proposition 1.2), and now, since everything is defined, I recommend proving what the units of the Gaussian integers are. We wrote them last time, but let's do something else now. *[erases upper middle board and writes "Ex. Prove units of $\mathbb{Z}[i]$ are $\{\pm 1, \pm i\}$"]*

This is an exercise: Prove the units of the Gaussian integers are $\pm 1, \pm i$. You need an argument for that, and that argument comes from the norm in complex analysis so that you prove it.

Okay, so, observation *[writes "Observation"]* — and it's really an important observation: If I take any ring R *[writes "R ring"]*, and let $\mathcal{U}_R$ *[writes "let $\mathcal{U}_R$ be the set of units of R"]* be the set of units of R.

Then this observation is: If I take the units, I take multiplication — that's multiplication in the ring — and I take the unit 1, this is a group *[writes "$(\mathcal{U}_R, \cdot, 1) \leftarrow$ Group"]*...

Exercise: Units of the Gaussian Integers

Exercise 2.69 (*Units of the Gaussian Integers*): Prove that the units of the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ are precisely:

$$\mathcal{U}_{\mathbb{Z}[i]} = \{\pm 1, \pm i\}. \tag{2.37}$$

The lecturer writes out this proof in Week 3 (Proposition 3.2).

Proof Strategy via the Multiplicative Norm: Consider the field norm $N \colon \mathbb{Q}(i) \to \mathbb{Q}$ defined by $N(a + bi) = a^2 + b^2$. For all $z, w \in \mathbb{Z}[i]$, the norm is multiplicative:

$$N(z \cdot w) = N(z) \cdot N(w).$$

If $z \in \mathbb{Z}[i]$ is a unit, there exists $w \in \mathbb{Z}[i]$ with $z \cdot w = 1$. Taking norms yields:

$$N(z) \cdot N(w) = N(1) = 1.$$

Since $a, b \in \mathbb{Z}$, $N(z) = a^2 + b^2$ is a non-negative integer. The only positive integer divisor of 1 in $\mathbb{Z}$ is 1, so $a^2 + b^2 = 1$. The only integer solutions are:

$$(a, b) \in \{(\pm 1, 0), (0, \pm 1)\} \implies z \in \{\pm 1, \pm i\}.$$

Spoken Transcript [00:29:52 - 00:30:47]

What does it say? If I take two units and I multiply them, I get a unit.

And that's clear because, you know, if I have two elements here, $u_1, u_2 \in \mathcal{U}_R$ *[writes "$u_1, u_2 \in \mathcal{U}_R$"]*, if I multiply them *[writes "$u_1 u_2$"]*, I have to show that's a unit. That means I have to show that if I multiply by some more stuff, I get 1.

But u_1 is a unit, so there's already its multiplicative inverse; and u_2 is a unit, so there's already its multiplicative inverse *[writes "$u_1 u_2 w_1 w_2 = 1$"]*. So, when I multiply these together, I get 1, because I can flip the order around. This says that the units are closed under multiplication.

And there's the neutral element; and then you have to show the existence of an inverse, but a unit is a unit because it has an inverse (Recall: Definition 2.66).

So, that's the proof of this. This is a good thing to think about: this is always a group.

The Group of Units

Proposition 2.70 (*Group of Units*): Let R be a (commutative) ring with unity 1_R, and let $\mathcal{U}_R$ (also denoted $U(R)$ or $R^\times$) be the set of units of R:

$$\mathcal{U}_R := \{u \in R \mid \exists w \in R \colon u \cdot w = 1_R\}. \tag{2.38}$$

Then $(\mathcal{U}_R, \cdot, 1_R)$ is a group (Definition 2.1) under the multiplication of R.

Proof

We verify the group axioms for $(\mathcal{U}_R, \cdot, 1_R)$:

- **Closure under Multiplication:** Let $u_1, u_2 \in \mathcal{U}_R$. By definition, there exist $w_1, w_2 \in R$ such that $u_1 w_1 = 1_R$ and $u_2 w_2 = 1_R$. Since R is commutative:
$$u_1 u_2 w_1 w_2 = (u_1 w_1)(u_2 w_2) = 1_R \cdot 1_R = 1_R.$$
Thus, $u_1 u_2$ has a two-sided multiplicative inverse in R, which shows $u_1 u_2 \in \mathcal{U}_R$.
- **Associativity:** Multiplicative associativity in $\mathcal{U}_R$ is directly inherited from the ring R.
- **Identity Element:** Since $1_R \cdot 1_R = 1_R$, the unity 1_R is its own inverse, so $1_R \in \mathcal{U}_R$. For all $u \in \mathcal{U}_R$, $1_R \cdot u = u$.
- **Existence of Inverses:** For any $u \in \mathcal{U}_R$, there exists $w \in R$ such that $uw = 1_R$. By symmetry of the relation $wu = 1_R$, w is also invertible with inverse u, so $w = u^{-1} \in \mathcal{U}_R$.

Hence, $(\mathcal{U}_R, \cdot, 1_R)$ is a group in the sense of Definition 2.1. If R is commutative, $\mathcal{U}_R$ is an abelian group. Q.E.D.

Units in Residue Class Rings $\mathbb{Z}/m\mathbb{Z}$

Characterization of Units via Coprimality

Spoken Transcript [00:30:47 - 00:32:27]

A very natural question, and one of the first things to think about regarding $\mathbb{Z}/m\mathbb{Z}$... If I take this ring $R = \mathbb{Z}/m\mathbb{Z}$ *[writes "$R = \mathbb{Z}/m\mathbb{Z}$"]* — and that was implicit there — I will always consider $m > 0$ here when I do this *[writes "$m > 0$"]*, because there's no reason to consider the negative ones, because it's just counting everything twice.

And then there's the confusing thing when $m = 0$, which is usually a special case, so I exclude it. Usually we allow $m = 1$, which is the zero ring, but anyway, this is kind of standard. When I write $\mathbb{Z}/m\mathbb{Z}$, I always mean this; and typically I also mean it's bigger than 1, but anyway.

So, the question *[writes "Question"]* is: What are the units *[writes "What are the units*

of $\mathbb{Z}/m\mathbb{Z}$?"] of $\mathbb{Z}/m\mathbb{Z}$?

Suppose we have $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$ *[writes "Suppose $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$"]*, and $\bar{x}$ is a unit *[writes "and $\bar{x}$ is a unit"]* — a unit. This means that there exists $\bar{y}$ such that *[writes "$\exists \bar{y}$ such that"]* $\bar{x}\bar{y} = \bar{1}$ in $\mathbb{Z}/m\mathbb{Z}$ *[writes "$\bar{x}\bar{y} = \bar{1} \in \mathbb{Z}/m\mathbb{Z}$"]*. That's what it means to be a unit; I'm just repeating it, right?

Spoken Transcript [00:32:27 - 00:33:32]

And then I can translate: What does this mean in this ring? What does it mean for the integers x, y, and 1?

This is equivalent to: xy is equal to 1 plus some multiple of m *[writes "$xy = 1 + dm$"]*. Do you agree with that? That's what it means, for some d. Okay?

And this, if you remember from the first week (Recall: Bézout's Identity), means that x and m have gcd equal to 1 *[writes "(x, m) have $\gcd = 1$"]*. The only integer that could divide both x and m would also have to divide 1. So, that's what we say: gcd equal to 1.

And in English, this condition is usually said as: x and m are relatively prime *[writes "x, m are relatively prime"]*. Those are just words that express $\gcd = 1$, okay?

Spoken Transcript [00:33:32 - 00:35:02]

We have now discovered what the units of $\mathbb{Z}/m\mathbb{Z}$ are: They're all the elements which are relatively prime. Because if it is a unit, it's the residue class of a relatively prime integer. If you have $\gcd = 1$, then you go backwards in this diagram.

If you write this as a result, then it says that the units of $\mathbb{Z}/m\mathbb{Z}$ *[writes "The units of $\mathbb{Z}/m\mathbb{Z}$"]* consist of all elements $\bar{x}$ *[writes "consist of $\{\bar{x} \mid$"]* where $\gcd(x, m) = 1$ *[writes "$\gcd(x, m) = 1\}$"]*.

Did I write GCD here? Maybe GCD. Okay? Is that clear?

And so, the thing we've learned is that this unit group, the units of $\mathbb{Z}/m\mathbb{Z}$, are a group *[writes "$\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$ are a group"]*. In particular, if you take $\mathbb{Z}/m\mathbb{Z}$ and you take the residue classes that are relatively prime to m, they form a group under multiplication.

Units of the Quotient Ring $\mathbb{Z}/m\mathbb{Z}$

Theorem 2.71 (*Units in $\mathbb{Z}/m\mathbb{Z}$*): Let $m \in \mathbb{Z}_{>0}$ be a positive integer. An element $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$ is a unit if and only if x and m are coprime (this does not depend on the representative x, since $\gcd(x + km, m) = \gcd(x, m)$):

$$\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}} = (\mathbb{Z}/m\mathbb{Z})^\times = \left\{\bar{x} \in \mathbb{Z}/m\mathbb{Z} \;\middle|\; \gcd(x, m) = 1\right\}. \tag{2.39}$$

The order of this group is given by Euler's totient function $\phi(m) := |\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}|$ (named by the lecturer in Week 3, Definition 3.5).

Proof

We establish the equivalence by unwinding the definition of divisibility:

- **Forward Direction ($\implies$):** Suppose $\bar{x} \in \mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$. Then there exists $\bar{y} \in \mathbb{Z}/m\mathbb{Z}$ such that:
$$\bar{x} \cdot \bar{y} = \bar{1} \in \mathbb{Z}/m\mathbb{Z}.$$
In terms of integer divisibility, this means $xy \equiv 1 \pmod{m}$. Hence, there exists an integer $d \in \mathbb{Z}$ such that:
$$xy = 1 + d \cdot m \iff xy - d \cdot m = 1.$$
Any common divisor $g \in \mathbb{Z}$ of x and m must divide the linear combination $xy - dm = 1$. Therefore, $g \mid 1$, which forces $\gcd(x, m) = 1$.
- **Reverse Direction ($\impliedby$):** Conversely, suppose $\gcd(x, m) = 1$. By Bézout's Identity (Fact 0.16; computed in Example 0.18), there exist integers $y, k \in \mathbb{Z}$ such that:
$$x \cdot y + m \cdot k = 1 \implies x \cdot y \equiv 1 \pmod{m}.$$
Projecting this relation into residue classes in $\mathbb{Z}/m\mathbb{Z}$, we obtain $\bar{x} \cdot \bar{y} = \bar{1}$. Thus, $\bar{x}$ is invertible, so $\bar{x} \in \mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$.

By Proposition 2.70, $(\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}, \cdot, \bar{1})$ forms a finite abelian group. Q.E.D.

The Units of $\mathbb{Z}/7\mathbb{Z}$ and Cyclic Generator Calculations

Spoken Transcript [00:35:02 - 00:36:08]

Let's do an example: $\mathbb{Z}/7\mathbb{Z}$ *[writes "$\mathbb{Z}/7\mathbb{Z}$"]*. That's actually a prime one, but let's do this example.

The elements of this are $\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}$ *[writes "$= \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$"]*, right? And which of these residue classes come here? They're classes of x where x and 7 are relatively prime.

So, the question is: What are the units? *[writes "Unit of $\mathbb{Z}/7\mathbb{Z} =$"]*

Of course, it's some of these. Well, which of these residue classes correspond to x and 7 being relatively prime? What would you say? As I said, if you don't know, guess.

All Elements Guess

All of them.

Spoken Transcript [00:36:11 - 00:36:14]

All of them? Well, what about this one? *[points at $\bar{0}$]*

Excluding Zero

Only 1, 1 to 6 are the units.

Spoken Transcript [00:36:16 - 00:36:50]

Exactly. The answer to this question is these: *[writes "$\{\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6}\}$"]*.

And the reason is because, look, this was the answer: It's the residue classes of those numbers that are relatively prime to 7. If you're $0 \pmod 7$, then you're not relatively prime to 7. As an example, $0 \pmod 7$ would be 14, and that's not relatively prime to 7. So, it's these.

Is that clear?

The Unit Group of $\mathbb{Z}/7\mathbb{Z}$

Example 2.72 (*Units of $\mathbb{Z}/7\mathbb{Z}$*): Consider the quotient ring $\mathbb{Z}/7\mathbb{Z}$. The elements of the ambient ring are:

$$\mathbb{Z}/7\mathbb{Z} = \{\bar{0},\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6}\}.$$

Since 7 is a prime number, every non-zero residue class $\bar{x} \in \{\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6}\}$ satisfies $\gcd(x,7) = 1$. The zero class $\bar{0}$ is not coprime to 7 (since $\gcd(0,7) = 7 \neq 1$, or for example $14 \equiv 0 \pmod 7$ where $\gcd(14,7) = 7$). Consequently, the group of units is:

$$\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}) = (\mathbb{Z}/7\mathbb{Z})^{\times} = \{\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6}\}. \tag{2.40}$$

The order of the unit group is $|\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})| = \phi(7) = 7 - 1 = 6$.

Spoken Transcript [00:36:50 - 00:37:28]

Okay, and the claim is this is a group under multiplication. Well, anyway, $\mathbb{Z}/7\mathbb{Z}$ is a group under addition; but if you take the relatively prime ones, this is a group under multiplication.

So, we should be able to figure out: What's the group?

What's the neutral element of this group? Yeah, 1 — it's going to be $\bar{1}$.

What is this group? It's a commutative group of order 6. And so, later we'll see that we know all commutative groups of order 6. We'd like to know which one this is. It's isomorphic to some group we know.

And so, it matters which element you pick. I propose we pick 3.

Spoken Transcript [00:37:28 - 00:38:51]

Let's say we take this element in this group, $a = \bar{3}$ *[writes "$a = \bar{3} \in \mathrm{Unit}\,\mathbb{Z}/7\mathbb{Z}$"]*. That's an element of this unit group.

And let's try to consider its powers. That's the neutral element *[writes "$\bar{1}$"]*, then I can take $a, a^2, a^3, a^4, a^5, a^6$ *[writes "$\bar{1}, a, a^2, a^3, a^4, a^5, a^6$"]*. You should know the powers of 3. Okay, let's see how many powers of 3 you know.

So, what is this one? I'm just going to do this: This is $\bar{1}$ *[writes $\bar{1}$ under $\bar{1}$]*.

What is a? Well, that's just $\bar{3}$ *[writes $\bar{3}$ under a]*.

What is this? That's the square of a. That's the residue class of 3^2, which is 9. So, I should put a 2 here, right? *[writes $\bar{2}$ under a^2]* Are you happy with this?

And this is a^3: 3^3 is 27. For 27, the residue modulo 7 is 6, right? *[writes $\bar{6}$ under a^3]* a^4 is 3^4, which is 81; and the residue modulo 7 of 81 is 4, I think *[writes $\bar{4}$ under a^4]*. 3^5 is 243, and its residue is going to be 5 *[writes $\bar{5}$ under a^5]*.

Spoken Transcript [00:38:51 - 00:39:45]

What do you think this is going to be? *[points at a^6]*

Do you know what 3^6 is? 3^6 is 729 *[writes "$729 = 3^6$"]*. That's 3^6. And if I look at its residue class modulo 7 — well, 7 is already here, and 28 is divisible — its residue class is 1 *[writes $\bar{1}$ under a^6]*. This is back to 1.

So, this is understanding the group. You should know the powers of 3; they're very useful.

This is what happens when you multiply in this group: You take a; a^2 is the next one; a^3 the next one; a^4 the next one; a^5 this one; and then when you go to the sixth, you go back to the identity.

What group do you know this happens in? Does this remind you of anything? Yeah.

Group Identification

It's a cyclic group.

Spoken Transcript [00:39:48 - 00:39:54]

It's $\mathbb{Z}$... The proposition that we've kind of proven here...

Powers of the Generator $a = \bar{3}$ in $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$

To examine the internal structure of the abelian group $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ of order 6, we choose the candidate generator $a := \bar{3} \in \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ and compute its successive powers:

Power	a^0	a^1	a^2	a^3	a^4	a^5	a^6
Symbolic	$\bar{1}$	a	a^2	a^3	a^4	a^5	a^6
Value in $\mathbb{Z}$	1	3	9	27	81	243	729
Residue modulo 7	$\bar{1}$	$\bar{3}$	$\bar{2}$	$\bar{6}$	$\bar{4}$	$\bar{5}$	$\bar{1}$

Modular Arithmetic Derivations for Powers of 3:

$$
\begin{aligned}
a^0 &= \bar{1}, \\
a^1 &= \bar{3}, \\
a^2 &= \overline{3^2} = \bar{9} = \bar{2} && (9 = 7 \cdot 1 + 2), \\
a^3 &= \overline{3^3} = \overline{27} = \bar{6} && (27 = 7 \cdot 3 + 6), \\
a^4 &= \overline{3^4} = \overline{81} = \bar{4} && (81 = 7 \cdot 11 + 4), \\
a^5 &= \overline{3^5} = \overline{243} = \bar{5} && (243 = 7 \cdot 34 + 5), \\
a^6 &= \overline{3^6} = \overline{729} = \bar{1} && (729 = 7 \cdot 104 + 1).
\end{aligned}
$$

Because the powers of $a = \bar{3}$ exhaust all 6 elements of $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ before returning to the identity $\bar{1}$ at exponent 6, the element $\bar{3}$ is a *"generator"* (or *"primitive root"* modulo 7). Consequently, $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ is a cyclic group of order 6 (Definition 0.24).

Isomorphism with the Additive Cyclic Group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$

Spoken Transcript [00:39:54 - 00:41:19]

We've proven... that calculation of powers of 3 — and I picked 3, you know, I picked the 3; if you picked another one, it would have maybe been slightly different — but that calculation proves a group isomorphism, which maybe I write here and we think about. *[erases lower left blackboard]*

You should check my powers of 3, and then after you take the power of 3, you have to take the residue modulo 7.

This great 20th-century Hungarian mathematician (Paul Erdős) said that if numbers aren't beautiful, I don't know what is. That's kind of the mathematical perspective on numbers.

We have here the units of $\mathbb{Z}/7\mathbb{Z}$ *[writes "$(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}), \cdot, \bar{1})$"]*, and then there's another group, which is $\mathbb{Z}/6\mathbb{Z}$ with $+$ and $\bar{0}$ *[writes "$(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$"]*, and I claim these are isomorphic *[writes "$\cong$"]*.

This is the units with multiplication and 1. This group is isomorphic to that group. And what is the isomorphism? It's written on the board.

***Insight:* Paul Erdős on the Beauty of Numbers**

The lecturer references the celebrated quote by the Hungarian mathematician Paul Erdős (1913–1996): *"If numbers are not beautiful, I don't know what is."* The unexpected isomorphism between the unit group $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ of the field $\mathbb{Z}/7\mathbb{Z}$ and the simple additive clock arithmetic of $\mathbb{Z}/6\mathbb{Z}$ is a classic embodiment of this aesthetic: multiplication modulo 7 transforms into addition modulo 6 via the powers of a single primitive root.

Spoken Transcript [00:41:19 - 00:42:48]

The isomorphism... This is a slightly painful way to write the isomorphism, but let's just be painful about it.

Here $\bar{1}$ is going to go to $\bar{0}$ *[writes "$\bar{1} \longrightarrow \bar{0}$"]*. And bar means something different: It means modulo 7 here, and modulo 6 here, right?

a — that's the 3 here, a is $\bar{3}$. So, the nice way to do it is: a goes to $\bar{1}$ *[writes "$a \longrightarrow \bar{1}$"]*, a^2 goes to $\bar{2}$ *[writes "$a^2 \longrightarrow \bar{2}$"]* — I guess I could write this in slightly better notation, but anyway, you get the pattern.

Well, there's not so many more: a^3, a^4, a^5 *[writes "a^3, a^4, a^5"]*. This goes to $\bar{3}, \bar{4}, \bar{5}$ *[writes "$\longrightarrow \bar{3}, \bar{4}, \bar{5}$"]*.

This is the simple way to write it. And the reason this is true is because on this side, when we multiply, we just add up the exponents; on this side, we add up those. And the important thing is the sixth one goes to zero, which is what we have here, okay?

And if you want to say this explicitly here, you have to write these a's in terms of the elements of the units, which I've done there. So, this is totally explicit.

Okay, that's that. *[lecture concludes; students applaud]*

Group Isomorphism: $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}) \cong (\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$

Proposition 2.73 (*Cyclic Group Isomorphism*): The multiplicative group of units $(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}), \cdot, \bar{1})$ is isomorphic to the additive cyclic group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$:

$$(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}), \cdot, \bar{1}) \cong (\mathbb{Z}/6\mathbb{Z}, +, \bar{0}). \tag{2.41}$$

Proof

Let $a = \bar{3} \in \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$. We define the mapping $\Phi\colon \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}) \to \mathbb{Z}/6\mathbb{Z}$ by sending powers of the generator a to their integer exponents modulo 6:

$$\begin{aligned}
\Phi\colon \mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}) &\to \mathbb{Z}/6\mathbb{Z},\\
\bar{1} = a^0 &\mapsto \bar{0},\\
\bar{3} = a^1 &\mapsto \bar{1},\\
\bar{2} = a^2 &\mapsto \bar{2},\\
\bar{6} = a^3 &\mapsto \bar{3},\\
\bar{4} = a^4 &\mapsto \bar{4},\\
\bar{5} = a^5 &\mapsto \bar{5}.
\end{aligned}$$

Homomorphism Property: For any exponents $j, k \in \{0, 1, 2, 3, 4, 5\}$, multiplication in the unit group corresponds to addition of exponents:

$$a^j \cdot a^k = a^{j+k} = a^{(j+k) \bmod 6},$$

since $a^6 = \bar{1}$. Applying the map Φ:

$$\Phi(a^j \cdot a^k) = \overline{j+k} = \bar{j} + \bar{k} = \Phi(a^j) + \Phi(a^k) \in \mathbb{Z}/6\mathbb{Z}.$$

Hence, Φ is a group homomorphism. Because Φ maps the generator a of order 6 to the generator $\bar{1}$ of order 6, Φ is a bijective homomorphism, establishing the isomorphism.

Q.E.D.

$$
\begin{array}{ccc}
(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}),\cdot) & & (\mathbb{Z}/6\mathbb{Z},+) \\
& \overset{\Phi}{} & \\
a^0 = \bar{1} & \longmapsto & \bar{0} \\
a^1 = \bar{3} & \longmapsto & \bar{1} \\
a^2 = \bar{2} & \longmapsto & \bar{2} \\
a^3 = \bar{6} & \longmapsto & \bar{3} \\
a^4 = \bar{4} & \longmapsto & \bar{4} \\
a^5 = \bar{5} & \longmapsto & \bar{5}
\end{array}
$$

Disambiguating Residue Class Notations Across Groups: The bar notation $\bar{\cdot}$ operates over two distinct moduli in this isomorphism:

- In the domain $(\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}),\cdot)$, bars denote residue classes modulo 7 under multiplication: $\bar{x} = x + 7\mathbb{Z}$.
- In the codomain $(\mathbb{Z}/6\mathbb{Z},+)$, bars denote residue classes modulo 6 under addition: $\bar{k} = k + 6\mathbb{Z}$.

The isomorphism Φ is the *"discrete logarithm"* with respect to the base 3 modulo 7.

Summary of Key Concepts

- **Category Data and Axioms:** A category $\mathcal{C}$ consists of objects $\mathrm{Obj}(\mathcal{C})$, morphism sets $\mathrm{Hom}(A,B)$, an associative composition law $\circ$, and identity morphisms $\mathrm{Id}_A \in \mathrm{Hom}(A,A)$ satisfying left and right identity laws.
- **Categorical Morphisms:**
  - **Isomorphism:** A morphism $f\colon A \to B$ admitting a two-sided inverse $g\colon B \to A$ satisfying $f \circ g = \mathrm{Id}_B$ and $g \circ f = \mathrm{Id}_A$.
  - **Monomorphism (Monic):** Left-cancellable morphism ($f \circ g_1 = f \circ g_2 \implies g_1 = g_2$), corresponding to injectivity in standard concrete categories.
  - **Epimorphism (Epic):** Right-cancellable morphism ($g_1 \circ f = g_2 \circ f \implies g_1 = g_2$), corresponding to surjectivity in **Set**, **Grp** and $\mathbf{Vect}_K$, but not in **Ring** (the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism).
- **Zero Divisors and Integral Domains:** An integral domain is a non-trivial commutative ring with identity ($R \neq \{0\}$) without non-zero zero divisors ($ab = 0 \implies a = 0 \vee b = 0$). In modular arithmetic, the residue class ring $\mathbb{Z}/m\mathbb{Z}$ ($m \in \mathbb{Z}_{>0}$) is an integral domain if and only if m is prime.
- **Divisibility and Units:** An element $u \in R$ is a unit if $u \mid 1$ (i.e., $\exists w \in R$ such that $uw = 1$). The units of any commutative ring R form an abelian group $(\mathcal{U}_R,\cdot,1)$ under ring multiplication.
- **Modular Units and Cyclic Generators:** The unit group $\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}}$ consists precisely of the residue classes coprime to m: $\mathcal{U}_{\mathbb{Z}/m\mathbb{Z}} = \{\bar{x} \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(x,m) = 1\}$. For $p = 7$, $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z}) = \{\bar{1},\ldots,\bar{6}\}$ is a cyclic group of order 6 generated by $a = \bar{3}$,

isomorphic to the additive cyclic group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$.

Tuesday 29 September 2026

Groups of Units, Fields of Fractions, and Polynomial Rings

Course Content Overview

This lecture investigates the transition from commutative rings to fields and polynomials, focusing on the invertible elements of rings, the localization process that produces the minimal ambient field of an integral domain, and the algebraic construction of polynomial rings:

- **The Group of Units** $\mathrm{Unit}(R)$**:** Structure of multiplicatively invertible elements, characterization of units in $\mathbb{Z}$ and the Gaussian integers $\mathbb{Z}[i]$ via the norm, and the group of units of residue class rings $(\mathbb{Z}/m\mathbb{Z})^*$.
- **Euler's Totient Function** $\phi(m)$**:** Definition as the order of $(\mathbb{Z}/m\mathbb{Z})^*$, with explicit computations for prime powers $\phi(p) = p-1$ and $\phi(p^2) = p(p-1)$.
- **Universal Property of** $\mathbb{Q}$**:** Rigorous formulation of $\mathbb{Q}$ as the smallest field containing $\mathbb{Z}$, and the unique extension of ring embeddings into fields.
- **Construction of the Field of Fractions** $\mathrm{Frac}(R)$**:** Generalization to arbitrary integral domains via equivalence classes of formal pairs $(x,y) \in R \times (R \setminus \{0\})$, verification of well-defined operations, and canonical embedding $i\colon R \hookrightarrow \mathrm{Frac}(R)$.
- **Concrete Quotient Fields:** Field of fractions of $\mathbb{Z}$, the Gaussian rationals $\mathbb{Q}(i) = \mathrm{Frac}(\mathbb{Z}[i])$, subrings of $\mathbb{Q}$ such as $\mathbb{Z}[\frac{1}{3}]$, and self-isomorphism for existing fields $\mathrm{Frac}(K) \cong K$.
- **Polynomial Rings** $R[x]$**:** Intuitive formulation via formal linear combinations, rigorous foundation using almost-zero sequences in $R^{\mathbb{N}_0}$, componentwise addition, and Cauchy product (convolution) multiplication.

Review: The Group of Units of a Ring

Spoken Transcript [00:00:00 - 00:01:40]

Okay, so, we start. About the quiz: as I said, I'm very much pro-guessing in mathematics — you're never penalized for getting the wrong answer. Lika said that there was some implementation problem and some people were penalized for getting the right answer, and she's going to fix that.

If you think the quiz was too easy, too hard, or too medium, tell us. As I said, I've never gone through this quiz thing before.

Last week we discussed rings, *[writes $(R,+,\cdot,0,1)$]* and associated to a ring, there is an additive group. *[draws arrow to $(R,+,0)$]* This is easy, this is part of the definition. But there's another group, which was also discussed in the last lecture (Recall: Proposition 2.70), which is the group of units. *[draws arrow and writes "the group of unit"]* And I remind you that this group... *[writes Unit_R]* Various people use different notation for describing it, but here I just say $\mathrm{Unit}(R)$, so there's no question: it's the group of units of the ring R.

It's those elements such that there exists somebody else where the product is 1 *[writes $= \{x \mid \exists y,\ xy = 1\}$]* — so, it's those elements with a multiplicative inverse. And this is a group; we discussed this last week. So, this group here is $\mathrm{Unit}(R)$, and it has, well, the units by this definition, multiplication, and 1. *[writes $\mathrm{Unit}_R = (Units,\cdot,1)$]*

Whenever you see a ring, you immediately have these two groups. And this one is often the more complicated one to think about, because it's not part of the definition in some sense.

The Group of Units

Recap: Units and the group of units were introduced in Week 2 as $\mathcal{U}_R$ (Definition 2.66, Proposition 2.70), with $\mathrm{Unit}(\mathbb{Z}/7\mathbb{Z})$ in the example; the lecturer now writes Unit_R, also $\mathrm{Unit}(R)$ below.

Let $(R,+,\cdot,0,1)$ be a ring with unity. Associated with R are two natural canonical groups:

(1) The Additive Group: $(R,+,0)$, which forms an abelian group by the ring axioms.

(2) The Group of Units: The set of multiplicatively invertible elements in R:

$$\mathrm{Unit}_R = \mathrm{Unit}(R) := \{x \in R \mid \exists y \in R \text{ such that } xy = 1\}. \tag{3.1}$$

Under ring multiplication, $(\mathrm{Unit}(R),\cdot,1)$ forms a group with identity element 1.

Multiplicative Group Structure: The group of units is closed under multiplication: if $x,z \in \mathrm{Unit}(R)$ with inverses $x^{-1},z^{-1} \in R$, then $(xz)(z^{-1}x^{-1}) = 1$, so $xz \in \mathrm{Unit}(R)$. Associativity is inherited from the ring, and the multiplicative identity 1 is invertible since $1 \cdot 1 = 1$.

Units in the Integers and the Gaussian Integers

Spoken Transcript [00:01:40 - 00:02:45]

We'd seen this — I just review, but you know this: if you take $\mathbb{Z}$, *[writes $\mathbb{Z} \to Units\ \{\pm 1\}$]* the units are ± 1. And I said this many times, but maybe since I didn't actually ever give a proof: *[writes $\mathbb{Z}[i] \to Units\ \{\pm 1, \pm i\}$]* the units are $\pm 1, \pm i$ (Recall: Exercise 2.69).

If you're taking complex analysis, you should be able to prove this. But let's actually write a proof; I don't think I ever wrote the proof for this.

So, proof. *[writes "Proof"]* This is proof that these are the only invertible elements in this ring. Is that clear what I'm trying to prove?

Proof: let $a + bi \in \mathbb{Z}[i]$ be a unit. *[writes "Let $a + bi \in \mathbb{Z}[i]$ be a unit."]*

Spoken Transcript [00:02:45 - 00:03:55]

When I write this, of course that means that a and b are integers, okay? *[writes $(a, b \in \mathbb{Z})$]* That's what this being in there means.

All right, so, then there exists somebody else, $c + di$, also here, *[writes Then $\exists\, c + di \in \mathbb{Z}[i]$]* such that $(a + bi)(c + di) = 1$. *[writes s.t. $(a + bi)(c + di) = 1$]* And this is multiplication of complex numbers. The whole idea of the proof is understanding multiplication of complex numbers.

If you've taken a complex numbers class, or you've met complex numbers, however you know: when you multiply complex numbers, there's a length of a complex number *[sketches axes and a complex number vector]* — the norm of the complex number. So, if you multiply two complex numbers, their angles add and their lengths multiply, right?

So, we just take the norm — the lengths — take the norms. *[writes "take the norms:"]* The norm here is the square root of $a^2 + b^2$, *[writes $\sqrt{a^2 + b^2}$]* that's the length of this one. I hope that's not controversial — that's Euclid or something. This one is the square root of $c^2 + d^2$, *[writes $\cdot\sqrt{c^2 + d^2}$]* and their norms multiply, so this has to be just 1. *[writes $= 1$]* Okay?

Spoken Transcript [00:03:55 - 00:05:14]

This is pretty restrictive: a, b, c, and d are all integers. So, we can square that equation: *[writes $(a^2 + b^2)(c^2 + d^2) = 1$]* $(a^2 + b^2)(c^2 + d^2) = 1^2$, which is 1.

Now you have an equation where you have integers all multiplying to 1, and you know the only way this can happen is if this integer itself is ± 1 — because you can't factor 1 in integers without having just $1 \cdot 1$ or $(-1)(-1)$. Is that clear?

So, how could this be 1? Well, a^2 for an integer is either 0, 1, or something bigger. So, the only way this could be ± 1 is if $a = 0$ or ± 1, and $b = 0$ or ± 1 *[writes $a = 0, \pm 1$ and $b = 0, \pm 1$]* — those are the only cases. If these are both positive integers, it's going to be bigger than 1. So, you only have to check these cases. Is that clear? And if you look at these cases, well, these are the answers. *[points at $\{\pm 1, \pm i\}$]*

Okay? So, that's a complete proof — that's as complete a proof as you get in this class. I think this proof was 100 percent complete. But that's a nice fact, and if you draw them, they look like this: *[draws a coordinate plane with four points at $1, -1, i, -i$]* Those are the four units in the Gaussian integers.

Classification of Units in the Gaussian Integers

Recap: The units of $\mathbb{Z}$ were determined in Week 2 (Example 2.67), and those of $\mathbb{Z}[i]$ were stated in Week 1 and set as an exercise (Exercise 2.69). The proof below settles that exercise.

Example 3.1 (*Units of the Integers*): For the ring of integers $\mathbb{Z}$:

$$\mathrm{Unit}(\mathbb{Z}) = \mathbb{Z}^* = \{1, -1\}.$$

Proposition 3.2 (*Units of the Gaussian Integers*): The group of units of the Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is:

$$\mathrm{Unit}(\mathbb{Z}[i]) = \{1, -1, i, -i\}.$$

Proof

Let $z = a + bi \in \mathbb{Z}[i]$ be a unit with $a, b \in \mathbb{Z}$. By definition, there exists $w = c + di \in \mathbb{Z}[i]$ $(c, d \in \mathbb{Z})$ such that:

$$(a + bi)(c + di) = 1. \tag{3.2}$$

Taking the complex modulus (Euclidean length) on both sides of (3.2):

$$|a + bi| \cdot |c + di| = |1| = 1 \implies \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2} = 1.$$

Squaring this relation yields an equation in non-negative integers:

$$(a^2 + b^2)(c^2 + d^2) = 1. \tag{3.3}$$

Since $a^2 + b^2 \in \mathbb{Z}_{\geq 0}$ and divides 1, we must have:

$$a^2 + b^2 = 1.$$

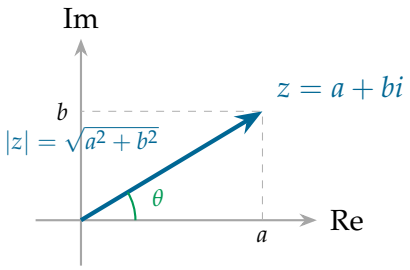
Because $a, b \in \mathbb{Z}$, the only integer solutions to $a^2 + b^2 = 1$ are:

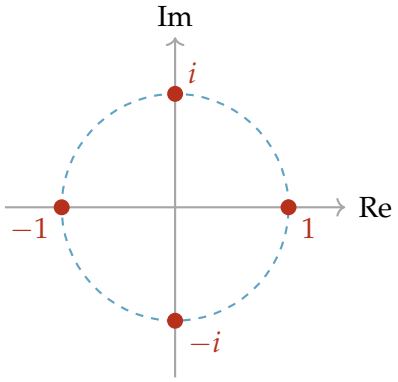
$$a = \pm 1, \quad b = 0 \implies z = \pm 1,$$
$$a = 0, \quad b = \pm 1 \implies z = \pm i.$$

Conversely, $1 \cdot 1 = 1$, $(-1)(-1) = 1$, and $i(-i) = 1$, so all four elements are invertible. Thus:

$$\mathrm{Unit}(\mathbb{Z}[i]) = \{1, -1, i, -i\}.$$

Q.E.D.





Algebraic Meaning of the Norm: The field norm $N(a+bi) = a^2 + b^2$ is multiplicative: $N(zw) = N(z)N(w)$. Since $N(z) \in \mathbb{Z}_{\geq 0}$ for any Gaussian integer, the norm of any invertible element must divide $N(1) = 1$, forcing $N(z) = 1$. The only lattice points on the circle of radius 1 are the four points shown above.

The Group of Units of $\mathbb{Z}/m\mathbb{Z}$

Spoken Transcript [00:05:14 - 00:07:18]

We did a much more interesting example last time, which was also with proof, but since various people asked me about it afterwards, I want to do just another example. And that was this ring: $\mathbb{Z}/m\mathbb{Z}$. *[writes $\mathbb{Z}/m\mathbb{Z}$]* That's a nice ring, and so it has a group of units: $\mathrm{Unit}(\mathbb{Z}/m\mathbb{Z})$. *[writes $\to$ Units $\mathbb{Z}/m\mathbb{Z}$]*

This was a result I proved last time (Recall: Theorem 2.71): the units, this is equal to those classes such that $\gcd(x,m) = 1$. *[writes $= \{\bar{x} \mid (x,m) = 1\}$]* And as I said, it's standard here to say m is at least 2, so we're not with the zero ring. *[draws a box around $m \geq 2$]* That's the units, and these are the residue classes which are relatively prime. And I did an example last time (Recall: $\mathbb{Z}/7\mathbb{Z}$, Example 2.72). This is a very interesting thing to study; it's just not a trivial thing.

In the sense that, one might say that if I take here the additive structure, *[writes $\mathbb{Z}/m\mathbb{Z} \to (\mathbb{Z}/m\mathbb{Z}, +, 0)$]* this is very simple to think about. It's just adding up integers mod$m$. In some sense, most people learn about this under one name or the other in school. But this thing — the units — are more complicated, are more interesting. *[writes "units are more interesting"]*

And so I wanted to do another... there were various people asking me some questions, so, let's try this: $10\mathbb{Z}$, just to say a little bit.

The standard notation for this is that: *[writes $(\mathbb{Z}/m\mathbb{Z})^*$]* This is the usual notation for units of $\mathbb{Z}/m\mathbb{Z}$, and also this notation is sometimes used in general. R^* sometimes denotes — could denote — units, depends on the author. So, I've just been writing Unit, so there's no question about what I mean. But in this case, it's so common to use the star here, I also use the star. Is that clear? So, this is it.

Example: Units of $\mathbb{Z}/10\mathbb{Z}$

Spoken Transcript [00:07:18 - 00:08:43]

How to think about this? Well, I'll just do an example because, as I said, there were various questions. This is all the residue classes that are relatively prime. So, you can just write them all first and cross them out: $\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}, \bar{8}, \bar{9}$. *[writes $\bar{0}, \bar{1}, \bar{2}, \ldots, \bar{9}$]* Maybe I should have picked something less than 10.

These are all the residue classes, and then we have to cancel all the ones that are not relatively prime.

Which one? Is $\bar{0}$ relatively prime? No, so, we cross that out. *[crosses out $\bar{0}$]* $\bar{1}$ is relatively prime — $\bar{1}$ is always relatively prime. What about $\bar{2}$? No. *[crosses out $\bar{2}$]* $\bar{3}$? Yes, so, we skip it. $\bar{4}$? That has a 2 in it and 10 has a 2, so, we cross that out. *[crosses out*

$\bar{4}$*]* $\bar{5}$ — 10 has a 5 in it, we cross that one out. *[crosses out* $\bar{5}$*]* Cross that one out, *[crosses out* $\bar{6}$ *and* $\bar{8}$*]* and what's left? Four, yeah, so, these are the four residue classes left. *[circles* $\bar{1},\bar{3},\bar{7},\bar{9}$*]*

The group of units here consists of these residue classes: *[writes* $\{\bar{1},\bar{3},\bar{7},\bar{9}\} \subset (\mathbb{Z}/10\mathbb{Z})^*$*]* $\{\bar{1},\bar{3},\bar{7},\bar{9}\}$. And the claim is: this is a group under multiplication. That's what that claim is.

For example, $\bar{3}$ is in this group, so, what is its inverse? *[writes* $\bar{3} \in (\mathbb{Z}/10\mathbb{Z})^*$*, what is its inverse?]*

Student Answer

7.

Spoken Transcript [00:08:44 - 00:09:45]

Seven, because $7 \times 3 = 21$, and so on. So anyway, you can play with this.

This is a more interesting object, and in fact you could ask: what is the order? This is a basic question in elementary number theory: what is the order of this group? *[writes "What is the order of* $(\mathbb{Z}/m\mathbb{Z})^*$*?"]*

And by order, this word means for groups the number of elements. *[writes "# of elements"]* So, when I say order of a group, if it's a finite group, I just mean the number of elements. What is the order of this? Sometimes this order is denoted by just straight lines — the bars: *[writes* $|(\mathbb{Z}/m\mathbb{Z})^*|$*]* You should be familiar with this set-theoretic notation; these two bars usually denote order. Is that clear? That's kind of standard set-theoretic notation.

First of all, let's do 10. What is this? Well, the answer is 4 — you just have to compute these. *[writes* $|(\mathbb{Z}/10\mathbb{Z})^*| = 4$*]*

Units in $\mathbb{Z}/m\mathbb{Z}$ and the Example $\mathbb{Z}/10\mathbb{Z}$

Recap: This characterization was proved in Week 2 (Theorem 2.71); the lecturer restates it with the notation $(\mathbb{Z}/m\mathbb{Z})^*$.

Proposition 3.3 (*Units in the Residue Class Ring*): Let $m \in \mathbb{Z}_{\geq 2}$. The group of units of the residue class ring $\mathbb{Z}/m\mathbb{Z}$ (frequently denoted by $(\mathbb{Z}/m\mathbb{Z})^*$ or R^* for general rings) consists of those residue classes whose representatives are coprime to m (on the board: $(x,m) = 1$):

$$(\mathbb{Z}/m\mathbb{Z})^* = \mathrm{Unit}(\mathbb{Z}/m\mathbb{Z}) = \{\bar{x} \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(x,m) = 1\}. \tag{3.4}$$

Example 3.4 (*Computing the Units of* $\mathbb{Z}/10\mathbb{Z}$): Consider the ring $\mathbb{Z}/10\mathbb{Z} = \{\bar{0},\bar{1},\bar{2},\bar{3},\bar{4},\bar{5},\bar{6},\bar{7},\bar{8},\bar{9}\}$. To find the units, we eliminate all classes that share a common divisor with 10 greater than 1:

- **Class** $\bar{0}$**:** $\gcd(0,10) = 10 \neq 1$ (eliminate $\bar{0}$),
- **Even classes:** $\gcd(2,10) = 2$, $\gcd(4,10) = 2$, $\gcd(6,10) = 2$, $\gcd(8,10) = 2$ (eliminate $\bar{2},\bar{4},\bar{6},\bar{8}$),

- **Class $\bar{5}$:** $\gcd(5,10) = 5$ (eliminate $\bar{5}$).

The remaining coprime residue classes form the group of units:

$$(\mathbb{Z}/10\mathbb{Z})^* = \{\bar{1}, \bar{3}, \bar{7}, \bar{9}\}.$$

This is an abelian group of order 4 under multiplication modulo 10. For instance, the inverse of $\bar{3}$ is $\bar{7}$, since:

$$\bar{3} \cdot \bar{7} = \overline{21} = \bar{1} \in \mathbb{Z}/10\mathbb{Z}.$$

Board Transition

The lecturer pushes the upper chalkboard up and erases the middle section to prepare space for defining Euler's totient function.

Euler's Totient Function

Spoken Transcript [00:10:36 - 00:12:03]

You can ask this question in general, and that's a fun thing to think about. We'll say more about this a bit later, but the definition can be given now: *[writes "Order of $(\mathbb{Z}/m\mathbb{Z})^* \stackrel{def}{=} \phi(m)$"]* the order of $(\mathbb{Z}/m\mathbb{Z})^*$, the group of units, has a name.

Do you know the name of this function? Yes, yeah, it's ϕ... It's Euler's totient function, that's just the definition. That's the definition of this function — it's just the order.

And you could ask: can you compute this? There's different ways to compute it, but it has to do with the factorization, and we will get to that later. So, you could say — this is maybe a question for you now: *[writes "Question: Try to compute $\phi(m)$"]* Question: Try to compute Euler's totient function.

Euler is like the most famous Swiss person that ever lived. A few years ago it might have been Federer, but now he's retired, so now it's Euler. And it turns out that Euler and Federer are both from Basel — you can check it.

You know, my first trip to Switzerland was when I was a kid in the '80s, and the Swiss still had Euler on the 10 Swiss franc note. I don't know if you've ever seen that; it's a beautiful note. You should go look it up.

Evaluating $\phi(p)$ and $\phi(p^2)$ for a Prime p

Spoken Transcript [00:12:03 - 00:12:15]

Okay, so, let's do one example to compute this. Well, let's do two examples. Suppose you have a prime p *[writes $\phi(p) =$]* — then what is this?

Student Answer

$p - 1$.

Spoken Transcript [00:12:16 - 00:12:20]

Yeah, $p-1$. And why is that?

Student Answer

Because any number less than p is coprime to p.

Spoken Transcript [00:12:22 - 00:13:48]

In other words, if you write this, you'll write p elements, and you'll cross out 0, and everybody else is relatively prime. So, it's $p-1$. *[writes $p-1$]*

And so, what about this one? *[writes $\phi(p^2) = ?$]* Do you know how to do this? What you do — I mean, of course once you do this, you can do the general case for powers of a prime — you write all the elements. This is a really primitive way to do it, but let's just do it: $0, 1, 2, \ldots, p^2-1$. *[writes $0, 1, 2, \ldots, p^2-1$]* That's what we did there, right?

So, it starts out, and then you cross out all of the... here you're going to cross out 0, so, it's p^2-1, but you have to take away... of course you don't want 0, but you have to take away all the multiples of p, right?

And so, now you have to concentrate really hard: how many multiples of p? Well, there's p here, there's $2p$ here, and what's the last one? The last one you have to cross out is $p(p-1)$ — that's the last one. *[writes $p, 2p, \ldots, p(p-1)$]* Is that clear?

So, we have to take away $p-1$, so, the answer is p^2-p. *[writes $(p^2-1)-(p-1) = p^2-p$]* The answer to this is p^2-p. Okay, that's the beginning; you can play with it a bit. All right?

Euler's Totient Function and Prime Power Formulas

Definition 3.5 (*Euler's Totient Function*)**:** The *"Euler totient function"* (or Euler's ϕ-function) $\phi\colon \mathbb{Z}_{\geq 1} \to \mathbb{Z}_{\geq 1}$ is defined as the order of the group of units of $\mathbb{Z}/m\mathbb{Z}$:

$$\phi(m) := |(\mathbb{Z}/m\mathbb{Z})^*| = \#\{x \in \{0, 1, \ldots, m-1\} \mid \gcd(x, m) = 1\}. \tag{3.5}$$

Proposition 3.6 (*Totient Values for Primes and Prime Squares*)**:** Let p be a prime number. Then:

$$\phi(p) = p-1, \tag{3.6}$$
$$\phi(p^2) = p^2-p = p(p-1). \tag{3.7}$$

Proof

We evaluate the number of coprime residue classes in each case:

- **For a prime p:** The representatives are $\{0, 1, 2, \ldots, p-1\}$. The only representative sharing a non-trivial factor with p is 0. Thus, the remaining $p-1$ elements are all coprime to p, giving $\phi(p) = p-1$ (e.g., $\phi(7) = 6$, as in Example 2.72).
- **For p^2:** The representatives are $\{0, 1, 2, \ldots, p^2-1\}$. An element x shares a non-

trivial factor with p^2 if and only if $p \mid x$. The multiples of p in this range are:

$$0,\ p,\ 2p,\ \ldots,\ (p-1)p.$$

There are exactly p such multiples. Subtracting these from the total of p^2 elements yields:

$$\phi(p^2) = p^2 - p = p(p-1).$$

Q.E.D.

Generalization to Prime Powers: For any prime power p^k with $k \geq 1$, an integer $x \in \{0, 1, \ldots, p^k - 1\}$ shares a factor with p^k if and only if $p \mid x$. The multiples of p are $0, p, 2p, \ldots, (p^{k-1} - 1)p$, totaling p^{k-1} elements. Hence, $\phi(p^k) = p^k - p^{k-1} = p^k(1 - 1/p)$.

Fields of Fractions

Spoken Transcript [00:13:48 - 00:15:30]

The next topic in the course has to do with fields of fractions. *[writes "Fields of fractions"]* And this is a peculiar subject.

It starts with this idea that $\mathbb{Z}$ is inside $\mathbb{Q}$. *[writes $\mathbb{Z} \subset \mathbb{Q}$]* So, $\mathbb{Z}$ is a ring, it's an integral domain *[writes arrow from $\mathbb{Z}$ to "int. domain"]* and $\mathbb{Q}$ is a field, *[writes arrow from $\mathbb{Q}$ to "field"]* and $\mathbb{Z}$ sits inside $\mathbb{Q}$. But the relationship is not just of any ring inside any field. In some sense, $\mathbb{Q}$ is the smallest field that contains $\mathbb{Z}$. This understanding will lead to the study of the field of fractions.

Let's try to first make this precise; let's try to make precise the qualitative statement that... *[writes "Make precise the statement that "$\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$""]* Here's the statement: "$\mathbb{Q}$ *is the smallest field which contains* $\mathbb{Z}$". That's a statement that probably you feel is true. It's not a 100 percent mathematical statement, so, let's try to make this precise.

You don't contradict the meaning of this statement, right?

Spoken Transcript [00:15:30 - 00:17:25]

[The lecturer moves the boards and begins erasing the lower board.] One thing that when one starts studying these fields of fractions and $\mathbb{Q}$'s, you get this kind of strange feeling that maybe you never checked everything in $\mathbb{Q}$ works. Have you checked everything in $\mathbb{Q}$ works?

It's kind of a strange feeling, because you learn about the rational numbers in school, but it's not like the definitions there are checked to work. You just accept them, which is pretty reasonable — they are correct.

And what do I mean by that? $\mathbb{Q}$ is a much more complicated thing than $\mathbb{Z}$ conceptually, because $\mathbb{Z}$ is really great: there are these objects and they have names, and everybody has one name. That's what's great about $\mathbb{Z}$.

The thing that's confusing about $\mathbb{Q}$ is that they're objects, but they have many

names. You know, you can have... *[writes $5/7 = 60/84$]* That's kind of confusing: all these numbers have many, many different names. And when I said, *"Have you ever checked it works?"*, it means that whatever operations you do, you have to be able to do them with any name, and it has to give the same answer, so to speak. That's what I mean. And so, the question is: have you ever checked that?

It's a weird question. But kind of the ideal math question is in first grade or second grade, whenever you're adding fractions, you should ask, say: *"But why is it well-defined?"* That's the question you should ask. And if you'd asked that question, then we wouldn't have to give this lecture today, actually.

Multiple Representations of Rational Numbers and Well-Definedness

A fundamental conceptual distinction between $\mathbb{Z}$ and $\mathbb{Q}$ is the uniqueness of representation:

- **Integers ($\mathbb{Z}$):** Each element has a unique canonical name: $0, 1, -1, 2, -2, \ldots$
- **Rationals ($\mathbb{Q}$):** Each number has infinitely many equivalent representations as a quotient of integers:

$$\frac{5}{7} = \frac{60}{84} = \frac{10}{14} = \ldots$$

Consequently, every operation on fractions (addition, multiplication, evaluation) must be proved to be *"well-defined"*, meaning the resulting value is strictly independent of the chosen fractional representative.

The Universal Property of $\mathbb{Q}$

Spoken Transcript [00:17:25 - 00:19:32]

I try to make precise the statement that $\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$. And there's different ways to do it, but one way is to say it like this:

Suppose I have $f\colon \mathbb{Z} \to F$, *[writes Suppose $f\colon \mathbb{Z} \to F$]* where F is a field *[writes arrow from F to "field"]* — this big F is a field, and this little f is an injection of rings, so, an injective homomorphism of rings. *[writes "injection of rings / homomorphism of rings"]*

So, I start with this: I take any possible field — maybe a field that I might think is smaller than $\mathbb{Q}$, or whatever field, take any field you want — and I start with a ring homomorphism which is injective. That means that I view this as $\mathbb{Z}$ sitting inside F, okay?

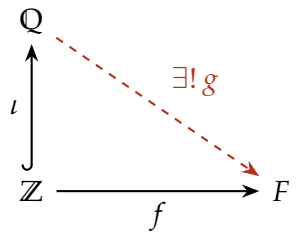
Then I claim... *[writes "Then I claim $\exists!\, g$"]* The typical way you write this is with a diagram: this is f, then I claim there exists a unique g... So, this is $\mathbb{Z}$ sitting inside $\mathbb{Q}$, and here there exists a unique g *[draws diagram with $\mathbb{Z} \to \mathbb{Q}$, $\mathbb{Z} \to F$, and dashed arrow $\mathbb{Q} \to F$ labeled g]* such that g is an injective homomorphism of fields. *[writes "such that g is inj. hom of fields"]*

And the diagram commutes, meaning I can take an integer here and go here, or I consider it in $\mathbb{Q}$ and go this way. We would say the diagram commutes: if you have x here, I can get to here going straight across, or I can go inside $\mathbb{Q}$ and come around, so, that would say that $f(x) = g(x)$. *[writes "s.t. $\forall x \in \mathbb{Z}$, $f(x) = g(x)$"]*

Universal Property of the Rational Field

Qualitative Observation: $\mathbb{Q}$ is the *"smallest"* field containing the integral domain $\mathbb{Z}$.

Theorem 3.7 (*Universal Property of* $\mathbb{Q}$): Let F be a field, and let $f\colon \mathbb{Z} \to F$ be an injective ring homomorphism (embedding of rings). Then there exists a unique field homomorphism $g\colon \mathbb{Q} \to F$ such that the following diagram commutes:



that is, $g|_{\mathbb{Z}} = f$, meaning $g(x) = f(x)$ for all $x \in \mathbb{Z}$. Furthermore, g is automatically injective.

Injectivity is automatic, because every ring homomorphism from a field to a non-zero ring is injective (see Theorem 1.32 and the steps after it, Week 1).

Universal Property as Categorical Minimality: In category theory, this universal mapping property identifies $\mathbb{Q}$ as the initial object among all field extensions receiving $\mathbb{Z}$. Any field F containing a copy of $\mathbb{Z}$ must necessarily contain a subfield isomorphic to $\mathbb{Q}$, showing that $\mathbb{Q}$ cannot be compressed or embedded into any strictly smaller field.

Construction and Well-Definedness of the Field Extension

Spoken Transcript [00:19:32 - 00:21:20]

I'll give a little proof of this. This is now totally precise mathematics; it makes that statement, which is somehow slightly qualitative, into a mathematical statement.

Do you accept that? When I say $\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$, it means that if I have any other candidate field F, I can actually squeeze $\mathbb{Q}$ into it. And that means that $\mathbb{Q}$ is smaller than F. Does that make sense?

Okay, but I haven't proven it, so, let's try to prove it. I won't prove the whole thing, because it's of course kind of painful.

So, let us see: I have to prove this, which means I get to start with the data I suppose. That's what *"suppose"* is about — *"suppose"* means I get to start with this data, which is here, all right? And then the whole claim is existence. To prove this, what I have to do is produce g.

Proof, more or less: *[writes "Proof I have to construct g"]* I have to construct g. So, I write down... *[self-correction]* a complex number, x over y, where x and y are integers and $y \neq 0$ — sorry, a rational number, that's what a rational number is: x/y, where $x, y \in \mathbb{Z}$ and $y \neq 0$. *[writes $g(x/y) =, x, y \in \mathbb{Z}, y \neq 0$]* I hope you're happy with that; that's what a rational number is.

This is my responsibility: I have to define this, or we define. *[writes "we define"]* So,

how am I going to define this?

Student Answer

$f(x)/f(y)$.

Spoken Transcript [00:21:23 - 00:21:51]

Yeah, so, formally, of course we'll write it the way you said, but formally we write it like this: *[writes* $= f(x)f(y)^{-1}$*]* $f(x)f(y)^{-1}$.

Because $f(x)$ is in the field F, and $f(y)$ is in the field F, and I want to take its inverse. This is the kind of field language. Why am I allowed to take its inverse? You cannot take inverses of everything.

Student Answer

Because F is a field and we chose, um... $y \neq 0$.

Spoken Transcript [00:21:58 - 00:23:11]

Yeah, but you have to check — and this is the whole thing, which is that you have to check. This is legal so long as $f(y) \neq 0$. *[writes "legal as long as* $f(y) \neq 0$*"]* And why is $f(y)$ not zero? It's because f is injective, and $y \neq 0$. *[writes "and this is because* f *is injective and* $y \neq 0$*"]* That's actually using this thing here. So, this is legal. Is that clear?

And as was implicit in the answer, we will often write this as that, *[writes* $= \frac{f(x)}{f(y)}$*]* because it's charming to do that. And in a field, if you see this, all that means is that the fellow downstairs can be put upstairs by multiplicative inverse. And you're only legally allowed to write it if you know this is not zero — just like in normal arithmetic, you can't divide by zero.

When we write this, it means various things: it means in particular this one's not zero, and you can put it up here, and you can put it on either side, it doesn't matter because our fields are commutative. Okay?

Candidate Definition of the Field Homomorphism g

Definition 3.8 (*Candidate Map for Field Embedding*): Let $f\colon \mathbb{Z} \to F$ be an injective ring homomorphism into a field F. For a rational number represented by $\frac{x}{y} \in \mathbb{Q}$ with $x, y \in \mathbb{Z}$ and $y \neq 0$, we define:

$$g\left(\frac{x}{y}\right) := f(x) \cdot f(y)^{-1} = \frac{f(x)}{f(y)} \in F. \tag{3.8}$$

Legitimacy of Inversion in the Field F: The inverse $f(y)^{-1}$ exists in F if and only if $f(y) \neq 0_F$:

(1) **Non-Zero Denominator:** In the rational representation $\frac{x}{y}$, we require $y \neq 0$ in $\mathbb{Z}$.
(2) **Injectivity of Ring Embedding:** Since $f\colon \mathbb{Z} \to F$ is an injective ring homomorphism, its kernel is trivial: $\ker(f) = \{0\}$.
(3) **Non-Vanishing Image:** It follows immediately that $y \neq 0 \implies f(y) \neq f(0) = 0_F$ (Proposition 1.19). Since F is a field, every non-zero element has a unique multiplicative inverse $f(y)^{-1} \in F$.

Spoken Transcript [00:23:11 - 00:25:18]

We managed to prove... But is this a definition? The answer is, of course, not: all you did was pick a particular name of this, and you gave a definition. You have to check that all the names give the same definition. That's the business about the rational numbers, which no one ever thinks about — and it's better not to think about it, but unfortunately today we have to think about it.

[The lecturer moves to the lower board to erase and write.]

What do we have to do? *[writes "We are done with the Def of g"]* We are done with the definition of g. But we must further check that if $x_1/y_1 = x_2/y_2$, *[writes "We must check that if $\frac{x_1}{y_1} = \frac{x_2}{y_2}$"]* then $g(x_1/y_1) = g(x_2/y_2)$. *[writes "then $g(\frac{x_1}{y_1}) = g(\frac{x_2}{y_2})$"]*

You understand we have to check this, because that's the slipperiness of the rational numbers: they don't have a unique name.

Spoken Transcript [00:25:18 - 00:27:08]

All right, but you accept we have to check this? Otherwise, if we can't check this, whatever was there was nonsense.

So, we have to check it. And how are we going to check it? Well, so, we have here: *[writes $g\left(\frac{x_1}{y_1}\right) = \frac{f(x_1)}{f(y_1)}$ and $g\left(\frac{x_2}{y_2}\right) = \frac{f(x_2)}{f(y_2)}$]*

And we want to figure out whether these things are equal: do we have... *[writes "do we have $\frac{f(x_1)}{f(y_1)} \stackrel{?}{=} \frac{f(x_2)}{f(y_2)}$"]* That's the question.

Using the fact that these are invertible, and we can multiply by them, this question is equivalent to what's called clearing the denominators. So, by clearing the denominators, this is equivalent *[draws $\iff$ and writes $f(x_1)f(y_2) = f(x_2)f(y_1)$]* to $f(x_1)f(y_2) = f(x_2)f(y_1)$ by just multiplying both sides by $f(y_1)f(y_2)$ *[writes $\cdot f(y_1)f(y_2)$]* — that's a legal move in fields. And since these are invertible, this implies that, and this implies the other way, because they're both non-zero. Everything here is invertible.

We're using all the time that when we write this, *[writes $y_1, y_2 \neq 0 \implies f(y_1), f(y_2) \neq 0$]* neither y_1 nor y_2 is zero, which implies $f(y_1)$ and $f(y_2)$ are not zero, okay?

Spoken Transcript [00:27:08 - 00:27:50]

And now, these things are all integers, and everything's upstairs — there's no division. This is if and only if *[draws $\Updownarrow$]* you can use the homomorphism property: *[writes*

$f(x_1y_2) = f(x_2y_1)$] $f(x_1y_2) = f(x_2y_1)$, right? Because f is a homomorphism.

And this is if and only if *[draws ⇳ and writes* $f(x_1y_2 - x_2y_1) = 0$*]* $f(x_1y_2 - x_2y_1) = 0$, okay?

Why would this be the case? Yeah?

Student Answer

We could rearrange...

Spoken Transcript [00:27:52 - 00:28:48]

Well, you have to say what this means. What does this mean? This goes to the heart of the issue about whether you learned the rational numbers correctly, and the answer is: this means *[points to* $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ *and writes below:* $x_1y_2 = x_2y_1$*]* that $x_1y_2 = x_2y_1$. It means nothing more or less than that; it just actually means that.

So, if we start with this, we have this equation: *[writes* $x_1y_2 - x_2y_1 = 0$ *in the integers]* $x_1y_2 - x_2y_1 = 0$ in the integers. That's what this means.

And very fortunately, that's what we need: we need this to be zero, so, this is true *[places a checkmark next to* $f(x_1y_2 - x_2y_1) = 0$*]* — and so, everything is true, and this is well-defined. All right?

That means I have defined a map, but there's a lot of things to check, which of course I'm not going to check. You have to do it, so to speak, on your own time.

Well-Definedness of the Extension Map g

Claim 3.9 (*Well-Definedness of g*): The map $g\colon \mathbb{Q} \to F$ defined in (3.8) is well-defined; that is, if $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ in $\mathbb{Q}$, then $g\left(\frac{x_1}{y_1}\right) = g\left(\frac{x_2}{y_2}\right)$ in F.

Proof

Let $\frac{x_1}{y_1}, \frac{x_2}{y_2} \in \mathbb{Q}$ with $y_1, y_2 \in \mathbb{Z} \setminus \{0\}$ such that $\frac{x_1}{y_1} = \frac{x_2}{y_2}$.

1. Meaning of Equality in $\mathbb{Q}$: By definition of the rational numbers as fractions, equality $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ means precisely that the cross-multiplied relation holds in $\mathbb{Z}$:

$$x_1y_2 = x_2y_1 \iff x_1y_2 - x_2y_1 = 0 \in \mathbb{Z}. \tag{3.9}$$

2. Algebraic Chain in F: We want to establish whether:

$$g\left(\frac{x_1}{y_1}\right) \stackrel{?}{=} g\left(\frac{x_2}{y_2}\right) \iff \frac{f(x_1)}{f(y_1)} \stackrel{?}{=} \frac{f(x_2)}{f(y_2)}.$$

Since $y_1, y_2 \neq 0$ and f is injective, $f(y_1) \neq 0$ and $f(y_2) \neq 0$. Multiplying both sides by

$f(y_1)f(y_2) \in F^\times$ (clearing denominators) yields an equivalent equality in F:

$$\begin{aligned}\frac{f(x_1)}{f(y_1)} = \frac{f(x_2)}{f(y_2)} &\iff f(x_1)\cdot f(y_2) = f(x_2)\cdot f(y_1)\\ &\iff f(x_1y_2) = f(x_2y_1) \qquad \text{(since } f \text{ is a ring homomorphism)}\\ &\iff f(x_1y_2) - f(x_2y_1) = 0_F\\ &\iff f(x_1y_2 - x_2y_1) = 0_F. \qquad (3.10)\end{aligned}$$

3. Conclusion: By (3.9), $x_1y_2 - x_2y_1 = 0 \in \mathbb{Z}$. Since any ring homomorphism maps 0 to 0 (Proposition 1.19):

$$f(x_1y_2 - x_2y_1) = f(0) = 0_F.$$

Thus, condition (3.10) holds, which proves that $g\left(\frac{x_1}{y_1}\right) = g\left(\frac{x_2}{y_2}\right)$. Q.E.D.

Core Pedagogical Principle: Non-Unique Names of Fractions: A rational number has infinitely many equivalent names (e.g., $\frac{1}{2} = \frac{2}{4} = \frac{-3}{-6}$). Any function defined on rational numbers using numerator and denominator symbols must be proven invariant under replacement of representatives. As shown above, this reduces entirely to the cross-multiplication identity in the underlying ring $\mathbb{Z}$.

Board Erasing and Transition

The lecturer erases the left and middle portions of the blackboard to write down the remaining checks required for g to be a field homomorphism.

Spoken Transcript [00:28:48 - 00:30:05]

I have now defined the map g, finally. Then you have to check that this map g is actually... I mean, all of the properties of g, that it's a homomorphism of fields. You have to check that it respects addition, and you have to check that it respects multiplication.

All of it is slightly painful, because at first blush it's obvious: you just pick representatives for $\mathbb{Q}$ as x_1/y_1, and then you just check it. But then you have to check that all of that stuff respects the different choices, the different names of elements of $\mathbb{Q}$, just like we did here.

And the checks are all the same, so you can do it yourself. That's the end of what I'm going to give as the proof of this; you can check the rest. We're going to have to do many more things like this, so...

Spoken Transcript [00:30:05 - 00:31:50]

We have now, successfully, defined g. Then we need to check that g is indeed a homomorphism of fields. *[writes "We have defined g. We need to check that g is indeed a hom. of fields."]* I leave this to you. We're going to have to do things like this anyway, and it's all just unraveling the definitions.

Meaning you have to check... *[writes "You have to check"]* You check something like:

[writes $g\left(\frac{x_1}{y_1}\right) + g\left(\frac{x_2}{y_2}\right) = g\left(\frac{x_1}{y_1} + \frac{x_2}{y_2}\right)$] $g(x_1/y_1) + g(x_2/y_2) = g(x_1/y_1 + x_2/y_2)$. You have to go check that; that's what you have to check.

And you can see when you write this, this side... Okay, I'm going to do part of it for you — I don't know why I'm feeling compassionate today here. This one is $f(x_1)/f(y_1) + f(x_2)/f(y_2)$. *[writes $= \frac{f(x_1)}{f(y_1)} + \frac{f(x_2)}{f(y_2)}$]* That's the definition here. And this has a different definition.

First of all, how do we do it? We first have to add this fraction, right? Because our definition only applies to names of rational numbers, and this is the sum. So, we first have to do this, but you know how to add fractions: this is equal to *[writes $= g\left(\frac{x_1y_2 + x_2y_1}{y_1y_2}\right)$]* g of $(x_1y_2 + x_2y_1)/(y_1y_2)$. At least I hope you know how to add fractions — at least I hope I know how to add fractions, right? Okay, so, that's the addition of the fraction.

Spoken Transcript [00:31:50 - 00:32:50]

And then this, by definition — use the definition — is $f(x_1y_2 + x_2y_1)/f(y_1y_2)$. *[writes $= \frac{f(x_1y_2 + x_2y_1)}{f(y_1y_2)}$]* Okay?

So, that's the one side to check, and that's supposed to equal this side, all right? And how are you going to check this? Well, you use the homomorphism property, you move everything upstairs, and then finally you're going to use the thing I erased — this, okay?

So, I leave it to you. There's nothing really I can add to this; you have to do it. But I do say it's painful — painful for me to write it on the board even.

Verification of the Field Homomorphism Axioms

Claim 3.10 (*Preservation of Field Operations*): The map $g\colon \mathbb{Q} \to F$ is a field homomorphism; that is, for all $q_1, q_2 \in \mathbb{Q}$:

$$g(q_1 + q_2) = g(q_1) + g(q_2), \tag{3.11}$$
$$g(q_1 \cdot q_2) = g(q_1) \cdot g(q_2), \tag{3.12}$$
$$g(1_{\mathbb{Q}}) = 1_F. \tag{3.13}$$

Proof

Let $q_1 = \frac{x_1}{y_1}$ and $q_2 = \frac{x_2}{y_2}$ with $x_i, y_i \in \mathbb{Z}$ and $y_i \neq 0$.

1. Verification of Additivity (3.11)**:** Adding fractions according to standard rational arithmetic:

$$q_1 + q_2 = \frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1y_2 + x_2y_1}{y_1y_2}.$$

Applying the definition of g and the ring homomorphism properties of f:

$$\begin{aligned} g(q_1 + q_2) &= g\left(\frac{x_1y_2 + x_2y_1}{y_1y_2}\right) \\ &= \frac{f(x_1y_2 + x_2y_1)}{f(y_1y_2)} \\ &= \frac{f(x_1)f(y_2) + f(x_2)f(y_1)}{f(y_1)f(y_2)}. \end{aligned}$$

On the other hand, evaluating the sum of images in the field F:

$$\begin{aligned} g(q_1) + g(q_2) &= \frac{f(x_1)}{f(y_1)} + \frac{f(x_2)}{f(y_2)} \\ &= f(x_1)f(y_1)^{-1} + f(x_2)f(y_2)^{-1}. \end{aligned}$$

Bringing both terms over the common denominator $f(y_1)f(y_2)$ in the field F:

$$\frac{f(x_1)}{f(y_1)} + \frac{f(x_2)}{f(y_2)} = \frac{f(x_1)f(y_2) + f(x_2)f(y_1)}{f(y_1)f(y_2)}.$$

Both expressions coincide identically, establishing (3.11). By Claim 3.9, this value is independent of the choice of fraction representatives.

2. Verification of Multiplicativity (3.12)**:** Multiplying fractions inside $\mathbb{Q}$:

$$q_1 \cdot q_2 = \frac{x_1}{y_1} \cdot \frac{x_2}{y_2} = \frac{x_1x_2}{y_1y_2}.$$

Applying the definition of g and the multiplicativity of f:

$$\begin{aligned} g(q_1 \cdot q_2) &= g\left(\frac{x_1x_2}{y_1y_2}\right) \\ &= \frac{f(x_1x_2)}{f(y_1y_2)} \\ &= \frac{f(x_1)f(x_2)}{f(y_1)f(y_2)} \\ &= \frac{f(x_1)}{f(y_1)} \cdot \frac{f(x_2)}{f(y_2)} \\ &= g(q_1) \cdot g(q_2), \end{aligned}$$

establishing (3.12).

3. Preservation of Multiplicative Identity (3.13)**:** Representing $1_{\mathbb{Q}} \in \mathbb{Q}$ as $\frac{1_{\mathbb{Z}}}{1_{\mathbb{Z}}}$:

$$g(1_{\mathbb{Q}}) = g\left(\frac{1_{\mathbb{Z}}}{1_{\mathbb{Z}}}\right) = \frac{f(1_{\mathbb{Z}})}{f(1_{\mathbb{Z}})}.$$

Since $f\colon \mathbb{Z} \to F$ is a unital ring homomorphism, $f(1_{\mathbb{Z}}) = 1_F$. Thus:

$$g(1_{\mathbb{Q}}) = \frac{1_F}{1_F} = 1_F,$$

establishing (3.13). Thus g is a field homomorphism. Q.E.D.

The Field of Fractions of an Integral Domain

Spoken Transcript [00:32:50 - 00:35:03]

That's the story about $\mathbb{Z}$. Their fates are really closely tied: $\mathbb{Z}$ and $\mathbb{Q}$. *[writes $\mathbb{Z} \subset \mathbb{Q}$]* $\mathbb{Q}$ is not just any field that $\mathbb{Z}$ is in — it's the smallest field it's in. So, we could ask: can we construct this relationship for other rings? Because we really like this relationship.

If you study the integers, it's very useful sometimes to be able to divide; so, that's the movement to the field. Let's try this with a ring in general.

Suppose R is an integral domain. *[writes "Suppose R is an integral domain"]* It turns out that this is extremely important, and we will see in the definition — it may not be clear from the discussion so far — but let's say R is an integral domain.

Can we find the analogue of $\mathbb{Q}$, which is the *"smallest"* field containing R? *[writes "Can we find "smallest" field containing R?"]* Already in this language, if we find a field containing R, *[writes $R \subset F$]* that means R has to be a domain (Recall: Definition 2.62), because a domain means that two elements that are non-zero, when you multiply them, stay non-zero. If you're inside a field, that's always the case, okay? This is one of the reasons why we're talking only about integral domains.

So, that's the question: if you find on the street an integral domain, can you put it in its smallest field? The answer is yes, and that construction is called the field of fractions. And I will show you how to do it.

In some ways, there's lots of conceptual nice things about it, but this pain, you can imagine, we're not going to get away from it — it's just going to be more abstract. One of the things you will see here is that when we define the field of fractions of a ring, it will have this property that the elements are not single objects, but equivalence classes. And you should expect that answer, because that already happens for the rational numbers.

Spoken Transcript [00:35:03 - 00:36:16]

By the way, for a rational number, the objection — and I'm surprised no one made it — is that you could say that you could fix the name by taking the numerator and denominator and cancelling all common factors. There's still a sign issue, but we leave that aside.

If you try to do arithmetic with that, it's super painful, because you have to do a lot of cancellation. So, that is the argument against it. The argument that that's kind of a lie is that that's not really how you do arithmetic in the rational numbers. But that is the argument against what I was saying about the rational numbers.

In a sense, if you do that, it's kind of even more painful, because then when you write these expressions, you can't write these expressions — you have to cancel everywhere, and you know...

Generalization from Integers to Integral Domains

Remark 3.11 (*Why Integral Domains?*): If a commutative ring R embeds into a field

F via an injective homomorphism $\iota\colon R \hookrightarrow F$, then R must necessarily have no zero divisors:

$$a, b \in R \setminus \{0\} \implies \iota(a), \iota(b) \neq 0 \implies \iota(ab) = \iota(a)\iota(b) \neq 0 \implies ab \neq 0.$$

Moreover, R is not the zero ring: $\iota(1) = 1 \neq 0 = \iota(0)$ in F, so $1 \neq 0$ in R. Thus, the property of being an *"integral domain"* (Definition 2.62) is a strictly necessary prerequisite for any ring to be embedded into a field of fractions.

The Goal: Field of Fractions

Given an arbitrary integral domain R, our goal is to construct the minimal field containing R, called the *"field of fractions"* (or quotient field) $\mathrm{Frac}(R)$, generalizing the classical relationship:

$$\mathbb{Z} \hookrightarrow \mathbb{Q} = \mathrm{Frac}(\mathbb{Z}).$$

Construction of the Set of Formal Fractions

Spoken Transcript [00:36:16 - 00:38:05]

Let's think about this in the abstract. The main thing is just the definition. Here is our ring R: we don't know what it is, it's just any ring, but it's an integral domain. *[writes R int. domain]* We want to construct a field containing R. *[writes "We want to construct a field containing R"]*

The field should have elements that are like x over y, where $x, y \in R$ and $y \neq 0$. *[writes $F \ni \frac{x}{y}$, $x, y \in R, y \neq 0$]* This should be the element of that field, because if it contains R, I should be able to take x and multiply it by the inverse of y in the field, so, it should have these elements in it. Is that clear? That's what happened with the rational numbers.

So, we can just define it that way. We start with: let S be the set of pairs. *[writes "Let $S = \{(x, y) \mid y \neq 0, x, y \in R\} \subset R \times R$"]* This whole thing is going to be a big disappointment if you think something fun is going to happen — I just tell you ahead of time, this is going to be a big disappointment. But some things in life are like that.

So, here: $y \neq 0$, and $x, y \in R$. Okay, that's the set of pairs. For the integers, if we start with the integers, we just get the set of pairs of integers where the second one is not zero, okay? But we can think about this set of pairs for any R. Is that clear? This is inside $R \times R$ — it's just some subset of pairs.

Digression: Equivalence Relations and Partitions

Spoken Transcript [00:38:05 - 00:40:21]

All right, and now the thing that will come is that there is an equivalence relation. *[writes "There is an equivalence relation on S"]* There is an equivalence relation on S, so maybe we call this equivalence relation... *[writes $\sim$, then changes to $\equiv$]* You can use various symbols for equivalence, that's kind of standard. How about three lines?

Do you know what an equivalence relation is? This was in your Foundations class. An equivalence relation on a set — maybe a little aside *[writes "Aside on T"]* — an equivalence relation on a set T tells you when two elements are related, and the rules for the equivalence relation are:

x is always related to itself — this is the reflexive property; *[writes $x \equiv x$, reflexive]*

if x is related to y, then y is related to x — this is the symmetric property; *[writes $x \equiv y \implies y \equiv x$, symmetric]*

and if you know the definition, what's the third one? Yeah?

Student Answer

Transitivity.

Spoken Transcript [00:40:21 - 00:41:00]

The transitive property. It says that if we have x related to y and y related to z, then x is related to z. *[writes $x \equiv y \wedge y \equiv z \implies x \equiv z$, transitive]* That's the whole thing of equivalence relations. It comes up many times in mathematics, and it will come up again in this course, so, if you have not seen it, think about it.

And the main thing we use about this equivalence relation is that if I have T with this equivalence relation, then I have another set — sometimes it's called this — which is the set of equivalence classes. *[writes $T_{/\equiv}$ = set of equivalence classes]* That's going to be very important for us.

Whenever we have a set with an equivalence relation, that equivalence relation divides the set up into equivalence classes — disjoint equivalence classes, okay? And this thing here is the set of equivalence classes. There's three objects in this kind of abstract situation: there's the set that we start with, there's the equivalence relation, and then there's the set of equivalence classes.

The Set of Pairs and General Equivalence Relations

Definition 3.12 (*The Set of Formal Fractions S*)**:** Let R be an integral domain. On the Cartesian product $R \times R$ (Notation 0.1), we define the set of formal numerator-denominator pairs S by:

$$S := \{(x, y) \in R \times R \mid y \neq 0\} \subset R \times R. \tag{3.14}$$

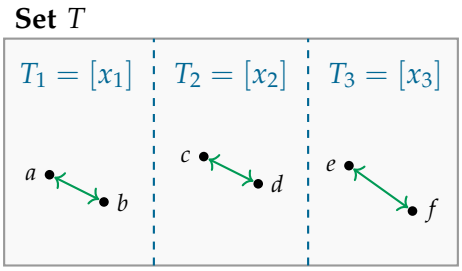
Remark 3.13 (*Review: Equivalence Relations and Partitions*)**:** An *"equivalence relation"* $\equiv$ on a set T is a binary relation satisfying three foundational axioms (Definition 0.8):

(i) Reflexivity: $\forall x \in T : x \equiv x$.

(ii) Symmetry: $\forall x, y \in T : x \equiv y \implies y \equiv x$.

(iii) Transitivity: $\forall x, y, z \in T : (x \equiv y \wedge y \equiv z) \implies x \equiv z$.

The quotient set $T/\equiv$ denotes the set of *"equivalence classes"* (Definition 0.9), which partitions T into mutually disjoint subsets.



Partition of T into disjoint equivalence classes $T/\equiv$

The Equivalence Relation on Formal Fractions

Spoken Transcript [00:41:00 - 00:42:06]

We have to define the equivalence relation. We now define an equivalence relation on our set S, which is the set of pairs *[writes "We now define $\equiv$ on S"]* — and you should think about these pairs as numerator and denominator, but those are just words here.

We need to define when (x_1, y_1) is equivalent to (x_2, y_2). *[writes $(x_1, y_1) \equiv (x_2, y_2)$]* We need a definition for this. What's the answer going to be? Yeah?

Student Answer

$x_1 y_2 = x_2 y_1$.

Spoken Transcript [00:42:06 - 00:44:08]

Okay, very good. This is the definition: *[writes $\overset{\text{def}}{\iff} x_1 y_2 = x_2 y_1$]* it's just exactly what you do with integer arithmetic. If you have a fraction, you say they're equal if $x_1 y_2 = x_2 y_1$, okay? That's the definition. There's nothing more or less than that — that's the whole definition.

And it's motivated by the integers. It's very important for this definition that this is only about R. *[writes "only about R"]* This whole discussion starts with R, so, we can only say things that are about R.

We can make these pairs — that's only about R. We want to define this equivalence relation. At the end, these things are only in R, and the definition is only about some equation in R, okay? We haven't gotten out of R in some sense in this whole discussion. Is that clear?

Now we come to an issue: why is this an equivalence relation? This is the question you should have asked your teacher in first grade or whatever it was. That's like the whole question about the rational numbers in some sense — the foundations.

[The lecturer washes and erases the left part of the bottom chalkboard.]

This is the first part where we have to do a little bit of algebra. The first two properties, $x = x$, that's just identically true, okay? That's not a problem. We have to check three properties. *[writes "3 properties"]*

Definition of the Equivalence Relation on S

Definition 3.14 (*Equivalence Relation on S*): On the set $S = \{(x, y) \in R \times R \mid y \neq 0\}$, we define the relation $\equiv$ by:

$$(x_1, y_1) \equiv (x_2, y_2) : \iff x_1 y_2 = x_2 y_1 \quad (\text{in } R). \tag{3.15}$$

Conceptual Motivation: Cross-Multiplication: This relation is the formal algebraic counterpart to equating cross-products of fractions:

$$\frac{x_1}{y_1} = \frac{x_2}{y_2} \iff x_1 y_2 = x_2 y_1.$$

Crucially, formula (3.15) involves only multiplication within the ring R, without invoking any external concept of division or fraction.

Spoken Transcript [00:44:08 - 00:45:05]

The first is: if I take (x_1, y_1), this is equivalent to (x_1, y_1). *[writes $(x_1, y_1) \equiv (x_1, y_1)$]* And why is this? Because by definition this is equal to $x_1 y_1 = x_1 y_1$, *[writes $\iff x_1 y_1 = x_1 y_1$ ✓]* so, we're very happy about that. There was nothing to do here.

You also have to check the symmetric property: that if we have $(x_1, y_1) \equiv (x_2, y_2)$, then the other way around is fine, and it'll just be the same equation. There's nothing to check about that. *[writes $x_1 y_2 = x_2 y_1 \iff x_2 y_1 = x_1 y_2$ ✓]*

The interesting one is the transitive property. *[writes "Transitive"]* This is the only fun one to check. We start with this one: *[writes $(x_1, y_1) \equiv (x_2, y_2)$ and $(x_2, y_2) \equiv (x_3, y_3)$]* these are the inputs.

And we must show — this is for us to do — that (x_1, y_1) is congruent (i.e., equivalent) to (x_3, y_3). *[writes $\implies (x_1, y_1) \equiv (x_3, y_3)$]*

The thing that's a little funny about this is... let's try to do this. We have to write the first equation. This is what we have: *[writes "What we have"]* that's what we've assumed, that's in the left column. *[The lecture bell rings.]*

Okay. You'll have to wait until after the break for this excitement to continue.

Verification of Reflexivity and Symmetry on S

Proposition 3.15 (*Reflexivity and Symmetry of $\equiv$*): The relation $\equiv$ defined on $S = \{(x, y) \in R \times R \mid y \neq 0\}$ by:

$$(x_1, y_1) \equiv (x_2, y_2) \iff x_1 y_2 = x_2 y_1 \tag{3.16}$$

is reflexive and symmetric.

Proof

Let $(x_1, y_1), (x_2, y_2) \in S$:

- **Reflexivity:** Trivially $x_1 y_1 = x_1 y_1$, which by (3.16) immediately implies:
$$(x_1, y_1) \equiv (x_1, y_1).$$
- **Symmetry:** Assume $(x_1, y_1) \equiv (x_2, y_2)$. Then $x_1 y_2 = x_2 y_1$. By symmetry of equal-

ity in R:

$$x_2y_1 = x_1y_2 \implies (x_2, y_2) \equiv (x_1, y_1).$$

Q.E.D.

Mid-Lecture Break

The bell rings, and the lecturer pauses the class for a short break. The lecture resumes at timestamp 00:45:05.

Spoken Transcript [00:45:05 - 00:45:33]

Okay, so, let's start. I should say that very few people have found my office, and it turns out I'll be in this visitor office for most of the semester, because they're clearing some kind of nuclear waste from my current office. So, my office now is HG G 65.1 *[writes "HG G 65.1"]* if you want to find me there. And yeah, I'm afraid I'll be there for most of the semester.

Administrative Announcement: Office Relocation

The lecturer writes *"HG G 65.1"* on the far right of the chalkboard and informs students that he will be stationed in this visitor office for the majority of the semester.

Spoken Transcript [00:45:33 - 00:47:00]

We come to this point that's kind of fun and interesting. We started with an integral domain, and led by the analogy with the rational numbers, we come to start defining this field as some equivalence class of pairs. And I should have said this — I did say it, but I didn't write it: the way to think about this *[writes "Think of as x_1/y_1 or x/y"]* — this is the mathematical text — you think of this pair in S as x/y, okay? Just like with the integers.

As an object, it's this formal object. And then there was this proposal, which was correct, of how to define the equivalence relation. And now we actually need to check something: is it an equivalence relation?

Here we have to prove the transitive property. And so, we start with: what do we have? We have that equivalence, so, if we translate that into equations, it says $x_1y_2 = x_2y_1$ — and you have to be very careful with the subscripts, otherwise you'll confuse yourself: *[writes $x_1y_2 = x_2y_1$]* $x_1y_2 = x_2y_1$. And then what we also have here is $x_2y_3 = x_3y_2$. *[writes $x_2y_3 = x_3y_2$]*

Spoken Transcript [00:47:20 - 00:48:40]

And what we want, *[writes "Want"]* we want this thing, which is $x_1y_3 = x_3y_1$. *[writes $x_1y_3 = x_3y_1$]*

The puzzling thing about this is: we have these quadratic equations and we need to get another quadratic equation. It seems a little bit unfortunate, because you might think you have to do some linear algebra with quadratic equations, and that's not going to work — you're not going to get a new quadratic equation from these.

The correct answer is we're supposed to multiply here. The clever thing is to multiply the top equation by y_3 *[writes "mult. by y_3"]* — y_3 is what we want. So, we get $x_1y_2y_3 = x_2y_1y_3$. *[writes $x_1y_2y_3 = x_2y_1y_3$]*

This is kind of good: we see the x_1y_3 here. Can you see what we have to do next? This is actually algebra. What?

[A student answers inaudibly.] Yeah, you could do that, but why don't we just take this y_2y_3 here and use this equation? It's the same thing. So, we take the y_2y_3 and use this equation to flip them. This is all about subscripts, and I should just tell you that subscripts have always been difficult for me. Do you accept that? Because I just took this and flipped them using this equation. *[writes $x_1y_3y_2 = x_3y_2y_1$]*

Spoken Transcript [00:48:40 - 00:50:02]

Now what do we do? Well, we have here x_1y_3 — that looks pretty good — and we have y_3y_1, but they're both multiplied by y_2, right?

Now we use the fact that in these pairs, the second one is always born non-zero: to be in this set, the second one has to be non-zero. And this is an integral domain. So, we have this equation — this one equals that one — and then we use the integral domain property. *[writes "use int. domain"]*

I said this is a place where it's extremely important it's an integral domain. Since $y_2 \neq 0$, *[writes "since $y_2 \neq 0$"]* we can cancel, *[writes "$\implies$ cancel"]* okay? Because an integral domain says that's the cancellation property. Is that okay?

If y_2 is not zero, we can cancel it. This is kind of dramatic, but we cancel it here. *[crosses out y_2 on both sides]* And that's what we want.

We have now proven transitivity. You see there was a little bit of cleverness here; it's easy to confuse yourself.

Proof of Transitivity and the Role of the Integral Domain

Proposition 3.16 (*Transitivity of $\equiv$ on S*): Let R be an integral domain. The relation $\equiv$ on S defined by $(x_1, y_1) \equiv (x_2, y_2) \iff x_1y_2 = x_2y_1$ is transitive. Consequently, $\equiv$ is an equivalence relation on S.

Proof

Let $(x_1, y_1), (x_2, y_2), (x_3, y_3) \in S$, which implies $y_1, y_2, y_3 \in R \setminus \{0\}$.

1. Assumptions: We assume the relations:

$$(x_1, y_1) \equiv (x_2, y_2) \implies x_1y_2 = x_2y_1, \tag{3.17}$$

$$(x_2, y_2) \equiv (x_3, y_3) \implies x_2y_3 = x_3y_2. \tag{3.18}$$

2. Target: We must prove that $(x_1, y_1) \equiv (x_3, y_3)$, which means showing that:

$$x_1 y_3 = x_3 y_1. \tag{3.19}$$

3. Derivation via Multiplication and Substitution: Multiplying equation (3.17) on both sides by $y_3 \in R$:

$$(x_1 y_2) y_3 = (x_2 y_1) y_3 \iff (x_1 y_3) y_2 = (x_2 y_3) y_1. \tag{3.20}$$

Substituting $x_2 y_3 = x_3 y_2$ from (3.18) into the right-hand side of (3.20):

$$(x_1 y_3) y_2 = (x_3 y_2) y_1 = (x_3 y_1) y_2. \tag{3.21}$$

Subtracting the right-hand side gives:

$$(x_1 y_3 - x_3 y_1) y_2 = 0. \tag{3.22}$$

4. Cancellation via the Integral Domain Property: By definition of S, $y_2 \neq 0$. Because R is an **integral domain**, it contains no zero divisors:

$$ab = 0 \text{ with } b \neq 0 \implies a = 0.$$

Applying this cancellation property to (3.22) with $b = y_2 \neq 0$:

$$x_1 y_3 - x_3 y_1 = 0 \implies x_1 y_3 = x_3 y_1.$$

Thus, $(x_1, y_1) \equiv (x_3, y_3)$, which completes the proof. **Q.E.D.**

Why Integral Domains Are Essential for Transitivity: Notice where the ring axioms alone would fail: equation (3.22) holds in any commutative ring. However, to deduce $x_1 y_3 = x_3 y_1$ from $(x_1 y_3 - x_3 y_1) y_2 = 0$, we must divide out and cancel the non-zero element y_2. If R has zero divisors, the relation $\equiv$ can **fail** to be transitive. For instance, in $\mathbb{Z}/6\mathbb{Z}$ (with $y_2 = \bar{2}$ and $x_1 y_3 - x_3 y_1 = \bar{3}$): $(\bar{3}, \bar{1}) \equiv (\bar{0}, \bar{2})$ since $\bar{3} \cdot \bar{2} = \bar{0} = \bar{0} \cdot \bar{1}$, and $(\bar{0}, \bar{2}) \equiv (\bar{0}, \bar{1})$ since $\bar{0} \cdot \bar{1} = \bar{0} = \bar{0} \cdot \bar{2}$, but $(\bar{3}, \bar{1}) \not\equiv (\bar{0}, \bar{1})$ since $\bar{3} \cdot \bar{1} = \bar{3} \neq \bar{0} = \bar{0} \cdot \bar{1}$.

Spoken Transcript [00:50:02 - 00:51:06]

We have come so far as to define S as a set with an equivalence relation. So, now we can define the points of the field that we're constructing, which will be the equivalence classes.

[The lecturer washes and erases the upper chalkboard.] This was almost like the first piece of algebra in this class, I think. I don't remember if we had any algebra before. By that I mean playing with equations. Which in the end — I'm saying a lot of words, but in the end the subject is about equations, you know.

Board Transition

The lecturer takes a sponge, washes the upper chalkboard clean, and dries it to prepare space for defining the field of fractions and its canonical inclusion.

The Field of Fractions and Its Canonical Embedding

Construction of the Field of Fractions

Spoken Transcript [00:51:06 - 00:52:35]

Where are we? We started with R as an integral domain. *[writes R int. domain]* We constructed S with an equivalence relation, *[writes $S = \ldots$ with $\equiv$]* and from this we define F. The name of this is the field of fractions of R. *[writes $\implies F := S_{/\equiv}$ "Field of Fractions of R"]* Field of fractions of R — that's the name of this object: F is equal to the set of equivalence classes of S. *[writes $F = S_{/\equiv}$]*

Is it completely clear what this thing is? We defined S, we defined an equivalence relation, and we took great pains to check all of its properties, so, we get the set of equivalence classes and that is F.

The Canonical Embedding of R into F

Spoken Transcript [00:52:35 - 00:53:14]

Then we need to do a lot of things. But let's start by defining the map which is the inclusion from R to F, *[writes $R \to F$]* because we said this F should be a field that contains R. That means we have to have some way of getting points from R to F. And what's that way going to be?

This ι, the inclusion: we have to put R in F. *[writes "We have to put R in F"]* This is by ι. Normally people don't even write it, but if I have an element of R, we have to put it into F. That means we have to assign it an equivalence class in this set. What are we going to do? Yeah?

Student Answer

The equivalence class of $(r, 1)$.

Spoken Transcript [00:53:15 - 00:54:22]

Yeah, that's a good idea: $(r, 1)$. *[writes $i(r) = (r, 1) \in S_{\equiv}$]* It has some advantages: $1 \neq 0$, and an integral domain is never the zero ring (Recall: Definition 2.62). So, this is legal: this is an element of S, and then we can take its equivalence class.

I'm not going to write equivalence classes because it becomes too complicated, but you could formally take its equivalence class like this. If you start doing this, you commit to writing things like $[(3, 4)]$ for rational numbers, and no one ever does that. We don't want to do it, but this would be kind of more logically correct.

Spoken Transcript [00:54:22 - 00:54:54]

When I write this, it's just an equivalence class. I looked at what the book does, and the book says that this is an element of S and that is its equivalence class. That's a very clever solution; I wouldn't say it's totally standard, but anyway. Standard is just to ignore the issue. So, this is the equivalence class here.

Is this reasonable? It better at least be... yeah.

Student Question

The set S, is this all of $R \times R$ or is it just any subset of $R \times R$?

Spoken Transcript [00:55:00 - 00:55:28]

No, the set S was very precisely defined on a previous board: S is equal to those elements in $R \times R$ where $y \neq 0$. *[writes $S = \{(x,y) \mid x,y \in R,\ y \neq 0\}$]* That's S.

And this is an element of S because $1 \neq 0$; and $1 \neq 0$ because R is an integral domain, and an integral domain is by definition not the zero ring. That's the full logic — that's the full logical certificate if you want it.

Spoken Transcript [00:55:28 - 00:57:05]

Still, we'd like more: we'd like this to be an injection. Is ι injective? *[writes "Is i injective?"]*

This means: when are $(r,1)$ and $(\hat{r},1)$ in the same equivalence class of S? *[writes "When are $(r,1)$ and $(\hat{r},1)$ in the same equiv. class of S?"]* That's what injectivity is about, right?

How do we check this? The answer is that these are in the same equivalence class if and only if that equivalence class identity holds — that one. *[points at $(x_1,y_1) \equiv (x_2,y_2) \iff x_1y_2 = x_2y_1$]*

It's if and only if $r \cdot 1 = \hat{r} \cdot 1$, *[writes $\iff r \cdot 1 = \hat{r} \cdot 1$]* which is if and only if $r = \hat{r}$, *[writes $\iff r = \hat{r}$]* which is great for us because it shows that this is an injection. *[places a checkmark]*

There's a lot of things to do in this construction — it's like a big house: you have to build all the pieces of it. So, now we not only have defined F, we've defined ι, and we've put R inside F, and this placement is injective. Okay?

Definition of the Field of Fractions and Canonical Embedding

Definition 3.17 (*Field of Fractions*): Let R be an integral domain, and let $S := \{(x,y) \in R \times R \mid y \neq 0\}$. The *"field of fractions"* (or quotient field) of R, denoted by $\mathrm{Frac}(R)$ or F, is defined as the quotient set:

$$F := S/\equiv = \{[(x,y)] \mid (x,y) \in S\}. \tag{3.23}$$

The equivalence class $[(x,y)]$ is commonly written as $\frac{x}{y}$.

Definition 3.18 (*Canonical Inclusion Map*): We define the canonical embedding

$i\colon R \to F$ by assigning to each $r \in R$ the equivalence class of the pair $(r,1)$:

$$i(r) := [(r,1)] \in F. \tag{3.24}$$

Proposition 3.19 (*Injectivity of the Canonical Inclusion*): The canonical map $i\colon R \to F$ is injective.

Proof

Let $r, \hat{r} \in R$. By definition of equality in the quotient set $F = S/\equiv$:

$$\begin{aligned} i(r) = i(\hat{r}) &\iff [(r,1)] = [(\hat{r},1)] \\ &\iff (r,1) \equiv (\hat{r},1) \\ &\iff r \cdot 1 = \hat{r} \cdot 1 \\ &\iff r = \hat{r}. \end{aligned}$$

Therefore, i is injective. Q.E.D.

Why $1_R \neq 0_R$ Is Guaranteed: For the pair $(r,1)$ to belong to S, the denominator must satisfy $1 \neq 0$. By condition (i) of Definition 2.62, an integral domain must satisfy $1_R \neq 0_R$ (i.e., it cannot be the trivial zero ring). Hence, $(r,1) \in S$ is unconditionally valid for every $r \in R$.

Field Operations on the Field of Fractions

Definition of Zero, One, and Addition on F

Spoken Transcript [00:57:05 - 00:58:38]

This is just the beginning of the story. R already is born with all of the data: R, $+$, multiplication, $0, 1$. *[writes $R, +, \cdot, 0, 1$]* We need to define all the structures on F and show this is a homomorphism also. *[draws arrow to F and gestures]*

[The lecturer pushes the upper chalkboard up to reveal the lower board.] We have to start talking about all the structures for F as a field. And for many of them we don't have any choice: the zero in F has to correspond to the zero in R for it to be a homomorphism. So, the zero in F has to be $(0,1)$, *[writes "Zero in F has to be $(0,1)$"]* okay? Also, the unit in F has to be $(1,1)$, *[writes "unit in F has to be $(1,1)$"]* because it has to correspond to the unit in R by the rule of getting R to F. We don't really have any choice in this matter.

But we have to define $+$ and multiplication on F. *[writes "We have to define $+$, $\cdot$ on F"]* That's something that we have to construct. We have to define it, check it's compatible with R, check it satisfies the field axioms — it's a lot of long and boring work, but the interesting part to start with is we have to define it.

If we have an element here, $(x_1, y_1) \in S$, *[writes $(x_1, y_1) \in S$]* and we have another one, $(x_2, y_2) \in S$, *[writes $(x_2, y_2) \in S$]* then we have to define this operation. This is up to us: we must define $(x_1, y_1) + (x_2, y_2) = \ldots$ *[writes $(x_1, y_1) + (x_2, y_2) =$] [writes "we must define"]*

That's the problem. How are we going to define this? What do you think? Tell me.

Student Answer

$x_1y_2 + x_2y_1$ over y_1y_2.

Spoken Transcript [00:58:39 - 01:00:40]

Yeah, so, we just use the standard definition for how you add fractions: this one times that one plus this one times that one over y_1y_2, okay? *[writes* $= (x_1y_2 + x_2y_1, y_1y_2)$*]*

Meaning you pretend that it's this problem, which it kind of is: *[writes* $\frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1y_2+x_2y_1}{y_1y_2}$*]* $\frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1y_2+x_2y_1}{y_1y_2}$. You know how to do this. I hope I got the subscripts right; you should tell me if I get the subscripts wrong.

This is the definition — that's just it, nothing more, nothing less.

Now, is it a legal definition? There's an easy part: for it to be legal, this better be non-zero, *[points at* y_1y_2 *and writes "non-zero"]* because in our set S, the denominator — that's the one in the basement — always has to be non-zero.

We have here: this guy is non-zero, that guy is non-zero, and that implies the product is non-zero, because R is an integral domain. *[writes "because* R *is an int. domain"]* We use it again here. Is that clear? Otherwise, if this were zero, this would not be legal.

Well-Definedness of Addition

Spoken Transcript [01:00:40 - 01:02:25]

That's the easy part. The hard part is: you have to check that while this is a definition on S, it's well-defined on equivalence classes.

That means if I trade this person for another one in the same equivalence class, and I trade this for another person in the same equivalence class, the result has to be in the same equivalence class. Is that clear? That's the check of rational arithmetic — the check that I claim you should have done in school to understand the rational numbers.

Otherwise, what are you doing? You're doing some random manipulation. So, you have to check this: *[writes "You have to check the indep. of choices of equiv. class"]* independence of choices of equivalence class.

And we've already done this kind of thing twice already in the lecture. If you just do it and follow your nose, you'll see it all unwinds. You just use the definition: you have to say formally that this result respects our definition of equivalence classes.

In various books they do it; mostly it's left as an exercise. It's really important, so, you should do it. If I do it, it's just a bunch of indices — it doesn't really help, and you have to go through it yourself.

Is it clear what the problem is? I recommend trying this yourself, and if you get stuck, look at the book — the book does it. It'll give you a line of things, and you can imagine it's probably true.

Neutral Elements and Addition on the Field of Fractions

Definition 3.20 (*Neutral Elements in F*): The zero element and multiplicative unit in F are defined by:

$$0_F := i(0_R) = [(0,1)], \tag{3.25}$$

$$1_F := i(1_R) = [(1,1)]. \tag{3.26}$$

Definition 3.21 (*Addition on F*): For any two equivalence classes $[(x_1,y_1)]$, $[(x_2,y_2)] \in F$, addition is defined by:

$$[(x_1,y_1)] + [(x_2,y_2)] := [(x_1y_2 + x_2y_1, y_1y_2)]. \tag{3.27}$$

Proposition 3.22 (*Legitimacy and Well-Definedness of Addition*): Addition on F as given in (3.27) is well-defined.

Proof

We verify two essential requirements:

(1) Legitimacy of the Denominator: For $(x_1y_2 + x_2y_1, y_1y_2)$ to lie in S, we must have $y_1y_2 \neq 0$. Since $(x_1,y_1),(x_2,y_2) \in S$, we know $y_1 \neq 0$ and $y_2 \neq 0$. Because R is an **integral domain**, the product of two non-zero elements is non-zero:

$$y_1 \neq 0 \wedge y_2 \neq 0 \implies y_1y_2 \neq 0.$$

Thus, $(x_1y_2 + x_2y_1, y_1y_2) \in S$.

(2) Independence of Representatives: Suppose $(x_1,y_1) \equiv (\hat{x}_1,\hat{y}_1)$ and $(x_2,y_2) \equiv (\hat{x}_2,\hat{y}_2)$, meaning:

$$x_1\hat{y}_1 = \hat{x}_1y_1 \quad \text{and} \quad x_2\hat{y}_2 = \hat{x}_2y_2. \tag{3.28}$$

We must show that $(x_1y_2 + x_2y_1, y_1y_2) \equiv (\hat{x}_1\hat{y}_2 + \hat{x}_2\hat{y}_1, \hat{y}_1\hat{y}_2)$, which requires:

$$(x_1y_2 + x_2y_1)(\hat{y}_1\hat{y}_2) \stackrel{?}{=} (\hat{x}_1\hat{y}_2 + \hat{x}_2\hat{y}_1)(y_1y_2).$$

Expanding the left-hand side and applying (3.28):

$$\begin{aligned}(x_1y_2 + x_2y_1)\hat{y}_1\hat{y}_2 &= (x_1\hat{y}_1)(y_2\hat{y}_2) + (x_2\hat{y}_2)(y_1\hat{y}_1)\\ &= (\hat{x}_1y_1)(y_2\hat{y}_2) + (\hat{x}_2y_2)(y_1\hat{y}_1)\\ &= (\hat{x}_1\hat{y}_2 + \hat{x}_2\hat{y}_1)y_1y_2.\end{aligned}$$

This establishes equivalence, confirming that addition is independent of the choice of representative.

Q.E.D.

Definition and Well-Definedness of Multiplication on F

Spoken Transcript [01:02:25 - 01:03:40]

Also you have to do it with multiplication. *[The lecturer washes the chalkboard to the left and writes]* Multiplication is a bit easier — that was the most painful one.

Also define multiplication, *[writes "Also define $\cdot$"]* which is $(x_1,y_1)\cdot(x_2,y_2)$ *[writes* $(x_1,y_1)\cdot(x_2,y_2) \stackrel{def}{=}$*]* — it will surprise no one that this is (x_1x_2, y_1y_2), okay? *[writes* (x_1x_2, y_1y_2)*]* That's the definition.

And again, next is plus check *[writes "+ check"]* of compatibility with the equivalence classes. *[writes "compatibility with ≡"]* And this check is easy, so, I'll do it since it's easy.

Spoken Transcript [01:03:40 - 01:05:45]

Suppose we have here $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$ *[writes $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$]* and $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$. *[writes $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$]* That's the input, right?

Then we must show *[writes "We must show"]* that (x_1x_2, y_1y_2) is in the same equivalence class as if we chose the other representatives: $(\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)$, *[writes $(x_1x_2, y_1y_2) \equiv (\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)$]* okay? That's what we have to do.

What we know is this: that equation that this is in the same equivalence class is $x_1\hat{y}_1 = \hat{x}_1y_1$, *[writes $x_1\hat{y}_1 = \hat{x}_1y_1$]* and this equation is $x_2\hat{y}_2 = \hat{x}_2y_2$ *[writes $x_2\hat{y}_2 = \hat{x}_2y_2$]* — all very confusing.

And now we just multiply them together! These are what we know, and having these two equations, we can multiply them all together, and we get $x_1x_2\hat{y}_1\hat{y}_2 = \hat{x}_1\hat{x}_2y_1y_2$. *[writes $x_1x_2\hat{y}_1\hat{y}_2 = \hat{x}_1\hat{x}_2y_1y_2$]* Now there's subscripts and hats.

And that equation is exactly the certificate that these things are in the same equivalence class, okay? *[draws double arrow $\iff$ and checkmark]* This implies that.

So, this shows that if I pick different equivalence classes here, the result doesn't matter. This one I did completely, because it's easy: all you do is multiply. For the other one, you have to do the kind of tricks we were doing before, where you have to clear the denominators, etc. You can do that, so to speak, on your own time.

Compatibility of the Operations with the Canonical Embedding

Spoken Transcript [01:05:45 - 01:07:25]

Then you have to check that these definitions are compatible with R. They're just all defined that way, meaning we have to check *[writes "Check $\iota\colon R \to F$ is a hom of rings"]* that this $\iota\colon R \to F$ is a homomorphism of rings.

Again, there's really nothing to this: it means that if you take $r + \hat{r}$ *[writes $r + \hat{r} \overset{\iota}{\mapsto} (r + \hat{r}, 1)$]* and you go by ι, you get $(r + \hat{r}, 1)$. Then you have to check that this thing is equal to $(r, 1) + (\hat{r}, 1)$, *[writes $= (r, 1) + (\hat{r}, 1)$]* and that's the same by the definition that we used there. But you have to check it.

You also have to check it for multiplication. So, it's just a bunch of checking. By here, all the cards are played, and you just have to check everything.

Multiplication on F and Ring Homomorphism Properties

Definition 3.23 (*Multiplication on F*): For any $[(x_1, y_1)], [(x_2, y_2)] \in F$, multiplication is defined by:

$$[(x_1, y_1)] \cdot [(x_2, y_2)] := [(x_1x_2, y_1y_2)]. \tag{3.29}$$

Proposition 3.24 (*Well-Definedness of Multiplication*): Multiplication on F is well-

defined and independent of the choice of representative.

Proof

The pair (x_1x_2, y_1y_2) lies in S, since $y_1y_2 \neq 0$ in the integral domain R (as for addition, Proposition 3.22).

Let $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$ and $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$ in S. By definition of $\equiv$:

$$x_1\hat{y}_1 = \hat{x}_1y_1 \quad \text{and} \quad x_2\hat{y}_2 = \hat{x}_2y_2. \tag{3.30}$$

Multiplying these two equations in the commutative ring R:

$$(x_1\hat{y}_1)(x_2\hat{y}_2) = (\hat{x}_1y_1)(\hat{x}_2y_2) \implies (x_1x_2)(\hat{y}_1\hat{y}_2) = (\hat{x}_1\hat{x}_2)(y_1y_2).$$

By (3.16), this algebraic identity is precisely the condition for equivalence:

$$(x_1x_2, y_1y_2) \equiv (\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2).$$

Thus, $[(x_1x_2, y_1y_2)] = [(\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)]$, proving well-definedness. Q.E.D.

Proposition 3.25 (*i is an Injective Ring Homomorphism*): The canonical inclusion $i\colon R \hookrightarrow F$ is an injective ring homomorphism:

$$i(r + \hat{r}) = i(r) + i(\hat{r}), \tag{3.31}$$

$$i(r \cdot \hat{r}) = i(r) \cdot i(\hat{r}), \tag{3.32}$$

$$i(1_R) = 1_F. \tag{3.33}$$

Proof

Evaluating the definitions directly:

- **Additivity:**
$$i(r) + i(\hat{r}) = [(r, 1)] + [(\hat{r}, 1)] = [(r \cdot 1 + \hat{r} \cdot 1, 1 \cdot 1)] = [(r + \hat{r}, 1)] = i(r + \hat{r}).$$
- **Multiplicativity:**
$$i(r) \cdot i(\hat{r}) = [(r, 1)] \cdot [(\hat{r}, 1)] = [(r\hat{r}, 1 \cdot 1)] = [(r\hat{r}, 1)] = i(r\hat{r}).$$
- **Identity:**
$$i(1_R) = [(1, 1)] = 1_F.$$

Injectivity was already established in Proposition 3.19. Q.E.D.

Multiplicative Inverses in F: To confirm that F is indeed a field, let $[(x, y)] \in F$ with $[(x, y)] \neq 0_F = [(0, 1)]$. Then $(x, y) \not\equiv (0, 1) \implies x \cdot 1 \neq 0 \cdot y \implies x \neq 0$. Thus, $(y, x) \in S$, and:

$$[(x, y)] \cdot [(y, x)] = [(xy, yx)] = [(1, 1)] = 1_F.$$

Therefore, every non-zero element in F possesses a multiplicative inverse $[(x, y)]^{-1} = [(y, x)]$.

Beyond the Lecture: The Remaining Field Axioms

The lecture checks that the operations are well-defined and that non-zero elements have inverses; the other axioms of a field reduce to computations in R on representatives. The key tool is that $(xs, ys) \equiv (x, y)$ for every $s \neq 0$, since $xs \cdot y = x \cdot ys$: a common non-zero factor can be cancelled.

- **Commutativity** of $+$ and $\cdot$ is visible in the formulas of Definition 3.21 and Definition 3.23, since R is commutative.
- **Associativity:** both ways of bracketing a sum of three classes give the same class,
$$\begin{aligned}([(x_1,y_1)]+[(x_2,y_2)])+[(x_3,y_3)] &= [(x_1y_2y_3+x_2y_1y_3+x_3y_1y_2,\ y_1y_2y_3)]\\ &= [(x_1,y_1)]+([(x_2,y_2)]+[(x_3,y_3)]),\end{aligned}$$
and both ways of bracketing a product give $[(x_1x_2x_3,\ y_1y_2y_3)]$.
- **Neutral elements and negatives:** $[(x,y)]+[(0,1)]=[(x,y)]$, $[(x,y)]\cdot[(1,1)]=[(x,y)]$, and $[(x,y)]+[(-x,y)]=[(0,y^2)]=[(0,1)]$.
- **Distributivity:** expanding both sides gives
$$[(x_1,y_1)]\cdot([(x_2,y_2)]+[(x_3,y_3)])=[(x_1x_2y_3+x_1x_3y_2,\ y_1y_2y_3)],$$
$$[(x_1,y_1)]\cdot[(x_2,y_2)]+[(x_1,y_1)]\cdot[(x_3,y_3)]=[(x_1x_2y_1y_3+x_1x_3y_1y_2,\ y_1^2y_2y_3)],$$
and the two right-hand sides are the same class after cancelling the factor $y_1\neq 0$.
- $0_F\neq 1_F$**:** $(0,1)\not\equiv(1,1)$, since $0\cdot 1\neq 1\cdot 1$ in the integral domain R.

Summary of the Field of Fractions Construction

Spoken Transcript [01:07:25 - 01:08:00]

That is the construction. When you understand it, you'll realize that actually you knew the whole thing anyway in your understanding of the rational numbers: you were subconsciously doing all of this. Not explicitly, but somehow underneath the surface, you were assuming all of these properties were true — it made the rational number arithmetic work.

There's different ways to try to learn this material, but the way that's foundational is you actually go yourself and check all the things that you're supposed to check. That takes a little bit of time and it's somewhat painful, but you only have to do it once in your life.

You can read the book too. I think that doesn't help so much, because then you just read what the author has written, and it's just a bunch of things with equalities — you can convince yourself of anything if you read what someone else did. The best thing is to go through all of it yourself; then you'll understand it once in your life.

And that's the construction. We're done with this construction.

Spoken Transcript [01:08:00 - 01:09:44]

The construction I just summarize here: *[writes "Construction"]* we start with R is an integral domain, *[writes "Start with R int. domain"]* and then we construct explicitly — that's as explicit a math construction as you get. You just take R, and you make some sets and equivalence classes and relations; it's totally explicit. Construct this F, which is the field of fractions. *[writes "Construct F = field of fractions"]*

And R sits inside F by this map: r goes to $(r,1)$. *[writes $R\hookrightarrow F$, $r\mapsto(r,1)$]*

When you get used to it, the way an algebraist always thinks about this is: you have elements in R, *[writes $r\in R$]* and the elements of F are just r over s. That's the element

of F; you just view them as kind of fractions in R. *[writes $\frac{r}{s} \in F$]*

This is a perfectly reasonable way to think about it. In our language, when someone writes this, what they really mean is the equivalence class of (r,s) in this set S. That's what it actually means. *[writes $(r,s) \in S$]*

And it's only legal if the bottom one is not zero, which you're not allowed to write anyway if the bottom one's not zero. And then you can add fractions, multiply fractions, just as you would normally do, and they exactly correspond to the rules that were prescribed here. So, that's the natural way to think about it.

Once all of this is done, you just start using it like you normally would anyway. But this is the underlying foundation.

Are there any conceptual questions about that?

Summary of the Construction of the Field of Fractions

Theorem 3.26 (*Existence and Structure of the Field of Fractions*): Let R be an integral domain. There exists a field F, called the *"field of fractions"* of R (denoted $\mathrm{Frac}(R)$), and an injective ring homomorphism:

$$i\colon R \hookrightarrow F, \quad r \mapsto [(r,1)],$$

such that:

(1) Field Structure: F is a field with zero $0_F = [(0,1)]$ and multiplicative identity $1_F = [(1,1)]$.

(2) Quotient Representation: Every element of F can be written as a quotient of elements in the image of R:

$$\forall q \in F\ \exists r,s \in R\ (s \neq 0) \quad \text{such that} \quad q = i(r)\cdot i(s)^{-1} = \frac{i(r)}{i(s)}. \tag{3.34}$$

Algebraic View: Working with Fractions

In mathematical practice, once the formal construction of $F = S/\equiv$ is established, we identify R with its image $i(R) \subseteq F$ and adopt standard fraction notation:

$$\frac{r}{s} := [(r,s)] \in \mathrm{Frac}(R) \quad (r,s \in R,\ s \neq 0). \tag{3.35}$$

Arithmetic in the Field of Fractions: Under this notation, the field operations on $\mathrm{Frac}(R)$ take the standard form familiar from elementary arithmetic:

$$\text{Equality:} \qquad \frac{r_1}{s_1} = \frac{r_2}{s_2} \iff r_1 s_2 = r_2 s_1, \tag{3.36}$$

$$\text{Addition:} \qquad \frac{r_1}{s_1} + \frac{r_2}{s_2} = \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}, \tag{3.37}$$

$$\text{Multiplication:} \qquad \frac{r_1}{s_1} \cdot \frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}, \tag{3.38}$$

$$\text{Inverse:} \qquad \left(\frac{r}{s}\right)^{-1} = \frac{s}{r} \quad (\text{provided } r \neq 0). \tag{3.39}$$

Examples of Fields of Fractions

Spoken Transcript [01:09:44 - 01:10:46]

If I take $\mathbb{Z}$, *[writes $\mathbb{Z} \to$ What is field of fractions?]* we can ask: what is the field of fractions?

The answer is going to be $\mathbb{Q}$, *[writes $\mathbb{Q}$]* which means that if you start with $\mathbb{Z}$ and you do this crazy construction, you get the rational numbers. I hope that's clear by now. All right?

What happens if I start with the ring of Gaussian integers? What is the field of fractions? *[writes $\mathbb{Z}[i] \to$ What is the field of fractions?]*

That's a strange question, because I've just explained what the field of fractions is: you take the Gaussian integers, you do this construction, that's the field of fractions. When I say *"what is the field of fractions?"*, what the question is actually asking is: what's another way of thinking about the answer?

Do you have a proposal for what that is? Yeah?

Student Answer

$\mathbb{Q}$ adjoined i.

Spoken Transcript [01:10:46 - 01:11:48]

Yeah, so, the answer is $\mathbb{Q}(i)$, *[writes $\mathbb{Q}(i) = \{a + bi \mid a, b \in \mathbb{Q}\}$]* which is equal to the set $a + bi$ where a and b are in $\mathbb{Q}$.

It's correct, but to check this is correct, you have to do a bunch of things: one is you have to show this is a field — you can invert — and then you have to show that this sits inside, and everything here is a ratio of that. It's all true. But that's the answer here, and requires some checking.

All right, this is correct.

Fields of Fractions of Intermediate Rings and Fields

Spoken Transcript [01:11:48 - 01:12:32]

What about this? We saw last week that there are some funny rings that sit inside $\mathbb{Z}$ and $\mathbb{Q}$. *[writes $\mathbb{Z} \subset \mathbb{Z}[\frac{1}{3}] \subset \mathbb{Q}$]*

This ring is the set $a/3^b$, where $a \in \mathbb{Z}$ and $b \geq 0$, right? *[writes $\{a/3^b \mid a \in \mathbb{Z}, b \geq 0\}$]*

We discussed this last week (Recall: Example 2.29): the subrings that sit in between. I hope you remember. Do you remember?

This is an integral domain also, *[writes "int. domain"]* because it sits inside $\mathbb{Q}$. What is the field of fractions of... *[writes "What is field of fractions of $\mathbb{Z}_{1/3}$?"]* I just gave it a name, $\mathbb{Z}_{1/3}$, because you're allowed to have threes in the bottom. There's two standard

names for this: one is $\mathbb{Z}_{1/3}$, and the other's... that's a better name, okay.

So, what is the field of fractions of this integral domain? Yeah?

Student Answer

$\mathbb{Q}$.

Spoken Transcript [01:12:33 - 01:13:22]

Also $\mathbb{Q}$. *[writes $\mathbb{Q}$]*

Nothing says it can't be $\mathbb{Q}$. It certainly sits inside $\mathbb{Q}$, and everything in $\mathbb{Q}$ is a ratio of numbers here — it's already a ratio of numbers here. So, it's just $\mathbb{Q}$! If you go through this definition, you get $\mathbb{Q}$.

As I said, you think about this field of fractions as the smallest field that contains the integral domain, and you can't really contain something smaller than $\mathbb{Q}$, because it already has $\mathbb{Z}$ in it, and $\mathbb{Z}$ already gives you everything in $\mathbb{Q}$.

So, what is the field of fractions of this? The answer: also $\mathbb{Q}$.

In fact, if you take any ring — any integral domain that sits in between $\mathbb{Z}$ and $\mathbb{Q}$ — the field of fractions is just $\mathbb{Q}$.

Examples: Fractions of the Integers, Gaussian Integers, and Subrings of $\mathbb{Q}$

Example 3.27 (*The Integers*): For the ring of integers $\mathbb{Z}$:

$$\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}.$$

Example 3.28 (*The Gaussian Integers*): For the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$:

$$\mathrm{Frac}(\mathbb{Z}[i]) = \mathbb{Q}(i) := \{a + bi \mid a, b \in \mathbb{Q}\}.$$

The field $\mathbb{Q}(i)$ is called the field of *"Gaussian rationals"*.

Example 3.29 (*Triadic Rationals*): Consider the ring of triadic fractions $\mathbb{Z}[\frac{1}{3}]$ (the case $k = 3$ of the generalization after Example 2.29), defined by:

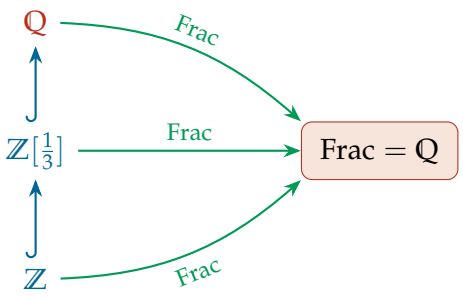
$$\mathbb{Z}\left[\frac{1}{3}\right] := \left\{ \frac{a}{3^b} \;\middle|\; a \in \mathbb{Z},\ b \in \mathbb{Z}_{\geq 0} \right\} \subset \mathbb{Q}.$$

Since $\mathbb{Z} \subset \mathbb{Z}[\frac{1}{3}] \subset \mathbb{Q}$, any fraction of elements from $\mathbb{Z}[\frac{1}{3}]$ is already a rational number, and every rational number $\frac{p}{q}$ can be written as a quotient of elements in $\mathbb{Z} \subset \mathbb{Z}[\frac{1}{3}]$. Thus:

$$\mathrm{Frac}\left(\mathbb{Z}\left[\frac{1}{3}\right]\right) = \mathbb{Q}.$$

Proposition 3.30 (*Intermediate Subrings of $\mathbb{Q}$*): Let R be any subring of $\mathbb{Q}$ with $\mathbb{Z} \subseteq R \subseteq \mathbb{Q}$. Then:

$$\mathrm{Frac}(R) = \mathbb{Q}. \tag{3.40}$$



Why All Intermediate Subrings Yield $\mathbb{Q}$: The universal property was only proved for $\mathbb{Z}$ (Theorem 3.7), so here is a direct argument. Since $R \subseteq \mathbb{Q}$, a formal fraction $\frac{r}{s} \in \mathrm{Frac}(R)$ can be read as an actual rational number, which gives a map

$$\Phi\colon \mathrm{Frac}(R) \to \mathbb{Q}, \qquad [(r,s)] \mapsto rs^{-1}.$$

It is well-defined: $(r,s) \equiv (r',s')$ means $rs' = r's$, hence $rs^{-1} = r's'^{-1}$ in $\mathbb{Q}$. It is a ring homomorphism, because the operations of $\mathrm{Frac}(R)$ (Definition 3.21, Definition 3.23) are the usual fraction rules of $\mathbb{Q}$. As a homomorphism out of a field into a non-zero ring it is injective (Theorem 1.32 and the steps after it). It is surjective, since every rational number is $\frac{p}{q}$ with $p, q \in \mathbb{Z} \subseteq R$, $q \neq 0$. Hence Φ is an isomorphism, $\mathrm{Frac}(R) \cong \mathbb{Q}$.

Chalkboard Transition & Anecdote on Guessing

The lecturer wipes the lower left chalkboard clean with a sponge while sharing an anecdote from an academic conference about his philosophy of teaching mathematics through guessing.

Spoken Transcript [01:13:22 - 01:14:49]

Now a stranger question: what happens if... well, yeah.

[The lecturer washes the board] You know, on the business of guessing, once I was in a conference, and the conference had some math researchers and some high school teachers. At dinner I was sitting next to some teachers, and I was trying to explain to them my theory of how you should teach children math. And the outcome was that they wanted me disbarred from academics! *[laughter]*

The theory, by the way, was that on exams you should make the kids just guess the answers: no calculations, just guess everything. That's maybe a little excessive, but maybe a compromise: like, half the exam they should just guess everything. But I think that would be really somehow more in the spirit of mathematics than...

[The lecturer finishes drying the board]

Spoken Transcript [01:14:49 - 01:15:25]

A stranger question is: suppose I start with a field. *[writes "Suppose R is a field"]* Then, of course, it's still an integral domain, *[writes "R is int. domain"]* because every

field is an integral domain (Recall: Example 2.61). Then what is the field of fractions? *[writes "What is the field of fractions?"]*

What do you think the field of fractions is going to be here? Yeah?

Student Answer

R itself.

Spoken Transcript [01:15:28 - 01:16:00]

Yes, R itself. And the answer is R itself. *[writes "R itself"]*

To prove that: again, this fits in this view that it's the smallest field. If it's already a field, how could there be a smaller field that contains it? It's already a field.

To prove it, you have to go through this construction and show that in the case where R is a field, that construction just reconstructs R. To do that, you have to go through that construction and see it. So, that's also an exercise for you.

Those are really the basic facts.

Proposition 3.31 (*Field of Fractions of a Field*): Let K be a field. Then the canonical inclusion $i\colon K \hookrightarrow \mathrm{Frac}(K)$ is an isomorphism of fields:

$$\mathrm{Frac}(K) \cong K. \tag{3.41}$$

Proof

By Proposition 3.25, $i\colon K \hookrightarrow \mathrm{Frac}(K)$ is an injective field homomorphism. To establish surjectivity, let $[(x,y)] \in \mathrm{Frac}(K)$ with $x, y \in K$ and $y \neq 0$. Since K is a field, y has a multiplicative inverse $y^{-1} \in K$. Then:

$$(x,y) \equiv (xy^{-1},1) \iff x \cdot 1 = (xy^{-1}) \cdot y = x.$$

Consequently:

$$[(x,y)] = [(xy^{-1},1)] = i(xy^{-1}).$$

Every element of $\mathrm{Frac}(K)$ is the image of the element $xy^{-1} \in K$. Hence, i is surjective and constitutes a field isomorphism. Q.E.D.

Polynomial Rings

Spoken Transcript [01:16:00 - 01:18:05]

The next topic this week is about polynomial rings. These are really fun. *[writes "Polynomial rings, Power series rings"]* We talk about polynomial rings, and in some sense even more fun than polynomial rings are power series rings. Again, in some sense you've been using all of this already in the first year, but here we take a formal approach.

The idea is: if I take R as a ring *[writes "R is a ring"]* — and now no assumptions,

meaning not an integral domain, so, this just means for us commutative ring with unit: *[writes "(comm. ring with unit)"]* that's our basic ring. I would say a ring with no assumptions: commutative ring with unit (Recall: Remark 1.17). In particular, everything today so far, R was an integral domain; now I don't care if it's an integral domain, just a commutative ring with unit.

Then I make here a new ring: $R[x]$. *[writes $R[x]$]* x is the variable, or sometimes it's called indeterminate *[writes "the indeterminate"]* — that's kind of a fancy name, but it's just x, right? *[writes "or the variable"]*

The name of this ring is the polynomial ring with coefficients in R. *[writes "(Poly. ring with the coeff's in R)"]*

Or if you want to be even more precise: the polynomial ring in one variable with coefficients in R. That's kind of a long sentence, but if you want to be totally precise. If you meet a mathematician and say, *"I'm going to consider the polynomial ring in one variable with coefficients in R"*, as long as you agree what a ring is, it's just absolutely unambiguous: the person will know exactly what you're talking about.

Informal Definition of Polynomials as Formal Expressions

Spoken Transcript [01:18:05 - 01:19:08]

What is this? Well, I have to tell you what it is as a set. So, the first thing is: what's the set? *[writes "What is the set?"]*

Here the definition is: it's the set of polynomials in x with coefficients in R. Polynomials look like this: $a_0 + a_1x + a_2x^2 + \dots$ *[writes $R[x] = \{a_0 + a_1x + a_2x^2 + \dots\}$]*

It's these symbols, and one of the properties of a polynomial is that it's not infinite: a polynomial is finite. That's part of the definition always in mathematics of a polynomial. That means we stop somewhere: some n, a_n. So, the elements of this ring are such symbols — just formal combinations of $a_0, \dots, a_n$, where these a_i's are in R, okay? *[writes $+a_nx^n \mid a_i \in R, n \in \mathbb{Z}_{\geq 0}\}$]*

And possibly different n's, and this n is some integer greater than or equal to zero.

Examples and Constant Polynomials

Spoken Transcript [01:19:08 - 01:20:48]

If I have $\mathbb{Z}[x]$, *[writes $\mathbb{Z}[x] \ni f(x)$]* the elements in here are sometimes written $f(x)$, because we like to write polynomials with the variable. It could be, for example, $f(x) = 3 + 2x + 15x^2 + 345x^{200}$. *[writes $= 3 + 2x + 15x^2 + 345x^{200}$]* This is a totally legitimate element of it. Other times you'll see people just write this as just f, and just suppress the variable. *[writes or f]* Those are two different ways you can write polynomials.

For $R[x]$, it's just the same thing: you just write an element here, $f(x)$ *[writes $R[x] \ni f(x) = a_0 + a_1x + \dots + a_nx^n$]* — you pick some elements of R, some finite number of elements of R as the coefficients, all the way to a_n, and then you just put the x's here.

x is a formal variable, so, x is just in some sense a placeholder. Those are the elements of $R[x]$.

And $R[x]$ has a confusing element, just like the integers; there's a confusing polynomial: the constant polynomial. If I put here 17, *[writes $17 \in \mathbb{Z}[x]$]* that is also here. That's the polynomial that only has a constant term: it just happens that there's no x. It can happen: there's no x in that. So, it's weird to write this polynomial as a function of x, but it is a function of x — it's a constant function.

In this ring there are also the constant polynomials — the ones with no x.

The Polynomial Ring $R[x]$: Informal Formulation

Recap: Polynomial rings were already used informally: $\mathbb{Z}[x] \subseteq \mathbb{Q}[x] \subseteq \mathbb{R}[x] \subseteq \mathbb{C}[x]$ in Week 1 (Example 1.23) and $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$ in Week 2 (Example 2.31). Here they are defined formally.

Definition 3.32 (*Polynomial Ring over a Commutative Ring*): Let R be a commutative ring with unit 1_R. The *"polynomial ring in one variable"* (or indeterminate) x with coefficients in R, denoted $R[x]$, is defined as the set of formal expressions:

$$R[x] := \left\{ \sum_{i=0}^{n} a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \;\middle|\; n \in \mathbb{N}_0,\ a_i \in R \right\}. \tag{3.42}$$

Example 3.33 (*Polynomials in $\mathbb{Z}[x]$*): In the ring $\mathbb{Z}[x]$, an element may be denoted $f(x)$ or simply f:

$$f(x) = 3 + 2x + 15x^2 + 345x^{200} \in \mathbb{Z}[x].$$

Constant elements such as $17 \in \mathbb{Z}$ correspond to polynomials of degree 0 with $n = 0$:

$$17 = 17 \cdot x^0 \in \mathbb{Z}[x].$$

The Indeterminate as a Formal Placeholder: Crucially, x is not an element of R, nor does it represent an unknown numerical value in this context. It is an *"indeterminate"* (formal variable) acting as a positional placeholder. The symbols $+$ and x^i in (3.42) are purely typographical markers separating the coefficients $a_i \in R$, rather than evaluations of operations.

Formal Rigorous Definition of $R[x]$ as Sequences

Spoken Transcript [01:20:48 - 01:22:05]

Is it clear what this set is? You take R and then you take an x that has nothing to do with R — it's just some kind of Platonic, ideal indeterminate — and then you just consider these formal linear combinations. That's what a polynomial is.

In particular, I don't mean really anything by this plus: it's just a formal linear combination. Some authors don't like to write this, because they think you might think that this plus here means something, which is a legitimate thing to think, actually.

So, if you look at books where they're really formally correct, *[pushes upper chalkboard upward]* they'll do some weird things just so you don't get confused.

I will explain it for a moment, but I don't do that because I think that it's a little dishonest. When I give these lectures, I try to actually give the lectures in the way I think about the subject: there's some kind of underlying honesty in the lecture.

Chalkboard Transition

The lecturer pushes the upper chalkboard upward and begins washing the lower chalkboard to formalize the rigorous definition of polynomials.

Spoken Transcript [01:22:05 - 01:23:58]

If I were to be dishonest and formally correct, the way you define these polynomials is the following:

[The lecturer washes and dries the board] You say a polynomial — an element of $R[x]$ *[writes "An element of $R[x]$ is a tuple of elements of R"]* — is a tuple of elements of R: $(a_0, a_1, a_2, \dots)$, *[writes $(a_0, a_1, a_2, \dots)$]* an infinite tuple such that there exists some big N so that for all $m > N$, the element in this tuple is zero, okay? *[writes "and that $\exists N\, \forall m > N,\ a_m = 0$"]*

That's a formal definition of a polynomial in one variable, where I just give you the coefficients. The coefficients are $a_0, a_1, a_2, \dots$ — they're all elements of R. *[writes $a_i \in R$]*

I just give you the coefficients, and they could be lots and lots of them — any finite number — but eventually this sequence of coefficients is all zero. That's the formal definition of a polynomial.

It's actually the same thing that I'm saying, but it's much more reasonable to think about it with the x's. This way is just so that you don't get confused formally.

That's the set. I hope there's no confusion about the set.

The Indeterminate and the Ring Operations on $R[x]$

Spoken Transcript [01:23:58 - 01:24:24]

In this abstract way, what is the variable x? *[writes $x = \dots$]*

You have five minutes to figure it out.

What's the variable x? In mine, there's an x. Here there's no x. So, what's x here? Yeah?

Student Answer

$(0, 1, 0, \dots)$

Spoken Transcript [01:24:27 - 01:25:20]

Yeah, so, x is this: *[writes $x = (0, 1, 0, \dots)$]* zero, one, and then all zeros. That's x if

you think about it this way, okay?

That's the set. Once you agree what the set is — and I gave you two ways to think about it: one way, which is a very simple way; this way is more formally correct — I need to give you the addition rule, the multiplication rule, zero, and one. *[writes $R[x], +, \cdot, 0, 1$]*

Zero is just the zero polynomial: in my case, just the polynomial zero. In this case, you have to write an infinite number of zeros, which takes a long time. *[writes 0 under 0, and $(0,0,\dots)$]*

And one is just the constant one polynomial. Here it's the $(1,0,0,\dots)$ polynomial. *[writes 1 and $(1,0,\dots)$]* Yeah?

Student Question

Is the plus in that definition not the plus from the ring?

Spoken Transcript [01:25:25 - 01:26:05]

It is eventually, but I haven't defined the plus for the ring. That's the reason why this definition is a little bit better; this is a good thing to think about.

I'm defining $R[x]$ as a set first. Then I'll define addition. When I finish, it will be, but when it's born, it's just a set. Is that clear?

[Student nods] It's a good question. That's the reason why thinking about it this way is slightly confusing. Although I encourage you to think about it this way, because you have a lot of experience with polynomials, and in the end, nothing will go wrong.

Formal Definition of Polynomials as Sequences with Finite Support

Definition 3.34 (*Rigorous Definition of $R[x]$ via Coefficient Tuples*): Let R be a commutative ring with unit. The polynomial ring $R[x]$ is formally defined as the set of sequences $(a_k)_{k\in\mathbb{N}_0}$ of elements in R (elements of $R^{\mathbb{N}_0}$, Notation 0.1) with *"finite support"* (almost-zero sequences):

$$R[x] := \left\{ (a_0, a_1, a_2, \dots) \in R^{\mathbb{N}_0} \;\middle|\; \exists N \in \mathbb{N}_0 \, \forall m > N \colon a_m = 0 \right\}. \tag{3.43}$$

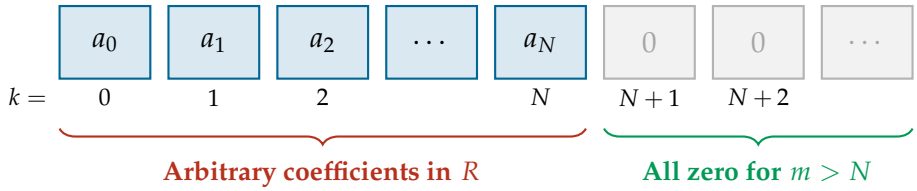
Proposition 3.35 (*Identification of the Indeterminate x and Constants*): Under the sequence model (3.43):

(1) Indeterminate x: The indeterminate x corresponds to the elementary sequence with 1 in position 1:

$$x := (0, 1, 0, 0, \dots). \tag{3.44}$$

(2) Constant Elements: For any $a \in R$, the constant polynomial corresponds to the sequence:

$$a := (a, 0, 0, 0, \dots). \tag{3.45}$$



Addition in $R[x]$

Spoken Transcript [01:26:05 - 01:26:28]

You have to agree what the set is, and I gave you two different ways to think about the set. Then you have to define addition. You can define addition in that way as addition of polynomials, which is very good, because I don't have to write it, because you already know how to add polynomials. Is that correct?

If I write it here, then I have to decide how to add polynomials, and that's not hard. How do you add polynomials here? Yeah?

Student Answer

Componentwise addition.

Spoken Transcript [01:26:30 - 01:27:08]

Componentwise addition. Which is also how you add polynomials there: componentwise addition.

And it will be the case there that if I take the constant polynomial and I take this polynomial and I add them, *[writes a_0, x, $a_0 + x$]* I get the thing I call this, which is also kind of nice. All right?

Addition is easy. Then they have to define multiplication. Multiplication is more of a pain. And if you think about it there, you already know how to multiply polynomials: you have to do this thing where you distribute and whatever — you know how to do it.

How do you define polynomial multiplication here? Let's try to do that.

Multiplication in $R[x]$ and the Cauchy Convolution

Spoken Transcript [01:27:08 - 01:27:43]

If I use this list of coefficients, here's my first polynomial, f, *[writes $f = (a_0, a_1, a_2, \dots)$]* and my second polynomial, g. *[writes $g = (b_0, b_1, b_2, \dots)$]* Then I have to define this polynomial, which is $f \times g$. *[writes $fg = ($]*

How do I do it? What's this coefficient going to be? What do you think? Yeah?

Student Answer

a_0b_0.

Spoken Transcript [01:27:44 - 01:28:55]

Yeah, because it's just the constant... The secret, the way to do this is you write this thing out... This is a definition, but the way to understand this definition is you write this out as a polynomial: *[writes $a_0 + a_1x + \dots$]* that's the easy way to think about it. And so on: write this as a polynomial, *[writes $b_0 + b_1x + \dots$]* and you do polynomial multiplication, and the constant term is that: a_0b_0. *[writes a_0b_0]*

What is this term? Well, it's going to be $a_1b_0 + a_0b_1$. *[writes $a_1b_0 + a_0b_1$]* The second term is going to be $a_2b_0 + a_1b_1 + a_0b_2$. *[writes $+a_2b_0 + a_1b_1 + a_0b_2, \dots$]*

Formally, this is the definition of polynomial multiplication. Is that clear?

Student Question

This is essentially like doing the Cauchy product?

Spoken Transcript [01:29:05 - 01:29:40]

Yeah, you do this thing. *[gestures diagonal grouping]* That's what you're saying? Yeah, that's what you do. That's how you multiply polynomials.

This is the same rule as you'd get here. Look: if you take the linear term, how do you get an x here? You have to take x times a constant — those are the two ways to get the x. So, you get this rule.

The Ring Structure of $R[x]$

Definition 3.36 (*Ring Operations on $R[x]$*): Let R be a commutative ring with unit 1_R. The polynomial ring $(R[x], +, \cdot, 0_{R[x]}, 1_{R[x]})$ is equipped with the following structures:

(1) Neutral Elements:

$$0_{R[x]} := (0_R, 0_R, 0_R, \dots), \tag{3.46}$$

$$1_{R[x]} := (1_R, 0_R, 0_R, \dots). \tag{3.47}$$

(2) Addition (Componentwise): For $f = (a_k)_{k\geq 0}$ and $g = (b_k)_{k\geq 0}$:

$$(a_0, a_1, a_2, \dots) + (b_0, b_1, b_2, \dots) := (a_0 + b_0,\ a_1 + b_1,\ a_2 + b_2,\ \dots). \tag{3.48}$$

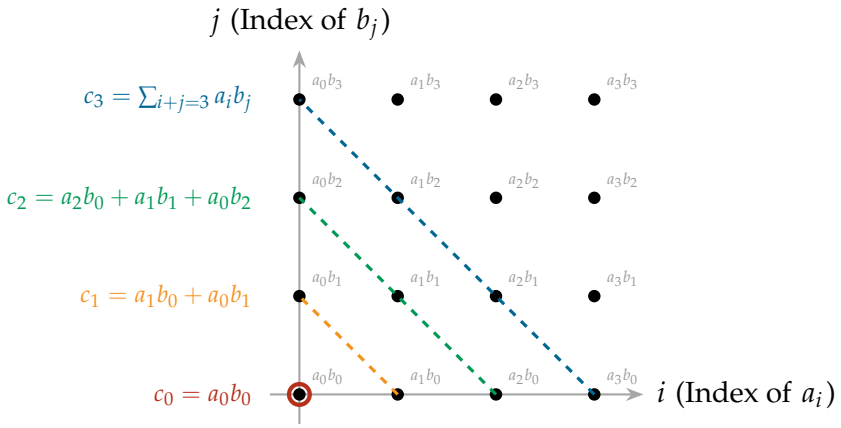
(3) Multiplication (Cauchy Product / Convolution): The product $f \cdot g = (c_k)_{k\geq 0}$ is defined by:

$$c_k := \sum_{i=0}^{k} a_i b_{k-i} = a_0b_k + a_1b_{k-1} + \cdots + a_kb_0. \tag{3.49}$$

Explicitly:

$$f \cdot g = (a_0b_0,\ a_0b_1 + a_1b_0,\ a_0b_2 + a_1b_1 + a_2b_0,\ \dots). \tag{3.50}$$

Both results are again almost-zero sequences: if $a_k = 0$ for $k > N$ and $b_k = 0$ for $k > M$, then $a_k + b_k = 0$ for $k > \max(N, M)$, and $c_k = 0$ for $k > N + M$ (each term $a_i b_{k-i}$ has $i > N$ or $k - i > M$).



Connection to the Cauchy Product and Discrete Convolution: The formula $c_k = \sum_{i=0}^{k} a_i b_{k-i}$ exactly mirrors:

(1) Distributive Expansion: In formal series notation:

$$\left(\sum_{i=0}^{n} a_i x^i\right)\left(\sum_{j=0}^{m} b_j x^j\right) = \sum_{i=0}^{n}\sum_{j=0}^{m} a_i b_j x^{i+j} = \sum_{k=0}^{n+m}\left(\sum_{i+j=k} a_i b_j\right) x^k.$$

The powers of the indeterminate x naturally collect all terms where the sum of indices equals k.

(2) Cauchy Product of Power Series: In analysis, the product of two absolutely convergent power series $\sum a_n x^n$ and $\sum b_n x^n$ is given by the exact same convolution formula (Fact 0.42).

Conclusion of the Lecture

Spoken Transcript [01:29:40 - 01:30:00]

That's the ring; we're going to talk about it more next time. It's a little bit of a subtle matter, and it's good to make sure you completely understand it. Given a ring, there's a polynomial ring. A polynomial is actually just a list of coefficients. It's very helpful to think about the polynomial that way, because that's how you normally think about it, and you won't go wrong thinking about it that way.

Okay, we'll stop. *[students applaud]*

Outlook: Formal Power Series in the Next Lecture

In the lecture introduction, the professor remarked that *"in some sense even more fun than polynomial rings are power series rings"*. This lecture formalized the polynomial ring $R[x]$ as sequences of coefficients with finite support (almost-zero sequences). Dropping the finite support requirement leads directly to the ring of *"formal power series"* $R[[x]]$, con-

sisting of arbitrary infinite sequences $(a_0, a_1, a_2, \dots)$ equipped with the same Cauchy product multiplication, which is systematically investigated at the start of the next lecture (Definition 3.40).

Summary of Key Concepts

- **The Group of Units $\mathrm{Unit}(R)$ (also denoted $U(R)$, $R^\times$, or R^*):** The set of multiplicatively invertible elements $R^\times = \{x \in R \mid \exists y \in R \colon xy = 1\}$ forms an abelian group under ring multiplication. Key examples include $\mathbb{Z}^* = \{\pm 1\}$, $\mathbb{Z}[i]^* = \{\pm 1, \pm i\}$ (characterized via the norm $N(a+bi) = a^2 + b^2 = 1$), and $(\mathbb{Z}/m\mathbb{Z})^* = \{\bar{x} \mid \gcd(x, m) = 1\}$.
- **Euler's Totient Function $\phi(m)$:** Defined as the order $|(\mathbb{Z}/m\mathbb{Z})^*|$, with explicit values $\phi(p) = p - 1$ and $\phi(p^k) = p^k - p^{k-1} = p^k(1 - 1/p)$ for prime powers.
- **Universal Property of $\mathbb{Q}$:** As the minimal field containing the integral domain $\mathbb{Z}$, any injective ring homomorphism $f \colon \mathbb{Z} \to F$ into a field F factors uniquely through an injective field homomorphism $g \colon \mathbb{Q} \to F$ defined by $g(x/y) = f(x)f(y)^{-1}$.
- **Field of Fractions $\mathrm{Frac}(R)$:** For any integral domain R, the field of fractions is constructed as the quotient of formal pairs $S = R \times (R \setminus \{0\})$ by the cross-multiplication equivalence relation $(x_1, y_1) \equiv (x_2, y_2) \iff x_1 y_2 = x_2 y_1$. Transitivity and the non-vanishing of the denominators $y_1 y_2 \neq 0$ crucially rely on the zero-divisor-free property of R.
- **Canonical Inclusion and Examples:** The map $i \colon R \hookrightarrow \mathrm{Frac}(R)$, $r \mapsto [(r, 1)]$, is an injective ring embedding. Examples include $\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}$, $\mathrm{Frac}(\mathbb{Z}[i]) = \mathbb{Q}(i)$, $\mathrm{Frac}(\mathbb{Z}[\frac{1}{3}]) = \mathbb{Q}$, and $\mathrm{Frac}(K) \cong K$ for any existing field K.
- **Polynomial Rings $R[x]$:** For a commutative ring with unity R, the polynomial ring $R[x]$ is formally defined as the ring of almost-zero sequences $(a_k)_{k \in \mathbb{N}_0}$ in $R^{\mathbb{N}_0}$. It is equipped with componentwise addition and Cauchy product (convolution) multiplication $c_k = \sum_{i=0}^{k} a_i b_{k-i}$, with indeterminate $x = (0, 1, 0, 0, \dots)$.

Wednesday 30 September 2026

Field of Fractions, Polynomial Rings, and Formal Power Series

Context

Continuing from the previous lecture, we review the construction and intuitive arithmetic of the field of fractions $\mathrm{Frac}(R)$ for an integral domain R. We then return to the polynomial ring $R[x]$ over an arbitrary commutative ring with identity, establish its universal mapping property, and determine its units when R is an integral domain. Finally, we introduce the formal power series ring $R[[x]]$, define its ring operations, and characterize its units via formal geometric series.

Field of Fractions and Ring Constructions

Field of Fractions of an Integral Domain

Spoken Transcript [00:00:00 - 00:01:00]

Can I start? Okay, we start.

The topic this week was... the first part was R is an integral domain *[writes R int domain]*, and then it's the construction of the field of fractions *[writes $\Longrightarrow$ $R \subset F$ and "field of fraction of R"]* of R (Recall: Definition 3.17). Sometimes this is denoted by $\mathrm{Frac}(R)$ *[writes $F = \mathrm{Frac}(R)$]*.

And then the other two topics are: if R is any ring *[writes R any ring]*, which for us is commutative with unit, then there's the construction of the polynomials *[writes $R[x]$]* — we started that, I'll say more — and also of the power series *[writes $R[[x]]$]*. This is the universal symbol for polynomial *[writes poly]*, and this is the universal mathematical way of describing power series, the power series ring *[writes power series]*.

So, I'll talk about these topics.

Topic Overview: Integral Domains and Ring Constructions

- **Field of Fractions:** If R is an integral domain, there exists an embedding into a field:

$$R \subset F := \mathrm{Frac}(R),$$

where F is called the *"field of fractions"* of R.

- **Polynomial and Power Series Rings:** If R is any commutative ring with identity 1, we construct:
  - $R[x]$: the *"polynomial ring"* in the indeterminate x with coefficients in R,
  - $R[[x]]$: the *"formal power series ring"* in the indeterminate x with coefficients in R.

Spoken Transcript [00:01:00 - 00:01:46]

I'll just say one more thing about the field of fractions. I gave you this very formal way to think about the field of fractions, which is you take some set S of pairs *[writes $S \subset R^2$]* — I won't review the whole thing now —, a set S of pairs, then you have an equivalence relation *[writes $\equiv$]*, and you look at the objects in this field of fractions as being equivalence classes in this equivalence relation *[writes $S_{\equiv} \ni f$]*. And then you can define everything at this formal level and then check all the properties.

And that is, you know, the underlying mathematical formalism of the field of fractions — meaning that if you're ever thinking about the field of fractions and you're wondering about something, you can always resolve it foundationally by saying, *"I know what the definition is. It's this, this, and this, precisely."*

Formal Construction of the Field of Fractions

Recap: The field of fractions was constructed in the previous lecture: the set of pairs (Definition 3.12), the equivalence relation (Definition 3.14, Proposition 3.16) and the quotient set (Definition 3.17). The lecturer summarizes it again.

Let R be an integral domain. To construct the field of fractions formally:

(1) Set of Pairs: Consider the Cartesian product $S := R \times (R \setminus \{0\}) \subset R^2$.

(2) Equivalence Relation: Define the relation $\equiv$ on S by:

$$(r, s) \equiv (t, w) \iff rw - st = 0.$$

(3) Equivalence Classes: The field of fractions is defined as the quotient set, written $S_{\equiv}$ on the board:

$$F = \mathrm{Frac}(R) := S/\equiv.$$

Spoken Transcript [00:01:46 - 00:03:25]

No one ever thinks about it that way. The way you think about it is: if R is an integral domain *[writes $R \ni r, s$]* — these are elements in R, if you have some elements r and s —, then for an element of the field of fractions *[writes $\mathrm{Frac}(R) =$]*, you just say this is r/s *[writes r/s]*. That's it, you know? And when you do this, you make sure s is not zero *[writes $s \neq 0$]*, because that was the second one there, just like you would with rational numbers.

And if you want to add, multiply, all of this, you just use the rules you did with rational numbers. Like if you want to add $r/s + t/w$ *[writes $\frac{r}{s} + \frac{t}{w}$]*, you know how to

add fractions: you just do it here. It's just the right answer; it's this *[writes* $= \frac{rw+st}{sw}$ *]*. And so on. If you multiply, then you multiply fractions; whatever rules you use to manipulate fractions, they're just correct here.

In particular, why is this a field? If I take r/s, if this is not zero *[writes* $\frac{r}{s} \neq 0$*]*, that implies that r is not zero *[writes* $\implies r \neq 0$*]*. If r were zero, then that's the zero element of this. And you can go check by the definitions that if r is zero, this is the zero element. So, if this is not zero, r must not be equal to zero. And then it has an inverse, and that's the multiplicative inverse *[writes* $\left(\frac{r}{s}\right)^{-1} = \frac{s}{r}$*]*. And it's legal because r's not zero, so I can put him downstairs.

Anyway, the way you think about it — I mean, you get to think about it however you want —, the way typically people think about the field of fractions is you just write the fractions. That's why it's called the field of fractions. And you just go forward. And the rules of this formalism are designed so that this actually works. Okay.

Intuitive Arithmetic in the Field of Fractions

Recap: These fraction rules were already listed in the previous lecture (Eqs. (3.36) to (3.39)).

In mathematical practice, elements of $\mathrm{Frac}(R)$ are represented as formal fractions $\frac{r}{s}$ with $r, s \in R$ and $s \neq 0$:

$$\mathrm{Frac}(R) = \left\{ \frac{r}{s} \;\middle|\; r, s \in R,\ s \neq 0 \right\}. \tag{3.51}$$

The field operations are defined analogously to elementary fraction arithmetic:

$$\textbf{Addition:} \quad \frac{r}{s} + \frac{t}{w} = \frac{rw + st}{sw}, \tag{3.52}$$

$$\textbf{Multiplication:} \quad \frac{r}{s} \cdot \frac{t}{w} = \frac{rt}{sw}, \tag{3.53}$$

$$\textbf{Multiplicative Inverse:} \quad \left(\frac{r}{s}\right)^{-1} = \frac{s}{r} \qquad \left(\text{valid whenever } \frac{r}{s} \neq 0, \text{ i.e., } r \neq 0\right). \tag{3.54}$$

Why the Integral Domain Assumption Is Essential: For the denominator sw in (3.52) and (3.53) to be non-zero, we must have:

$$s \neq 0 \text{ and } w \neq 0 \implies sw \neq 0.$$

This property holds if and only if R has no zero divisors, which is precisely the defining condition of an **integral domain** (Definition 2.62). Furthermore, the zero element of $\mathrm{Frac}(R)$ is $\frac{0}{1} = \frac{0}{s}$ (Definition 3.20). Thus, $\frac{r}{s} \neq 0$ implies $r \neq 0$, guaranteeing that $\left(\frac{r}{s}\right)^{-1} = \frac{s}{r}$ is a well-defined fraction with non-zero denominator.

Spoken Transcript [00:03:25 - 00:04:45]

This is a basic issue that comes up very often in math education about how to think. The first time this comes up, which you've already passed, is the real numbers. To define the real numbers is complicated: you have to either define them as Cauchy sequences or Dedekind cuts. And that's pretty complicated to define, and then you have to check that it satisfies the properties. That's all fine — that's kind of like some logical underlayer —,

but no one thinks about real numbers that way.

And it's just that that's part of learning mathematics: that there's some kind of logical underlayer, and then there's the way you think about it. And usually they have nothing to do with each other. Okay, that's maybe an exaggeration, but... *[laughter]*

Actually, about the real numbers, I can't help now but comment that Dedekind cuts was discovered by Richard Dedekind, who had moved to Zurich in the second part of the 19th century. And he's an algebraist; he's a very famous mathematician. He's, of course, passed away by now. And he was assigned to teach analysis. He writes it very beautifully in this book that he wrote about the real numbers: he was assigned to teach analysis, and he's an algebraist, and he said, *"Well, I had a difficult time teaching analysis because I don't know what a real number is."* And then he spends the class constructing the real numbers. It's kind of amazing to read the introduction to that. You can find it.

Insight: The Logical Underlayer vs. Working Mathematical Intuition

The lecturer highlights a fundamental pedagogical dichotomy in abstract mathematics: the distinction between the **foundational underlayer** and the **working mathematical intuition**.
Whether constructing $\mathbb{R}$ via Dedekind cuts or Cauchy sequences (Fact 0.36), or constructing $\mathrm{Frac}(R)$ via quotient sets of ordered pairs $S/\equiv$ (Definitions 0.9 and 3.14), the foundational machinery serves solely to establish rigorous existence and soundness. Once established, mathematicians operate with the intuitive rules (e.g., standard fraction manipulations $\frac{r}{s}$) without thinking of the underlying equivalence classes.

Beyond the Lecture: Dedekind's Book on the Real Numbers

The book the lecturer refers to is Richard Dedekind's **Stetigkeit und irrationale Zahlen** (*"Continuity and Irrational Numbers"*, 1872). In its preface Dedekind recalls teaching the differential calculus at the Polytechnikum in Zurich (today's ETH) in 1858 and feeling the lack of a truly rigorous foundation of arithmetic; the book introduces what are now called Dedekind cuts.

Polynomial Rings

Definition and Structure of $R[x]$

Spoken Transcript [00:04:45 - 00:06:30]

Okay, so we go to polynomials.

I explained yesterday (Recall: Definition 3.32) that the way to think about polynomials is just the way you always think about polynomials. Elements of this ring, I advise you to think about this way: *[writes* $R[x] = \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_i \in R\}$*]*

But in all things, you get your own choice how to think about them. So, they're polynomials that have this formal variable x that's denoted here, and you consider polynomials in x with coefficients in R.

And this polynomial ring, that's the set, and it has addition; and the addition is you add it like polynomials. So, whatever power, you keep the power the same, and you add the coefficients of the power. You know how to add polynomials, so you just do it here.

This is more complicated, multiplying polynomials, but you already know how to multiply polynomials, and you just do it here. You can define it formally, which I did last time (Recall: Definition 3.36), but you just do it.

And it has the zero polynomial and the one polynomials *[writes $(R[x], +, \cdot, 0, 1)$ ring: poly in x with coef in R]* — ring of polynomials in the variable x with coefficients in R.

So, we can do this for any ring — I said that. This is important to keep track of: sometimes we make assumptions on rings, sometimes not. Here we needed the assumption (i.e., R an integral domain), that's kind of crucial; here we don't need the assumption.

The Polynomial Ring over a Commutative Ring

Recap: The polynomial ring was introduced in the previous lecture, informally (Definition 3.32) and as sequences of coefficients (Definition 3.34, Definition 3.36). The lecturer recalls it.

Definition 3.37 (*Polynomial Ring*): Let R be an arbitrary commutative ring with identity 1. The *"polynomial ring"* in the variable x with coefficients in R, denoted $R[x]$, is the set of formal sums:

$$R[x] := \left\{ a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \mid n \in \mathbb{N},\ a_i \in R \right\}. \tag{3.55}$$

Equipped with the standard polynomial addition and Cauchy multiplication:

$$\left(\sum_{i=0}^{n} a_i x^i \right) + \left(\sum_{i=0}^{n} b_i x^i \right) := \sum_{i=0}^{n} (a_i + b_i) x^i, \tag{3.56}$$

$$\left(\sum_{i=0}^{n} a_i x^i \right) \cdot \left(\sum_{j=0}^{m} b_j x^j \right) := \sum_{k=0}^{n+m} \left(\sum_{i+j=k} a_i b_j \right) x^k, \tag{3.57}$$

the structure $(R[x], +, \cdot, 0, 1)$ forms a commutative ring with identity $1 \in R \subset R[x]$ and zero element $0 \in R \subset R[x]$.

Generality of the Polynomial Construction: Unlike the construction of the field of fractions $\mathrm{Frac}(R)$, which strictly requires R to be an integral domain, the polynomial ring $R[x]$ is well-defined for **any** commutative ring with identity R. Zero divisors in R do not obstruct the definition of polynomial addition or multiplication.

The Universal Property of Polynomial Rings

Spoken Transcript [00:06:30 - 00:09:00]

Okay, so properties of this polynomial ring. The first is — this is somehow the most basic property of polynomial rings:

Suppose R and S are rings *[writes Suppose R, S are rings]* — which I say commutative rings with unit, just for us —, and I have an element $y \in S$ *[writes $y \in S$]*. Okay?

Then, given any homomorphism *[writes Then given $\phi\colon R \to S$ homomorphism of rings]* — so you have some homomorphism from R to S, a homomorphism of rings —, there exists a unique extension *[writes $\exists!$ extension]* — I will say what this means in a moment — from the polynomial ring to S *[writes $\hat{\phi}\colon R[x] \to S$]* which satisfies these properties *[writes which satisfies 2 properties]*, two properties:

Property **(1)** is that if I take a constant polynomial *[writes $\bullet\ \hat{\phi}(r) = \phi(r)$ (const)]*... So, this is just the constant polynomial, which is to say that R sits inside $R[x]$ as the constants *[writes $R \subset R[x]$]*, okay? If I take a constant polynomial, this should just be the value that I've already specified here, okay?

And the second property is: I have to say what happens to x, because that's not specified here. And I just say x goes to y *[writes $\bullet\ \hat{\phi}(x) = y$]*.

Okay, so that's the claim. If I start with these two rings — there's the ring where my maps are starting, and the ring where the target map, where my maps are ending, the homomorphisms are ending — and I pick an element y, I can take any homomorphism from R to S and I can extend it to the polynomial ring by just setting x to y.

The Universal Property of Polynomial Rings

Theorem 3.38 (*Universal Mapping Property of $R[x]$*): Let R and S be commutative rings with identity, let $\phi\colon R \to S$ be a ring homomorphism, and let $y \in S$. Then there exists a unique ring homomorphism $\hat{\phi}\colon R[x] \to S$ extending ϕ such that:

(1) Extension of Constants: For every constant polynomial $r \in R \subset R[x]$:

$$\hat{\phi}(r) = \phi(r). \tag{3.58}$$

(2) Evaluation at y: The indeterminate x is mapped to y:

$$\hat{\phi}(x) = y. \tag{3.59}$$

Evaluation Homomorphism and Factorization: Going beyond the lecture: the map $\hat{\phi}$ is often called the *"evaluation homomorphism"* associated with ϕ and y. In the special case where $S = R$ and $\phi = \mathrm{id}_R$, $\hat{\phi}$ is simply the evaluation map $\mathrm{ev}_y\colon R[x] \to R$ given by $f(x) \mapsto f(y)$.

The universal property states that $R[x]$ is the *"freest"* ring containing R and an additional generator x: mapping $R[x]$ to any ring S requires only choosing a base ring homomorphism $\phi\colon R \to S$ and choosing where to send x. It is the same kind of statement as the universal property of $\mathbb{Q}$ from the previous lecture (Theorem 3.7): a homomorphism defined on a smaller ring extends uniquely to the bigger one (for $R[x]$, once the image of x has been chosen).

Proof of Theorem 3.38

Spoken Transcript [00:09:00 - 00:09:48]

Okay, and the proof of this is just by the formula. So, the proof: *[writes Proof]* I have to define $\hat{\phi}$ on any polynomial. So, a polynomial looks like this: *[writes Def* $\hat{\phi}(a_0 + a_1 x + \cdots + a_n x^n) =$*]*

And I have to define it. Formally, I would have to first define it, then check that it's a homomorphism — so I can say it extends to a homomorphism: I have to define it, check it's a homomorphism, then show it satisfies these two properties. That's the task at hand, so to speak.

So, how am I going to define this?

Student Answer

Evaluate the coefficients, and then replace each x with y.

Spoken Transcript [00:09:53 - 00:10:45]

Yeah. So, this is the definition: I have to somehow take this data and get to S. Any element here — so these a_i's, that's part of the definition of polynomial, they're in R *[writes* $a_i \in R$*]*. And I only have on the board one way to ship elements from R to S: it's here. So, I can't make up another one, that's, you know.

So, all these coefficients I just move by the only way I know how to get to S. So, that's the constant coefficient, then I move this coefficient to S, because this result has to be in S. And then the question is: how do I move x to S? Well, I know one element of S, that's y. And then I do this: $\phi(a_2)y^2$ and so on, to y^n. *[writes* $\stackrel{def}{=} \phi(a_0) + \phi(a_1) \cdot y + \phi(a_2)y^2 + \cdots + \phi(a_n)y^n$*]*

Spoken Transcript [00:10:45 - 00:12:20]

So, I just define that. The first thing is you have to accept this definition makes sense. And, you know, when you define a map, usually if I define it so that if you give me anything, I tell you what the answer is, that's the definition of the map — I mean, as long as it takes any possibility here, which it does.

In the case when you have equivalence classes, the matter is a little subtle, because, you know, it could be that in equivalence classes things have different names, and you have to be careful. But that's not what's going on here. This is not equivalence classes: the field of fractions was equivalence classes, but the polynomial ring, that is not equivalence classes. I tell you specifically who is a polynomial, and everyone has just one name.

So, that's the definition. Okay. But is that the end? No, I have to check that it satisfies this property, on constant polynomials. But for constant polynomials, well, it does satisfy that property: that would just mean there's none of this, so just by definition it satisfies that property.

So, I have to check constants *[writes "• Check consts"]*, and I have to check x *[writes "• Check x"]*. What is $\hat{\phi}(x)$ by this definition? Well, that would be just one x here and everything else is zero or one. Formally, I'd have to put like $\phi(0)$ and $\phi(1)$. I mean, yeah, I'm not going to write that, but you could write that. But $\phi(0) = 0$ and $\phi(1) = 1$, so you don't have to write it. So, this is just y *[writes $\hat{\phi}(x) = y$]*.

Okay, so I have now checked these things, but the one thing I haven't checked is that it's a homomorphism. Let's think about how you'd check that.

Definition of $\hat{\phi}$ and Verification of Conditions

1. Definition of the Candidate Map: Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \in R[x]$ with $a_i \in R$. We define the map $\hat{\phi}\colon R[x] \to S$ by:

$$\hat{\phi}(a_0 + a_1x + \cdots + a_nx^n) \stackrel{\text{def}}{=} \phi(a_0) + \phi(a_1) \cdot y + \phi(a_2)y^2 + \cdots + \phi(a_n)y^n. \tag{3.60}$$

Since every polynomial in $R[x]$ has a unique representation as a formal sum of powers of x (unlike equivalence classes in $\mathrm{Frac}(R)$, there are no ambiguous representatives), $\hat{\phi}$ is unconditionally well-defined.

2. Verification of Properties (1) and (2):

- **Constant Polynomials:** If $f(x) = r \in R$, then $a_0 = r$ and $a_i = 0$ for all $i \geq 1$. Hence:
$$\hat{\phi}(r) = \phi(r).$$
- **Image of x:** For $f(x) = x$, we have $a_0 = 0$, $a_1 = 1$, and $a_i = 0$ for all $i \geq 2$. Since $\phi(0) = 0$ and $\phi(1) = 1$:
$$\hat{\phi}(x) = \phi(0) + \phi(1)y = 0 + 1 \cdot y = y.$$

Spoken Transcript [00:12:20 - 00:14:25]

To check it's a homomorphism, you have to check some things about zero going to zero, and one goes to one, which it certainly does. So, the main thing to check is that $\hat{\phi}$ of two polynomials — so f and g now are my polynomials: *[writes $\hat{\phi}(f+g) = \hat{\phi}(f) + \hat{\phi}(g)$, $f, g \in R[x]$]*.

Okay? For this, you need to know... sorry, these are f, right? So, you need to know how to add the polynomials here, and then you need to know how to add elements of S. So, this addition is addition of polynomials *[writes addition of polys]*, and this addition is addition in S *[writes addition in S]*, right?

And the way this thing works: what are we going to do here? Let's say that we take this $\hat{\phi}$; if f is given by a's, it's $a_0 + a_1x$, let's say — let's say that's the f, just a linear one —, plus $b_0 + b_1x$ *[writes $\hat{\phi}(\underbrace{a_0 + a_1x}_{f} + \underbrace{b_0 + b_1x}_{g})$]*.

If we add them first in polynomials, then this is $(a_0 + b_0) + (a_1 + b_1)x$ *[writes $\hat{\phi}((a_0 + b_0) + (a_1 + b_1)x)$ below]*.

And then, if we go over by the definition, then this is $\phi(a_0 + b_0) + \phi(a_1 + b_1)y$ *[writes $\Longrightarrow \phi(a_0 + b_0) + \phi(a_1 + b_1)y$]*, right?

Spoken Transcript [00:14:25 - 00:16:30]

And this, I can now use the homomorphism property of R and write as $\phi(a_0) + \phi(a_1)y + \phi(b_0) + \phi(b_1)y$ *[writes* $= \phi(a_0) + \phi(a_1)y + \phi(b_0) + \phi(b_1)y$*]*. I don't know if this is really worth doing all out, but...

And that, at the end, is $\hat{\phi}(a_0 + a_1x) + \hat{\phi}(b_0 + b_1x)$ *[writes* $= \hat{\phi}(a_0 + a_1x) + \hat{\phi}(b_0 + b_1x)$*]*.

Okay. So, what did I do here? I just expanded out by the definition, and at a certain point I have to use the homomorphism property of R. Nothing happened here, actually; it was just a lot of waste of chalk. But I just follow... at a certain point I have to use the homomorphism property of R. There. And I did this with linear polynomials, but it's just the same with more coefficients.

And then you have to check something that's more complicated with the indices: that if I take $f \cdot g$, this is equal to $\hat{\phi}(f) \cdot \hat{\phi}(g)$ *[writes* $\hat{\phi}(fg) = \hat{\phi}(f) \cdot \hat{\phi}(g)$*]*. And, again, this follows from the same expansion.

So, I tried to do some checking for you, but in the end I don't know how successful that was. The fact of the matter is: I give you the definition, and then you have to check the compatibility; and when you check the compatibility, all you use is the homomorphism property of this.

That's one way, one abstract property of polynomial rings: that if you can map R to something, then you can pick any place, any landing point for x, and it gives you a homomorphism. Sometimes this is called the universal property of polynomial rings.

Verification of Additivity

3. Verification of Additivity ($\hat{\phi}(f+g) = \hat{\phi}(f) + \hat{\phi}(g)$): Let $f, g \in R[x]$. We verify additivity explicitly for linear polynomials $f(x) = a_0 + a_1x$ and $g(x) = b_0 + b_1x$:

$$\begin{aligned}
\hat{\phi}(f+g) &= \hat{\phi}\big((a_0 + a_1x) + (b_0 + b_1x)\big) \\
&= \hat{\phi}\big((a_0 + b_0) + (a_1 + b_1)x\big) && \text{(addition in } R[x]) \\
&= \phi(a_0 + b_0) + \phi(a_1 + b_1)y && \text{(by definition of } \hat{\phi}) \\
&= \big(\phi(a_0) + \phi(b_0)\big) + \big(\phi(a_1) + \phi(b_1)\big)y && (\phi\colon R \to S \text{ is a homomorphism}) \\
&= \big(\phi(a_0) + \phi(a_1)y\big) + \big(\phi(b_0) + \phi(b_1)y\big) && \text{(reordering in ring } S) \\
&= \hat{\phi}(a_0 + a_1x) + \hat{\phi}(b_0 + b_1x) \\
&= \hat{\phi}(f) + \hat{\phi}(g). && (3.61)
\end{aligned}$$

The same argument works for polynomials of arbitrary degree: writing $f = \sum_{i=0}^{n} a_i x^i$ and $g = \sum_{i=0}^{n} b_i x^i$ (padding the shorter one with zero coefficients),

$$\hat{\phi}(f+g) = \sum_{i=0}^{n} \phi(a_i + b_i)\, y^i = \sum_{i=0}^{n} \phi(a_i)\, y^i + \sum_{i=0}^{n} \phi(b_i)\, y^i = \hat{\phi}(f) + \hat{\phi}(g).$$

Verification of Multiplicativity and Uniqueness

4. Verification of Multiplicativity ($\hat{\phi}(fg) = \hat{\phi}(f) \cdot \hat{\phi}(g)$): Let $f(x) = \sum_{i=0}^{n} a_i x^i$ and

$g(x) = \sum_{j=0}^{m} b_j x^j$. Using the Cauchy product in $R[x]$:

$$\begin{aligned}
\hat{\phi}(f \cdot g) &= \hat{\phi}\left(\sum_{k=0}^{n+m}\left(\sum_{i+j=k} a_i b_j\right) x^k\right) \\
&= \sum_{k=0}^{n+m} \phi\left(\sum_{i+j=k} a_i b_j\right) y^k && \text{(by definition of } \hat{\phi}\text{)} \\
&= \sum_{k=0}^{n+m}\left(\sum_{i+j=k} \phi(a_i)\phi(b_j)\right) y^k && (\phi \text{ is a ring homomorphism}) \\
&= \left(\sum_{i=0}^{n} \phi(a_i) y^i\right) \cdot \left(\sum_{j=0}^{m} \phi(b_j) y^j\right) && \text{(distributivity and Cauchy product in } S\text{)} \\
&= \hat{\phi}(f) \cdot \hat{\phi}(g). && (3.62)
\end{aligned}$$

Furthermore, $\hat{\phi}(1) = \phi(1) = 1_S$. Thus, $\hat{\phi}$ is a ring homomorphism.

5. Uniqueness: Any ring homomorphism $\psi\colon R[x] \to S$ satisfying conditions **(1)** and **(2)** of Theorem 3.38 must preserve sums and products:

$$\psi\left(\sum_{i=0}^{n} a_i x^i\right) = \sum_{i=0}^{n} \psi(a_i)\psi(x)^i = \sum_{i=0}^{n} \phi(a_i) y^i = \hat{\phi}\left(\sum_{i=0}^{n} a_i x^i\right).$$

Hence $\psi = \hat{\phi}$, proving uniqueness.

Q.E.D.

Units in the Polynomial Ring over an Integral Domain

Spoken Transcript [00:16:30 - 00:18:16]

[The lecturer erases the lower board.]

So, let's think about... We know that if you have rings, you can think about their units. So, we can ask: What are the units in a polynomial ring? That's a very reasonable question. And that question actually turns out to have some wrinkles.

Let's start with: Suppose R is an integral domain *[writes Suppose R is an integral domain]*. Then the question is: What are the units in the polynomial ring? *[writes What are the units in R[x]?]* Which is to say: which elements of this polynomial ring have a multiplicative inverse?

So, what do you think the answer to this is? Well, let's say if I have some f in the polynomial ring, it's $a_0 + a_1 x + \cdots + a_n x^n$ *[writes $f = a_0 + a_1 x + \cdots + a_n x^n$]*. And if it's a unit, that means that there exists a multiplicative inverse — there exists $g = b_0 + b_1 x + \cdots +$ some different number, $b_m x^m$ *[writes $\exists\, g = b_0 + b_1 x + \cdots + b_m x^m$]* —, and I multiply them so that I get 1, the constant polynomial *[writes $fg = 1$]*.

That would mean that if f is a unit, then there exists a g such that when I multiply them I get the unit, the constant polynomial 1.

How could this happen?

Yeah?

Student Question

If there is an element in R, you can plug it in and then it goes to zero?

Spoken Transcript [00:18:21 - 00:18:25]

There exists an element of... Sorry, I didn't... You have to speak louder.

Student Answer

When there exists an inverse of...

Spoken Transcript [00:18:27 - 00:18:31]

Well, this x stays. You can't replace x. We're talking about this polynomial ring.

Student Answer

These are the units...

Spoken Transcript [00:18:33 - 00:18:56]

Yeah, these are... I mean, but actually the discussion is kind of more simple. Just if I multiply this f with g, I get another polynomial. And for it to be 1, all of its higher terms have to somehow cancel, right? So, the question is: can you arrange it so all the higher terms cancel? That's the real question.

Yeah?

Student Answer

I suppose if f is a constant term?

Spoken Transcript [00:18:59 - 00:20:09]

Yeah, so, one way to do it is: the answer is that a unit... if I take the polynomial to be a constant, f equals a constant polynomial *[writes $f = a$]*, and this happens to be a unit for R *[completes the line to $f = a \in \mathrm{Unit}(R)$]*.

Right? So, that's a good... So, he's saying that just forget it: we could have just the constants, take that to be a constant polynomial, and if I further choose that to be a unit, then I know it has a friend, because it has a unit in R.

This part, I hope, is good: that the units of the original ring sit inside the units of this polynomial ring, and they sit inside as the constant polynomials *[writes $\mathrm{Unit}(R) \subset \mathrm{Unit}(R[x])$ with "Const Polys" below]*.

Okay. I hope that's clear.

So, the question is: could there be any more units here?

And you should, in the way I presented this, be sensitive to the fact that I didn't take any ring, I took an integral domain. So, you might want to think: why did I do that?

Yeah?

Constant Units in the Polynomial Ring

Let R be a commutative ring with identity, and consider $R[x]$:

- A polynomial $f(x) \in R[x]$ is a *"unit"* (Definition 2.66) if there exists $g(x) \in R[x]$ such that:
$$f(x) \cdot g(x) = 1. \tag{3.63}$$
- Any unit $u \in \mathrm{Unit}(R)$ has an inverse $u^{-1} \in R$. Considered as constant polynomials in $R[x]$:
$$u \cdot u^{-1} = 1 \implies u \in \mathrm{Unit}(R[x]).$$
Thus, the units of R always embed into the units of $R[x]$:
$$\mathrm{Unit}(R) \subset \mathrm{Unit}(R[x]) \quad \text{(as constant polynomials)}. \tag{3.64}$$

Student Answer

There are no more units?

Spoken Transcript [00:20:12 - 00:20:37]

Yes, so, the answer here is that there's no more. It's actually equal. So, the answer is that there are no more units; the units in R equal the units in $R[x]$ *[writes "Ans. $\mathrm{Unit}(R) = \mathrm{Unit}(R[x])$"]*.

And I just remind you, because otherwise this is going to be wrong: R is an integral domain *[writes R is an int domain]*.

And how are you going to use this condition?

Student Answer

So, we have the general form of f and g, and if we multiply them, then the term with the highest power...

Spoken Transcript [00:20:45 - 00:22:05]

Yeah, this is the right answer. When you multiply polynomials — and as I said, you must have had some years of experience multiplying polynomials, and you should not forget these years of experience —, if you multiply polynomials, in the middle is a complicated mess of things: you have to add up all the things, it's some complicated mess. But there are two terms which are very simple.

If you want the constant polynomial of the product, that's the product of the constants, right? *[writes* $\text{Const}(fg) = a_0 b_0$ *]*

And if you want the highest term in the product, that's the product of the highest terms. *[writes "highest term ="]*

So, with polynomials, if you're multiplying them, before you do all the multiplication, you should think: is what I'm thinking about resolved by understanding what happens to the constants or the highest term? Because those are the easiest ones, because there's not a big sum there.

As I said, you know polynomials: the constant term of fg is equal to just $a_0 b_0$.

And the highest term, meaning the highest-degree term in x...

Leading and Constant Terms in Polynomial Multiplication

Let R be a commutative ring with identity, and let $f(x), g(x) \in R[x]$ be non-zero polynomials represented in canonical form:

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n, \quad a_n \neq 0 \ (\deg(f) = n), \tag{3.65}$$

$$g(x) = b_0 + b_1 x + \cdots + b_m x^m, \quad b_m \neq 0 \ (\deg(g) = m). \tag{3.66}$$

When multiplying $f(x)$ and $g(x)$, the extremal coefficients take an uncomplicated form:

Constant Term: $$\text{Const}(fg) = a_0 b_0, \tag{3.67}$$

Highest Degree Term: $$\text{highest}(fg) = (a_n x^n) \cdot (b_m x^m) = a_n b_m x^{n+m}. \tag{3.68}$$

Spoken Transcript [00:22:05 - 00:23:22]

...the highest term comes from the highest term of f times the highest term of g.

So, if I write it like this, when I take this highest term... Usually when they write polynomials — but you have to be careful about the conventions and the convention of whatever author you're reading —, the typical thing is when you write a polynomial like this, you could say a_n is not zero *[writes* $a_n \neq 0$*]*, because if it were zero, then you wouldn't have written it. So, it's kind of standard that when you write a polynomial like this, the highest term you say is not zero, because as I said, if it were zero, you didn't have to write it.

And this one *[points at* b_m*]* is not zero *[writes* $b_m \neq 0$*]*, because otherwise I wouldn't have to write it.

So, when I take these polynomials, then this is the highest term, and that's the highest term. So, this would be the highest term here: $a_n x^n$. That would be the highest term there: $b_m x^m$. And the highest term would be $a_n \times b_m \times x^{n+m}$ *[writes* $(a_n x^n) \cdot (b_m x^m)$*, with "highest of* f*" and "highest of* g*" below, and* $a_n \cdot b_m \, x^{n+m}$*]*: $n + m$, that would be the highest term.

And if we're in an integral domain, this product, if they start out to be not zero, they end up to be not zero. So, this would still be non-zero if R is an integral domain *[writes* $\neq 0$ *if* R *int domain]*.

Units of the Polynomial Ring over an Integral Domain

Theorem 3.39 (*Units in $R[x]$ for an Integral Domain R*): Let R be an integral domain. Then the units of the polynomial ring $R[x]$ are precisely the units of the base ring R, regarded as constant polynomials:

$$\mathrm{Unit}(R[x]) = \mathrm{Unit}(R). \tag{3.69}$$

Proof

The inclusion $\mathrm{Unit}(R) \subset \mathrm{Unit}(R[x])$ holds trivially because any unit $u \in \mathrm{Unit}(R)$ has an inverse $u^{-1} \in R$, which satisfies $u \cdot u^{-1} = 1$ in $R[x]$ as constant polynomials.

Conversely, suppose $f(x) \in \mathrm{Unit}(R[x])$. Then there exists an inverse polynomial $g(x) \in R[x]$ such that:

$$f(x) \cdot g(x) = 1. \tag{3.70}$$

Both f and g are non-zero, since $fg = 1 \neq 0$ (an integral domain has $1 \neq 0$). Write $f(x) = \sum_{i=0}^{n} a_i x^i$ with $a_n \neq 0$ and $g(x) = \sum_{j=0}^{m} b_j x^j$ with $b_m \neq 0$. The leading term of the product is given by:

$$\mathrm{highest}(f \cdot g) = a_n b_m x^{n+m}.$$

Because R is an **integral domain**, the absence of non-trivial zero divisors ensures:

$$a_n \neq 0 \text{ and } b_m \neq 0 \implies a_n b_m \neq 0 \quad \text{in } R.$$

On the other hand, the constant polynomial 1 has degree 0. Thus, comparing degrees on both sides of (3.70) yields:

$$n + m = \deg(fg) = \deg(1) = 0.$$

Since $n, m \in \mathbb{N} = \{0, 1, 2, \dots\}$, this forces:

$$n = 0 \quad \text{and} \quad m = 0.$$

Therefore, $f(x) = a_0$ is a constant polynomial. The constant term relation gives:

$$\mathrm{Const}(fg) = a_0 b_0 = 1 \implies a_0 \in \mathrm{Unit}(R).$$

This proves that every unit in $R[x]$ is a constant polynomial coming from $\mathrm{Unit}(R)$.

Q.E.D.

Degree Formula in Integral Domains: The proof relies on the fundamental degree formula:

$$\deg(f \cdot g) = \deg(f) + \deg(g),$$

which holds for all non-zero polynomials $f, g \in R[x]$ if and only if R has no zero divisors. If R possessed non-trivial zero divisors, the leading coefficient $a_n b_m$ could vanish, causing $\deg(f \cdot g) < \deg(f) + \deg(g)$.

Spoken Transcript [00:23:22 - 00:25:19]

This means that if you ever take a polynomial which has a positive-degree part — meaning it's not the constant polynomial, meaning one of these coefficients with x's is not zero —, you will never be able to find an inverse, because when you multiply it by somebody else, you'll always have a term in $n + m$; and if one of them is ≥ 1 (i.e., $n \geq 1$), this will never be a constant term.

Does that make sense? In other words, if your ring is an integral domain, when you multiply two polynomials with x's, the x's will never go away, because you can always look at the highest one, and they will persist.

So, the answer is: if it's an integral domain, you'll have no more units.

Of course, this begs the question: What happens when it's not? *[writes "What happen when R is not an int domain."]*

Okay, so I leave this to you to think about. That's kind of a fun thing; maybe we'll discuss it later.

You should either prove that this argument somehow can be fixed, or you should find a ring that's not an integral domain where there are more units. Those are the two possible solutions.

But my memory from school — which was really a long time ago — was there were a lot of polynomials in school, weren't there? Huh?

So, you know, you should have some experience with adding, subtracting, multiplying polynomials; and as I said, all that experience is directly applicable here.

The Non-Integral Domain Question

Question posed by the lecturer: What happens when R is not an integral domain?

Beyond the Lecture: Non-Constant Units in $R[x]$

When R is not an integral domain, $R[x]$ can have non-constant units. In general, a polynomial $f(x) = a_0 + a_1x + \cdots + a_nx^n \in R[x]$ is a unit if and only if $a_0 \in \mathrm{Unit}(R)$ and all higher coefficients $a_1, \ldots, a_n \in R$ are *"nilpotent"*, that is, $a_i^k = 0$ for some $k \geq 1$. For a concrete counterexample, consider the ring $R = \mathbb{Z}/4\mathbb{Z}$ (which is not an integral domain since $2 \cdot 2 = 0$). In $(\mathbb{Z}/4\mathbb{Z})[x]$, consider the linear polynomial $f(x) = 1 + 2x$. We compute:

$$(1 + 2x)^2 = 1 + 4x + 4x^2 \equiv 1 \pmod 4.$$

Thus $f(x) = 1 + 2x$ is a unit of degree 1 with $f(x)^{-1} = 1 + 2x$.

Formal Power Series Rings

Definition and Ring Operations in $R[[x]]$

Spoken Transcript [00:25:19 - 00:27:02]

[The lecturer moves to the lower left chalkboard and writes "Power Series".]

Okay, so the next thing is power series. And this is maybe... You may have less experience manipulating power series, I don't know. But it's a beautiful thing.

Here, I have to define for you the power series ring *[writes $R[[x]] \ni$]*. And the reason there are two brackets here, somehow, is because there are a lot more x's, you know? *[laughter]*

An element of the power series ring... And, of course, we now have, as with polynomial rings, two ways to think about it: the formal way, or the way people actually think about it.

Now, the formal way: we had before, a polynomial was given by a sequence of elements of R *[writes $(a_0, a_1, a_2, \dots, a_n]$*, and that sequence had a highest term and then had an infinite number of zeros afterwards *[continues with $0, 0, 0]$*. This was the definition of polynomial, if you remember from yesterday (Recall: Definition 3.34). You could add these, subtract these, multiply them, and it would just be the same thing as the x's, but this is the abstract discussion.

From this point of view, power series are easy: you just don't put this condition of zeros. It's just infinite elements. *[erases the zeros and continues the sequence with further coefficients]*

So, from this abstract point of view, elements of the power series ring are infinite lists — infinite vectors — of elements of R *[writes Elements of $R[[x]]$ are infinite vectors of elements of R]*. Before, we had lists that eventually become all zero, and now we don't; we just lift that condition. You can do anything you want.

Formal Definition of Power Series as Sequences

Definition 3.40 (*Formal Power Series Ring*): Let R be a commutative ring with identity 1.

(1) Sequence Perspective: Formally, a *"formal power series"* with coefficients in R is an infinite sequence of elements of R:

$$(a_0, a_1, a_2, \dots, a_n, \dots) \in R^{\mathbb{N}}. \tag{3.71}$$

In contrast to polynomials in $R[x]$ (sequences $(a_0, a_1, \dots, a_n, 0, 0, 0, \dots)$ with $a_k = 0$ for all sufficiently large k), no vanishing condition is imposed on the terms of a formal power series.

(2) Notation: The set of all formal power series in the indeterminate x with coefficients in R is denoted by:

$$R[[x]]. \tag{3.72}$$

Spoken Transcript [00:27:02 - 00:29:00]

Now, the way to think about it as power series: all this means is that before, for polynomials, we had the condition that eventually you stop — that's a polynomial *[points to the sequence]*. And for power series, you just don't stop: you can go forever *[writes $a_0 + a_1 x + \cdots + a_n x^n + \dots$ after $R[[x]] \ni$]*.

It's hard to go forever, so typically you would write the following thing: the elements of the power series ring are the sum from $i = 0$ to infinity of $a_i x^i$ *[writes $R[[x]] = \sum_{i=0}^{\infty} a_i x^i, \ a_i \in R$]*, where the a_i's are in R.

You can see this is exactly the same data as that. The x, as I said, is just the variable; it's not influencing the a's. Like here, it just makes clear that x is just a placeholder, right? And that's all it's doing here also.

Okay, so that's power series. You must have had whole classes on power series —

which is good, because I'm not going to teach you about them this way.

So, do you know how to add power series, multiply power series?

The zero is just the zero power series, that's where every coefficient forever is zero. The one power series is just a one and then everybody else is zero *[writes $(R[[x]],+,\cdot,0,1)$ and below it $(0,0,0,\dots)$, $(1,0,0,0,\dots)$]*.

Do you know how to add power series? I say that you do know how to add power series: in every analysis class, I think you add power series.

Here's the rule, which will surprise no one: *[writes $\sum_{i=0}^{\infty} a_i x^i + \sum_{i=0}^{\infty} b_i x^i \overset{def}{=} \sum_{i=0}^{\infty}(a_i + b_i)x^i$]*.

It's a lot of work to write all of these things, so it's... yeah.

Okay, so that's the exciting rule: it's component-wise addition.

Representation as Formal Sums

Every formal power series $(a_n)_{n\in\mathbb{N}} \in R[[x]]$ is represented as a formal infinite sum:

$$a_0 + a_1 x + a_2 x^2 + \cdots = \sum_{i=0}^{\infty} a_i x^i, \quad a_i \in R. \tag{3.73}$$

Algebraic Meaning of the Infinite Sum: In algebra, the expression $\sum_{i=0}^{\infty} a_i x^i$ is purely a formal symbol representing the sequence of coefficients $(a_0, a_1, a_2, \dots)$. There are no questions of convergence, metrics, or limits involved, and the symbol x serves solely as an algebraic placeholder tracking positions.

Ring Structure and Addition in $R[[x]]$

The formal power series ring is equipped with a ring structure $(R[[x]],+,\cdot,0,1)$:

- **Additive Identity (0):**

$$0_{R[[x]]} := (0,0,0,\dots) = \sum_{i=0}^{\infty} 0 \cdot x^i. \tag{3.74}$$

- **Multiplicative Identity (1):**

$$1_{R[[x]]} := (1,0,0,\dots) = 1 + \sum_{i=1}^{\infty} 0 \cdot x^i. \tag{3.75}$$

- **Addition:** Defined componentwise:

$$\sum_{i=0}^{\infty} a_i x^i + \sum_{i=0}^{\infty} b_i x^i \overset{\text{def}}{=} \sum_{i=0}^{\infty}(a_i + b_i)x^i. \tag{3.76}$$

Spoken Transcript [00:29:00 - 00:30:25]

More complicated is: how do you multiply power series?

Again, that's a definition, by the way: *[writes $\sum_{i=0}^{\infty} a_i x^i \cdot \sum_{i=0}^{\infty} b_i x^i \overset{def}{=}$]*.

Here... I must say that I really dislike indices in mathematics, somehow. Okay.

So, we have to define this. How are we going to define this? As I said, this is how we define it: you define it to be the sum from $i = 0$ to infinity of $c_i x^i$ *[writes* $\sum_{i=0}^{\infty} c_i x^i$*]*, and then I have to define for you the c_i's — I have to define what c_i is.

Let's say c_k is... You can write this as a sum from $i = 0$ to k, and I write $a_{k-i} b_i$ *[writes* $c_k = \sum_{i=0}^{k} a_{k-i} b_i$*]*.

Okay. All right, I did it.

But you know this: this is how you multiply power series.

Cauchy Multiplication in $R[[x]]$

Definition 3.41 (*Cauchy Product of Power Series*): Let $\sum_{i=0}^{\infty} a_i x^i$ and $\sum_{i=0}^{\infty} b_i x^i$ be formal power series in $R[[x]]$. Their product is defined by:

$$\left(\sum_{i=0}^{\infty} a_i x^i \right) \cdot \left(\sum_{i=0}^{\infty} b_i x^i \right) \stackrel{\text{def}}{=} \sum_{k=0}^{\infty} c_k x^k, \tag{3.77}$$

where each coefficient $c_k \in R$ is given by the Cauchy convolution formula:

$$c_k := \sum_{i=0}^{k} a_{k-i} b_i = \sum_{i+j=k} a_i b_j. \tag{3.78}$$

Finiteness of Each Coefficient Calculation: Notice that to determine any individual coefficient c_k of the product, only the finite collection of coefficients $\{a_0, \ldots, a_k\}$ and $\{b_0, \ldots, b_k\}$ is involved. The sum in (3.78) has exactly $k+1$ terms in R, meaning that no infinite sums in the base ring R are required to define the product (unlike the Cauchy product of analytic power series in $\mathbb{C}$, Fact 0.42, no convergence issues arise).

Spoken Transcript [00:30:25 - 00:31:28]

That is a ring. So, is it a ring? Well, then you have to check.

Formally, you have to check all the axioms. The specific axiom that probably you don't see in your head immediately, unless you've thought about it, is... Like, it's kind of obvious this is associative, commutative, everything, because all we're doing is component-wise; really nothing's happening here.

This one may be more interesting if you haven't thought about it: this is somehow a slightly strange rule that tells you how to get the new power series from the old power series. You have to check that this is associative, et cetera.

Commutative is kind of obvious, even though, you know, I've broken the symmetry here; but if you think about it, that's kind of clear that it's commutative. But then you have to check it's associative. And, you know, formally you have to check it, unless somehow you already know that it's true.

And the formal way to check it is you have to multiply three things, and you multiply them in two different ways, and then you go look at the coefficients, and it's an identity, and it will work out.

So, you know, you have to decide how you interact with math; but formally, if you want to say you understand what a power series ring is, you have to check that the

multiplication is associative.

Ring Structure of $R[[x]]$

Equipped with addition (3.76) and multiplication (3.77), the tuple $(R[[x]], +, \cdot, 0, 1)$ forms a commutative ring with unit $1_{R[[x]]}$.

Verification of Ring Axioms:

- **Abelian Group under Addition:** Follows immediately from component-wise addition inheriting associativity, commutativity, and inverses from $(R, +)$.
- **Commutativity of Multiplication:** By substituting $j = k - i$:
$$c_k = \sum_{i=0}^{k} a_{k-i} b_i = \sum_{j=0}^{k} b_{k-j} a_j.$$
- **Associativity of Multiplication:** For three series $f = \sum a_i x^i$, $g = \sum b_j x^j$, $h = \sum d_l x^l$ in $R[[x]]$, the n-th coefficient of both $(f \cdot g) \cdot h$ and $f \cdot (g \cdot h)$ evaluates to:
$$\sum_{i+j+l=n} a_i b_j d_l,$$
which establishes $((f \cdot g) \cdot h)_n = (f \cdot (g \cdot h))_n$ for all $n \in \mathbb{N}$.

Spoken Transcript [00:31:28 - 00:32:38]

[The lecturer pushes the power series board up; the board with the units of $R[x]$ comes down, and he erases it.]

One of the issues about the power series ring is: if you want to communicate to someone else an element of the power series ring, it's a lot harder than the polynomial ring. The polynomial ring, you just write it: it's a finite amount of data, you know, you just write like seven coefficients, that's it.

With the power series ring, a priori, there's like an infinite amount of data there. So, if you want to communicate it, you know, you have to give some rule, or something like that. We'll do some examples eventually, but it's, you know, somehow a more complicated ring.

When you encounter power series in analysis, you know, they could be like the Taylor expansion of trigonometric functions, and they have rules; that's a way to explain it.

Anyway, if you have not encountered these before, you should make yourself comfortable with the idea of polynomials in R, power series in R, how to add, multiply, how to manipulate them.

Units in the Formal Power Series Ring

Spoken Transcript [00:32:38 - 00:35:07]

So, let's try to ask this question: What are the units of this ring? *[writes "What are unit $R[[x]]$?"]*

So, again, that means we start with a particular power series f: $a_0 + a_1x + a_2x^2 + \dots$ *[writes $f = a_0 + a_1x + a_2x^2 + \dots$]*.

And it goes forever, in principle. Of course, it could also all be zero: a polynomial is a power series, just one that doesn't last too long, because zero is fine.

And g: $b_0 + b_1x + b_2x^2 + \dots$ *[writes $g = b_0 + b_1x + b_2x^2 + \dots$]*.

Okay, and for this to be a unit — if f is a unit —, then there exists g such that $f \times g = 1$ *[writes "If f is a unit, $\exists g \implies f \cdot g = 1$"]*.

And to be 1 is dramatic! It means when you multiply this, the rule in principle gives you infinitely many terms. If it's 1, it means almost all of them are zero.

If we multiply them, $f \times g$, we get: the first term is a_0b_0 *[writes $fg = a_0b_0+$]*. The second term is $(a_1b_0 + a_0b_1)x$ *[writes $(a_1b_0 + a_0b_1)x+$]*.

You can tell I've been spending too much time in Europe because I say *"nought"* all the time. *[laughter]*

Here, a_2... That was a complicated joke. *[writes $(a_2b_0 + a_1b_1 + a_0b_2)x^2 + \dots$]*

Okay, so that's how the power series multiplication works, right? How is it possible that this could be 1?

Well, could this be 1? 1, I remind you, is $1 + 0 \times x + 0 \times x^2$, and so on *[writes 1, $0 \cdot x$, $0 \cdot x^2$ under the terms of fg]*. So, the only way it can be 1 is that this is 1 *[points to a_0b_0]*. So, this would imply that $a_0b_0 = 1$ *[writes $\implies a_0b_0 = 1$]*. This means that the constant term of f is a unit *[writes $\implies$ Const(f) is a unit]*. Right?

What about all this other stuff? So, what's your feeling about that?

Setting up the Invertibility Condition in $R[[x]]$

Let $f(x) \in R[[x]]$ be a formal power series:

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots = \sum_{i=0}^{\infty} a_i x^i. \tag{3.79}$$

By definition, $f(x)$ is a *"unit"* in $R[[x]]$ if and only if there exists an inverse series $g(x) = \sum_{j=0}^{\infty} b_j x^j \in R[[x]]$ such that:

$$f(x) \cdot g(x) = 1_{R[[x]]} = 1 + 0 \cdot x + 0 \cdot x^2 + \dots \tag{3.80}$$

The Constant Term Invertibility Condition

Multiplying $f(x)$ and $g(x)$ via Cauchy multiplication:

$$\begin{aligned} f(x) \cdot g(x) &= a_0b_0 + (a_1b_0 + a_0b_1)x + (a_2b_0 + a_1b_1 + a_0b_2)x^2 + \dots \\ &\stackrel{!}{=} 1 + 0 \cdot x + 0 \cdot x^2 + \dots \end{aligned} \tag{3.81}$$

Equating coefficients power by power, the constant term yields:

$$a_0 b_0 = 1 \implies a_0 = \text{Const}(f) \in \text{Unit}(R). \tag{3.82}$$

Thus, a necessary condition for $f(x)$ to be invertible in $R[[x]]$ is that its constant term a_0 is a unit in the base ring R.

Spoken Transcript [00:35:07 - 00:36:51]

Of course, it's also true that the constants — if we take the units of R — sit inside the units of the power series *[writes* $\text{Unit}(R) \subset \text{Unit}(R[[x]])$*]*, by just this unit a going to the constant power series a *[writes* $a \mapsto a$*]*. That's also true, right?

So, the same question as above: Is this an equality?

And now we come to something interesting, because what was the argument before (i.e., the proof of Theorem 3.39)? The argument before was that — at least when R was an integral domain — you go to the two polynomials, take their highest piece, and multiply them together. That was a great argument for polynomials; it absolutely fails for power series, because power series, you know, have no highest piece. So, this is problematic from the point of view of using that argument.

Maybe instead of... Since we don't have so much time, I'll tell you the answer, which I'm always reluctant to do — this is part of my job, actually, but I'm still...

Something that's kind of surprising here is that f is a unit in $R[[x]]$ if and only if the constant term is a unit in R *[writes "*f *is a unit in* $R[[x]] \iff \text{Const}(f) \in$ *Unit in* R*"]*.

All I have to do to check whether it's a unit is whether the constant term is a unit.

So, that's a weird thing, isn't it?

Characterization of Units in $R[[x]]$

As for polynomials, the units of R sit inside as constant series: $\text{Unit}(R) \subset \text{Unit}(R[[x]])$, $a \mapsto a$. Unlike for $R[x]$, there are many more:

Theorem 3.42 (*Units in the Formal Power Series Ring*): Let R be an arbitrary commutative ring with identity 1. A formal power series $f(x) = \sum_{i=0}^{\infty} a_i x^i \in R[[x]]$ is a unit if and only if its constant term $a_0 = \text{Const}(f)$ is a unit in R:

$$f(x) \in \text{Unit}(R[[x]]) \iff a_0 \in \text{Unit}(R). \tag{3.83}$$

Striking Contrast with Polynomial Rings: Recall from Theorem 3.39 that in $R[x]$ (over an integral domain), a polynomial is a unit if and only if it is a constant unit: no positive-degree terms may appear.

In radical contrast, in $R[[x]]$, a power series can have completely arbitrary non-zero coefficients $a_1, a_2, a_3, \ldots$ for all higher powers of x without restriction; as long as the single scalar a_0 is invertible in R, the entire infinite series $f(x)$ possesses an exact multiplicative inverse in $R[[x]]$!

Spoken Transcript [00:36:51 - 00:37:45]

It shouldn't surprise you if you've played with polynomials.

Well, I wanted to give two proofs; we will see. You should always be suspicious when a mathematician says that they will give you two proofs: it usually means both will be sloppy. *[laughter]*

By the way, since people didn't seem to recognize this Euler note (Recall: the old 10-franc note from the previous lecture), I brought some copies, so you can see if you're interested: the Swiss franc Euler note from the previous century *[holds up copies of the former Swiss 10-franc banknote featuring Leonhard Euler]*. When I came to Switzerland, I went to a used coin shop and bought a booklet of them.

Beyond the Lecture: The Euler Banknote

The banknote is the 10-franc note of the sixth series of Swiss banknotes, in circulation from 1979 to 2000. Its front shows Leonhard Euler (born 1707 in Basel); its back shows a water turbine, the solar system and the path of light through lenses, three subjects of his work.

Proof of Theorem 3.42

Spoken Transcript [00:37:45 - 00:39:07]

Okay, so I'll try to prove this. I want to prove this statement at the end.

Suppose f has the form $a_0 + a_1x + a_2x^2 + \dots$, where $a_0 \in R$ is a unit *[writes Suppose $f = a_0 + a_1x + a_2x^2 + \dots$, $a_0 \in R$ is a unit]*. That means that there exists some $y \in R$ such that $a_0y = 1$ in R *[writes $\exists y \in R$, $a_0y = 1$ in R]*, right?

So, the first thing I can do is just multiply f by y *[writes $f \cdot y =$]*.

Is that clear? Whenever I think about this, y is now the constant power series. Are you happy with this?

And who knows what I get here, but I know one thing: it starts with 1, because when I multiply by a constant power series, it just multiplies all these by y, right? So, whatever it is, it starts with 1, and then it's $(a_1y)x$ *[writes $= 1 + (a_1y)x + (a_2y)x^2 + \dots$]* — y is an element of the ring, it has nothing to do with x —, $(a_2y)x^2$, and so on.

Okay, so the first step is: I use the fact it's a unit, I multiply by something in R and make this 1.

Step 1: Normalization to Constant Term 1

1. Forward Implication ($\Rightarrow$): As established in (3.82), if $f(x) \cdot g(x) = 1$, then $\mathrm{Const}(f) \cdot \mathrm{Const}(g) = a_0b_0 = 1$, forcing $a_0 \in \mathrm{Unit}(R)$.

2. Reverse Implication ($\Leftarrow$) — Normalization: Suppose $a_0 \in \mathrm{Unit}(R)$. Then there exists an inverse element $y \in R$ such that:

$$a_0y = 1 \quad \text{in } R. \tag{3.84}$$

Multiplying $f(x)$ by the constant power series $y \in R \subset R[[x]]$:

$$\begin{aligned} f(x) \cdot y &= (a_0 + a_1 x + a_2 x^2 + \dots) \cdot y \\ &= a_0 y + (a_1 y)x + (a_2 y)x^2 + \dots \\ &= 1 + (a_1 y)x + (a_2 y)x^2 + \dots \end{aligned} \tag{3.85}$$

Because $y \in \mathrm{Unit}(R) \subset \mathrm{Unit}(R[[x]])$, $f(x)$ is invertible if and only if the normalized series $f(x) \cdot y$ is invertible. Indeed, if $(f(x) \cdot y)^{-1}$ exists, then $f(x)^{-1} = y \cdot (f(x) \cdot y)^{-1}$.

Spoken Transcript [00:39:07 - 00:40:25]

Okay? Then it's enough to invert this, because inversion means that there's somebody else I multiply to get 1.

So, then it's sufficient to find $g \in R[[x]]$ *[writes Sufficient to find $g \in R[[x]]$]* such that g times this result, $1 + \dots$ *[writes $g \cdot (1 + b_1 x + b_2 x^2 + \dots) = 1$]*. And now I just make these coefficients b's, because I don't want to write $a_1 y$, and I don't know anything about them anyway, so I might as well make it b.

So, it's sufficient to find a g such that whenever I have something of this form — which is what I have there —, this is equal to 1.

Meaning I only have to invert such things. All I've done is use the fact it's a unit to change the first coefficient to 1. And then it's sufficient to invert that; because once I invert that, then I can just multiply it by y, and then, you know...

So, I have to invert this: $(1 + b_1 x + b_2 x^2 + \dots)^{-1}$ *[writes $(1 + b_1 x + b_2 x^2 + \dots)^{-1}$]*.

Okay, well, you never finish, so maybe like this: I have to invert this.

Step 2: Reduction to Inverting $1 + \gamma$

Setting $b_k := a_k y \in R$ for all $k \geq 1$, the normalized series (3.85) reads $1 + b_1 x + b_2 x^2 + \dots$. It is therefore sufficient to find $g \in R[[x]]$ such that:

$$g \cdot (1 + b_1 x + b_2 x^2 + \dots) = 1, \tag{3.86}$$

that is, to show that the inverse $(1 + b_1 x + b_2 x^2 + \dots)^{-1}$ exists in $R[[x]]$. Then $f^{-1} = y \cdot g$.

Spoken Transcript [00:40:25 - 00:41:31]

Now, you know how to do this, right? You definitely know how to do this. It's, you know, somehow in your brain; you have to bring it out.

So, you write this whole thing, let's say as γ *[writes $\gamma = b_1 x + b_2 x^2 + \dots$]*: γ is equal to this power series. And this power series has x's in it, okay?

So, you can say: I want to invert this, $1 + \gamma$ *[writes $(1 + \gamma)^{-1}$]*. And if I write this as $1/(1 + \gamma)$ *[writes $= \frac{1}{1+\gamma}$]*, surely you know how to invert this: it's equal to $1 - \gamma + \gamma^2 - \gamma^3$, and so on *[writes $= 1 - \gamma + \gamma^2 - \gamma^3 + \dots$]*.

That's the answer! This actually proves it's invertible. And you have to think: why does it prove...? You know this, this is like some identity from your previous life. I just

write it, and I claim this is the formula for the inverse.

Step 3: The Formal Geometric Series

Define the auxiliary power series $\gamma(x) \in R[[x]]$ consisting of all higher-degree terms:

$$\gamma := b_1 x + b_2 x^2 + b_3 x^3 + \cdots = \sum_{k=1}^{\infty} b_k x^k. \tag{3.87}$$

The expression to invert is simply $1+\gamma$. Inspired by the classical geometric series $\frac{1}{1-z} = \sum_{k=0}^{\infty} z^k$ (Fact 0.42), we substitute $z = -\gamma$ to propose the candidate inverse formula:

$$(1+\gamma)^{-1} \overset{\text{formal}}{=} \frac{1}{1+\gamma} = 1 - \gamma + \gamma^2 - \gamma^3 + \gamma^4 - \cdots = \sum_{k=0}^{\infty} (-1)^k \gamma^k. \tag{3.88}$$

Spoken Transcript [00:41:31 - 00:42:50]

And the reason is that this thing makes sense. It looks like an infinite sum, but this γ has x's in it. γ is divisible by x, so x divides γ *[writes $x \mid \gamma$]*. That means x^2 divides γ^2 *[writes $x^2 \mid \gamma^2$]*, and x^k divides γ^k *[writes $x^k \mid \gamma^k$]*.

So, when I expand this, if I want the coefficient of the power series in any particular x, only finitely many of these terms enter.

The whole point of this argument is that this thing is a well-defined element of the power series ring *[writes "well defined elem of $R[[x]]$"]*.

That is what you have to convince yourself of: just that it makes sense to write this. And I gave you the argument, but that's different than you accepting the argument. But it just says that if you do this, every one of these terms is well-defined. And you're not allowed to add infinitely many things in algebra, but since the powers of x get bigger and bigger, you don't have to do it.

So, maybe we'll talk a little bit more about that next time. *[lecturer claps hands and turns toward desk, students applaud]*

Step 4: Well-Definedness via Divisibility by Powers of x

Why the Infinite Sum of Power Series Is Algebraically Sound: In an arbitrary ring, infinite sums are not defined. However, because γ has no constant term (its lowest degree term is $b_1 x$), x divides γ:

$$x \mid \gamma \implies x^2 \mid \gamma^2 \implies \cdots \implies x^k \mid \gamma^k. \tag{3.89}$$

Consequently, the lowest power of x appearing with non-zero coefficient in γ^k is at least x^k:

$$\gamma^k = (b_1 x + b_2 x^2 + \dots)^k = b_1^k x^k + (\text{terms of degree } \geq k+1).$$

Now consider the coefficient of x^N in the formal sum $\sum_{k=0}^{\infty} (-1)^k \gamma^k$:

- For every $k > N$, every term in γ^k has degree at least $k > N$.
- Therefore, terms with $k > N$ contribute strictly 0 to the coefficient of x^N!

Hence, for any fixed power x^N, only the finitely many terms $k \in \{0, 1, \dots, N\}$ con-

tribute to its coefficient:

$$[x^N]\left(\sum_{k=0}^{\infty}(-1)^k\gamma^k\right) = [x^N]\left(\sum_{k=0}^{N}(-1)^k\gamma^k\right) \in R. \tag{3.90}$$

Because every individual coefficient of the resulting power series is computed as a finite sum of elements in R, the series:

$$g(x) := \sum_{k=0}^{\infty}(-1)^k\gamma(x)^k$$

is a rigorous, well-defined element of $R[[x]]$. Multiplying by $(1+\gamma)$ explicitly cancels all terms by telescoping:

$$(1+\gamma)\cdot\sum_{k=0}^{\infty}(-1)^k\gamma^k = \sum_{k=0}^{\infty}(-1)^k\gamma^k + \sum_{k=0}^{\infty}(-1)^k\gamma^{k+1} = 1.$$

This manipulation of infinite sums is legitimate because it can be checked coefficient by coefficient: for the coefficient of x^N, only the finitely many terms with $k \le N$ contribute, and for these the identity is the finite telescoping sum $(1+\gamma)\sum_{k=0}^{N}(-1)^k\gamma^k = 1+(-1)^N\gamma^{N+1}$, whose last term has no x^N component. Thus, $(1+\gamma)$ is invertible, completing the proof that $f(x)$ is a unit in $R[[x]]$.

Q.E.D.

Summary of Key Concepts

- **Field of Fractions:** For an integral domain R, $\mathrm{Frac}(R)$ is formally the set of equivalence classes of pairs $(r,s) \in R \times (R\setminus\{0\})$ with $(r,s) \equiv (t,w) \iff rw = st$. In practice one computes with fractions $\frac{r}{s}$ by the usual rules; the integral domain property guarantees that the denominators sw stay non-zero, and every $\frac{r}{s} \neq 0$ has the inverse $\frac{s}{r}$.
- **Polynomial Rings:** $R[x]$ exists for every commutative ring R with unit. Its universal property: for every ring homomorphism $\phi\colon R \to S$ and every $y \in S$, there is a unique ring homomorphism $\hat{\phi}\colon R[x] \to S$ with $\hat{\phi}|_R = \phi$ and $\hat{\phi}(x) = y$, namely $\hat{\phi}\left(\sum a_i x^i\right) = \sum \phi(a_i)y^i$.
- **Units of $R[x]$:** If R is an integral domain, then $\mathrm{Unit}(R[x]) = \mathrm{Unit}(R)$, because the leading coefficient of a product is the product of the leading coefficients. Without this assumption there can be more units, e.g., $1+2x$ in $(\mathbb{Z}/4\mathbb{Z})[x]$.
- **Formal Power Series:** $R[[x]]$ consists of all sequences $(a_0, a_1, a_2, \dots)$ of elements of R, written $\sum_{i\ge 0} a_i x^i$, with componentwise addition and the Cauchy product $c_k = \sum_{i=0}^{k} a_{k-i}b_i$; every coefficient of a product is a finite sum.
- **Units of $R[[x]]$:** For every commutative ring R with unit, $f \in R[[x]]$ is a unit if and only if its constant term is a unit in R. After normalizing the constant term to 1, the inverse of $1+\gamma$ is the formal geometric series $\sum_{k\ge 0}(-1)^k\gamma^k$, which is well-defined because $x^k \mid \gamma^k$.

Tuesday 6 October 2026

Polynomial Rings, Laurent Series, Quotient Rings, and the First Isomorphism Theorem

Course Content Overview

This lecture explores fundamental algebraic ring constructions, bridging polynomial extensions, localized series, and residue structures:

- **Polynomial and Power Series Rings:** Multivariable polynomial rings, the binomial theorem in commutative rings, and the Frobenius identity $(x+y)^p = x^p + y^p$ over $\mathbb{Z}/p\mathbb{Z}$.
- **Fields of Fractions and Laurent Series:** The field of rational functions $R(x)$ and formal Laurent series, which over a field K give $K((x)) \cong \mathrm{Frac}(K[[x]])$.
- **Ideals and Quotient Rings:** Review of ideals and kernels, the linear algebra analogy, and the construction of the quotient ring R/I from scratch (highlighting how ideal absorption ensures well-defined coset multiplication).
- **The First Isomorphism Theorem for Rings:** Formulation and complete three-step proof of $R/\ker(f) \cong \mathrm{Im}(f)$.
- **Ideals Generated by Elements and Polynomial Quotients:** Contrasting the quotients $\mathbb{C}[x]/(x^2+1)$ (which possesses zero divisors) and $\mathbb{Q}[x]/(x^2+1) \cong \mathbb{Q}[i]$ (which forms the field of Gaussian rationals).

Review: Polynomial Rings and Power Series

Spoken Transcript [00:00:00 - 00:01:55]

Okay, so, we start, and I wanted to start with some things from last week. *[walks to the board and writes on the blackboard]* If R is a ring, we had defined this polynomial ring in one variable, and also the power series ring. *[writes $R \to R[x], R[[x]]$]* And just some comments: the polynomial ring is a subring of the power series ring. *[writes $R[x] \subset R[[x]]$]* That's important to note, because every polynomial you can view as a finite power series.

So, this is one comment. The second comment is that — and people were asking me after class —: what happens if I have two variables or three variables? *[writes $\to R[x,y]$]* There's two ways to approach this. One is you could just define this in the same way as we define this, *[writes $\to$ defined]* which is: it's the set of finite sums $a_{i,j}x^iy^j$, where these indices go over some finite set, and the coefficients are in R, *[writes $\{\sum_{0\le i,j\in\mathbb{N}} a_{i,j}x^iy^j \mid$]*

$a_{i,j} \in R\}$] and it's just the same definition as before. So, that's one way to do it. I hope that's clear — it's just multiplying with more polynomials.

But there's a more formal way, which is that R is a ring, so then we've already defined this, that's where we add one variable, *[writes $R \Rightarrow R[x]$]* and so this is a ring, so we can also define this: this is a ring, and have one more variable. *[writes $\Rightarrow R[x][y]$]* And you could take this to be the definition of $R[x,y]$. *[writes $= R[x,y]$]* So, it's just: we add the variables one at a time, as this is a polynomial ring over the polynomial ring over R.

So, it's another way. And of course, if you have two different paths, then you have the burden of proving they're both the same, etc., and I leave this burden for you to carry, all right? So, and these are the same, maybe I'll just say that: *[draws a double arrow between the two definitions and writes "SAME"]* same.

Polynomial Rings in One and Several Variables

Let R be a commutative ring with identity. Recall the construction of the *"polynomial ring"* $R[x]$ (Definition 3.32) and the *"formal power series ring"* $R[[x]]$ (Definition 3.40). Every polynomial can be naturally identified with a formal power series having only finitely many non-zero coefficients, yielding the subring inclusion:

$$R[x] \subseteq R[[x]].$$

Definition 4.1 (*Polynomial Ring in Two Variables*): For two indeterminates x and y, the polynomial ring $R[x,y]$ can be defined in two equivalent ways:

(1) Explicit Definition (Finite Linear Combinations):

$$R[x,y] := \left\{ \sum_{i,j\geq 0}^{\text{finite}} a_{i,j} x^i y^j \;\middle|\; a_{i,j} \in R \right\}.$$

(2) Iterative Definition (Polynomials over Polynomials): Viewing $R[x]$ as the coefficient ring, we form the univariate polynomial ring in y:

$$R \longrightarrow R[x] \longrightarrow (R[x])[y] =: R[x,y].$$

Equivalence of Definitions: Going beyond the lecture, which leaves this check to the students: both constructions yield canonically isomorphic rings:

$$(R[x])[y] \cong R[x,y] \cong (R[y])[x].$$

In the iterative perspective, an element is a polynomial $P(y) = \sum_{j=0}^{m} p_j(x) y^j$, where each coefficient $p_j(x) = \sum_{i=0}^{k_j} a_{i,j} x^i \in R[x]$. Expanding via the distributive law expresses $P(y)$ uniquely as the finite double sum $\sum_{i,j} a_{i,j} x^i y^j$. By induction, this extends to n variables:

$$R[x_1, \ldots, x_n] \cong (R[x_1, \ldots, x_{n-1}])[x_n].$$

The Binomial Formula in Commutative Rings

Spoken Transcript [00:01:55 - 00:03:48]

Okay, so, then there's some comments from essentially... Well, something you should know is if I have this polynomial ring in two variables, *[writes $R[x,y]$ on the board]* and I take $x+y$, that's in this polynomial ring of two variables. And I can raise it to the n, *[writes $(x+y)^n =$]* and this is something you know: this is this binomial expansion, or I don't know how it's called, binomial formula... *[writes "binomial expansion"]*

This is equal to the sum — so, this is from $a=0$ to n — *[writes $\sum_{a=0}^{n}$]* of $x^{n-a}y^a\binom{n}{a}$. *[writes $x^{n-a}y^a\binom{n}{a}$]*

Okay? That's the binomial formula. Are you familiar with this binomial formula? So, for example, $(x+y)^5$ *[writes $(x+y)^5 =$]* is equal to $1x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5$. *[writes $x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5$]* I guess I should have done a smaller one.

Okay, so, that's an example. And this holds in every ring, because these elements x and y are here, and we've already discussed: these numbers make sense in any ring, they're just how you add up 1 (Recall: Definition 1.16). Is that clear? So, this formula holds in any ring.

And to prove it, it's just the same as by induction we do. If you wanted to prove this formula in any ring, you just write it out as $(x+y)(x+y)$ and use the distributive law, and you get x^2, and you regroup everything, and you get $xy+xy+y^2$, and that gives you the formula; and then you prove it by induction. *[writes $(x+y)^2=(x+y)(x+y)=x^2+xy+xy+y^2$]*

The Binomial Expansion

Let R be any commutative ring with identity 1_R. For integers $k\in\mathbb{Z}$, the integer scalar $k\cdot 1_R$ is canonically defined by repeated addition or negation of 1_R (Definition 1.14). Consequently, the binomial coefficients $\binom{n}{a}\in\mathbb{N}$ act canonically as coefficients in R.

Theorem 4.2 (*Binomial Formula in Commutative Rings*): In the polynomial ring $R[x,y]$, or for any two commuting elements $x,y\in R$ ($xy=yx$), the following identity holds for all $n\in\mathbb{N}$:

$$(x+y)^n=\sum_{a=0}^{n}x^{n-a}y^a\binom{n}{a}. \tag{4.1}$$

Example 4.3 (*Expansion for $n=5$ and $n=2$*): For $n=5$:

$$(x+y)^5=x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5.$$

For $n=2$, distributivity and commutativity ($yx=xy$) give:

$$(x+y)^2=(x+y)(x+y)=x^2+xy+xy+y^2=x^2+2xy+y^2.$$

Proof by Induction: The general identity follows by induction on n. For $n=0$,

both sides equal $1 = \binom{0}{0}$. For the induction step:

$$\begin{aligned}(x+y)^{n+1} &= (x+y)\sum_{a=0}^{n}\binom{n}{a}x^{n-a}y^a \\ &= \sum_{a=0}^{n}\binom{n}{a}x^{n+1-a}y^a + \sum_{a=0}^{n}\binom{n}{a}x^{n-a}y^{a+1} \\ &= x^{n+1} + \sum_{k=1}^{n}\left[\binom{n}{k} + \binom{n}{k-1}\right]x^{n+1-k}y^k + y^{n+1} \\ &= \sum_{k=0}^{n+1}\binom{n+1}{k}x^{n+1-k}y^k,\end{aligned}$$

using Pascal's identity $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$ (Fact 0.13).

Characteristic p and the Frobenius Identity

Spoken Transcript [00:03:48 - 00:05:44]

Okay, so, there's one comment here, which is kind of a fun comment. That holds in any ring, that binomial formula. But what happens if we take this ring $\mathbb{Z}/p\mathbb{Z}$, *[writes $\mathbb{Z}/p\mathbb{Z}$ on the bottom blackboard]* where p is a prime? *[writes "prime" under it, with an arrow from the p]* So, for example, and then in this ring, you could take this ring $(\mathbb{Z}/p\mathbb{Z})[x,y]$, and the fun thing to consider there is $(x+y)^p$. *[writes $(x+y)^p$]*

So, that's an example. And so, if I take this formula in $\mathbb{Z}/p\mathbb{Z}$ — $\mathbb{Z}/5\mathbb{Z}$ in this case, where it's the same exponent as here — something kind of nice happens: all these internal coefficients here, they all have a 5 in it, so they all go away. And this is true just in general. In $\mathbb{Z}/p\mathbb{Z}$, if you take $(x+y)^p$, you get this formula that you secretly wanted anyway, which is this: *[writes $= x^p + y^p$]* $x^p + y^p$. It's a simple formula.

And the reason is because for all of the binomial coefficients, $\binom{p}{a}$, p divides $\binom{p}{a}$ *[writes $p \mid \binom{p}{a}$]* when a goes from 1 to $p-1$. *[writes when $1 \le a \le p-1$]* All the internal coefficients, p divides that. And why is that? It's because this binomial starts with a $p!$, *[writes $\frac{p!}{a!(p-a)!}$]* and there's no p's here to cancel the p in the $p!$, because these are all lesser numbers. Is that clear?

So, you'll always have... if you look at the exponent being prime in this binomial theorem, all of the middle coefficients here will all have that prime in it. So, if I look at it in $\mathbb{Z}/p\mathbb{Z}$, I get this great formula where there's no... So, this is a well-known formula, it will come up. Okay. So, that's an aside about the binomial formula.

The Freshman's Dream in Characteristic p

Lemma 4.4 (*Divisibility of Binomial Coefficients by a Prime*): Let p be a prime number. For every integer $a \in \{1, 2, \ldots, p-1\}$, the prime p divides the binomial coefficient $\binom{p}{a}$:

$$p \,\Big|\, \binom{p}{a}. \tag{4.2}$$

Proof

Recall the algebraic formula:

$$\binom{p}{a} = \frac{p!}{a!(p-a)!} \implies a!(p-a)!\binom{p}{a} = p! = p \cdot (p-1)!.$$

Thus, p divides the product $a!(p-a)!\binom{p}{a}$. Since $1 \leq a \leq p-1$, every factor occurring in the factorials $a! = 1 \cdot 2 \ldots a$ and $(p-a)! = 1 \cdot 2 \ldots (p-a)$ is strictly less than p. Because p is prime, Euclid's lemma (Fact 0.17), applied factor by factor, shows that p does not divide $a!(p-a)!$. By Euclid's lemma once more, p must divide the remaining factor $\binom{p}{a}$. Q.E.D.

Proposition 4.5 (*Frobenius Identity*): Let p be a prime. In the polynomial ring $(\mathbb{Z}/p\mathbb{Z})[x,y]$, we have:

$$(x+y)^p = x^p + y^p. \tag{4.3}$$

Proof

Expanding $(x+y)^p$ via Theorem 4.2:

$$(x+y)^p = x^p + \sum_{a=1}^{p-1} \binom{p}{a} x^{p-a}y^a + y^p.$$

By Lemma 4.4, $p \mid \binom{p}{a}$ for all $1 \leq a \leq p-1$, so the coefficient $\binom{p}{a} \cdot 1 = \overline{\binom{p}{a}}$ is $\bar{0}$ in $\mathbb{Z}/p\mathbb{Z}$. Hence each intermediate term vanishes, leaving $(x+y)^p = x^p + y^p$. Q.E.D.

Beyond the Lecture: Characteristic p and the Frobenius Endomorphism

The proof of Proposition 4.5 only uses that $p \cdot 1 = 0$ in the coefficient ring. A non-zero commutative ring R with $p \cdot 1_R = 0$ is said to have *"characteristic"* p (the characteristic is the least $n \geq 1$ with $n \cdot 1_R = 0$); examples are $\mathbb{Z}/p\mathbb{Z}$ and the polynomial rings over it. In every such ring, $(a+b)^p = a^p + b^p$ for all $a, b \in R$, by the same proof. Together with $1^p = 1$ and $(ab)^p = a^p b^p$, this says that $a \mapsto a^p$ is a ring homomorphism $R \to R$, the *"Frobenius endomorphism"* of R.

Integral Domains, Fields of Fractions, and Laurent Series

Polynomial Rings and Power Series Rings as Integral Domains

Spoken Transcript [00:05:44 - 00:06:39]

And then there were some definitions: if I take R now to be an integral domain, *[writes R int. domain]* we had defined here $R[x]$, that's polynomials in one variable, this is also an integral domain. *[writes R[x] also int. domain]* We discussed that (Recall: the highest-term argument in Theorem 3.39). You use the highest term, and if R is an integral domain, when you multiply polynomials, the highest term lives, so it stays an integral domain.

But what I didn't say is this is also an integral domain: $R[[x]]$. *[writes R[[x]] also*

int. domain] And one needs a different argument here, because power series don't have highest terms. And so, the answer is, well, what do you think the answer is? You can't use the highest term here, so what should you use?

Student Answer

Lowest term.

Spoken Transcript [00:06:41 - 00:07:07]

Lowest term, exactly. You could have used the lowest term here, but psychologically with polynomials, you go to the highest term. But it turns out lowest term is a better idea here, because for a power series, a power series does have a lowest term. And when you multiply the two lowest terms of power series, it's the lowest term of the product. So, it's the same argument here, it just uses the lowest term, which is, you know, psychologically a little strange. You could have used the lowest term here, but you didn't.

But, okay, so, this is also an integral domain.

Integral Domain Properties: Degrees vs. Orders

Theorem 4.6 (*Preservation of the Integral Domain Property*): If R is an integral domain, then:

(1) The polynomial ring $R[x]$ is an integral domain.

(2) The formal power series ring $R[[x]]$ is an integral domain.

Proof

Since $1 \neq 0$ in R, also $1 \neq 0$ in $R[x]$ and in $R[[x]]$; it remains to show that there are no zero divisors. Let f, g be non-zero elements in the respective ring.

(1) For $R[x]$ (Highest Degree Term): Let $f = a_n x^n + \cdots + a_0$ and $g = b_m x^m + \cdots + b_0$ with $a_n \neq 0$ and $b_m \neq 0$. As in (3.68), the leading term of the product fg is:

$$a_n b_m x^{n+m}.$$

Since R is an integral domain, $a_n b_m \neq 0$. Thus, $fg \neq 0$ and $\deg(fg) = \deg(f) + \deg(g)$.

(2) For $R[[x]]$ (Lowest Degree / Order Term): A formal power series does not have a maximal degree, but every non-zero series $f = \sum_{k=0}^{\infty} a_k x^k$ has a well-defined *"order"* (minimal non-zero index):

$$\operatorname{ord}(f) := \min\{k \in \mathbb{N} \mid a_k \neq 0\}.$$

We write $f = x^n(a_n + a_{n+1}x + \dots)$ with $a_n \neq 0$, and $g = x^m(b_m + b_{m+1}x + \dots)$ with $b_m \neq 0$. The lowest non-vanishing term of the Cauchy product (Definition 3.41) is:

$$a_n b_m x^{n+m}.$$

Since R is an integral domain, $a_n b_m \neq 0$. Consequently, $fg \neq 0$ with $\operatorname{ord}(fg) = \operatorname{ord}(f) + \operatorname{ord}(g)$.

Q.E.D.

Fields of Fractions: Rational Functions and Laurent Series

Spoken Transcript [00:07:07 - 00:08:35]

And the point I wanted to make here is that the main construction last week was the field of fractions (Recall: Theorem 3.26). *[writes "field of fractions" over an arrow]* We'll have a different construction this week, but that was the field of fractions, and the important thing about a field of fractions: you can only build that on an integral domain. So, if I have an integral domain, I could take the field of fractions of that integral domain. That was a construction.

But I could also take the field of fractions of polynomials, because that's an integral domain, and that is called... well, the symbol for that is R with parentheses, that's brackets: *[writes $R(x)$]* $R(x)$. That's the name of the field of fractions of the polynomial ring when R is an integral domain. And it's called the field of rational functions. *[writes "field of rational functions"]*

So, maybe rational functions with coefficients in R. This is polynomials with coefficients in R, that's the field of rational functions. And you know how to think about this, because we've thought about the field of fractions: an element here in rational functions is a polynomial divided by another polynomial. So, both of these, f and g, are polynomials, and of course g is not allowed to be zero. *[writes $R(x) \ni \frac{f(x)}{g(x)}$, $f, g \in R[x]$, $g \neq 0$]* So, that's how to think about an element here, and you know, that's that. Okay.

The Field of Rational Functions

Recall that for any integral domain D, its *"field of fractions"* $\mathrm{Frac}(D)$ consists of equivalence classes of formal quotients $\frac{a}{b}$ with $a, b \in D$ and $b \neq 0$ (Definition 3.17).

Definition 4.7 (*Field of Rational Functions $R(x)$*): Let R be an integral domain, so that $R[x]$ is an integral domain (Theorem 4.6). The field of fractions of the polynomial ring $R[x]$ is denoted by round parentheses $R(x)$ and is called the *"field of rational functions"*:

$$R(x) := \mathrm{Frac}(R[x]) = \left\{ \frac{f(x)}{g(x)} \;\middle|\; f(x), g(x) \in R[x],\ g(x) \neq 0 \right\}. \tag{4.4}$$

Spoken Transcript [00:08:35 - 00:10:15]

Now, power series are also an integral domain, so we can again consider the field of fractions, and the symbol for that is two parentheses: $R((x))$. *[writes $R((x))$]* All right? So, this is kind of universal math notation. There's the integral domain, you have the polynomial ring, power series, field of fractions, and this one has a special name. Does anyone know what the name of this thing is? It's called the field of Laurent series. *[writes "field of Laurent series"]*

Actually, I'm having some doubts about how to spell Laurent. Is it with an a? No, I think it's with an e. I'm not 100% sure about that.

So, this is the field of Laurent series, and this will come up in your complex analysis classes for sure, but with a $\mathbb{C}$ here, and usually the argument is called z in complex analysis. So, the field of Laurent series — this is actually something... you could say we've already constructed the field of fractions, so why do I have to talk about it here? You could say this field of Laurent series we think about the same way as $f/g(x)$, where f and g are power series. *[writes* $R((x)) = \frac{f(x)}{g(x)}$, $f, g \in R[[x]], g \neq 0$*]* And this would be absolutely correct. That would be correct, and g is not equal to zero. It's just there's a nicer way to think about it, and that has to do with the last thing I said yesterday... not yesterday, last week (Recall: Theorem 3.42).

The Field of Fractions of Formal Power Series

Definition 4.8 (*Field of Fractions* $R((x))$): Let R be an integral domain. Since $R[[x]]$ is an integral domain (Theorem 4.6), we define its field of fractions, denoted with double round parentheses $R((x))$:

$$R((x)) := \mathrm{Frac}(R[[x]]) = \left\{ \frac{f(x)}{g(x)} \;\middle|\; f(x), g(x) \in R[[x]],\ g(x) \neq 0 \right\}. \tag{4.5}$$

It is called the *"field of Laurent series"*: when $R = K$ is a field, it is isomorphic to the field of formal Laurent series over K (Theorem 4.11).

Board Transition

The lecturer moves the board with the fields of fractions up and washes the panel below it for the definition of Laurent series.

Formal Laurent Series with Finitely Many Negative Powers

Spoken Transcript [00:10:48 - 00:12:45]

Have you seen Laurent series already in the complex analysis class? No? All right, well, we'll see.

So, you won't go wrong if you think about it this way — that's the definition, that's how we constructed it. But there's another way, which is why it's called the field of Laurent series. So, what is a Laurent series? A Laurent series is a power series where you are allowed to have finitely many negative coefficients, that's the definition.

So, I can say this: *[writes on the board: Laurent series in x]* Laurent series in a variable x, this is a power series with — and this is the crucial thing — *[writes "is a power series with"]* finitely many negative coefficients... *[writes "finitely many negative powers"]* You can have as many positive coefficients, positive degree coefficients as you want, it's a power series, but only finitely many negative powers. So, that's the crucial thing for a Laurent series.

So, what is a power series? A power series might be like the power series for e^x,

[writes $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \in R[[x]]$] we've seen these also anyway. So, that's a power series; these are elements of the ring, the power series ring.

In Laurent series, you're allowed to have some negative, so you could have some $x^{-12} + 3x^{-2} + x^{-1}$, then the constant term... I got too excited about the constant term! $17 + x + x^{15} + \ldots$. *[writes $x^{-12} + 3x^{-2} + x^{-1} + 17 + x + x^{15} + \ldots$]* So, that is a Laurent series. So, have you seen these objects in mathematics? Okay, well, maybe it's the first time, but you can imagine: you just take power series, you're allowed to have negative terms, but only finitely many. And it's extremely important you have only finitely many.

Formal Definition and Arithmetic of Laurent Series

Definition 4.9 (*Formal Laurent Series*): A *"formal Laurent series"* in the indeterminate x over a ring R is a formal sum of the form:

$$f(x) = \sum_{n=-N}^{\infty} a_n x^n = a_{-N}x^{-N} + a_{-N+1}x^{-N+1} + \cdots + a_0 + a_1 x + a_2 x^2 + \ldots \tag{4.6}$$

where $N \in \mathbb{Z}$ and $a_n \in R$ for all $n \geq -N$. That is, $f(x)$ contains at most **finitely many negative powers** of x.

Example 4.10 (*Power Series vs. Laurent Series*):

- **Power Series:** Elements of $R[[x]]$ contain only non-negative powers:
$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \in R[[x]].$$
(For this series, R must contain $\mathbb{Q}$, e.g. $R = \mathbb{R}$, so that the coefficients $\frac{1}{k!}$ lie in R.)
- **Laurent Series:** Elements can have negative exponents up to some lowest power:
$$x^{-12} + 3x^{-2} + x^{-1} + 17 + x + x^{15} + \cdots \in R((x)).$$

Spoken Transcript [00:12:45 - 00:14:25]

And you'll find: if you have only finitely many, you can add them, and when you add them, you'll still only have finitely many because two of them only had finitely many. When you multiply them, the lowest term now — you know, if I multiply this with something with lowest term -7 *[writes x^{-7}]*, its lowest term will be x^{-19} — so, it will still have only finitely many. Is that clear?

And here's the great thing: if you're non-zero here, you can divide. That's why the Laurent series turns out to be a field. And this is something you should think about. We discussed this last time. So, if you have a power series... maybe I just say this, because this is maybe now departing a little bit from the course, but it is the case that if I take this field of Laurent series, *[writes "field of Laurent series"]* this is isomorphic to $R((x))$, *[writes $\cong R((x))$]* the field of fractions of the power series. And what you have to show there is: if you have Laurent series, you can divide. That's the interesting thing.

And that has to do with the last thing I said last week, which is that it's pretty easy to divide power series: all you have to do is find the first non-zero coefficient, then use this rule I explained last week (Recall: Theorem 3.42), and then you can play with this power to take it out. So, I leave this for you.

Spoken Transcript [00:14:25 - 00:15:20]

I'll say one thing: why don't we allow power series with infinitely many negative terms? Because if you try to multiply them, then you have to do infinite sums. And we don't do infinite sums in algebra, we only do finite sums. Because if you do infinite sums, you need convergence, and you can try to do that, and that's an interesting subject, but in this class, we're not doing infinite sums. And if you have finitely many negative terms, when you multiply, you see you'll never have to do infinite sums. It's just the same.

So, anyway, that's just this comment here. This field of Laurent series, it turns out, is just a nice way of thinking about this field of fractions for power series. And you can imagine, it's nicer to think about this kind of object than the ratio. And this second way of looking at things is not available for polynomials. There's only one way to think about it there: as ratios of polynomials. Okay.

Isomorphism between Laurent Series and the Field of Fractions

Theorem 4.11 (*Algebraic Characterization of $R((x))$*)**:** When K is a field, the set of formal Laurent series with finitely many negative powers forms a field under formal addition and multiplication, and is naturally isomorphic to the field of fractions of $K[[x]]$:

$$\mathrm{Frac}(K[[x]]) \cong K((x)) = K[[x]][x^{-1}]. \tag{4.7}$$

Why Invertibility Holds and Why Infinitely Many Negative Terms Fail:

(1) Invertibility in $K((x))$: Any non-zero Laurent series can be factored as $f(x) = x^k u(x)$, where $k \in \mathbb{Z}$ is the minimal index with $a_k \neq 0$, and $u(x) = a_k + a_{k+1}x + \cdots \in K[[x]]$ with $a_k \neq 0$. Since K is a field, the leading coefficient a_k is a unit in K, so $u(x)$ is an invertible unit in the power series ring $K[[x]]$ (Theorem 3.42). Its multiplicative inverse is:

$$f(x)^{-1} = x^{-k}u(x)^{-1} \in K((x)).$$

(2) Failure of Infinitely Many Negative Terms: If series with infinitely many negative powers were permitted ($\sum_{n=-\infty}^{\infty} a_n x^n$), the coefficient of x^k in the product of two such series would require evaluating the infinite sum:

$$c_k = \sum_{j=-\infty}^{\infty} a_j b_{k-j}.$$

Without topological convergence, such an infinite sum is undefined in pure algebra. Restricting to finitely many negative powers ensures that for each k, only finitely many non-zero products $a_j b_{k-j}$ occur, keeping all algebraic operations purely finite.

Beyond the Lecture: Why K Must Be a Field

For an integral domain R that is not a field, the Laurent series with coefficients in R do not form a field, so they cannot be all of $R((x)) = \mathrm{Frac}(R[[x]])$. For $R = \mathbb{Z}$, the constant series 2 has no inverse among the Laurent series over $\mathbb{Z}$: comparing constant terms

in $2 \cdot g = 1$ would give $2c_0 = 1$ with $c_0 \in \mathbb{Z}$. Yet $\frac{1}{2}$ lies in $\mathrm{Frac}(\mathbb{Z}[[x]])$. In general, the Laurent series over R form a subring of $\mathrm{Frac}(R[[x]])$, via $x^{-N}g \mapsto \frac{g}{x^N}$, and the two coincide precisely when R is a field.

Ideals and Ring Homomorphisms

Definition and Axioms of an Ideal

Transition to the Main Weekly Topic

The lecturer takes a sponge and cleans the boards, erasing the Laurent series overview and preparing the blackboards for the introduction of ideals and quotient rings.

Spoken Transcript [00:15:33 - 00:16:29]

Okay, so, the main topic this week is about ideals and quotient rings, and the main construction is that of a quotient ring.

If you like polynomial rings, the class will talk about results of polynomial rings, but mostly polynomial rings in one variable. The subject of polynomial rings in several variables is kind of very important in mathematics, and there's a whole class about it more or less, which is... the class is called commutative algebra.

Spoken Transcript [00:16:29 - 00:18:33]

Okay, so, now we finish these constructions and we start something new. R is our ring, *[writes R ring]* and the first thing: we've already defined I is an ideal. *[writes $I \subset R$ is an ideal]* So, we've already defined this (Recall: Definition 1.30), but I'll just review it briefly because now it's crucial for this week.

I is an ideal if it satisfies that, if I take R plus and 0... this is a subgroup of the additive part of the ring. *[writes $\bullet (I,+,0) \subset (R,+,0)$ subgroup]* And the second is: if I take anybody in the ring and anybody in the ideal, then their product lies in the ideal. *[writes $\bullet \ \forall r \in R, \forall a \in I$ then $r \cdot a \in I$]* This says that the ideal is an additive subgroup under addition, but there's another condition. The ideal has kind of two constraints: one is the addition constraint, and the other is the multiplication constraint. They're just different, because they're talking about two different operations.

And our example we had seen... we did some examples anyway, but a place where we know we always get ideals: if I take a homomorphism... *[writes $f : R \to S$]* maybe I should put it here... so, f is a homomorphism of rings, *[writes f is a hom of rings]* then whenever you have a homomorphism of rings, you get an ideal for free. Sorry. *[chalk slips]*

And we discussed this also before: then the kernel of f, which is defined to be all the people in the ring such that $f(a) = 0$, is an ideal. *[writes $\implies \ker(f) = \{a \in R \mid f(a) = 0\}$ is an ideal]* So, we proved this in one of the early weeks (Recall: Proposition 1.31).

Definition of Ideals and the Kernel of a Homomorphism

Recap: Ideals and the kernel of a ring homomorphism were defined in Week 1 (Definition 1.30, Definition 1.27), where the kernel was also shown to be an ideal (Proposition 1.31); both statements were restated in Week 2 (Definition 2.24, Proposition 2.25). The lecturer reviews them because the quotient ring construction of this lecture rests on them.

Let R be a commutative ring with identity.

Definition 4.12 (*Ideal*): A subset $I \subseteq R$ is called an *"ideal"* of R if it satisfies the following two conditions:

(1) **Additive Subgroup Condition:** $(I, +, 0)$ is a subgroup of the additive group $(R, +, 0)$. That is, $0 \in I$, and for all $x, y \in I$, $x - y \in I$.

(2) **Absorption / Multiplication Constraint:** I absorbs multiplication by arbitrary elements of the ambient ring R:

$$\forall r \in R, \quad \forall a \in I \implies r \cdot a \in I.$$

Proposition 4.13 (*Kernel of a Ring Homomorphism as an Ideal*): Let $f\colon R \to S$ be a homomorphism of commutative rings. The *"kernel"* of f, defined by:

$$\ker(f) := \{a \in R \mid f(a) = 0_S\}, \tag{4.8}$$

is an ideal of R.

Proof

We verify the two defining conditions of Definition 4.12:

(1) **Additive Subgroup:** Since $f(0_R) = 0_S$, $0_R \in \ker(f)$. For $x, y \in \ker(f)$, additivity of f yields $f(x - y) = f(x) - f(y) = 0_S - 0_S = 0_S$, so $x - y \in \ker(f)$.

(2) **Multiplicative Absorption:** For any $r \in R$ and $x \in \ker(f)$, multiplicativity gives:

$$f(r \cdot x) = f(r) \cdot f(x) = f(r) \cdot 0_S = 0_S.$$

Thus, $r \cdot x \in \ker(f)$.

Q.E.D.

The Linear Algebra Analogy: Kernels, Images, and Isomorphisms

Spoken Transcript [00:18:33 - 00:20:12]

So, in general, just when you study algebra, even in linear algebra, I remind you in the study of linear algebra... *[writes "Reminder" on the middle board]* this is a reminder: if you have a vector space V and you have a linear transformation, let's call it ϕ... *[writes $\phi\colon V \to W$]* so, if these are two vector spaces — so, V, W are vector spaces over some field, *[writes V, W vector spaces $/\mathbb{C}$]* it's not relevant for this example, I don't know, say complex numbers — and this is a linear transformation — I don't know what you called homomorphisms of vector spaces, linear transformation. *[writes ϕ linear transformation]*

So, in the abstract study of linear algebra, when you have such a linear transformation, you automatically get two other vector spaces, so to speak for free. So, the

two vector spaces are the kernel — this is a subspace of V — and the image, this is a subspace of W. *[writes $\ker(\phi) \subset V$, $\mathrm{Im}(\phi) \subset W$]* And there is a theorem that links this data. If you remember, it says that if I take V and I quotient by the kernel... I hope you remember this, it's kind of crucial here... if you take V and you quotient by the kernel, this is isomorphic as vector spaces to the image. *[writes $V/\ker(\phi) \cong \mathrm{Im}(\phi)$]*

All right, so, this is kind of a model. It's some aspect of your... Did you learn it in these terms? I've never taught linear algebra here. Okay.

Linear Algebra Analogy: The First Isomorphism Theorem

For a linear transformation $\phi: V \to W$ between vector spaces over a field $\mathbb{F}$:

- **Kernel:** $\ker(\phi) = \{v \in V \mid \phi(v) = 0\} \leq V$ is a vector subspace of V.
- **Image:** $\mathrm{Im}(\phi) = \{\phi(v) \mid v \in V\} \leq W$ is a vector subspace of W.
- **First Isomorphism Theorem for Vector Spaces:**

$$\frac{V}{\ker(\phi)} \cong \mathrm{Im}(\phi).$$

In ring theory, we seek an identical structural correspondence for a ring homomorphism $f: R \to S$.

Spoken Transcript [00:20:12 - 00:21:10]

Yeah, I haven't taught linear algebra for... I don't know, it's possible not in this century. But it's a great class, linear algebra. I used to tell students it's like the most important math class, linear algebra, because humans are actually pretty good at solving linear equations and pretty terrible at solving any other kind of equation.

That's not true, I did teach it. *[cleans and wipes the lower board]*

Ring Homomorphisms vs. Linear Transformations

Spoken Transcript [00:21:10 - 00:23:08]

Okay, so, we would like to think a little bit about this kind of construction for rings. And we see a lot of it to start.

If I have this ring homomorphism... so, these are both rings now, so, I try to make the parallel here: R and S are rings, *[writes $f: R \to S$, R, S are rings]* and I remind you, they're commutative rings with unit, and f always takes unit... I don't remind you of all the definitions, but it's consistent with our previous... f takes the 1 to 1, that's the definition of a ring (Recall: Definition 1.18). And f is a homomorphism. *[writes f hom]* So, we just want to complete that board.

So, we get now in this situation, we get two objects for free. Or I don't know, free might be the wrong word, but anyway, this gives us two objects: one is the kernel inside R, *[writes $\to \ker(f) \subset R$]* but this is not... so, there, everything was vector spaces. But this is not a ring now, this is an ideal. *[writes "ideal"]*

All right, but we also get the image of f inside S, *[writes $\mathrm{Im}(f) \subset S$]* and that is a subring. *[writes "subring"]* All right, so, we saw that was a subring, the image (Recall: Proposition 2.27).

So, that's kind of the parallel there. And then we're going to work on this quotient business, but you see the matter is slightly... you know, in the vector space world, everything is a vector space. But in the ring world, sometimes you get rings, and sometimes you get ideals, and they're slightly different objects, so, we have to come to terms with that.

So, let me just say that the statements there that you know... so, basic statements: *[writes "STATEMENT"]* One is that f is surjective if and only if this subring is the whole ring. *[writes (i) f is surjective $\iff$ $\mathrm{Im}(f) = S$]* That's almost like a tautology, right? That's what surjective means: you get everybody here, and if the image is everybody, it's surjective. So, that's essentially a tautology.

The Morphism Data for Rings: Ideals vs. Subrings

When comparing linear transformations $\phi\colon V \to W$ and ring homomorphisms $f\colon R \to S$, a crucial categorical divergence arises:

Structure	Linear Algebra ($\phi\colon V \to W$)	Ring Theory ($f\colon R \to S$)
Domain Sub-object	$\ker(\phi) \leq V$ (Vector Subspace)	$\ker(f) \subset R$ (**Ideal**, not a subring with 1_R)
Codomain Sub-object	$\mathrm{Im}(\phi) \leq W$ (Vector Subspace)	$\mathrm{Im}(f) \leq S$ (**Subring**, contains $1_S = f(1_R)$)

Proposition 4.14 (*Surjectivity Criterion*): Let $f\colon R \to S$ be a ring homomorphism. Then f is surjective if and only if:

$$\mathrm{Im}(f) = S. \tag{4.9}$$

Injectivity and the Zero Ideal

Spoken Transcript [00:23:08 - 00:24:34]

The second statement — and that's also true there — the second statement: *[writes (ii)]* f is injective if and only if... *[writes f is injective $\iff$]* so, what's the statement going to be? It's going to be about the kernel: the kernel of f is the zero ideal. *[writes $\ker(f) = (0)$ zero ideal]*

This is the standard terminology for the zero ideal, the standard notation for the zero ideal. The zero ideal is the ideal that has zero and nothing else. If you look at the definition of an ideal, the ideal that has the single element zero... So, this is going to be used a lot, so, it should be completely clear. *[moves to the right and writes: Zero ideal, (0), 0, $\{0\}$]*

The zero ideal: sometimes this is denoted like this: (0), sometimes it's just denoted

0, but the zero ideal is the ideal that contains one element: zero. So, it's definitely a subgroup, because it's just zero. Zero, when you add it, stays zero; it has inverses; there's nothing wrong with zero. And the second one is: it's closed under multiplication, because if you take zero in the ideal and multiply by anybody, you stay zero. So, that's the zero ideal.

And this last statement says f is injective if and only if the kernel is the zero ideal.

Injectivity and the Zero Ideal

Definition 4.15 (*The Zero Ideal*): In any ring R, the singleton set consisting solely of the additive neutral element:

$$(0) = \{0\} \tag{4.10}$$

is an ideal of R, called the *"zero ideal"*. It is also written simply as 0.

Proposition 4.16 (*Injectivity and the Kernel*): Let $f\colon R \to S$ be a ring homomorphism. Then f is injective if and only if its kernel is trivial:

$$f \text{ is injective} \iff \ker(f) = (0). \tag{4.11}$$

Proof of Proposition 4.16

Spoken Transcript [00:24:34 - 00:25:38]

So, let's try to see what the argument for that is; that's almost a tautology. So, proof of (2). *[writes Proof (ii).]*

We have to do two things. Suppose f is injective. *[writes Suppose f is injective]* Then let a in R be not zero. *[writes let $a \in R, a \neq 0$]* Then $f(a)$ is not equal to $f(0)$, because injective means you take different elements, they go to different places — that's what injective means. So, if it's injective, this implies that a... oh, and $f(0) = 0$, so, that means $f(a)$ is not zero, *[writes then $f(a) \neq f(0) = 0$]* so, this implies a is not in the kernel of f. *[writes $\implies a \notin \ker(f)$]*

So, if we have f is injective, nobody who's not zero is in the kernel of f, so, this implies the kernel of f equals zero. *[writes $\implies \ker(f) = 0$]*

So, that's one half of it. I assume it's injective, and I deduce the kernel of f is zero; there can only be one element in it.

Forward Direction: Injectivity Implies Trivial Kernel

Forward Implication ($\implies$)

Suppose f is injective. Let $a \in R$ with $a \neq 0$. Since f is injective and $a \neq 0$:

$$f(a) \neq f(0).$$

Since f is a ring homomorphism, $f(0) = 0_S$ (Proposition 1.19). Thus:

$$f(a) \neq 0_S \implies a \notin \ker(f).$$

Consequently, no non-zero element of R belongs to $\ker(f)$. Since $0 \in \ker(f)$ always holds, we conclude:

$$\ker(f) = (0).$$

Q.E.D.

Spoken Transcript [00:25:38 - 00:26:43]

The other way around is: suppose the kernel of f is zero, *[writes Suppose* $\ker(f) = 0$*]* and as I said, it's just standard to use both things, depending on what you want to emphasize.

Suppose the kernel of f is zero; then let $f(a) = f(b)$. *[writes let* $f(a) = f(b)$*]* We're assuming it's zero, and then we pick two elements a, b in the ring that go to the same place.

So, this implies, by the homomorphism property, $f(a) - f(b) = f(a - b)$... *[writes* $\implies f(a - b) = f(a) - f(b) = 0$*]* which equals zero, because I'm assuming they're the same. So, this implies that $a - b$ is in the kernel, *[writes* $\implies a - b \in \ker(f)$*]* but we're assuming the kernel is zero, so, this implies $a - b = 0$, which implies $a = b$. *[writes* $\implies a - b = 0 \implies a = b$ *and boxes* $a = b$*]*

So, that's the second implication.

That's a basic implication, and actually, if you look at this proof, it's the same proof as you had in linear algebra. There's actually no difference.

Reverse Direction: Trivial Kernel Implies Injectivity

Reverse Implication ($\Longleftarrow$)

Suppose $\ker(f) = (0)$. Let $a, b \in R$ such that $f(a) = f(b)$. Using the additive homomorphism property of f:

$$f(a) - f(b) = f(a - b) = 0_S.$$

By definition of the kernel, this implies:

$$a - b \in \ker(f).$$

Since $\ker(f) = (0) = \{0\}$ by hypothesis, it follows that:

$$a - b = 0_R \implies a = b.$$

Thus, f is injective, completing the proof of Proposition 4.16. Q.E.D.

Q.E.D.

Spoken Transcript [00:26:43 - 00:28:05]

So, we want now... for this linear algebra picture, we've recovered all the pieces except this one, *[pushes up the lower blackboard and points at* $V/\ker(\phi) \cong \mathrm{Im}(\phi)$ *on the*

upper board] which says that V mod the kernel is the image, and we now want to recover that. And for that, we need a construction.

And there was a construction already in linear algebra you needed, which was the notion of a quotient vector space, which is, if you remember that, it's going to be very helpful.

The idea of a math education is that, actually, you're not allowed to forget anything you've learned. That's one theory. The other theory is that education is what's left after you've forgotten everything you've learned. You can decide which theory of education you'd like to subscribe to.

Board Cleaning and Pedagogical Aside

The lecturer erases the lower blackboard to prepare for the construction of quotient rings, sharing an aside on two contrasting theories of education.

The Construction of Quotient Rings

Spoken Transcript [00:28:05 - 00:29:59]

Okay, so, the task at hand is to understand this quotient. And now we do this not just for the kernel — we'll get back to the kernel — but for any ideal. So, we let I in R be an ideal. *[writes on the lower left board: Let $I \subset R$ be an ideal]*

And now, this is the main construction for this week: we will construct the quotient ring. *[writes: we will construct the quotient ring]* So, that's the name: quotient ring. And the symbol is R mod I. *[writes R/I and circles it]* That's the symbol. So, we have to construct it.

And what's the parallel to linear algebra, of course, is Rmod the kernel; that's the quotient ring. So, we'll get back to that in a moment. We forget about this linear algebra analogy, etc., and we go for this construction. We need to construct this.

So, is that clear? We have I is an ideal, so, our hypothesis is: R is a ring, I is an ideal, and we need to construct the quotient. But we need to construct this object as a ring. That means, as usual, it's a multi-step construction: we have to first construct it as a set, then we have to find zero in it, we have to find one in it, *[writes $(R/I, +, \cdot, 0, 1)$]* zero and one, and we have to find an addition and multiplication. So, you have to construct the whole thing. That's just what you have to do; there's no other way around it.

So, Step 1... there's a lot of steps, and as usual, I don't do all of them... *[writes: Step 1: What is the set R/I?]*

The Quotient Ring Construction: Roadmap

Let R be a commutative ring with identity 1, and let $I \subset R$ be an ideal. Our objective is to construct a new ring:

$$R/I = (R/I, +, \cdot, 0_{R/I}, 1_{R/I}), \tag{4.12}$$

called the *"quotient ring"* (or residue class ring) of R modulo I.

The Five Stages of Constructing a Ring from Scratch: To endow the quotient R/I with a complete ring structure, we must rigorously execute the following stages:

Step 1: Underlying Set: Define an equivalence relation on R and take R/I to be the set of its equivalence classes.

Step 2: Distinguished Elements: Identify the additive zero $0_{R/I}$ and multiplicative identity $1_{R/I}$.

Step 3: Addition Operation: Define binary addition $+$ on equivalence classes and prove it is **well-defined**.

Step 4: Multiplication Operation: Define binary multiplication $\cdot$ on equivalence classes and prove it is **well-defined** using the ideal absorption property.

Step 5: Ring Axioms Verification: Verify associativity, commutativity, distributivity, and existence of inverses.

Step 1: The Underlying Set and Equivalence Relation Modulo an Ideal

Spoken Transcript [00:29:59 - 00:31:57]

Step 1: What is the set R mod I? And here the notation is totally standard. We have I and R, R mod I. Everybody uses just this notation. What is the set? And this should, now if you remember linear algebra, that should help you now.

So, the step for this is: we define an equivalence relation *[writes: We define an equivalence relation $\equiv$]* on R. Sometimes this equivalence relation could be denoted by this, but you have to carry around a lot of stuff, so, I'm just going to leave that I out. Going to define an equivalence relation on R.

Here's the definition: we say that if I have x and y in R, then x is equivalent to y *[writes: Def: $x, y \in R$, $x \equiv y$]* — and as I said, if you like, you can put I here because the equivalence relation depends on I — if and only if $x - y$ is in the ideal I. *[writes: $\iff x - y \in I$]* That's the definition.

And in words — so, as I said, if you want to be really precise, so, this is an aside, *[writes (Aside $\equiv_I$,]* if you want to be really precise, you could define it like this, because this thing depends on I — in words, this is usually put: x is equivalent to y mod I. *[writes "x is equivalent to y mod I")]* This is very standard language: mod I. x is equivalent to y mod I. That's what this equivalence relation is. It's just like for the integers.

Anyway, that's the symbol. And so, formally now, you have to check it's an equivalence relation. And to do that, you have to remember the three properties of an equivalence relation (Recall: Remark 3.13). And the three properties are: x is equivalent to x, which is zero being in the ideal; the other property is: if x is equivalent to y, then y is equivalent to x, and that would just put a sign here, but we know that I is a subgroup under addition, so, it's closed under the sign; and the last one is transitive. Maybe I do the last one: the transitive property.

The Equivalence Relation Modulo an Ideal

Definition 4.17 (*Congruence Modulo an Ideal*): Let R be a ring and $I \subset R$ an ideal.

For elements $x, y \in R$, we define the relation of *"congruence modulo I"* (denoted $x \equiv_I y$ or simply $x \equiv y$) by:

$$x \equiv y \iff x - y \in I. \tag{4.13}$$

In words, we say that *"x is equivalent to y modulo I"*.

Connection to Modular Arithmetic in $\mathbb{Z}$: This directly generalizes congruence modulo n in the integers:

$$x \equiv y \pmod{n} \iff x - y \in n\mathbb{Z},$$

where $I = n\mathbb{Z} = (n)$ is the principal ideal generated by n (Definition 2.33).

Spoken Transcript [00:31:57 - 00:33:28]

[writes: Transitive:] The transitive property says: if x is equivalent to y and y is equivalent to z, then x is equivalent to z. *[writes: If $x \equiv y$ and $y \equiv z$, then $x \equiv z$]*

x is equivalent to y means, if I translate this now in the definition, $x - y$ is in I, *[writes $x - y \in I$]* and this is $y - z$ is in I. *[writes $y - z \in I$]*

What I need to show is: $x - z$ is in I. But these two elements are in I, so, I can add them. *[writes $(x - y) + (y - z) \in I$]* These two assumptions, since I is closed under addition, I can add them. Is that clear?

And then happily the y falls out, so, you get $x - z$ is in I, *[writes $x - z \in I$]* and we're happy about that, okay? So, that's the proof of the transitivity.

Verification: Equivalence Relation Properties

Proposition 4.18 ($\equiv_I$ *is an Equivalence Relation*): Let $I \subset R$ be an ideal. The relation $\equiv_I$ is an equivalence relation on R.

Proof

We verify the three defining axioms:

(1) Reflexivity ($x \equiv x$): For any $x \in R$, $x - x = 0 \in I$ because $(I, +, 0)$ is a subgroup. Thus, $x \equiv x$.

(2) Symmetry ($x \equiv y \implies y \equiv x$): If $x \equiv y$, then $x - y \in I$. Since $(I, +, 0)$ contains additive inverses:

$$y - x = -(x - y) \in I \implies y \equiv x.$$

(3) Transitivity ($x \equiv y \wedge y \equiv z \implies x \equiv z$): Suppose $x \equiv y$ and $y \equiv z$. By definition:

$$x - y \in I \quad \text{and} \quad y - z \in I.$$

Since I is closed under addition, their sum lies in I:

$$(x - y) + (y - z) \in I \implies x - z \in I \implies x \equiv z.$$

Q.E.D.

Subgroup Property Sufficiency: Notice that verifying that $\equiv_I$ is an equivalence relation relies **exclusively** on the fact that $(I, +, 0)$ is an additive subgroup of $(R, +, 0)$. The multiplicative absorption condition $R \cdot I \subseteq I$ has not yet been utilized!

Spoken Transcript [00:33:28 - 00:34:47]

If you think about what we've used: we've used zero is in I, we've used the minus is in I, and we've used I is closed under addition, so, we've kind of used all the group conditions for I. We couldn't get away with less for the ideal.

What we have not used is the multiplicative condition. That's going to come later. Because an ideal has two assumptions: one is the additive structure, it's a subgroup, and we've used it all here. But we have not touched the multiplicative condition; that'll come in a moment.

Okay, so, if it's an equivalence relation, then you can imagine that we're going to take its equivalence classes.

Spoken Transcript [00:34:47 - 00:36:03]

Now we can define the set: R mod I as a set is equal to the set of equivalence classes *[writes: R/I = Set of equivalence classes]* of this relation, let us say, on R. *[writes: of $\equiv_I$ on R]*

Okay, so, I've defined this as a set. And now the question of names: how do I name things? And that's a separate discussion. It's the same discussion we had about the rational numbers, the same for the field of fractions, that one of the conceptual issues about the quotient is that people do not have unique names. And this is just, you know, something you have to live with.

So, the question is: how do I name an element? *[writes: How do I name an element in R/I?]*

Definition 4.19 (*The Quotient Set R/I*): The underlying set of the quotient ring R/I is defined as the set of all equivalence classes of R with respect to the equivalence relation $\equiv_I$:

$$R/I := R/\equiv_I = \{[x] \mid x \in R\}. \tag{4.14}$$

Naming Conventions for Equivalence Classes (Cosets)

Spoken Transcript [00:36:03 - 00:37:33]

How do I name an element in R mod I? One way to do it is: you take x in R, *[writes: $x \in R$]* then x is in R, so, it determines an equivalence class: the equivalence class which x lies in. And that is usually called $\bar{x}$. *[writes: $\bar{x} \in R/I$, the equiv. class that x lies in]* This is the equivalence class that x lies in — the equivalence class that x lies in. Okay, that's not beautiful English, but anyway, that's what it is. You take x, you get an equivalence class; that's usually denoted $\bar{x}$. That's what we do with integers also. Is that okay?

Another standard way to denote it... there's kind of three ways to denote this: if x is in R, we can take $\bar{x}$ in R mod I; *[writes: $\bar{x} \in R/I$]* if you really want to write things out really explicitly, you can write it like that: *[writes: $x + I \in R/I$]* because the equivalence class of x is the set of things where you take... the equivalence class of x is equal to all

those things $x+i$ where i is in I. *[writes $\bar{x} = \{x+i \mid i \in I\}$]* Is that clear? Because if you're equivalent to x, your difference is in I. So, this is explicitly what the equivalence class is. And since it's x plus something in I, some people will write it like this: *[points to $x+I$]* okay? It's kind of messy to keep that around, but in some arguments it's nice, so, that's a second way to write it.

The third way to write it, which is by far the most common, is you just call it x. *[writes: $x \in R/I$]* That's by far the most common, because you can see that... then if you use this, you have to be very careful. Sometimes, like in this, when x was born, it was an element, and then when you put it here, it becomes an equivalence class, and you have to keep track of the data type of x in your head. But I would say that that's kind of the three ways to do it. They all have their own advantages. Well, this one has limited advantage, but okay.

Three Standard Notations for Residue Classes (Cosets)

An element of R/I is an equivalence class containing elements of R. Because representations are not unique (different representatives can specify the exact same class), three standard notations are commonly employed:

(1) Overline Notation ($\bar{x}$): Denoted $\bar{x} \in R/I$. This is compact and mirrors the notation used for residue classes $\mathbb{Z}/n\mathbb{Z}$ (Definition 1.24).

(2) Coset Notation ($x+I$): Explicitly writing the equivalence class as an affine shift of the ideal:

$$[x] = x + I := \{x+i \mid i \in I\} \subseteq R. \tag{4.15}$$

Since $y \equiv x \iff y - x = i \in I \iff y = x+i$, the equivalence class of x is literally the subset $x+I$.

(3) Bare Representative Notation (x or $[x]$): Writing simply $[x]$ or, by abuse of notation, $x \in R/I$. While by far the most common in advanced algebra, this requires carefully tracking whether x represents an individual ring element in R or an entire equivalence class in R/I (the *"data type"* in one's head).

Step 2 & 3: Distinguished Elements and the Well-Definedness of Addition

Spoken Transcript [00:37:33 - 00:39:08]

All right, so, since we've defined the set, we need to now define addition. So, let's do the zero and one. First, we define zero in R mod I: *[writes $0 \in R/I$, Def: $0 = \bar{0}$]* 0 is equal to $\bar{0}$, okay? Because this is the equivalence class of zero. Zeros are always called zeros.

So, that's the zero here: it's the equivalence class of zero, which is not a surprise. And similarly, the definition of one is: it's the equivalence class of one. *[writes Def: $1 = \bar{1}$]* So, now we have at least this R mod I, and we have some distinguished elements: zero, one.

Okay, so, now let us define addition. *[writes: Let us define $+$]* And it's still called plus. Very formally, you shouldn't call the binary operation in two different rings the same thing, because they're kind of different. But no one does that; everyone calls it plus,

and it makes following the definitions... it makes you have to concentrate to follow the definitions.

So, let's define plus. What does that mean? To define plus, I take an element — this is an element of R mod I — and I take another element of R mod I, and I have to define this plus. I have to define it: *[writes: $\bar{x}+\bar{y} \underset{DEF}{=} \overline{x+y}$]* okay? This equation here is going to define it. This is the definition. And how am I going to define it? Well, I only know how to add x and y one way, because they were born in R, and it's the only thing I know how to do. So, this is the only thing possibly I could write.

Distinguished Elements and the Definition of Addition

Definition 4.20 (*Zero and Identity in R/I*): The distinguished neutral elements in the quotient ring R/I are defined by the classes of the respective identities in R:

$$0_{R/I} := \bar{0} = 0 + I = I, \tag{4.16}$$

$$1_{R/I} := \bar{1} = 1 + I. \tag{4.17}$$

Definition 4.21 (*Addition on R/I*): For any two equivalence classes $\bar{x}, \bar{y} \in R/I$, their sum is defined by adding representatives in R:

$$\bar{x} + \bar{y} := \overline{x+y}, \quad \text{or equivalently} \quad (x+I)+(y+I) := (x+y)+I. \tag{4.18}$$

Spoken Transcript [00:39:08 - 00:40:48]

So, that's the definition, and as I said, to read it, you have to concentrate on what's being defined in this whole thing. This is being defined; that was already previously known, because it was in R. All right? So, this whole equation is a definition of plus, and it's not a full definition, because you have to check it's well-defined, because these things could have different names. So, that's kind of a pain.

So, let's check it's well-defined. *[writes Let us check $+$ is well defined in R/I]* What could we do? We could have $\bar{x}$ is equal to some $\overline{x'}$, *[writes: $\bar{x} = \overline{x'}$]* where x' is equal to $x+i$, *[writes: $x' = x+i$]* where i is in I. *[writes: $i \in I$]* So, we could have just had a different name for the same equivalence class. And if we had a different name, what we know is that it will just be some shift of x by I, because if they're in the same equivalence class, their difference is an element of I — that's by definition. Okay?

Similarly for y, we could have a different name here: *[writes: $\bar{y} = \overline{y'}$, $y' = y+j$, $j \in I$]* j, where j is in I.

So, now we have to test that this definition is the same. We already have the formula for the names x and y, so, let's try to see what happened if we use these names. We better get the same answer.

Spoken Transcript [00:40:48 - 00:42:05]

If we use these names, this definition would give us $\overline{x'}+\overline{y'}$; *[writes: $\overline{x'}+\overline{y'} = \overline{x'+y'}$]* then we use the formulas for $x'+y'$, and then I regroup them — so, this is... well, okay, maybe I don't... *[writes $= \overline{x+i+y+j}$]* $x+i$, $y+j$, and then I regroup it: $x+y+i+j$,

[writes $= \overline{x+y+(i+j)}$*]* but now $i+j$ has to be in the ideal I because I is closed under addition.

And so, this equivalence class is actually the same as that equivalence class, because it's the equivalence class of the same thing shifted by an element of I.

The answer is: we have checked that if I use different names and combined them with addition, we get the same equivalence class on the right side. So, this proves it's well-defined, okay?

And if you remember, I said from linear algebra, you've done all of this before.

Proof that Addition is Well-Defined on R/I

Proposition 4.22 (*Well-Definedness of Addition*): The addition operation defined in Definition 4.21 is well-defined: it is independent of the choice of representatives.

Proof

Suppose $\bar{x} = \bar{x}'$ and $\bar{y} = \bar{y}'$ in R/I. By Definition 4.17, there exist elements $i, j \in I$ such that:

$$x' = x + i, \tag{4.19}$$

$$y' = y + j. \tag{4.20}$$

Evaluating the sum using the primed representatives yields:

$$\begin{aligned} \bar{x}' + \bar{y}' &= \overline{x' + y'} \\ &= \overline{(x+i)+(y+j)} \\ &= \overline{(x+y)+(i+j)}. \end{aligned}$$

Since $(I, +, 0)$ is an additive subgroup, $i, j \in I$ implies $i + j \in I$. Therefore, $(x+y)+(i+j) \equiv (x+y) \pmod{I}$, which implies:

$$\overline{(x+y)+(i+j)} = \overline{x+y} = \bar{x} + \bar{y}.$$

Thus, the sum is independent of representatives. Q.E.D.

The Quotient Additive Group: Since addition is well-defined, associativity, commutativity, the identity $\bar{0}$, and inverses $-\bar{x} = \overline{-x}$ all inherit directly from $(R, +, 0)$. Hence, $(R/I, +, \bar{0})$ forms an abelian group.

Spoken Transcript [00:42:05 - 00:43:13]

Then formally, we have to show... now I've defined addition and the zero, you have to now formally check this addition is associative, and that zero plays the role of a zero, and there's additive inverses, and I leave all of this to you.

But there is multiplication, and that's worth doing carefully, because so far we have not used the multiplicative part of the hypothesis on the ideal, and it comes out now, and it's kind of crucial. So, I will try to do this carefully, maybe after the break, but let me just say for...

Board Transition and Preparation for Multiplication

The lecturer cleans the middle blackboard to prepare for the definition of multiplication on R/I and the proof of its well-definedness.

Step 4: Multiplication and the Crucial Role of the Ideal Absorption Property

Spoken Transcript [00:43:13 - 00:45:11]

The next step is define multiplication on R mod I. *[writes: Define $\cdot$ on R/I:]* That means if we have this x, it's exactly as before, we need to define this: *[writes $\bar{x}\cdot\bar{y} \underset{DEF}{=} \overline{xy}$]* and this is going to be the answer, this is the definition. And how are you going to define it? You don't know... you only know how to do one thing to define things. So, you really have no choice except this. Okay? I could put the dot there too if I want.

This just says that if I want to multiply two equivalence classes, I take one element of one equivalence class, another element of the other equivalence class, I multiply them in the ring, and then take the resulting equivalence class. Is that clear?

And so, then I have to prove this is well-defined. *[writes: Prove well defined:]*

So, I use the same different choices for x, x'. Suppose I have... if I pick these different names, *[writes: $\overline{x'}\cdot\overline{y'} = \overline{x'\cdot y'}$]* and I use the definition, then I get... and then I use their formulas: $(x+i)(y+j)$, *[writes: $= \overline{(x+i)(y+j)}$]* and when I multiply them out, this is the crucial part: if I multiply them out, I get... sorry, there's no primes here, I get xy, and then I get $i\cdot y + j\cdot x$ — it's commutative, so, I can write it either way — plus ij. *[writes: $\overline{xy+iy+jx+ij}$]*

This is with one name, I get that; with the other name, I get this, and the difference is this. So, the only way this is well-defined is if this whole thing is in I. *[underlines $iy+jx+ij$ and writes "is in I"]* That's the only way it could be well-defined.

And it is in I, and here finally we use the fact that if you have an element of I and multiply it by anything, you stay in I. So, that's why it's used here. Okay, so...

Definition of Multiplication on R/I

Definition 4.23 (*Multiplication on R/I*): For equivalence classes $\bar{x}, \bar{y} \in R/I$, their product is defined by multiplying representatives in R:

$$\bar{x}\cdot\bar{y} := \overline{x\cdot y}, \quad \text{or equivalently} \quad (x+I)\cdot(y+I) := (x\cdot y)+I. \tag{4.21}$$

Proof that Multiplication is Well-Defined on R/I

Proposition 4.24 (*Well-Definedness of Multiplication*): The multiplication operation defined in Definition 4.23 is well-defined.

Proof

Let $\bar{x}' = \bar{x}$ and $\bar{y}' = \bar{y}$. Then $x' = x + i$ and $y' = y + j$ for some $i, j \in I$. Expanding the product of the alternative representatives:

$$\begin{aligned} \bar{x}' \cdot \bar{y}' &= \overline{x' \cdot y'} \\ &= \overline{(x+i)(y+j)} \\ &= \overline{xy + iy + jx + ij}. \end{aligned}$$

To establish that $\overline{(x+i)(y+j)} = \overline{xy}$, we must verify that the difference belongs to I:

$$(xy + iy + jx + ij) - xy = iy + jx + ij \stackrel{?}{\in} I.$$

We examine each of the three cross-terms:

- $i \in I$ and $y \in R \implies i \cdot y \in I$ by the **multiplicative absorption property** of ideals ($I \cdot R \subseteq I$).
- $j \in I$ and $x \in R \implies j \cdot x \in I$ by the **multiplicative absorption property** ($R \cdot I \subseteq I$).
- $i, j \in I \implies i \cdot j \in I$ (in fact, $I \cdot I \subseteq I$).

Since $(I, +, 0)$ is closed under addition:

$$iy + jx + ij \in I \implies \overline{xy + iy + jx + ij} = \overline{xy} = \bar{x} \cdot \bar{y}.$$

Therefore, multiplication on R/I is strictly well-defined. Q.E.D.

Why Subrings Fail and Ideals are Necessary: If I were merely an additive subgroup or a subring (closed under multiplication within itself, but not absorbing arbitrary elements of R), the terms iy and jx would generally fail to lie in I, causing the multiplication on equivalence classes to collapse! **This step is the sole reason why ideals, rather than general subrings, are required to define quotient rings.**

Mid-Lecture Break

The lecturer takes a short pause for the mid-lecture break. The lecture resumes at timestamp 00:45:11 with a pedagogical recap of the essential role played by the ideal axioms.

Recap: The Crucial Role of the Ideal Axioms

Spoken Transcript [00:45:11 - 00:47:25]

Okay, so, we start, and the last moment in the first half of the class was about this observation, and it's a really important thing, so, I repeat it now. This is the only time...

If you remember, the definition of an ideal *[writes on the right board: Def of Ideal]* had two clauses: the first is that *[writes • $(I, +, 0) \subset (R, +, 0)$ Subgroup]* this is an additive subgroup, and we've used that repeatedly. This has been used repeatedly in the definition of the equivalence relation, in the definition of addition, if you wanted to prove associativity and you wanted to prove all of those properties that I didn't prove, you'd be using them here.

But it has a second property, which was the multiplicative property, which is that for

every element of the ring and for every a in the ideal, the product is in the ideal. *[writes* $\bullet\ \forall r \in R, \forall a \in I \implies ra \in I$*]* That's just a totally separate property, because this one doesn't have anything to do with multiplication, this one has to do with multiplication, and this is the first time we use it.

And it's used to show that this definition of multiplication — which I tried to say philosophically you didn't have very much choice: if you want to multiply two equivalence classes, you pick an element in one equivalence class, you pick an element in the other, you multiply those elements, and you take the equivalence class. That's pretty much the only choice you have to define it, because you start with just an abstract ring.

And to show this is well-defined, which I did here quickly at the end with different elements in those equivalence classes, you come to this issue: you're taking the equivalence class of this element, it has to be the same as the equivalence class of that element. And the only way it is the same is if the extra terms, when you combine them, are all in I. And to see that: this is a product of two elements of I. So, that's okay, because it's any element of R and any element of I — you only need one element of I. So, this one is in I, and these two have one element of I and one element of R each, because this x and y are arbitrary elements of R. So, these are three times we apply this rule.

Review: The Two Defining Axioms of an Ideal

Recap: This is the definition from the start of this lecture (Definition 4.12), which the lecturer writes on the board again.

Definition 4.25 (*Ideal Axioms Reviewed*): A subset $I \subseteq R$ is an ideal if and only if:

(1) Additive Subgroup Axiom:

$$(I, +, 0) \subset (R, +, 0) \quad \text{is an additive subgroup.}$$

(2) Multiplicative Absorption Axiom:

$$\forall r \in R, \quad \forall a \in I \implies r \cdot a \in I.$$

Division of Labor Between the Two Axioms:

- Axiom **(1)** governs all additive aspects: that $\equiv_I$ is an equivalence relation (Proposition 4.18), well-definedness of addition, additive identity, and additive inverses in R/I.
- Axiom **(2)** is reserved solely for multiplication: ensuring that products of cosets do not depend on the representative chosen.

Completion of the Quotient Ring Structure

Spoken Transcript [00:47:25 - 00:48:47]

Okay, so, that proves the... *[walks across the front]* And this is one aspect of the construction which is not in your linear algebra class. Although it's not far from what happened in your linear algebra class when you considered the problem of scalar multiplication on the quotient, which I don't repeat. But anyway, even that shouldn't be

totally foreign to you.

So, now we've constructed this: we have plus, we have multiplication, we have zero and one. *[writes $(R/I, +, \cdot, \bar{0}, \bar{1})$]* The part that requires the foundational work is the definition as equivalence classes and making sure these are well-defined. The rest just follow the nose, that the identities... I mean, you can imagine: you have to check this times that, so, by definition this is that, and so, that's that, so, that checks one of the identity properties; and there's like 10 other properties you have to check, and they're all checked like this. *[writes $\bar{1} \cdot \bar{x} = \overline{1 \cdot x} = \bar{x}$]*

So, that's the construction, and that's the main construction this week. Last week we had the construction of the field of fractions, which you should understand completely, and now there is this construction, which is the construction of the quotient ring, which is your task to understand really completely this week: the construction.

Okay, so, are there any questions about this?

The Complete Quotient Ring Structure

Theorem 4.26 (*The Quotient Ring $(R/I, +, \cdot, \bar{0}, \bar{1})$*): Let R be a commutative ring with identity, and let $I \subset R$ be an ideal. The quintuple:

$$(R/I, +, \cdot, \bar{0}, \bar{1}) \tag{4.22}$$

equipped with the coset operations:

$$\bar{x} + \bar{y} := \overline{x + y}, \qquad \bar{x} \cdot \bar{y} := \overline{x \cdot y}, \qquad 0_{R/I} := \bar{0}, \qquad 1_{R/I} := \bar{1},$$

forms a commutative ring with identity.

Proof

Having proven that addition and multiplication are well-defined, the remaining ring axioms follow directly from the ring axioms of R applied to representatives:

- **Multiplicative Identity:** $\bar{1} \cdot \bar{x} = \overline{1 \cdot x} = \bar{x}$ for all $\bar{x} \in R/I$.
- **Associativity:** $(\bar{x} \cdot \bar{y}) \cdot \bar{z} = \overline{(xy)z} = \overline{x(yz)} = \bar{x} \cdot (\bar{y} \cdot \bar{z})$.
- **Distributivity:** $\bar{x} \cdot (\bar{y} + \bar{z}) = \overline{x(y+z)} = \overline{xy + xz} = (\bar{x} \cdot \bar{y}) + (\bar{x} \cdot \bar{z})$.
- **Commutativity:** $\bar{x} \cdot \bar{y} = \overline{xy} = \overline{yx} = \bar{y} \cdot \bar{x}$.

Q.E.D.

Spoken Transcript [00:48:47 - 00:50:58]

So, people were asking me: what's the relationship of this class to the book? That's a really good question.

Every week I picked out some... there were some pages, and that book is a huge book. And somehow the class covers a very small part of the book, so, it's not your goal to read that book from cover to cover, although it's not a bad thing to do.

So, every week I pick out some topics there, and my lectures are roughly on those topics. Now, my notation is sometimes different, the presentation is sometimes different in the book. The book is useful if you want... I skip some details in lectures; if you want

all the details, sometimes they're in the book.

The book also... that selection for the whole semester covers the actual curriculum I'm required to cover more or less for this course, so, it's helpful for me. But I would say: use the book if it's helpful for you. If not, you can use the lectures. What happened when I taught this course at ETH in 2018 was that some students wrote some notes, and they're really great notes. Of course, they sent it to me afterwards, but it's 8 years ago, I couldn't find them. But you know, probably some of your colleagues write some good notes, you can also use that. The lectures are recorded, so, you can also look at that.

Another option is that I gave some really good references, some really classic algebra references. They have the unfortunate quality of not being... their PDF file is not available through the library. But you can look at those books also. But if you look at the other books, the data won't be exactly in the order I'm giving you. If you want a book where it's in the order I'm giving you, then you should follow those page numbers in the actual reference. And that book is freely available on the ETH website.

So, that's a discussion about the book. Use it if it's helpful. For the exam material and things, I will only use material that I lectured in class, not some exotic topic in the book that I didn't lecture.

Okay.

Course References and Exam Scope

The lecturer answers student inquiries regarding the course literature:

- **Textbook Alignment:** Weekly reading assignments correspond to specific sections of the comprehensive course textbook (freely accessible to students via the ETH website). The textbook provides supplementary proofs and alternative presentations.
- **Exam Scope:** The examinations will strictly test the concepts, definitions, and theorems presented during the lectures, rather than extraneous textbook topics.

The First Isomorphism Theorem for Rings

Spoken Transcript [00:50:58 - 00:52:37]

All right, so, we go back to this issue now. We've made some progress.

If I have R to S... *[writes $f: R \to S$, R, S rings, f is a hom]* So, this R and S are rings, f is a homomorphism. Now, this is sometimes called the First Isomorphism Theorem, *[writes First Isom. Thm]* and it plays the role of the missing piece that we had in linear algebra that I wanted to develop here, and that says that this ring, Rmod the kernel *[writes $R/\ker(f)$]* — because when you have here, we have the kernel is an ideal; if we have an ideal, we can take the quotient, because that's what the last 45 minutes were about — this ring is canonically isomorphic to the image. *[writes $\cong \mathrm{Im}(f)$]* Okay?

So, that plays the role of the statement in vector spaces. Is that clear? That's the replacement. So, we have to prove this.

So, let us see what we have to do. We start with this f from R to S, *[writes We start*

$f\colon R \to S$] and we will define a homomorphism [*writes we will define a hom*] $\bar{f}$ from Rmod the kernel of f to S, [*writes* $\bar{f}\colon R/\ker(f) \to S$] just to start.

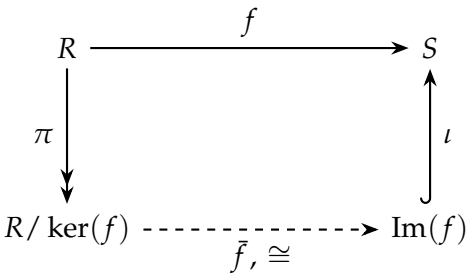
And how do you think we're going to define this?

Statement of the First Isomorphism Theorem for Rings

Theorem 4.27 (*First Isomorphism Theorem for Rings*)**:** Let $f\colon R \to S$ be a homomorphism of commutative rings. Then the kernel $\ker(f)$ is an ideal of R (Proposition 4.13), the image $\mathrm{Im}(f)$ is a subring of S (Proposition 2.27), and there exists a canonical ring isomorphism:

$$\bar{f}\colon R/\ker(f) \xrightarrow{\cong} \mathrm{Im}(f), \qquad \bar{x} \mapsto f(x). \tag{4.23}$$

In particular, if f is surjective, then $R/\ker(f) \cong S$ (Proposition 4.14).



Board Transition and Wet Washing

The lecturer pauses as students reflect on the proposed definition of $\bar{f}$. From timestamp 00:52:37 to 00:53:31, he takes a sponge and cleans the lower blackboard to clear working space for the step-by-step proof of the First Isomorphism Theorem.

Proof of the First Isomorphism Theorem

Step 1: Construction and Well-Definedness of the Induced Map $\bar{f}$

Spoken Transcript [00:53:31 - 00:54:07]

Okay, so, we have to define $\bar{f}$. What does it mean? We have to define this, this is the definition. So, define $\bar{f}$: it is a homomorphism that starts from R mod $\ker(f)$. So, what this thing eats is elements of R mod $\ker(f)$. [*writes* $\bar{f}(\bar{x})$] We have to define this function, and it eats equivalence classes. [*writes* $\bar{x} \in R/\ker(f)$]

What do you think we're going to do here?

Student Answer

Send it to $f(x)$.

Spoken Transcript [00:54:08 - 00:55:09]

Yeah. I mean, in this kind of generality, there's very little you can do, because we start... you can imagine R is like one country, S is another, there's only one flight here between them, and you've got to take it. There's no other way. There's just no other way. So, to get in here, you've got to get on f somehow, and that's the only way.

So, that's the definition. *[writes Def: $\bar{f}(\bar{x}) = f(x) \in S$]* When you define it like this, as I said, it's the standard pain about this naming business. And you know, I say it slightly facetiously, but this name is an incredibly crucial thing in this. If you study mathematics more and more, you'll see this idea of what you can name, what you can name precisely, how you can name it, turns out to be an incredibly important, subtle point. So, it's good to get used to thinking about that.

Anyway, so, the point: we have to show this is well-defined. *[writes Show $\bar{f}$ is well defined]*

Spoken Transcript [00:55:09 - 00:56:41]

So, what does that mean? If we take another name with somebody else in the equivalence class, it satisfies this equation; *[writes $x' = x + i$, $i \in \ker(f)$, $\bar{x'} = \bar{x}$]* then we must check that $\bar{f}$ of this is equal to $\bar{f}$ of that. *[writes We must check $\bar{f}(\bar{x'}) = \bar{f}(\bar{x})$]* We have to check it.

Well, let's see. What is this? $\bar{f}$ of this — here we use the definition — that's f of x', *[writes $\bar{f}(\bar{x'}) \overset{def}{=} f(x')$]* and then we use the definition of x', that's f of $x + i$. *[writes $= f(x + i)$]* Okay, now what do we do?

I should have said here very carefully that i is in the kernel. *[adds $i \in \ker(f)$]* Yeah, so, what do we do here? Well, maybe you should expand this as $f(x) + f(i)$, *[writes $= f(x) + f(i)$]* because f itself is a homomorphism. Now i is in the kernel, so, this is equal to $f(x)$, *[writes $= f(x)$]* because $f(i)$ is zero. *[writes $f(i) = 0$ underneath]* And that's what we wanted, right? And this is equal to $\bar{f}$ of $\bar{x}$. *[writes $= \bar{f}(\bar{x})$]* Okay, yeah.

Student Question

Don't you need to first show that $\bar{f}$ is a homomorphism, because we are using the argument that $\bar{f}(\bar{i})$ which is $\bar{f}(\bar{0})$ is zero, but...

Spoken Transcript [00:56:55 - 00:57:33]

No, I'm just showing it's well-defined. I only want to show this is that. *[points at $\bar{f}(\bar{x'}) = \bar{f}(\bar{x})$]*

All this f stuff is fine, because f is a homomorphism. Remember that? There, f is a homomorphism. *[points at the top board]*

We have to do all the things you're saying, but there's a certain order you do it. So, I define f — well, this is part of the definition of $\bar{f}$. I propose a definition here. The problem with this definition is it may depend on which element I check. And now I

check that it doesn't depend. But I haven't proven this is a homomorphism yet. I will have to do that.

Student Question

Where do we know that i is in the kernel?

Spoken Transcript [00:57:37 - 00:57:45]

Well, that's here. *[points at $x' = x + i$]* For another name, that's what it means: it's in the kernel.

Student Answer

Oh yeah, makes sense.

Spoken Transcript [00:57:48 - 00:58:44]

Okay, so, maybe if that were confusing, I say it again, because it's kind of important.

I just propose this as a definition, but it has the flaw that I have to check this definition does not depend on which element of the equivalence class I chose. And this is another element of the equivalence class. Every element of the equivalence class... their difference is an element of the kernel. So, that's the equation there. I use this equation to show this thing is well-defined.

And actually I wrote out all the steps. In my experience in teaching, it doesn't help writing all the steps. It just doesn't help. But I did it, maybe it was a mistake. Because there's a lot of steps, and every time you have to check what I'm using at every equality. And you have to do that. I already know, but you have to do it, and if I just write it, it doesn't always help. But...

Definition of $\bar{f}$ and Proof of Well-Definedness

We define the induced map:

$$\bar{f}\colon R/\ker(f) \longrightarrow S, \qquad \bar{f}(\bar{x}) := f(x). \tag{4.24}$$

Claim 4.28 ($\bar{f}$ *is Well-Defined*): The map $\bar{f}$ is independent of the choice of representative in the coset $\bar{x}$.

Proof

Suppose $\bar{x}' = \bar{x}$ in $R/\ker(f)$. By definition of congruence modulo $\ker(f)$, there exists an element $i \in \ker(f)$ such that:

$$x' = x + i.$$

Applying the definition of $\bar{f}$ and the additive homomorphism property of f:

$$\begin{aligned}
\bar{f}(\bar{x}') &= f(x') && \text{(by definition of } \bar{f}) \\
&= f(x+i) && \text{(since } x' = x+i) \\
&= f(x) + f(i) && \text{(since } f \text{ is a ring homomorphism)} \\
&= f(x) + 0_S && \text{(since } i \in \ker(f) \implies f(i) = 0_S) \\
&= f(x) \\
&= \bar{f}(\bar{x}).
\end{aligned}$$

Thus, $\bar{f}(\bar{x}') = \bar{f}(\bar{x})$, proving that $\bar{f}$ is well-defined as a set-theoretic function.

Q.E.D.

Spoken Transcript [00:58:44 - 01:00:17]

So, that was a little painful. But we're not done with the pain: this just shows it's well-defined. So, we have $\bar{f}\colon R/\ker(f) \to S$ is a function now. *[writes $\bar{f}\colon R/I \to S$ function]* And we have to show all the properties to show it's a ring homomorphism.

So, we'd have to show, for example, $\bar{f}(\bar{0})$... well, what is this? We use our definition, and that's equal to $f(0)$, that's what that definition says, and $f(0)$ has to be zero because f itself is a homomorphism. *[writes $\bar{f}(\bar{0}) \overset{def}{=} f(0) = 0$]* So, that's good: it's sending zero to zero. That's maybe what you were worried about. Now we've shown it.

Well, we also have to show $\bar{f}(\bar{x}+\bar{y}) = \bar{f}(\bar{x}) + \bar{f}(\bar{y})$. *[writes $\bar{f}(\bar{x}+\bar{y}) = \bar{f}(\bar{x}) + \bar{f}(\bar{y})$]* We also have to show this. And now you just have to expand these sides completely, and if you do that, you'll get a chain like this, more or less similar to that, and you'll find that it's compatible with the definition. Okay, so, expand yourself. *[writes "expand yourself"]* As I said, you'll get a chain where...

Board Transition and Cleaning

The lecturer pulls down the chalkboard and cleans the panel from timestamp 01:00:17 to 01:01:06 to prepare for the multiplicative verification of $\bar{f}$.

Spoken Transcript [01:01:06 - 01:01:57]

That's the additive structure. Then you also have to show that $\bar{f}$ respects the multiplicative structure: *[writes Check $\bar{f}(\bar{x}\cdot\bar{y}) \overset{?}{=} \bar{f}(\bar{x})\cdot\bar{f}(\bar{y})$]*

Check this, and if we do that, this is defined to be... first we have the definition of the product, that's this, *[writes $\bar{f}(\overline{xy})$]* and then this is the definition of $\bar{f}$, that's $f(xy)$, *[writes $= f(xy)$]* and if we expand this side, this is $f(x)\cdot f(y)$. *[writes $= f(x)f(y)$]* So, you see, and that's true because f is a homomorphism. *[writes True because f is hom $R \to S$.]*

So, true because f was born a homomorphism. Okay.

Spoken Transcript [01:01:57 - 01:02:41]

So, you have to concentrate with the bars, but really the proofs of this is just unraveling definitions.

So, that's the first step in the proof. I remind you here: we have this $R \xrightarrow{f} S$, *[writes $f\colon R \to S$]* and the kernel of f is my ideal, *[writes $\ker(f) \subset R$]* and we've defined this $\bar{f}$ from Rmod the kernel of f to S. *[writes $\bar{f}\colon R/\ker(f) \to S$]* Okay? And this is a homomorphism, *[writes is a hom]* meaning that we know this is a ring — that was the first part of this class — we've defined $\bar{f}$, and then the claim is that's a ring homomorphism.

Proof that $\bar{f}$ is a Ring Homomorphism

Claim 4.29 ($\bar{f}$ *is a Ring Homomorphism*): The well-defined map $\bar{f}\colon R/\ker(f) \to S$ preserves zero, identity, addition, and multiplication.

Proof

Using the coset definitions in $R/\ker(f)$ and the ring homomorphism properties of $f\colon R \to S$:

(1) Preservation of Zero:

$$\bar{f}(\bar{0}) = f(0_R) = 0_S.$$

(2) Preservation of Identity:

$$\bar{f}(\bar{1}) = f(1_R) = 1_S.$$

(3) Preservation of Addition:

$$\begin{aligned} \bar{f}(\bar{x}+\bar{y}) &= \bar{f}(\overline{x+y}) && \text{(by definition of addition in } R/\ker(f)) \\ &= f(x+y) && \text{(by definition of } \bar{f}) \\ &= f(x)+f(y) && \text{(since } f \text{ is a ring homomorphism)} \\ &= \bar{f}(\bar{x})+\bar{f}(\bar{y}). && \text{(by definition of } \bar{f}) \end{aligned}$$

(4) Preservation of Multiplication:

$$\begin{aligned} \bar{f}(\bar{x}\cdot\bar{y}) &= \bar{f}(\overline{x\cdot y}) && \text{(by definition of multiplication in } R/\ker(f)) \\ &= f(x\cdot y) && \text{(by definition of } \bar{f}) \\ &= f(x)\cdot f(y) && \text{(since } f \text{ is a ring homomorphism)} \\ &= \bar{f}(\bar{x})\cdot\bar{f}(\bar{y}). && \text{(by definition of } \bar{f}) \end{aligned}$$

Therefore, $\bar{f}$ is a homomorphism of commutative rings. Q.E.D.

Step 2: Surjectivity of $\bar{f}$ onto $\mathrm{Im}(f)$

Spoken Transcript [01:02:41 - 01:04:17]

All right, so, that's the first step. Next, Step 2: what is the image of $\bar{f}$? *[writes Step 2: What is $\mathrm{Im}(\bar{f})$?]*

That means, where do we go here? And the answer is: it's the same as the image of

f. So, actually $\bar{f}$ goes from R mod $\ker(f)$ to the image of f. *[writes $\bar{f}\colon R/\ker(f) \to \mathrm{Im}(f)$]* This is some subring of S. *[writes $\cap$ and S under $\mathrm{Im}(f)$]*

Why? Because the definition just says that if I take an equivalence class, I choose an element, I go to $f(x)$. So, everything I get is in the image of f. On the other hand, if I'm in the image of f, that means that I come from some element of R, and I can take its equivalence class. So, I get exactly the image of f.

That's Step 2, and this is a surjection. So, I call $\bar{f}$ the same thing, so, $\bar{f}$ is surjective onto the image of f, okay? *[writes $\bar{f}$ is surjective onto the image of f]* Sometimes in math, you write this kind of two arrows for surjectivity; do you follow that kind of stuff or not? Okay, I won't give you more notation now.

Surjectivity of the Induced Homomorphism onto the Image

Claim 4.30 (*Surjectivity of $\bar{f}$ onto $\mathrm{Im}(f)$*): The image of $\bar{f}$ is precisely the subring $\mathrm{Im}(f) \subseteq S$. Consequently, the restricted codomain map:

$$\bar{f}\colon R/\ker(f) \twoheadrightarrow \mathrm{Im}(f) \tag{4.25}$$

is a surjective ring homomorphism.

Proof

We verify mutual inclusion of sets:

- $\mathrm{Im}(\bar{f}) \subseteq \mathrm{Im}(f)$: For any coset $\bar{x} \in R/\ker(f)$, $\bar{f}(\bar{x}) = f(x) \in \mathrm{Im}(f)$ by definition of the image of f.
- $\mathrm{Im}(f) \subseteq \mathrm{Im}(\bar{f})$: Let $y \in \mathrm{Im}(f)$. By definition of the image, there exists an element $x \in R$ such that $f(x) = y$. Considering the coset $\bar{x} \in R/\ker(f)$, we have:

  $$\bar{f}(\bar{x}) = f(x) = y.$$

  Thus, $y \in \mathrm{Im}(\bar{f})$.

Hence, $\mathrm{Im}(\bar{f}) = \mathrm{Im}(f)$, proving that $\bar{f}$ maps $R/\ker(f)$ surjectively onto the subring $\mathrm{Im}(f)$. Q.E.D.

Step 3: Injectivity and Trivial Kernel of $\bar{f}$

Spoken Transcript [01:04:17 - 01:06:42]

So, first we define this $\bar{f}$, and $\bar{f}$ is defined in like the only possible way you can. The second step is we realize it's surjective onto the image of f.

And the third step — and this one requires some real concentration — is: what is the kernel? What is the kernel of $\bar{f}$? *[writes Step 3: What is the $\ker(\bar{f})$?]*

Let's try to think about that. That means: suppose I have $\bar{x}$ in R mod I... *[writes Suppose $\bar{x} \in R/I$ and $\bar{f}(\bar{x}) = 0 \in S$, $\bar{x} \in \ker(\bar{f})$]* This is another way of saying $\bar{x}$ is in the kernel of $\bar{f}$. Suppose I have that.

Then what can I conclude? Well, I can use the definition: the definition of this is $f(x)$, right? So, then this implies that actually $f(x)$ is equal to zero in S, right? *[writes*

Then $f(x) = 0 \in S$] And this implies x is in the kernel of f. *[writes $\implies x \in \ker(f)$]*

All right? So, what have we done here? We said: we start with any equivalence class; if it's in the kernel of $\bar{f}$, then that element has to be in the kernel of f. And that means that equivalence class is the zero equivalence class, because it's equivalent to zero. This means that x is equivalent using the kernel relation to zero. *[writes $\implies x \equiv_{\ker(f)} 0$]*

And this means that actually, you know, x is equal to zero in R mod I. *[writes $\implies \bar{x} = \bar{0} \in R/I$]*

So, that proves that the kernel of $\bar{f}$ is zero. So, the conclusion here — let me sorry... that's Step 3 — the conclusion is the kernel of $\bar{f}$ is zero.

Injectivity of $\bar{f}$ via Triviality of the Kernel

Claim 4.31 ($\bar{f}$ *is Injective*): The kernel of the homomorphism $\bar{f}\colon R/\ker(f) \to \mathrm{Im}(f)$ is the zero ideal of the quotient ring:

$$\ker(\bar{f}) = (\bar{0}). \tag{4.26}$$

Consequently, $\bar{f}$ is injective.

Proof

Let $\bar{x} \in \ker(\bar{f}) \subseteq R/\ker(f)$. By definition of the kernel:

$$\bar{f}(\bar{x}) = 0_S.$$

Using the definition of the map $\bar{f}(\bar{x}) := f(x)$, this means:

$$f(x) = 0_S \implies x \in \ker(f).$$

By definition of cosets modulo $\ker(f)$, an element belongs to the ideal if and only if its coset coincides with the zero coset:

$$x \in \ker(f) \iff x \equiv_{\ker(f)} 0 \iff \bar{x} = \bar{0} \in R/\ker(f).$$

Hence, the only element in $\ker(\bar{f})$ is the zero coset $\bar{0}$, establishing:

$$\ker(\bar{f}) = (\bar{0}).$$

By the injectivity criterion established earlier in Proposition 4.16, a ring homomorphism is injective if and only if its kernel is trivial. Therefore, $\bar{f}$ is injective. Q.E.D.

Conclusion of the Isomorphism: Combining the three steps:

- $\bar{f}\colon R/\ker(f) \to \mathrm{Im}(f)$ is a well-defined ring homomorphism (Step 1).
- $\bar{f}$ is surjective onto $\mathrm{Im}(f)$ (Step 2).
- $\bar{f}$ is injective, having trivial kernel (Step 3).

A bijective ring homomorphism is an isomorphism (Definition 2.23). Thus, $R/\ker(f) \cong \mathrm{Im}(f)$, which completes the proof of the First Isomorphism Theorem (Theorem 4.27).

Q.E.D.

Recapitulation and the Anatomy of the Theorem

Spoken Transcript [01:06:43 - 01:07:37]

So, those are the three steps. You have to define $\bar{f}$, you show it's surjective onto the image, and you show its kernel is zero. And the reason why the last one is confusing is: you have to understand who is zero where. The kernel of f is not zero, the kernel of f is the kernel of f. But we quotient by the kernel of f, so, the kernel of $\bar{f}$ is zero.

So, that's an admittedly confusing state of affairs, but you have to somehow come to terms with that.

Board Transition and Washing

The lecturer pulls down the upper blackboard and thoroughly cleans the lower right panel with a wet sponge from timestamp 01:07:37 to 01:07:47 to prepare space for the formal synthesis of the First Isomorphism Theorem.

Spoken Transcript [01:07:47 - 01:09:13]

So, I say it in words because it's confusing. We start with f from R to S. This has a kernel, which we call the kernel of f, *[writes $f\colon R \to S$ and "This has kernel $\ker(f)$"]* which could be anything — maybe zero, maybe not, could be anything in the world.

Then from this, we make $\bar{f}$ from R mod $\ker(f)$ to S — well, it goes to the image of f, that subring inside S. *[writes: $\bar{f}\colon R/\ker(f) \to \mathrm{Im}(f) \subset S$]* So, we study this, and the last step is that the kernel of $\bar{f}$...

This is an ideal in R mod $\ker(f)$, right? *[writes: $\ker(\bar{f}) \subset R/\ker(f)$]* Because it's an ideal here in this ring. And this ideal is the zero ideal — the zero ideal in this quotient ring. *[writes: $\ker(\bar{f}) = (0) \subset R/\ker(f)$]*

So, the kernel of f may not be zero, but when I consider the kernel of $\bar{f}$ in the quotient ring, it's zero. And the way to think about that is kind of — and we'll discuss this more tomorrow — is that you can write this kernel of $\bar{f}$ as kind of whatever was in the kernel before, but now we quotient out by f, and then that's just the zero group. *[writes: $\ker(\bar{f}) = \frac{\ker(f)}{\ker(f)} = 0$]*

That's a nice way to think about it. I'll explain tomorrow how this kind of logic makes sense. But roughly speaking: there was a kernel, but when we quotient out, we quotient out by that kernel, so, after we quotient out, we have no kernel. Okay?

Spoken Transcript [01:09:13 - 01:10:35]

So, these three steps, if you take Steps 1 to 3 together — because first we defined $\bar{f}$, we show it's surjective onto the image of f, and now we show the kernel is zero, so, that means it's also injective. I remind you: the kernel being zero implies that $\bar{f}$ is injective. *[writes: $\implies$ $\bar{f}$ is injective]* We started the lecture with that, I think (Recall: Proposition 4.16).

So, Steps 1 to 3 show that $\bar{f}$ from R mod $\ker(f)$ to the image of f is well-

defined, injective, and surjective, so, $\bar{f}$ is an isomorphism of rings. *[writes STEP 1–3: $\bar{f}\colon R/\ker(f) \to \mathrm{Im}(f)$, $\bar{f}$ is isom.]*

So, that's the end of the proof of the First Isomorphism Theorem. Okay.

There was a lot of chalk on the board, but if you really understand this, you'll see there's actually nothing happening in this proof. It's just somehow you have to follow the definitions at every stage. I think it's similar to when you really understand that linear algebra result: the fact that the image is the space mod the kernel, there's almost nothing happening in that proof. You almost... the definitions make it so, and that's what happens here also.

Summary: Anatomy of the First Isomorphism Theorem

The architecture of the First Isomorphism Theorem relies on distinguishing the three rings and their respective kernels:

Mapping	Domain / Codomain	Kernel
f	$R \longrightarrow S$	$\ker(f) \subset R$ (Arbitrary ideal of R)
$\bar{f}$	$R/\ker(f) \xrightarrow{\cong} \mathrm{Im}(f)$	$\ker(\bar{f}) = (\bar{0}) \subset R/\ker(f)$ (The zero ideal in $R/\ker(f)$)

$$f\colon R \longrightarrow S \qquad \text{with kernel } \ker(f) \subset R, \tag{4.27}$$

$$\bar{f}\colon R/\ker(f) \xrightarrow{\sim} \mathrm{Im}(f) \subset S \qquad \text{with kernel } \ker(\bar{f}) = (0) \subset R/\ker(f). \tag{4.28}$$

Heuristically, the kernel of f has been quotiented out by itself (the next lecture makes this precise):

$$\ker(\bar{f}) = \frac{\ker(f)}{\ker(f)} = 0.$$

Ideals Generated by Elements

Spoken Transcript [01:10:37 - 01:12:08]

So, let's do an example.

Maybe I have some room here, I can say: one of the things to think about rings and their ideals and quotient rings is that we need to have some way of writing down ideals, otherwise we can't have too much fun with examples.

A standard way of writing an ideal is: let x_1 to x_n be elements of the ring R, *[writes on the board: Let $x_1, \ldots, x_n \in R$]* then... they could also be infinite, but in this class typically it's finite, but anyway...

Then if you have some list of elements in R, the standard notation is you put parentheses around them, and you define an ideal. This is the definition of this: it is the ideal defined by this equation, *[writes then $(x_1, \ldots, x_n) \subset R$ is the ideal defined by]* so, I have to just tell you who is in this ideal: this is equal to those elements that are of the form

a_1x_1... in language you could say this is R-linear combinations of these elements: a_nx_n, where the a_i's are in R. *[writes* $(x_1,\dots,x_n)=\{a_1x_1+\cdots+a_nx_n \mid a_i\in R\}$*]* So, this is a particular set.

Spoken Transcript [01:12:08 - 01:13:39]

As I said, you could say in language: they're R-linear combinations of the x_i. *[writes: R-linear combs. of the x_i]* Other language is that this is the ideal generated by the x_i. That's also language used: ideal generated by the x_i. *[writes: ideal generated by the x_i]*

Those are two different ways to express the same thing. And it's just this set. So, in order for you to accept that this is an ideal, you have to show that... well, zero's here — of course zero's here if I take all these elements to be zero; you have to show if something is here, then its minus is here — I just flip all the signs of all the a_i's; you have to show if something's here, when I add it to something else, it's still here — and then I just add the coefficients of the linear combination; and finally, I have to show the multiplicative property: if something's here and I multiply it by an r, then it's still there — but view that r as a scalar, and it just multiplies all these coefficients. Okay?

So, you have to check it. Actually, I just checked it for you, but, you know.

So, this is a very nice way: if you take a ring, you can just pick any elements, and you get an ideal. It's the ideal generated by that. Are there questions about this? This is kind of a really fundamental way of generating ideals.

Definition of Ideals Generated by Elements

Let R be a commutative ring with identity 1.

Definition 4.32 (*Ideal Generated by Finitely Many Elements*): Let $x_1,\dots,x_n\in R$. The *"ideal generated by"* $x_1,\dots,x_n$, denoted by $(x_1,\dots,x_n)$, is the set of all R-linear combinations of the generators:

$$(x_1,\dots,x_n):=\left\{\sum_{i=1}^{n}a_ix_i \;\middle|\; a_1,\dots,a_n\in R\right\}. \tag{4.29}$$

When $n=1$, the ideal $(x)=\{ax \mid a\in R\}=xR$ is called a *"principal ideal"* (for $R=\mathbb{Z}$, these are the ideals $n\mathbb{Z}$ of Definition 2.33).

Verification of the Ideal Axioms for $(x_1,\dots,x_n)$

Proposition 4.33: For any elements $x_1,\dots,x_n\in R$, the set $(x_1,\dots,x_n)$ defined in Definition 4.32 is indeed an ideal of R.

Proof

We check the two defining conditions of an ideal:

(1) Additive Subgroup:

- Setting $a_1=\cdots=a_n=0$ yields $0=\sum_{i=1}^{n}0\cdot x_i\in(x_1,\dots,x_n)$.
- For any element $u=\sum_{i=1}^{n}a_ix_i$, its additive inverse is $-u=\sum_{i=1}^{n}(-a_i)x_i\in$

$(x_1, \ldots, x_n)$.

- If $u = \sum_{i=1}^{n} a_i x_i$ and $v = \sum_{i=1}^{n} b_i x_i$, their sum is:

$$u + v = \sum_{i=1}^{n} (a_i + b_i) x_i \in (x_1, \ldots, x_n).$$

(2) Multiplicative Absorption: For any ring element $r \in R$ and $u = \sum_{i=1}^{n} a_i x_i \in (x_1, \ldots, x_n)$:

$$r \cdot u = r \sum_{i=1}^{n} a_i x_i = \sum_{i=1}^{n} (r a_i) x_i.$$

Since R is a ring, each coefficient $r a_i \in R$, so $r \cdot u \in (x_1, \ldots, x_n)$.

Q.E.D.

Beyond the Lecture: The Smallest Ideal Containing the x_i

The name *"ideal generated by the x_i"* reflects a minimality property: $(x_1, \ldots, x_n)$ is the smallest ideal of R containing $x_1, \ldots, x_n$. It contains each $x_i = 1 \cdot x_i$. Conversely, an ideal J containing $x_1, \ldots, x_n$ contains every $a_i x_i$ by absorption, hence every sum $\sum_{i=1}^{n} a_i x_i$ by additive closure, so $(x_1, \ldots, x_n) \subseteq J$.

Quotients of Univariate Polynomial Rings

Spoken Transcript [01:13:39 - 01:14:41]

I have a few minutes, let's do some examples.

So, now you have a very flexible way of generating really a lot of rings. And these rings already — the rings that you have the power to generate — have all sorts of interesting, different behavior. And part of the course is exploring some of that structure, by far not all of it.

Board Transition and Cleaning

The lecturer takes a wet sponge and washes the left panel of the lower chalkboard from timestamp 01:14:12 to 01:14:41 to prepare space for working out concrete polynomial quotient ring examples.

Spoken Transcript [01:14:41 - 01:16:10]

Okay, so, for example, I can take my ring R to be polynomials in one variable: *[writes: $R = \mathbb{C}[x]$]* $R = \mathbb{C}[x]$. You should be totally happy with this ring: it's polynomials with complex coefficients, all right?

And I can take an element here, f, to be like $x^2 + 1$. *[writes: $f = x^2 + 1 \in R$]* That's a polynomial. And then, I'm using the notation there: if you have one element, that's an element; if I put parentheses around it, this is the ideal. *[writes $(f) \subset R$ with an arrow to "ideal"]*

Very concretely, using that definition, this is just the ideal of f times a polynomial where the polynomial is in the ring. *[writes: $(f) = \{f \cdot p \mid p \in R\}$]* That's what it is: it's just one linear combination. So, all this is is polynomials that have an $x^2 + 1$ factor. Okay?

What we've done today is that we have a new ring now: Rmod this ideal. *[writes: $R/(f)$]* Rmod this ideal, this is the quotient ring. Are you happy to think about that? We take this polynomial ring, we take this ideal, and we have this quotient ring. I should put here $\mathbb{C}[x]$... *[writes: $\mathbb{C}[x]/(f)$]* Okay.

Do you accept that all the symbols on this board have been defined?

Spoken Transcript [01:16:10 - 01:16:41]

So, I can ask you now any question about this ring. For example: is this an integral domain? *[writes: Is this an int. domain?]*

What's your opinion about that?

Student Answer

No, because $(x - i)(x + i)$...

Spoken Transcript [01:16:43 - 01:17:31]

Yeah. So, the answer is no, *[writes "No!" and circles it]* and the reason is: I can take $x + i$ times $x - i$, because $x + i$ and $x - i$ are here, so, I can put bars, they're in the quotient ring. *[writes: $\overline{x + i} \cdot \overline{x - i}$]*

And the claim is that this is equal to, by definition, that, *[writes: $= \overline{x^2 + 1}$]* and then this is zero, *[writes: $= \bar{0}$]* right? This is the beginning of your argument. That's a totally solid start.

But to be an integral domain says that if these are zero... you know, what if one of these are zero? To be an integral... so, the answer is no, then you have to find two non-zero things that multiply to zero. You found two things that multiply to zero. Why are these non-zero?

Student Answer

That would be like their difference to zero... if you multiply $x^2 + 1$ by a polynomial, it must have degree at least 2.

Spoken Transcript [01:17:43 - 01:18:56]

Okay, that's the correct argument. So, now, this is the beginning to say it's no, then we have to show that $x + i$ is not equal to zero, and $x - i$ is not equal to zero. *[writes: But $\overline{x + i} \neq \bar{0}$ and $\overline{x - i} \neq \bar{0}$]*

If we have these three statements — that equals zero, the product was zero, and they're both not equal to zero — then it's not an integral domain. You agree with that?

And he gave the correct argument for that: because if $x+i$ were equivalent to zero, *[writes If $x+i \equiv 0$]* this means that $x+i$ is in this ideal generated by x^2+1, *[writes $\iff x+i \in (x^2+1)$]* but the ideal generated by x^2+1 has x^2+1 times things in it, times polynomials, and you can never lower the degree to a linear polynomial by taking a quadratic polynomial and multiplying it by another polynomial.

So, that's the full argument, the full painful argument that says the answer is no. Okay.

Zero Divisors in the Quotient Ring $\mathbb{C}[x]/(x^2+1)$

Example 4.34 (*Non-Domain Quotient Ring*): Consider the polynomial ring $R = \mathbb{C}[x]$ and the principal ideal generated by $f(x) = x^2+1$:

$$I = (x^2+1) = \{(x^2+1)p(x) \mid p(x) \in \mathbb{C}[x]\}.$$

The quotient ring $\mathbb{C}[x]/(x^2+1)$ is **not an integral domain**.

Proof

In $\mathbb{C}[x]$, the polynomial x^2+1 factors into linear factors:

$$x^2+1 = (x+i)(x-i).$$

In the quotient ring $\mathbb{C}[x]/(x^2+1)$, consider the cosets $\overline{x+i}$ and $\overline{x-i}$:

(1) Their Product is Zero:

$$\overline{x+i} \cdot \overline{x-i} = \overline{(x+i)(x-i)} = \overline{x^2+1} = \bar{0} \in \mathbb{C}[x]/(x^2+1).$$

(2) Neither Factor is Zero: Suppose $\overline{x+i} = \bar{0}$. Then $x+i \in (x^2+1)$, which means there exists $p(x) \in \mathbb{C}[x]$ such that:

$$x+i = (x^2+1)p(x).$$

If $p(x) = 0$, the right-hand side is $0 \neq x+i$. If $p(x) \neq 0$, comparing degrees gives:

$$1 = \deg(x+i) = \deg(x^2+1) + \deg(p) = 2 + \deg(p) \geq 2,$$

a contradiction. Thus, $\overline{x+i} \neq \bar{0}$. By identical degree considerations, $\overline{x-i} \neq \bar{0}$.

Since $\overline{x+i}$ and $\overline{x-i}$ are non-zero elements whose product is zero, they are zero divisors. Hence, $\mathbb{C}[x]/(x^2+1)$ is not an integral domain.

Q.E.D.

The Quotient $\mathbb{Q}[x]/(x^2+1)$ and the Gaussian Rationals

Spoken Transcript [01:18:56 - 01:20:11]

What about this one? *[writes $\mathbb{Q}[x]/(x^2+1)$]*

This is kind of cutting closer to what this class is about, actually. Is this an integral domain?

His argument you cannot use exactly here, because i is not a rational number, right? You agree you can't use the same argument. But I can ask the same question, though: what do you think the answer to that is?

This is really a question that the last part of the class is about. But I'm going to give you a proof of this. The answer is: this is an integral domain. *[writes is an int. dom.]* Actually, more than that: *[writes Field]* actually, it's a field!

It doesn't look like it immediately here, but I will try to give you a proof in the next few minutes about this.

There's a lot of ways to prove it, and I said a certain part of the class is about such things called field extensions. And if you take the successor in the spring, that's like entirely about this kind of stuff — at least Galois theory, etc. But it's a very good example to think about.

Spoken Transcript [01:20:11 - 01:20:56]

So far, the only thing you're supposed to understand is that the argument proposed for the complex numbers does not work here. But let's try to find an argument that does work, that's related to this lecture.

In some sense, that question is kind of like searching for the complex numbers. Because x^2+1, you know the roots, so, somehow built into that question is this quest for the complex numbers.

The Rational Quotient $\mathbb{Q}[x]/(x^2+1)$

Question 4.35: Is the quotient ring $\mathbb{Q}[x]/(x^2+1)$ an integral domain?

Answer 4.36: Yes! In fact, it is a field. Notice that the factorization $x^2+1=(x+i)(x-i)$ used over $\mathbb{C}$ is unavailable in $\mathbb{Q}[x]$, because $i \notin \mathbb{Q}$. In fact, x^2+1 is *"irreducible"* over $\mathbb{Q}$: it is not a product of two non-constant polynomials in $\mathbb{Q}[x]$, because such a factorization would consist of two linear factors and give a rational root of x^2+1. We will prove that $\mathbb{Q}[x]/(x^2+1)$ is isomorphic to the field of Gaussian rationals $\mathbb{Q}[i]$.

Construction of the Evaluation Homomorphism at i

Spoken Transcript [01:20:56 - 01:22:36]

We know the complex numbers, so, we write it down. *[writes: $\mathbb{C}$]* And I take $\mathbb{Q}$... *[writes $\mathbb{Q} \to \mathbb{C}$]* $\mathbb{Q}$ embeds in the complex numbers. There is no doubt about that. And now I look at the polynomial ring. *[writes $\cap$ and $\mathbb{Q}[x]$ under $\mathbb{Q}$]*

And we proved that if I can choose any value of x, I get a homomorphism from the polynomial ring (Recall: Theorem 3.38). Do you remember that?

So, I'm going to choose a homomorphism f that takes... for constants, it just commutes, but f on x is i. *[draws an arrow f from $\mathbb{Q}[x]$ to $\mathbb{C}$ and writes $f(x)=i$]* I just do that. I get to do it. Because I told you: for the polynomial ring, if I know how the original ring maps to any other ring, I can always extend the map to anything I want for

x. So, I get to just do this. It's somehow my right from the previous lectures.

In other words, I define a homomorphism f from polynomials to the complex numbers, *[writes: $f\colon \mathbb{Q}[x] \to \mathbb{C}$]* where f of a polynomial equals the polynomial evaluated at i. *[writes $f(p(x)) = p(i)$]* Okay?

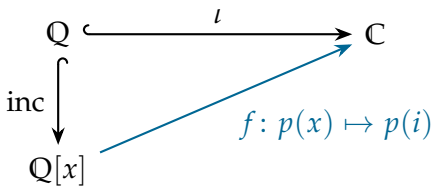
Do you accept this?

And you can, even if you forgot that, you can just check that this is a legal definition: given any polynomial, I just evaluate at i, get a complex number, and it's a ring homomorphism. *[writes: Ring hom.]*

So, as I said, you can just prove that from scratch in your head here, because I gave you the formula; or you can remember it has to be true because I said you could always do that — I can just take this x going to anywhere. Okay.

The Evaluation Homomorphism at i

Let $\iota\colon \mathbb{Q} \hookrightarrow \mathbb{C}$ denote the canonical inclusion of fields. By the universal property of polynomial rings (Theorem 3.38), specifying the image of the indeterminate x uniquely determines a ring homomorphism extending ι.



Definition 4.37 (*Evaluation Homomorphism at i*): We define the *"evaluation homomorphism"* at the imaginary unit $i \in \mathbb{C}$ by:

$$f\colon \mathbb{Q}[x] \longrightarrow \mathbb{C}, \qquad f(p(x)) := p(i). \tag{4.30}$$

Explicitly, if $p(x) = \sum_{k=0}^{m} a_k x^k \in \mathbb{Q}[x]$, then $f(p(x)) = \sum_{k=0}^{m} a_k i^k \in \mathbb{C}$.

Homomorphism Property Verification: For any $p(x), q(x) \in \mathbb{Q}[x]$:

$$\begin{aligned} f(p(x)+q(x)) &= (p+q)(i) = p(i)+q(i) = f(p(x))+f(q(x)), \\ f(p(x)\cdot q(x)) &= (p\cdot q)(i) = p(i)\cdot q(i) = f(p(x))\cdot f(q(x)), \\ f(1) &= 1. \end{aligned}$$

The middle equalities, such as $(p \cdot q)(i) = p(i) \cdot q(i)$, are not automatic: they hold because f is the homomorphism $\hat{\iota}$ of Theorem 3.38 for $\phi = \iota$ and $y = i$, whose proof checks them for arbitrary polynomials. Thus, f is a well-defined homomorphism of commutative rings with identity.

Image of the Evaluation Homomorphism and the First Isomorphism Theorem

Spoken Transcript [01:22:36 - 01:23:27]

What is the image of this? *[writes: What is* $\mathrm{Im}(f) = ?$*]*

What is the image, do you know what that is? Yeah?

Student Answer

I mean, if you consider polynomials of degree 1 with the constants, you get the Gaussian integers?

Spoken Transcript [01:23:27 - 01:24:50]

Yeah, and you don't get any more than that, because if you take a polynomial, all the quadratic terms give you -1, and all you get is i. So, the answer is: the image of this is what we'd call the Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$)... *[writes:* $\mathrm{Im}(f) = \mathbb{Q}[i]$*]*

If you take a polynomial with rational coefficients and you evaluate it at i, all the even terms just give you rational numbers, and all the odd terms give you i's, and the coefficients are all rational numbers. And you can get anyone you want here.

So, that's kind of interesting. That says that... so, now we'll use our First Isomorphism Theorem! *[writes: First Isom. Thm]*

That says that $\mathbb{Q}[x]$mod the kernel of f is isomorphic to the image of f, which is the Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$), which, by the way, is a field! *[writes:* $\mathbb{Q}[x]/\ker(f) \cong \mathrm{Im}(f) = \mathbb{Q}[i]$*]* It's a domain, and it's a field.

So, now it would be nice to know what this kernel is. What is the kernel? *[writes: What is the* $\ker(f)$*?]*

Image of f and Application of the First Isomorphism Theorem

Proposition 4.38 (*Image of the Evaluation Map*): The image of the evaluation homomorphism $f\colon \mathbb{Q}[x] \to \mathbb{C}$ is the field of Gaussian rationals (Example 3.28):

$$\mathrm{Im}(f) = \mathbb{Q}[i] := \{a + bi \mid a, b \in \mathbb{Q}\} \subset \mathbb{C}. \tag{4.31}$$

Proof

Let $p(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Q}[x]$. Since $i^2 = -1$, every even power $i^{2k} = (-1)^k \in \mathbb{Q}$ and every odd power $i^{2k+1} = (-1)^k i$. Collecting the even and odd terms yields:

$$p(i) = \left(\sum_{k\geq 0} a_{2k}(-1)^k\right) + \left(\sum_{k\geq 0} a_{2k+1}(-1)^k\right) i = a + bi,$$

where $a, b \in \mathbb{Q}$. Conversely, for any $a + bi \in \mathbb{Q}[i]$, the linear polynomial $p(x) = bx + a \in \mathbb{Q}[x]$ satisfies $f(p(x)) = a + bi$. Thus, $\mathrm{Im}(f) = \mathbb{Q}[i]$. Q.E.D.

Theorem 4.39 (*First Isomorphism Theorem Application*): Applying the First Isomorphism Theorem (Theorem 4.27) to $f\colon \mathbb{Q}[x] \to \mathbb{C}$:

$$\frac{\mathbb{Q}[x]}{\ker(f)} \cong \mathrm{Im}(f) = \mathbb{Q}[i]. \tag{4.32}$$

Since $\mathbb{Q}[i]$ is a subfield of $\mathbb{C}$ (Definition 0.25), the quotient $\mathbb{Q}[x]/\ker(f)$ is an integral domain and a field.

Terminology Note: Gaussian Rationals vs. Gaussian Integers

In the lecture audio, the professor refers to $\mathbb{Q}[i] = \{a + bi \mid a, b \in \mathbb{Q}\}$ as the *"Gaussian integers"*. In standard algebraic terminology, the Gaussian integers are $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ (Definition 1.1), while $\mathbb{Q}[i] = \mathrm{Frac}(\mathbb{Z}[i])$ is the field of *"Gaussian rationals"* (Example 3.28), often written $\mathbb{Q}(i)$. The blackboard notation $\mathbb{Q}[i]$ is preserved verbatim.

Computing the Kernel via Polynomial Long Division

Spoken Transcript [01:24:50 - 01:26:25]

Well, one thing that's definitely in the kernel is this polynomial: $x^2 + 1$ is definitely in the kernel. *[writes: $x^2 + 1 \in \ker(f)$]* Because if I take this polynomial — that's the P — if I take f... I write it here, it's closer to the definition: $f(x^2 + 1)$, by definition this is equal to $i^2 + 1$, which is zero, *[writes: $f(x^2 + 1) = i^2 + 1 = 0$]* right?

So, whatever the kernel is, it definitely has this polynomial. Okay?

Then we'd like to see whether there's anything else in the kernel. Once this is in the kernel, then the whole ideal generated by this polynomial is also in the kernel. *[writes $(x^2 + 1) \subset \ker(f)$]* Because if I take this polynomial and multiply it by something else, it'll still be in the kernel.

So, the question now is: is there anything else? The question here is: is the ideal generated by this polynomial equal to the kernel of f? *[writes Is $(x^2 + 1) = \ker(f)$?]*

Okay? If the answer is yes, then I've answered that question! Because here it is, it's $\mathbb{Q}[x]$ mod ... and spoiler: it's going to be yes. If it were yes, then I will have answered this question, because I've computed the kernel here, and I've showed it's isomorphic to the ring of Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$).

Actually, that is a way of writing the ring of Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$) in slightly hidden form. All I have to do is prove this thing is the kernel.

Containment of the Principal Ideal $(x^2 + 1)$ in the Kernel

Lemma 4.40 (*Ideal Containment in the Kernel*): The principal ideal $(x^2 + 1)$ is contained in $\ker(f)$:

$$(x^2 + 1) \subseteq \ker(f). \tag{4.33}$$

Proof

Evaluating f on the generator x^2+1:

$$f(x^2+1) = i^2+1 = -1+1 = 0 \implies x^2+1 \in \ker(f).$$

Since $\ker(f)$ is an ideal of $\mathbb{Q}[x]$ (Proposition 4.13), it must absorb multiplication by any polynomial $q(x) \in \mathbb{Q}[x]$:

$$\forall q(x) \in \mathbb{Q}[x] \implies q(x)(x^2+1) \in \ker(f).$$

Hence, $(x^2+1) \subseteq \ker(f)$. Q.E.D.

Board Transition and Washing

The lecturer takes a wet sponge and cleans the lower blackboard from timestamp 01:26:25 to 01:27:00 to prepare room for polynomial long division.

Spoken Transcript [01:26:25 - 01:28:30]

Let us try to think really what we have to do to prove it's the kernel. You have to actually understand the definitions of the class.

We have f from $\mathbb{Q}[x]$ to $\mathbb{C}$, *[writes: $f: \mathbb{Q}[x] \to \mathbb{C}$]* and f eats polynomials — that's an element here — and it gives back a complex number, which is $p(i)$. *[writes: $f(p(x)) = p(i)$]*

And the kernel of f: a polynomial is in the kernel of f if and only if $p(i)$ is zero. *[writes $\ker(f) \ni p(x) \iff p(i) = 0$]* In real life, you say i is a root of p, *[writes "i is root of p"]* right? So, the ones in the kernel are those polynomials where i is a root. Okay?

So, we have to see how that's related to this x^2+1.

There's different ways to say this, but we could do the following thing: we try to divide $p(x)$ by x^2+1, *[writes: $x^2+1 \mid p(x)$]* right? This is some polynomial, and we try to divide it, and everything's happening in this algebra of polynomials $\mathbb{Q}[x]$. *[writes: $\mathbb{Q}[x]$]*

This is what's called polynomial long division, which you should have learned in school: as you have a polynomial, you can start dividing it. And the answer is that this will be equal to some polynomial $q(x)$ times x^2+1 plus a remainder term. *[writes $p(x) = q(x) \cdot (x^2+1) + r(x)$]*

Do you remember polynomial long division? You just keep taking out the highest powers all the way down. I don't know, did you learn this stuff?

If you do polynomial long division, you start dividing, and you'll eventually get some polynomial times this, plus a remainder term. And this remainder term, since this is a quadratic polynomial, this remainder term is a linear polynomial! *[writes: linear poly.]* So, this remainder term is some $r_1x + r_2$. *[writes $r_1x + r_2$]* Of course, these could be zero. All right.

Polynomial Division by x^2+1

Lemma 4.41 (*Division Algorithm in $\mathbb{Q}[x]$*): For any polynomial $p(x) \in \mathbb{Q}[x]$, there exist unique polynomials $q(x), r(x) \in \mathbb{Q}[x]$ such that:

$$p(x) = q(x)(x^2+1) + r(x), \quad \text{with } r = 0 \text{ or } \deg(r) < 2. \tag{4.34}$$

In particular, the remainder $r(x)$ is a linear polynomial with rational coefficients:

$$r(x) = r_1 x + r_2, \qquad r_1, r_2 \in \mathbb{Q}. \tag{4.35}$$

This is the case $K = \mathbb{Q}$, $g = x^2+1$ of division with remainder for polynomials (Fact 0.31); the lecture uses it without proof.

Spoken Transcript [01:28:30 - 01:29:15]

That's polynomial long division, that's something you learn in school, and you should remember it now.

Now, I just stick i in this equation: *[writes: $p(i) = q(i)(i^2+1) + r(i)$]* $p(i)$... I put i's everywhere: $q(i) \cdot (i^2+1) + r(i)$.

And now, the assumption is $p(i) = 0$, so, this side is zero. *[writes 0 under $p(i)$]* This term is also zero, because i^2+1 is zero. *[crosses out (i^2+1)]*

That means that the conclusion is $r(i)$ is zero! *[writes: $\implies r(i) = 0$]*

How could that be? r is a linear polynomial, the remainder is a linear polynomial; how could it be zero? There's only one way it could be zero. Which is what? Yeah?

Student Answer

Both coefficients must be zero.

Spoken Transcript [01:29:20 - 01:30:14]

Yeah, the only way, because i is not... if this were a non-zero polynomial, these are rational numbers, the only way when you stick in i this is zero is if this is the zero polynomial! Because i is not a rational number.

So, this implies that r is the zero polynomial, *[writes $\implies r = 0$]* so, actually it's not here, and this p is in the ideal! *[crosses out $r(x)$ in the equation]*

And that finishes the argument, actually.

So, the conclusion is that $p(i) = 0$ implies $p(x)$ is in this ideal. *[writes: $p(i) = 0 \implies p(x) \in (x^2+1)$]*

And the final conclusion, using the First Isomorphism Theorem, is this: $\mathbb{Q}[x]$ mod (x^2+1) is isomorphic to this ring of Gaussian integers (i.e., Gaussian rationals $\mathbb{Q}[i]$). *[writes: $\mathbb{Q}[x]/(x^2+1) \cong \mathbb{Q}[i]$]*

Okay. Made it in time. That's it. *[puts down the chalk and concludes the lecture]*

Proof that $\ker(f) = (x^2+1)$ and Final Isomorphism

Theorem 4.42 (*Equality of Kernel and Principal Ideal*)**:** The kernel of the evaluation homomorphism $f\colon \mathbb{Q}[x] \to \mathbb{C}$ at i is precisely the principal ideal generated by x^2+1:

$$\ker(f) = (x^2+1). \tag{4.36}$$

Proof

We established in Lemma 4.40 that $(x^2+1) \subseteq \ker(f)$. We now prove the reverse inclusion $\ker(f) \subseteq (x^2+1)$.

Let $p(x) \in \ker(f)$. By definition, $p(i) = 0$. By Lemma 4.41, divide $p(x)$ by x^2+1:

$$p(x) = q(x)(x^2+1) + r(x), \quad \text{where } r(x) = r_1x + r_2 \text{ with } r_1, r_2 \in \mathbb{Q}.$$

Evaluating both sides at $x = i$:

$$p(i) = q(i)(i^2+1) + r(i).$$

Since $p(i) = 0$ and $i^2+1 = -1+1 = 0$, we obtain:

$$0 = q(i)\cdot 0 + r(i) \implies r(i) = 0.$$

Substituting the linear form of $r(x)$:

$$r(i) = r_1 i + r_2 = 0 \implies r_2 + r_1 i = 0.$$

Since $r_1, r_2 \in \mathbb{Q}$ and $i \notin \mathbb{Q}$ (so 1 and i are linearly independent over $\mathbb{Q}$), this forces:

$$r_1 = 0 \quad \text{and} \quad r_2 = 0 \implies r(x) = 0.$$

Therefore, the remainder vanishes identically, and:

$$p(x) = q(x)(x^2+1) \in (x^2+1).$$

Thus, $\ker(f) \subseteq (x^2+1)$, which completes the proof that $\ker(f) = (x^2+1)$.

Q.E.D.

Corollary 4.43 (*Realization of the Gaussian Rationals as a Quotient Ring*)**:** Combining Theorem 4.39 and Theorem 4.42, there is a canonical isomorphism of fields:

$$\frac{\mathbb{Q}[x]}{(x^2+1)} \xrightarrow{\cong} \mathbb{Q}[i], \qquad \overline{p(x)} \mapsto p(i). \tag{4.37}$$

In particular, $\mathbb{Q}[x]/(x^2+1)$ is an integral domain and a field.

Summary of Key Concepts

- **Several Variables and the Binomial Formula:** $R[x,y]$ consists of the finite sums $\sum a_{i,j}x^i y^j$, or equivalently $R[x,y] = (R[x])[y]$. The binomial formula $(x+y)^n = \sum_{a=0}^{n} x^{n-a}y^a\binom{n}{a}$ holds in every commutative ring; since $p \mid \binom{p}{a}$ for $0 < a < p$, it becomes $(x+y)^p = x^p + y^p$ over $\mathbb{Z}/p\mathbb{Z}$.
- **Polynomial and Power Series Rings:** The rings $R[x]$ and $R[[x]]$ preserve the integral domain property of R, relying on the highest-degree terms and lowest-degree orders, respectively.
- **Laurent Series and Rational Functions:** The field of rational functions $R(x)$ is $\mathrm{Frac}(R[x])$, while over a field K the field of formal Laurent series $K((x))$ (finitely many negative powers) is naturally isomorphic to $\mathrm{Frac}(K[[x]])$.

- **Ideals and Well-Defined Quotients:** An ideal $I \subset R$ requires additive closure and multiplicative absorption ($R \cdot I \subseteq I$). In constructing the quotient ring R/I, the additive subgroup condition guarantees well-defined addition, while the absorption property is strictly required to ensure well-defined coset multiplication.
- **The First Isomorphism Theorem for Rings:** Every ring homomorphism $f\colon R \to S$ induces a canonical ring isomorphism $\bar{f}\colon R/\ker(f) \xrightarrow{\cong} \mathrm{Im}(f)$ via $\bar{x} \mapsto f(x)$.
- **Generated Ideals and Polynomial Quotients:** Elements $x_1, \ldots, x_n$ generate the ideal $(x_1, \ldots, x_n)$ of their R-linear combinations. Modding out by x^2+1 over $\mathbb{C}$ produces zero divisors via $(x+i)(x-i) \equiv 0$, whereas over $\mathbb{Q}$, evaluation at i, the First Isomorphism Theorem and division with remainder by x^2+1 give $\mathbb{Q}[x]/(x^2+1) \cong \mathbb{Q}[i]$, the field of Gaussian rationals.

Wednesday 7 October 2026

Quotient Rings, the Correspondence Theorem for Ideals, and Maximal Ideals

Context and Overview

In this lecture, we explore the algebraic architecture of quotient rings and the structural behavior of ideals under ring homomorphisms:

- **Review of Quotient Rings and the Canonical Projection:** We revisit the coset construction of R/I for an ideal $I \subset R$, the well-definedness of its operations, and the canonical quotient homomorphism $q\colon R \to R/I$ with $\ker(q) = I$. We also recall the First Isomorphism Theorem for rings.
- **Pullbacks vs. Pushforwards of Ideals:** We prove that the preimage (pullback) $f^{-1}(J)$ of an ideal $J \subset S$ under any ring homomorphism $f\colon R \to S$ is always an ideal of R, whereas forward images generally fail to be ideals unless the map is surjective.
- **The Correspondence Theorem for Ideals:** We formulate and rigorously prove the Lattice Isomorphism Theorem for ideals, establishing an inclusion-preserving bijection between the ideals of the quotient ring R/K and the ideals of R containing the modulo ideal K.
- **Maximal Ideals and Fields:** We introduce maximal proper ideals, examine concrete examples in $\mathbb{Z}$, and establish the characterization that a ring R is a field if and only if the zero ideal (0) is maximal.

Review: Ideals and the Quotient Ring Construction

Spoken Transcript [00:00:05 - 00:01:10]

Okay, so we start. And I remind you of yesterday — it wasn't a long time ago, so you probably haven't forgotten.

So, if we have an ideal inside a ring... Typically, when I write R, it's a ring; I is an ideal, so it's an ideal *[writes $I \subset R$, with an arrow from I to "ideal" and one from R to "ring"]* in a ring.

Then I want to say there are three main things to take away from yesterday. The first *[writes (i) Construction of R/I]* is the construction of the quotient R/I (Recall: Theorem 4.26) — and we went through this rather carefully, not fully; there were things to check that are left to you, but this is equal to, as a set, the set of equivalence classes of this

relation $\equiv_I$ *[writes = set of equiv. classes of $\equiv_I$]*. And then we defined on top of that — I don't review it here, but — we defined addition, multiplication; there's a zero element and a one element (Recall: Definitions 4.20, 4.21 and 4.23). *[writes $(R/I, +, \cdot, \bar{0}, \bar{1})$]*

Review: Construction of the Quotient Ring

Recap: The quotient ring was constructed in the previous lecture: congruence modulo I (Definition 4.17, Proposition 4.18), the quotient set (Definition 4.19), the operations and their well-definedness (Definition 4.21, Proposition 4.22, Definition 4.23, Proposition 4.24), and the resulting ring (Theorem 4.26). The lecturer reviews it as the first of three main points.

Definition 4.44 (*Quotient Ring*): Let R be a ring and $I \subset R$ an ideal. The quotient ring R/I is defined as the set of equivalence classes under the congruence relation modulo I:

$$x \equiv_I y \iff x - y \in I.$$

The equivalence class of an element $x \in R$ is the coset:

$$\bar{x} := x + I = \{x + a \mid a \in I\}.$$

As a set:

$$R/I := \{\bar{x} \mid x \in R\}.$$

Equipped with the operations:

$$\bar{x} + \bar{y} := \overline{x + y}, \tag{4.38}$$

$$\bar{x} \cdot \bar{y} := \overline{x \cdot y}, \tag{4.39}$$

additive identity $\bar{0} = 0 + I = I$, and multiplicative identity $\bar{1} = 1 + I$, the algebraic structure $(R/I, +, \cdot, \bar{0}, \bar{1})$ forms a well-defined ring.

Well-Definedness of Operations on Cosets: The well-definedness of addition and multiplication relies directly on I being an ideal:

- **Addition:** If $\bar{x} = \bar{x}'$ and $\bar{y} = \bar{y}'$, then $x - x' \in I$ and $y - y' \in I$. Since I is an additive subgroup, $(x + y) - (x' + y') = (x - x') + (y - y') \in I$, so $\overline{x + y} = \overline{x' + y'}$.
- **Multiplication:** We have $xy - x'y' = (x - x')y + x'(y - y')$. Since I absorbs elements of R from both sides, $(x - x')y \in I$ and $x'(y - y') \in I$, which implies their sum lies in I, hence $\overline{xy} = \overline{x'y'}$.

The Canonical Quotient Map

Spoken Transcript [00:01:10 - 00:02:45]

So, then something I didn't say as explicitly, but will be the topic today, is that there is a map called the quotient map. We'll discuss this today. I didn't say this yesterday, but: *[writes (ii) $q\colon R \to R/I$]* R to R/I, and this map takes an element here to its equivalence class: *[writes $q(x) = \bar{x}$]* $q(x) = \bar{x}$.

Okay? So, it was implicit last time, but now it'll be explicit. So, after we've made this construction, there's a rule that if I take an element here, I can take it to its equivalence

class. You accept that, right? It's the most basic way.

And the claim, which is implicit in the construction, but now I make explicit, is that this q — which is called the quotient map, that's why I called it q — is a ring homomorphism. *[writes Claim: q is ring hom.]* These are both rings; this is a ring homomorphism. And why is that? That's exactly just by the understanding of the definition: how we define addition here and multiplication here.

So, to check this, we'd have to check something like: what is $q(x+y)$? *[writes $q(x+y) = \overline{x+y}$]* This is, by this map, that. And what is $q(x)+q(y)$? That's, by this map, that *[writes $q(x)+q(y) = \bar{x}+\bar{y}$]*. And why are they equal? Well, that's how we define addition here. *[underlines the two terms]* Okay? Is that clear?

And the same is for multiplication. This is in some sense a review of the definition: what is $q(xy)$? By this map, it's the equivalence class of xy. *[writes $q(xy) = \overline{xy}$]* What's $q(x)q(y)$? *[writes $q(x)\cdot q(y) = \bar{x}\cdot\bar{y}$]* It's that. And why are these equal? Because that's how we define multiplication in the quotient.

Definition 4.45 (*Quotient Map*): Let R be a ring and $I \subset R$ an ideal. The *"canonical quotient map"* (or projection map) is defined by:

$$q\colon R \to R/I, \quad x \mapsto q(x) := \bar{x} = x + I. \tag{4.40}$$

Claim 4.46 (*Homomorphism Property of the Quotient Map*): The projection map $q\colon R \to R/I$ is a ring homomorphism.

Proof

By definition of the algebraic operations in the quotient ring R/I:

$$q(x+y) = \overline{x+y} = \bar{x}+\bar{y} = q(x)+q(y), \tag{4.41}$$

$$q(x\cdot y) = \overline{x\cdot y} = \bar{x}\cdot\bar{y} = q(x)\cdot q(y). \tag{4.42}$$

Furthermore, $q(1) = \bar{1} = 1 + I$. Thus, q preserves addition, multiplication, and the multiplicative identity, so it is a ring homomorphism (Definition 1.18). Q.E.D.

Spoken Transcript [00:02:45 - 00:04:13]

So, it's another way to think about our construction: it is defined so that the map that takes x to its equivalence class is a ring homomorphism. That's another way to think about it.

What is that? Is that my phone? *[microphone emits loud electronic static and whistling noise; lecturer checks pocket and audio pack]* What aspect of me is at fault? *[checks transmitter]* Ah! I think it was just not twisted correctly... I'm not sure it was my fault, but... *[fiddles with audio jack]*

Anyway, so that's another way of thinking about what we did yesterday: is that we take R, we have the set of equivalence classes, and we want to define a ring structure on it for which this quotient — this map, x goes to its equivalence class — is a ring homomorphism. And we just define all the structure so it is. That's another way of thinking about what happened yesterday.

But anyway, the outcome is that we've constructed it, and this quotient map is a ring homomorphism. And just so we're somehow on the same page here, if it's a ring homomorphism...

Now it's turned it off... *[audio receiver clicks off]* Okay, it's turned it off. This is not good news.

Technical Difficulties: Microphone Interference

The wireless microphone emits feedback and electronic static. The lecturer pauses the mathematical exposition to troubleshoot the audio transmitter and cable connection.

Spoken Transcript [00:04:21 - 00:05:34]

Let's try this one. *[untangles replacement microphone cable and clips it on]* Okay, now I switch... Can you hear me?

We had this problem before, right? Could it be my phone? It can't be the phone... *[checks phone in pocket]* Does anyone have an idea? Now it's off. Maybe they were both on, that might have been the problem. All right.

How's that? We will see. So, maybe the solution was turning this one off, maybe.

Technical Difficulties Resolved

The lecturer replaces the faulty microphone transmitter with a secondary unit and confirms the audio is functioning properly before resuming the lecture.

Spoken Transcript [00:05:35 - 00:06:17]

Okay, so this is all review from yesterday. So, if we're on the same page, let's say this q, this ring homomorphism, is surjective *[writes q is surjective]*. That's clear, because everything here is the equivalence class of somebody. Okay?

So, that's one of the properties of the quotient map: it's always surjective.

What is the kernel of q? *[writes $\ker(q) = ?$]* If I have a ring homomorphism, it always has a kernel; the kernel is always an ideal (Recall: Proposition 1.31). So, when you answer this question, you have to give me an ideal in R. And what's the answer to this question?

Yeah?

Student Answer

I.

Spoken Transcript [00:06:17 - 00:07:05]

I. It kind of has to be I *[writes I]* because, you know, there's no other ideal in the picture. Maybe you could say it could be zero or the full ring, but it's I. And the proof of this — you should really make sure that you understand it, because this is really about whether you understand the construction.

For this, we have to find those elements *[writes Find those $x \in R$ such that]* — those x in R — such that $q(x)$, which equals $\bar{x}$, is equal to zero *[writes $q(x) = \bar{x} = \bar{0}$]*; and zero is the equivalence class of I. So, this is the same thing as x being in I *[writes $\bar{x} = \bar{0} \iff x \in I$]*. So, it's I. It's good to understand that.

Properties of the Canonical Quotient Map

Proposition 4.47 (*Surjectivity and Kernel of the Quotient Map*): Let R be a ring, $I \subset R$ an ideal, and $q\colon R \to R/I$ the canonical quotient homomorphism. Then:

- **(i)** q is surjective.
- **(ii)** $\ker(q) = I$.

Proof

We verify each statement directly:

- **(i) Surjectivity:** Every element of R/I is of the form $\bar{x} = x + I$ for some $x \in R$. By definition, $q(x) = \bar{x}$, so q maps R onto R/I.
- **(ii) Kernel:** By definition of the kernel of a ring homomorphism (Definition 1.27):

$$\ker(q) = \{x \in R \mid q(x) = \bar{0}\}.$$

Since the zero element in R/I is the coset $\bar{0} = 0 + I = I$, we have:

$$q(x) = \bar{0} \iff \bar{x} = \bar{0} \iff x - 0 \in I \iff x \in I,$$

where the middle equivalence is the definition of congruence modulo I (Definition 4.17). Thus, $\ker(q) = I$.

Q.E.D.

Review: The First Isomorphism Theorem for Rings

Spoken Transcript [00:07:05 - 00:08:15]

All right, so the third property is the one we discussed carefully last time (Recall: Theorem 4.27), which is that if I have a homomorphism *[writes (iii) $R \xrightarrow{f} S$, f is a hom.]* — $R \xrightarrow{f} S$, where it's a homomorphism —, then we have this data, which is somehow analogous to the linear algebra situation: that this kernel here is an ideal *[writes $\ker(f) \subset R$ ideal]*, and the image of f is a subring *[writes $\mathrm{Im}(f) \subset S$ subring]*, such that the image is canonically — and that's a complicated word in mathematics, *"canonically"*, but anyway, I just wrote it down explicitly — this image is isomorphic to R mod $\ker(f)$ *[writes $\mathrm{Im}(f) \cong R/\ker(f)$]*.

So, that's like the three basic properties of the quotient construction, and you should really make sure that you understand them. In some sense, that's the main thing this week.

Recap: The First Isomorphism Theorem was stated and proved in the previous lecture (Theorem 4.27); parts (i) and (ii) are Proposition 4.13 and Proposition 2.27. The lecturer recalls it as the third main point.

Theorem 4.48 (*First Isomorphism Theorem for Rings*): Let $f\colon R \to S$ be a homomorphism of rings. Then:

(i) $\ker(f)$ is an ideal of R.

(ii) $\mathrm{Im}(f)$ is a subring of S.

(iii) There is a canonical ring isomorphism $\mathrm{Im}(f) \cong R/\ker(f)$, given by:

$$\bar{f}\colon R/\ker(f) \xrightarrow{\sim} \mathrm{Im}(f), \quad \bar{x} \mapsto f(x). \tag{4.43}$$

Pullbacks of Ideals under Ring Homomorphisms

Spoken Transcript [00:08:05 - 00:10:25]

So, we'll explore **(ii)** (i.e., the quotient map $q\colon R \to R/I$, Definition 4.45) a bit more today. The first statement is: let f from R to S be any homomorphism of rings *[writes Let $f\colon R \to S$ be any hom. of rings]*, and then let J in S be an ideal *[writes Let $J \subset S$ be an ideal]*.

All right, so what we have here in the picture is that inside here is J, which is an ideal. Then the claim, so the proposition — and this is kind of an important proposition — is: if I take the inverse image of J *[writes Proposition: $f^{-1}(J) \subset R$ is an ideal]*... Are you happy with this set-theoretic terminology? This is just a map: if I take a set here, I can take its inverse image. The inverse image lives here. This is inside R. That's clear — that would not be a proposition, that's just, you know, by definition... It is an ideal.

So, the statement here is: if I have a homomorphism and if I have an ideal here, I can always bring it back here, and it'll be an ideal.

I am not saying the other way around. If I have an ideal here, I can also take the image of that ideal. I'm not saying that's an ideal, and it may not be an ideal. An example, for example: if you take $\mathbb{Z}$ goes to $\mathbb{Q}$ *[writes $\mathbb{Z} \hookrightarrow \mathbb{Q}$]*, and I take the ideal $2\mathbb{Z}$ *[writes $2\mathbb{Z}$]*, that's an ideal here. If I view it here — this is an injection, this is still $2\mathbb{Z}$, so to speak *[writes $2\mathbb{Z}$]* —, that's not an ideal here. Is that clear?

Because if it were an ideal here, it would have to be closed under multiplication by any element of $\mathbb{Q}$; in particular, it would have to be closed under multiplication by a half. And even numbers are not closed under multiplication by a half.

So, there's kind of a difference: if you have an ideal here, you can bring it back; if you have an ideal here, you're not allowed, without any conditions, to get an ideal there as an image. All right? But anyway, we're not proving the thing that's wrong; we're proving the thing that's correct.

Preimage of an Ideal under a Homomorphism

Proposition 4.49 (*Preimage of an Ideal is an Ideal*): Let $f: R \to S$ be a homomorphism of rings, and let $J \subset S$ be an ideal. Then the preimage (or pullback) of J under f:

$$f^{-1}(J) := \{x \in R \mid f(x) \in J\} \tag{4.44}$$

is an ideal of R.

Example 4.50 (*Direct Images of Ideals Need Not Be Ideals*): Consider the canonical inclusion homomorphism:

$$\iota: \mathbb{Z} \hookrightarrow \mathbb{Q}, \quad n \mapsto n.$$

In the domain $\mathbb{Z}$, the set of even integers $2\mathbb{Z}$ is an ideal (Example 2.28). However, its direct image $\iota(2\mathbb{Z}) = 2\mathbb{Z}$ is **not** an ideal in $\mathbb{Q}$.

Why the Direct Image Fails the Ideal Property: For $2\mathbb{Z}$ to be an ideal of the field $\mathbb{Q}$, it would have to absorb multiplication by all rational scalars:

$$\forall q \in \mathbb{Q},\ \forall x \in 2\mathbb{Z} \implies q \cdot x \in 2\mathbb{Z}.$$

However, taking $q = \frac{1}{2} \in \mathbb{Q}$ and $x = 2 \in 2\mathbb{Z}$, we obtain:

$$q \cdot x = \frac{1}{2} \cdot 2 = 1 \notin 2\mathbb{Z}.$$

Thus, direct images under ring homomorphisms generally fail to absorb external ring elements unless the map is surjective.

Spoken Transcript [00:10:33 - 00:12:45]

Okay, so let's try to prove this. Let x and y be in $f^{-1}(J)$, right? *[writes Proof: Let $x, y \in f^{-1}(J)$]* This means that $f(x)$ is in J and $f(y)$ is in J *[writes ⇕ and below it $f(x) \in J, f(y) \in J$]*. That's what it means to be in $f^{-1}(J)$.

And to check that $f^{-1}(J)$ is an ideal, I have to show that if I pick any two elements in $f^{-1}(J)$, then certain properties are closed. There's a list of properties, and we'll start them.

So then, for example, if I take $x + y$ and I take f of it, that's equal to $f(x) + f(y)$ — that's using the homomorphism property of f, right? *[writes $f(x+y) = f(x) + f(y)$]*

Now, both of these are in J, so that means — and then using the fact that J is an ideal — this is in J *[writes $\in J$]*. So, that implies that $x + y$ is in $f^{-1}(J)$ *[writes $\implies x + y \in f^{-1}(J)$]*.

So, that's, you know, the beginning of checking that this set $f^{-1}(J)$ is an ideal. I've checked already for you that if you pick any two elements in it, their sum is in it, which is a good start.

Spoken Transcript [00:12:45 - 00:14:42]

You could also check whether zero is in it, but... let me do something a little fancier. So, I checked that it's closed under addition. Do you accept that?

Let us check also that: let r be any element of R *[writes Let $r \in R$]*, and let x be an element of $f^{-1}(J)$ *[writes $x \in f^{-1}(J)$]*. Now, we have to consider $f(r \cdot x)$ *[writes $f(r \cdot x)$]*. Using the homomorphism property of f, this is $f(r) \cdot f(x)$ *[writes $= f(r) \cdot f(x)$]*. Now, $f(x)$ is in J, and using that J is an ideal, if I take someone in J and multiply it by any person, it stays in J. So, this stays in J *[writes $\in J$]*.

Okay? So, this shows that $r \cdot x$ is in $f^{-1}(J)$ *[writes $\implies r \cdot x \in f^{-1}(J)$]*.

So, we have these two properties of this set: if I take any two elements, it's closed under addition; and the other is, if I take any element and any element of R, it's closed under the product. And the claim is that that's actually all you have to check to be an ideal — only these two properties. The other ones follow from it.

For example, we'd have to check: if x is in $f^{-1}(J)$, we would have to check $-x$ is in $f^{-1}(J)$ *[writes If $x \in f^{-1}(J) \implies -x \in f^{-1}(J)$]*? But $-x$ is $(-1) \cdot x$. So, it's checked by this property. Is that okay? I mean, so we use here that $-x$ is $(-1) \cdot x$ *[writes $-x = (-1) \cdot x$]*, so we use the second property. That already gets negative.

Once you have x and negative, you can get zero. Zero's in it anyway. Yeah, and then you can add x and $-x$ and get zero. I think formally we should check zero's in it, because otherwise this whole thing could be empty. But anyway. So, let me say here: we also know that $f(0) = 0 \in J$, because zero goes to zero *[writes $0 \in f^{-1}(J)$ above the proof]*.

Okay? So, anyway, the precise statement is: you need to check zero's in it, these two properties, and then you can check the definition. You can also just check the definition from the beginning.

So, this is a complete argument that if I take J here and I consider $f^{-1}(J)$, this is an ideal if we started with an ideal there.

Proof of Proposition 4.49

Let $f\colon R \to S$ be a ring homomorphism and $J \subset S$ an ideal. To prove that $f^{-1}(J)$ is an ideal of R, we verify the ideal axioms (Definition 1.30):

1. Closure under Addition: Let $x, y \in f^{-1}(J)$. By definition of the preimage (Definition 0.5):

$$f(x) \in J \quad \text{and} \quad f(y) \in J.$$

Evaluating f on the sum $x + y$:

$$f(x + y) = f(x) + f(y).$$

Since J is an ideal, it is closed under addition, so $f(x) + f(y) \in J$. Therefore:

$$f(x + y) \in J \implies x + y \in f^{-1}(J).$$

2. Ideal Absorption (Multiplication by Ring Elements): Let $r \in R$ and $x \in f^{-1}(J)$. Then $f(x) \in J$. Using the multiplicativity of f:

$$f(r \cdot x) = f(r) \cdot f(x).$$

Because J is an ideal in S and $f(r) \in S$, the ideal absorbs multiplication:

$$f(r) \cdot f(x) \in J \implies r \cdot x \in f^{-1}(J).$$

3. Closure under Negatives: For any $x \in f^{-1}(J)$, we have $-x = (-1) \cdot x$ (Lemma 1.13). This is an element of R times an element of $f^{-1}(J)$, so Step 2 with $r = -1$ gives $-x \in f^{-1}(J)$.

4. The Zero Element: Since f is a ring homomorphism, $f(0) = 0$ (Proposition 1.19), and $0 \in J$ because J is an ideal. Thus, $0 \in f^{-1}(J)$; in particular, $f^{-1}(J)$ is not empty. Hence, $f^{-1}(J)$ is an ideal of R. Q.E.D.

The Correspondence Theorem for Ideals

Spoken Transcript [00:14:42 - 00:16:01]

Okay, so then we're going to look at this in the case at hand, and then there's something that's kind of pretty that happens there. *[lecturer erases the right and bottom boards]*

So, you know, we have zero in it, it's closed under addition, and closed under minus, and closed under multiplication. I guess I checked them all, the properties there.

So, now we're going to study this pullback of ideals. Often, if we start with J, if you want to say what that $f^{-1}(J)$ is in English, you'd say it's the pullback of J under f — you know, the inverse image or the pullback, something like that.

So, we want to study this for the quotient map, and there we learn something that's kind of fundamental in the behavior of ideals in algebra in general. So, we start with some definitions.

Sets of Ideals and Notation

Spoken Transcript [00:16:01 - 00:18:40]

So, we consider now the situation of the quotient map $q\colon R \to R/I$ *[writes $q\colon R \to R/I$]*. So, we start with some... I is an ideal in R, we construct the quotient, and we consider the quotient map. And I want to now start pulling back ideals here to here.

Since we have I's in there, let's just — to avoid the notation — call this one K, because we can use I and K *[corrects notation on board: changes I to K in $q\colon R \to R/K, K \subset R$]*. Sorry about that. So, we start with an ideal K, because I want to keep I and J for my ideals that I play with, rather than fix it.

So, we have an ideal K inside R, and I consider this quotient map. And I want to understand here — and this part of the quotient map is crucial to understand —: I want to understand *[writes I want to understand the pullback of ideals here]* what this pullback of ideals looks like here in this situation.

For that, I need some definitions. **Definition 1** *[writes Definition I]*: let's look at ideals here. So, let Id of this ring *[writes $\mathrm{Id}(R/K)$]* be the set of ideals. So, this equals the set of ideals of this ring *[writes $=$ Set of ideals of R/K]*. This is a ring, it has some set of ideals; just let this be the set of ideals of that ring. It could be infinite, could be small, but anyway, it's a set of ideals.

And **Definition 2** *[writes Definition II]* is: let $\mathrm{Id}_K(R)$ *[writes $\mathrm{Id}_K(R)$]* be the set of ideals of R which contain K *[writes $=$ Set of ideals of R which contain K]*.

So, these definitions are kind of important. Let me just say it, and I'll write them again in case you don't: $\mathrm{Id}(R/K)$ equals all J in R/K where J is an ideal *[writes*

$\mathrm{Id}(R/K) = \{J \subset R/K \mid J \textit{ is an ideal}\}$]. This is just writing slightly more mathematically what I said there.

And this $\mathrm{Id}_K(R)$ is equal to the set of I, ideals in R, such that K is inside I and I is an ideal [*writes* $\mathrm{Id}_K(R) = \{I \subset R \mid K \subset I, I \textit{ ideal}\}$]. Okay? So, I just said it again, but the definitions are crucial.

Notation 4.51 (*Ideal Sets for the Quotient Setting*): Let R be a ring and $K \subset R$ an ideal. We consider the quotient map:

$$q\colon R \to R/K, \quad x \mapsto \bar{x} = x + K.$$

We define two collections of ideals:

(I) Ideals of the Quotient Ring:

$$\mathrm{Id}(R/K) := \{J \subset R/K \mid J \text{ is an ideal of } R/K\}. \tag{4.45}$$

(II) Ideals Containing the Kernel:

$$\mathrm{Id}_K(R) := \{I \subset R \mid K \subset I \text{ and } I \text{ is an ideal of } R\}. \tag{4.46}$$

Statement of the Lattice Correspondence

Spoken Transcript [00:18:40 - 00:20:07]

So, is there any confusion about these definitions? I'm going to consider any ideal here, that's that set; and I consider here not any ideal, just ideals that contain the original K. Is that okay?

So, the proposition, which is kind of fundamental here, is the relationship. And the proposition is that there is a bijection [*writes There is a bijection*], and this bijection is between, on the left side, $\mathrm{Id}_K(R)$ [*writes* $\mathrm{Id}_K(R)$] — so those are ideals in R which contain K —; they're in bijective correspondence — this is the bijection [*draws double arrow* $\longleftrightarrow$ *with label bijection*] — with this $\mathrm{Id}(R/K)$ [*writes* $\mathrm{Id}(R/K)$]. That's what we're going to prove.

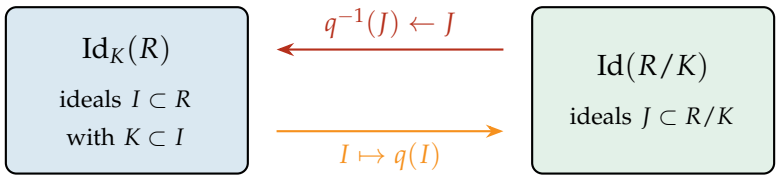
So, we're going to prove this statement.

This is some kind of very nice way of thinking about ideals. We start with R — maybe it's a ring you know, you're comfortable with —, you can make a quotient. And for this quotient ring, you know, it's maybe sometimes more complicated to think about, an interesting object; but if you want to answer the question: *"What do the ideals look like here?"*, this bijection tells you how to think about the ideals in this quotient ring: they're just the same as ideals before you quotient, but the ideals before you quotient have to contain K.

Proposition 4.52 (*Correspondence Theorem / Lattice Theorem for Ideals*): Let R be a ring and $K \subset R$ an ideal. There is a canonical bijection:

$$\mathrm{Id}_K(R) \longleftrightarrow \mathrm{Id}(R/K). \tag{4.47}$$

That is, the ideals of the quotient ring R/K are in exact one-to-one correspondence with the ideals of R that contain K.



Conceptual Meaning of the Bijection: This theorem reveals the internal structure of quotient rings:

- **Ideals in R/K Are Pre-Existing Ideals in R:** One does not need to search for new, exotic ideals in the quotient ring R/K. Every ideal in R/K arises directly from an ideal of R that happens to contain the modulo ideal K.

Proof of the Correspondence Theorem for Ideals

Well-Definedness of the Correspondence Maps

Spoken Transcript [00:20:07 - 00:22:05]

All right, so on this bijection, to prove it, I have to give you maps. If I have J here *[writes $J \in \mathrm{Id}(R/K)$]*, in R/K, I have to take it to get an ideal here, right? *[writes $J \mapsto \ldots$]* And the map this way is going to be $f^{-1}(J)$ — this is an ideal here *[writes $q^{-1}(J) \in \mathrm{Id}_K(R)$]* —, okay?

Let's first understand it. First, we said that if you take any ideal here, when you take f^{-1}, it is an ideal (Recall: Proposition 4.49). I'm saying more now: not only is it an ideal, but it also contains K. And why does it contain K? Because if I take zero here, the inverse image of zero is all of K.

So, this requires some justification that you have to think about. The claim here is that f^{-1} — it's not f here, it's q, sorry: *[corrects f^{-1} to q^{-1} on board]* q^{-1} for the quotient map —, $q^{-1}(J)$ has to contain this ideal K *[writes Claim: $q^{-1}(J) \supset K$]*.

And why is that? Well, we've already seen that $q^{-1}(0)$, that's the kernel, and that equals K (Recall: Proposition 4.47) *[writes $q^{-1}(0) = K$]*. And this J contains zero *[writes $0 \in J$]*. So, $q^{-1}(J)$ has to contain K.

Are you happy with this statement? It's kind of important, because, you know, it's saying that we're trying to prove this bijection between all ideals here and ideals that contain K. And the claim is that if I take any ideal here and I take this q^{-1} to get it to this side, it will always contain K, because this q^{-1} certainly contains all the things that go to zero, because zero is in here, and the things that go to zero are exactly K.

The Leftward Map: Pullback under the Quotient Map

Claim 4.53 (*Well-Definedness of the Pullback Map*)**:** For any ideal $J \in \mathrm{Id}(R/K)$, its preimage under q:

$$q^{-1}(J) = \{x \in R \mid q(x) \in J\} \tag{4.48}$$

is an ideal of R that contains K, i.e., $q^{-1}(J) \in \mathrm{Id}_K(R)$.

Proof

By Proposition 4.49, the preimage of any ideal under a ring homomorphism is an ideal of the domain, so $q^{-1}(J)$ is an ideal of R. It remains to show that $q^{-1}(J) \supset K$: Since J is an ideal in R/K, it contains the additive identity $\bar{0} \in J$. By the definition of preimage and the kernel of q (Proposition 4.47):

$$K = \ker(q) = q^{-1}(\bar{0}) \subset q^{-1}(J).$$

Hence, $K \subset q^{-1}(J)$, which proves that $q^{-1}(J) \in \mathrm{Id}_K(R)$. Q.E.D.

Spoken Transcript [00:22:05 - 00:23:35]

That's one direction. Now we have to do the other direction. If I have an ideal I here *[writes $I \in \mathrm{Id}_K(R)$]* — I have an ideal here, I —, I have to be able to go this way and get an ideal of R/K *[writes $I \mapsto \ldots$]*.

And the natural thing to do is just to take q of this *[writes $q(I)$]*.

But I had just told you: in general, if you start with an ideal, you can't just take q of it and get an ideal. But in this case — in the case of a quotient — you can! So, this requires some justification also *[writes requires justification]*, which we will do in a second.

As I said, if you take an ideal in the domain of a ring homomorphism, you can't always just take the image and have it be an ideal. We did an example where that failed (Recall: $\mathbb{Z} \hookrightarrow \mathbb{Q}$ and $2\mathbb{Z}$, Example 4.50). But for the quotient map, it is true.

So, these are the two motions. I mean, it's a bijection, so you have to say how to go left and how to go right, and here they are explicitly. And so we have to prove some things about them being well-defined and why this is an ideal, etc.; and then I have to prove that the compositions are the same.

Anyway, that's the task at hand.

The Explicit Bijection Maps

To establish the correspondence $\mathrm{Id}_K(R) \longleftrightarrow \mathrm{Id}(R/K)$, we define two mutually inverse maps:

$$\Phi\colon \mathrm{Id}(R/K) \to \mathrm{Id}_K(R), \quad J \mapsto q^{-1}(J), \tag{4.49}$$

$$\Psi\colon \mathrm{Id}_K(R) \to \mathrm{Id}(R/K), \quad I \mapsto q(I) = \{\bar{x} \in R/K \mid x \in I\}. \tag{4.50}$$

Why the Forward Image $q(I)$ is an Ideal for the Quotient Map: Earlier, Example 4.50 demonstrated that forward images of ideals generally fail to be ideals. However, q is **surjective** (Proposition 4.47). For any ideal $I \subset R$, $q(I)$ is an additive subgroup of R/K because q is a homomorphism. To check ideal absorption: Let $\bar{r} \in R/K$ and $\bar{x} \in q(I)$ with $x \in I$. Since q is surjective, $\bar{r} = q(r)$ for some $r \in R$. Then:

$$\bar{r} \cdot \bar{x} = q(r) \cdot q(x) = q(r \cdot x).$$

Since I is an ideal of R, $r \cdot x \in I$. Thus, $q(r \cdot x) \in q(I)$, showing that $q(I)$ absorbs multiplication from all elements of R/K.

Spoken Transcript [00:23:35 - 00:25:06]

To prove the left arrow — this arrow *[points at ⟵ on the board]* — is well-defined *[writes To prove the ⟵ is well defined]*, which we already did (Recall: Claim 4.53), but I'll just write it formally: we must show that K is inside $q^{-1}(J)$ *[writes we must show that $K \subset q^{-1}(J)$]*.

And the proof of that, the argument for that, is: well, J is an ideal, so 0 is in J *[writes $0 \in J$ so]*. So $q^{-1}(0)$ is in $q^{-1}(J)$ *[writes $q^{-1}(0) \subset q^{-1}(J)$]*, and $q^{-1}(0)$ is exactly K *[writes K]* — that's the kernel of the quotient map. So that's the argument there.

Next, to prove this arrow *[points at ⟶ on the board]* is well-defined *[writes To prove ⟶ is well defined]*, we just have to prove that *[writes we have to prove that]* $q(I)$ inside this quotient is an ideal *[writes $q(I) \subset R/K$ is an ideal]*. That's all we have to do.

All right, so how are we going to prove it's an ideal? You know, we have to check these properties in general. So we just...

Verification: Well-Definedness of Both Directions

1. Left Arrow (⟵ is Well-Defined): To show $\Phi(J) = q^{-1}(J) \in \mathrm{Id}_K(R)$, we must show that $K \subset q^{-1}(J)$:

$$\bar{0} \in J \implies q^{-1}(\bar{0}) \subset q^{-1}(J).$$

Since $q^{-1}(\bar{0}) = \ker(q) = K$ (Proposition 4.47), we obtain $K \subset q^{-1}(J)$.

2. Right Arrow (⟶ is Well-Defined): To show $\Psi(I) = q(I) \in \mathrm{Id}(R/K)$, we must show that for any ideal $I \in \mathrm{Id}_K(R)$:

$$q(I) \subset R/K \quad \text{is an ideal.}$$

Spoken Transcript [00:25:06 - 00:26:54]

So, if I have an element here *[writes $R \to R/K$]* — R/K, and inside here we have $q(I)$ *[writes $q(I)$]*, and here's the map q *[writes q]*...

If we pick two elements here... I want to show it's an ideal, so in particular I have to show it's closed under addition. First of all, let's just get zero out of the way. Zero is definitely in $q(I)$, because zero is in I, and if I apply q to zero, I get zero, right? *[writes $\bar{0} \in q(I)$]* Is this clear? Because 0 is in I, so $q(0)$ is in $q(I)$, and $q(0) = 0$ *[writes Because $0 \in I \implies q(0) \in q(I)$, and $\bar{0}$ above $q(0)$]*. So, zero is definitely here.

Then I have to check: if two people are here, that their sum is here, so it's closed under addition. So, let us pick some people — how are we going to say who's here? Let's pick two elements. Well, everything that's in here is in the image of I, so I can pick elements in here such that x and y are in I, so that two general elements here are those that come from I. So, I can write them like this: *[writes $q(I) \ni \bar{x}, \bar{y}$ and $x, y \in I$]*.

So, I pick some elements here; both of them come from I. Since I is an ideal *[writes I ideal]*, this means $x + y$ is in I *[writes $\implies x + y \in I$]*, and then that means that that is in $q(I)$ *[writes $\implies \overline{x+y} \in q(I)$]*.

Okay? So, that shows that $q(I)$ has zero and it's closed under addition.

Proof that $q(I)$ is an Additive Subgroup

We verify that $q(I)$ is an additive subgroup of R/K:

Zero Element: Since I is an ideal of R, we have $0 \in I$. Thus:

$$q(0) \in q(I) \implies \bar{0} \in q(I).$$

Closure under Addition: Let $\bar{x}, \bar{y} \in q(I)$. Since elements of $q(I)$ are images of elements in I, there exist $x, y \in I$ such that:

$$q(x) = \bar{x} \quad \text{and} \quad q(y) = \bar{y}.$$

Because I is an ideal, it is closed under addition:

$$x + y \in I.$$

Applying the homomorphism q:

$$q(x+y) \in q(I) \implies \overline{x+y} = \bar{x} + \bar{y} \in q(I).$$

Thus, $q(I)$ contains $\bar{0}$ and is closed under addition. (Negatives need no separate check: $-\bar{x} = (-\bar{1}) \cdot \bar{x}$ lies in $q(I)$ by the absorption shown next, just as in the proof of Proposition 4.49.)

Spoken Transcript [00:26:54 - 00:28:26]

So, finally, we have to do the multiplication part.

Suppose we have to take an element x — that's an arbitrary element here, where x is in I — and we take its equivalence class; and then we have to multiply it by any element of the quotient ring, any single element, nothing to do with I, right? Well, that's some element, but that element is still some element of R.

So, in this multiplication case, we have to show that this set is closed by picking any element in that set — that's here, $\bar{x} \in q(I)$ *[writes $\bar{r} \cdot \bar{x}$ and below it $\bar{x} \in q(I)$]* — and multiplying it by any element. But for both of those, we can find preimages in R, because the quotient map is surjective. So, whatever element here is, I can write it as $\bar{r}$ of something; and here, since it's in $q(I)$, I can write $\bar{x}$ where $x \in I$.

Okay? And then, since I is an ideal, $r \cdot x$ is in I *[writes $r \in R, x \in I \implies rx \in I$]*, and that implies that this product is in $q(I)$ *[writes $\implies \bar{r} \cdot \bar{x} \in q(I)$]*.

So, now I've checked it. I've checked that this is well-defined. So, they're both well-defined. That's a good start. But then I have to show that if you go one way and back, or the other way and back, you get the identity. As usual, there's a bunch of steps.

Ideal Absorption under Multiplication

Let $\bar{r} \in R/K$ be an arbitrary element of the quotient ring, and let $\bar{x} \in q(I)$ with $x \in I$. Because the canonical quotient map $q\colon R \to R/K$ is **surjective**, the element $\bar{r}$ possesses a preimage $r \in R$ such that $q(r) = \bar{r}$.

Since I is an ideal of R, it absorbs multiplication by any ring element:

$$r \in R,\ x \in I \implies r \cdot x \in I.$$

Applying the quotient homomorphism q:

$$q(r \cdot x) \in q(I) \implies \overline{r \cdot x} = \bar{r} \cdot \bar{x} \in q(I).$$

Similarly, $\bar{x} \cdot \bar{r} = \overline{x \cdot r} \in q(I)$. Hence, $q(I)$ satisfies all ideal axioms, proving that $q(I) \subset R/K$ is an ideal.

Board Preparation

The lecturer pauses to slide and erase the central and upper chalkboards, preparing the surface to demonstrate that the compositions $\Psi \circ \Phi$ and $\Phi \circ \Psi$ equal the identity.

Bijectivity of the Correspondence

Spoken Transcript [00:28:35 - 00:30:18]

And the trick now is to try to prove these steps without erasing the claim.

So, I'm going to erase the definitions of this that you're supposed to remember (Recall: Notation 4.51). *[erases Def 1 and Def 2 from the center chalkboard]*

To prove it's a bijection, we should prove that if I go here and then back, I get the same thing. That's the first thing; and then I have to do it the other way, you know.

So now... *[writes Now proof of the bijective property]* proof of the bijective property. There are two steps. The first step, maybe **Step (A)** *[writes (A)]*, is: start with an ideal J in the quotient — we start there *[writes Start with $J \subset R/K$]*.

And then I have to show that if I take the pullback and then I take q of that, this equals J *[writes $q(q^{-1}(J)) = J$]*, okay? That's what I have to show. That's saying that if I do the motion left and then the motion right, I have altogether done nothing.

And the claim is: this is clear *[writes This is clear]*. This is clear because q is surjective *[writes because q is surjective]*.

Okay, and I'm going to draw you a picture. So, here's R/K, here's R...

Spoken Transcript [00:30:18 - 00:31:26]

And it's surjective. That means that every single point — if you take a point here — is always hit, maybe by several, but it's definitely always hit, right?

So, if I take a point that happens to be in J, then I go over here, that's a point in $q^{-1}(J)$. If I do q, I just go back to J, and in fact I go back to the point I started with. Because every point is hit. So, I take any point in J, it's hit by somebody, so that gives me some point here; and then when I do q, I go back to it.

What would have been a problem is if there was a point here that nobody hit; then when I went back here and forward, I would not get it. But since this is surjective, this is clear just for set-theoretic reasons, not algebra-theoretic reasons. So, this is true. *[writes (TRUE) after $q(q^{-1}J) = J$]*

So, we're happy about that. Okay, then we have one more thing to check.

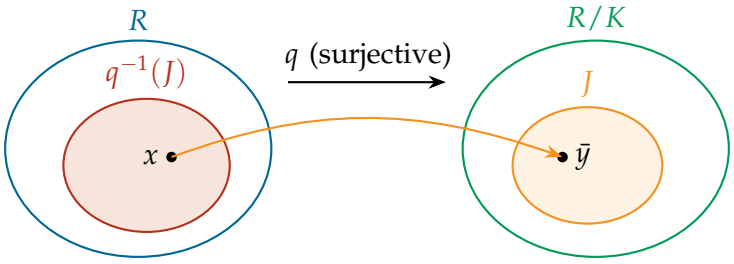
Bijective Property Part (A): $q(q^{-1}(J)) = J$

Let $J \in \mathrm{Id}(R/K)$ be an ideal of the quotient ring. We show:

$$q(q^{-1}(J)) = J. \tag{4.51}$$

This equality holds purely set-theoretically because the quotient map $q\colon R \to R/K$ is **surjective**:

- **Inclusion ($\subseteq$):** The relation $q(q^{-1}(J)) \subseteq J$ always holds by the general definition of the preimage (Fact 0.7).
- **Inclusion ($\supseteq$):** Conversely, let $\bar{y} \in J$. By surjectivity of q, there exists $x \in R$ such that $q(x) = \bar{y}$. Thus $x \in q^{-1}(J)$, which implies $\bar{y} = q(x) \in q(q^{-1}(J))$.



Every element $\bar{y} \in J$ has a preimage $x \in q^{-1}(J)$ with $q(x) = \bar{y}$, so $q(q^{-1}(J)) = J$.

Spoken Transcript [00:31:26 - 00:32:06]

[erases the upper chalkboard] Okay, I'm going to erase this board.

The usual thing I tell students is that all the stuff I talk about, it's much easier to learn than German. German is a much harder language to learn than this, yeah. And I can tell you that from personal experience, yeah.

Spoken Transcript [00:32:06 - 00:34:06]

So, we need now to prove... We started on the right-hand side, so we need to start on this side now. We start with some I that contains K, and this is an ideal *[writes $K \subset I \subset R$]* — so it's an ideal that contains that ideal K. And then we go with I to the right-hand side, so we just take $q(I)$; and then we have to go back to q^{-1}, so we take $q(I)$ and then we take the pullback, and we need to show that this is I, okay? *[writes $I \longrightarrow q^{-1}(q(I)) \stackrel{?}{=} I$]*

And now it's not true for set-theoretic reasons. So, with the same diagram *[draws two blobs for R and R/K]*... Then we have some I here, and here is R and R/K. And when we go forward, we just go with whatever q; so q sends points all over here, right? So it just sends points to some image, that's $q(I)$ *[draws arrows from I to $q(I)$]*. Okay?

And now you see the problem, because then when we go back, there could be a point here: there's some point x here that goes to $\bar{x}$, but then there could be a y that also goes to that. And we get nervous now, because I want this to be I, but this y makes me nervous, because these two things could have gone here, so this y is definitely in

$q^{-1}(q(I))$ *[writes $y \in q^{-1}(q(I))$]*, okay? And it makes me very nervous.

So, I need to prove: whenever this happens, actually y was already in I. I need to prove that. But it's not true for set-theoretic reasons. That one (i.e., Eq. (4.51)) was true for set-theoretic reasons, but this one is not. Because the q map is not a bijection; you know, it puts things together.

Spoken Transcript [00:34:06 - 00:34:45]

All right, so how could this happen? We have this: x is in I *[writes $x \in I$]*, and so then on this side we have $\bar{x}$ *[writes $\bar{x}$]*, and we have some y in R *[writes $y \in R$]*. And what we have here is that $\bar{y}$ — that's what q does, it takes y to $\bar{y}$ —, $\bar{y}$ might equal $\bar{x}$ *[writes $\bar{y} = \bar{x}$]*. That's the situation we have, okay?

And we need to now do something about it. And you see, that gives us a piece of information. What does it mean that $\bar{y} = \bar{x}$? It only means one thing, which you should now repeat to me.

Student Answer

Their difference is in K.

Spoken Transcript [00:34:49 - 00:36:36]

Yeah, so what's the equation? *[student speaks]*

Well, the equation is: this is if and only if there exists some element z in K such that $x - y = z$, right? *[writes $\exists z \in K\colon x - y = z$]* That's exactly the equivalence. Or I could write it the other way... Let's write it... okay, anyway, let's write it the other way: $y - x = z$ *[corrects to $y - x = z$]*, because it doesn't matter, it's symmetric. Okay?

So, that's the crucial thing. Since their equivalence classes are the same, it means their difference is in K. So that means that y — if I just now logically conclude this —: $y = x + z$ *[writes $y = x + z$]*, okay?

But x is in I and z is in K, and so finally we use this funny condition that we're always assuming: K is inside I. So here we have x is in I, z is in K, but K is inside I, so they're both inside I *[writes $x \in I, z \in K \subset I$]*. So, this thing is in I *[writes $\implies \in I$]*. That means this funny y is actually in I *[writes $y \in I$]*.

That's the argument. So, what happens: it looks like this can happen, and it can happen set-theoretically, but the algebraic hypothesis says that it actually does not happen. And that's the argument: if it were to happen, they'd have the same equivalence class. If they had the same equivalence class, their difference must be in K, and then I can write y as something in I plus something in K; but since K is in I, that whole thing is still in I. So, that's the argument.

So, now I've proven that bijection. All right.

Bijective Property Part (B): $q^{-1}(q(I)) = I$

Let $I \in \mathrm{Id}_K(R)$ be an ideal containing K (so $K \subset I \subset R$). We show:

$$q^{-1}(q(I)) = I. \tag{4.52}$$

The inclusion $I \subseteq q^{-1}(q(I))$ always holds set-theoretically (Fact 0.7). For the reverse inclusion, let $y \in q^{-1}(q(I))$. By definition of the preimage:

$$q(y) \in q(I) \implies \bar{y} \in q(I).$$

Thus, there exists some $x \in I$ such that:

$$\bar{y} = \bar{x}.$$

By definition of congruence modulo K (Definition 4.17):

$$\bar{y} = \bar{x} \iff y - x \in K \iff \exists z \in K \text{ such that } y - x = z.$$

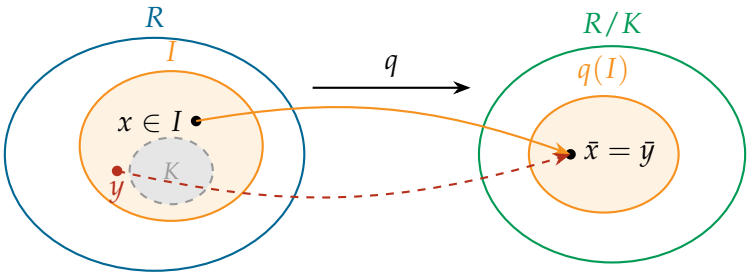
Rearranging yields:

$$y = x + z.$$

Since $x \in I$ and $z \in K$, and by hypothesis $K \subset I$, we have $z \in I$. Because I is an ideal (hence closed under addition):

$$x \in I \quad \text{and} \quad z \in I \implies y = x + z \in I.$$

Therefore, $q^{-1}(q(I)) \subseteq I$, which proves $q^{-1}(q(I)) = I$.



For $y \in q^{-1}(q(I))$: since $\bar{y} = \bar{x}$, we have $y - x = z \in K$.
Since $K \subset I$, $y = x + z \in I$, so y is forced into I.

Together, Steps (A) and (B) establish that Φ and Ψ are mutually inverse bijections, completing the proof of the Correspondence Theorem (Proposition 4.52).

Q.E.D.

Beyond the Lecture: The Correspondence Preserves Inclusions

The bijection of Proposition 4.52 respects inclusions in both directions, which is why it is also called the lattice theorem. For ideals $I_1, I_2 \in \mathrm{Id}_K(R)$, $I_1 \subseteq I_2$ clearly gives $q(I_1) \subseteq q(I_2)$; conversely, $q(I_1) \subseteq q(I_2)$ gives $I_1 = q^{-1}(q(I_1)) \subseteq q^{-1}(q(I_2)) = I_2$ by Step (B) (Eq. (4.52)). In the same way, $J_1 \subseteq J_2$ holds in $\mathrm{Id}(R/K)$ if and only if $q^{-1}(J_1) \subseteq q^{-1}(J_2)$, using Step (A) (Eq. (4.51)).

Spoken Transcript [00:36:36 - 00:37:56]

And this is kind of uncharacteristic: I really proved every single step of it. I don't know whether you like that or not. Normally, in the middle I would start waving my hands, but this time I proved every single step of it.

[lecturer erases the boards]

But it's very good to understand this, because it shows some non-trivial aspect of this construction. And it really allows you to grasp what the ideals of the quotient are. I wanted to say some more things about fields; I don't have so much time, unfortunately. Anyway, maybe I can start.

So, the next topic, related to the ideas here, is the notion of a maximal ideal. Now we've finished two themes in the class; the third one is the notion of a maximal ideal.

Maximal Ideals

Spoken Transcript [00:37:56 - 00:40:01]

Okay, so there are different ways to define a maximal ideal. The most simple one is that *[writes Notion of a maximal ideal* $\mathfrak{m} \subset R$*]*:

(i) $\mathfrak{m}$ is an ideal, of course. *[writes (i)* $\mathfrak{m}$ *is ideal]* This is going to be a definition: maximal ideal. To be a maximal ideal, of course, you have to be an ideal.

(ii) This ideal is not allowed to be the whole thing. So this normally is said that you should be a proper ideal *[writes (ii)* $\mathfrak{m} \neq R$ *(proper ideal)]*. Normally you'd say that in one: $\mathfrak{m}$ is a proper ideal, which means it's an ideal that's not the whole thing. Because the whole thing is cheating, it turns out, for a maximal ideal.

(iii) And this is the maximal property: if I have another ideal, another proper ideal *[writes (iii) If* $\hat{\mathfrak{m}} \subset R$*,* $\hat{\mathfrak{m}}$ *proper ideal]*, which has the property that $\mathfrak{m}$ is inside it... *[writes which has the property* $\mathfrak{m} \subset \hat{\mathfrak{m}}$*]*

When I use this notation, I hope this is clear: this means that this is a subset of that. It's just silent on whether it's equal or not. Is that how you use it? You know, so this means that this is a subset of that. It could be equal; in fact, it's okay. If you wanted to be specific, you could say like that — that means it's not equal. Anyway, it's important here. So, which has the property that whenever this is inside that, that actually means they're equal *[writes* $\implies \mathfrak{m} = \hat{\mathfrak{m}}$*]*.

That's the maximality property. It's saying that, well, first of all, it's a proper ideal; and the second one says there's nobody above it that's a proper ideal. Is that clear? So, the reason why you want to rule out R always is because R is the biggest ideal, and no one ever beats R because it's the whole ring. So it kind of has an unfair role in that.

Spoken Transcript [00:40:01 - 00:40:55]

So, for a maximal ideal... It's a confusing thing, and, you know, it's a point that's related to the confusion about definitions of what a zero ring is, or whether a field is trivial or not; it's the same kind of confusion there. And we will discuss that in the next

lecture. But it's important to understand: it has to be proper, and nobody's bigger.

So, for example, if you take the integers: the multiples of 4 — sometimes we write that as $4\mathbb{Z}$ (Recall: Definition 2.33) —, this is definitely an ideal; is it a maximal ideal? *[writes $\mathbb{Z}$, then $(4) \subset \mathbb{Z}$ and $4\mathbb{Z} \subset \mathbb{Z}$]*

Student Answer

No.

Spoken Transcript [00:40:57 - 00:41:18]

No, because it sits inside multiples of 2 *[writes $4\mathbb{Z} \subset 2\mathbb{Z}$]*. And this is not equal *[writes $\neq$ under the $\subset$, giving $4\mathbb{Z} \subsetneq 2\mathbb{Z}$]*. So it would be contradicted by this fact: if there's somebody bigger that's not the full ring, then that's not a maximal ideal.

So, what are the maximal ideals for $\mathbb{Z}$?

Student Answer

Prime numbers.

Spoken Transcript [00:41:21 - 00:43:42]

Yeah, so if you have $p\mathbb{Z}$ inside $\mathbb{Z}$, where p is prime *[writes If $p\mathbb{Z} \subset \mathbb{Z}$ where p is prime]*, then these are the maximal ideals *[writes then these are max ideals]*. This is if and only if... yeah, these are the maximal ideals.

In fact, you can kind of draw... Or maybe I'll do it next time; I'll talk about the pictures of this.

If you have a field — so if F is a field *[writes F is a field]* —, what are the maximal ideals there?

Well, if you remember, a field only has two ideals (Recall: Theorem 1.32) *[writes has only 2 ideals: $0, F$]*: it has the zero ideal and the whole thing. And when we talk about maximal ideals, we don't talk about the whole thing. So, it only has one proper ideal, and that proper ideal is maximal.

So, you can say: field F has 0 as a maximal ideal *[writes Field F has 0 as a max. ideal]*.

And the converse is maybe the last thing I say today: let R be a ring *[writes Let R be a ring]* with 0 in R a maximal ideal *[writes $0 \subset R$ a max. ideal]*. So, we take any ring, and suppose you find out, one way or the other, that 0 is a maximal ideal.

Then what am I going to write here? Well, R is a field *[writes $\implies$ R is a field]*.

And the reason for this is: first of all, you know R is not the zero ring. A field is not the zero ring (i.e., $1 \neq 0$, Definition 2.9). So you know R is not the zero ring because zero is a maximal ideal, and zero in the zero ring is not a maximal ideal because it's the whole thing. So, this hypothesis secretly says that R is not the zero ring. And then I will explain this next time: that has to be a field. *[video ends, students applaud]*

Definition of Maximal Ideals and Examples

Definition 4.54 (*Maximal Ideal*): Let R be a ring. A subset $\mathfrak{m} \subset R$ is called a *"maximal ideal"* if:

(i) $\mathfrak{m}$ is an ideal of R.

(ii) $\mathfrak{m} \neq R$ ($\mathfrak{m}$ is a *"proper ideal"*).

(iii) Whenever $\hat{\mathfrak{m}} \subset R$ is a proper ideal of R with $\mathfrak{m} \subset \hat{\mathfrak{m}}$, then:

$$\mathfrak{m} = \hat{\mathfrak{m}}.$$

In other words, $\mathfrak{m}$ is a maximal element of the set of proper ideals of R, ordered by inclusion (Definition 0.10).

Notation: As the lecturer explains here, $A \subset B$ on the board says only that A is a subset of B and allows $A = B$, like $A \subseteq B$. For a strict inclusion, he writes $A \subsetneq B$.

Examples 4.55 (*Maximal Ideals in $\mathbb{Z}$*): Consider the ring of integers $\mathbb{Z}$:

- **The Ideal $4\mathbb{Z}$ is Not Maximal:** We have the chain of inclusions:

  $$4\mathbb{Z} \subsetneq 2\mathbb{Z} \subsetneq \mathbb{Z}.$$

  Since $2\mathbb{Z}$ is a proper ideal strictly containing $4\mathbb{Z}$, the ideal $4\mathbb{Z} = (4)$ (Definition 4.32) is not maximal.

- **The Ideals $p\mathbb{Z}$ for Primes p Are Maximal:** For any prime number p, the principal ideal $p\mathbb{Z} = (p)$ is a maximal ideal:

  $$p\mathbb{Z} \subseteq I \subsetneq \mathbb{Z} \implies I = p\mathbb{Z}.$$

  Indeed, every ideal of $\mathbb{Z}$ is $I = n\mathbb{Z}$ with $n \geq 0$ (Theorem 2.35), and $p\mathbb{Z} \subseteq n\mathbb{Z}$ means $n \mid p$, so $n = 1$ (then $I = \mathbb{Z}$ is not proper) or $n = p$. Thus, the maximal ideals of $\mathbb{Z}$ are precisely the ideals $p\mathbb{Z}$ for prime numbers p: the remaining ideals are not maximal, since $m\mathbb{Z} \subsetneq a\mathbb{Z} \subsetneq \mathbb{Z}$ for $m = ab$ with $1 < a < m$, and $0\mathbb{Z} \subsetneq 2\mathbb{Z} \subsetneq \mathbb{Z}$, while $1\mathbb{Z} = \mathbb{Z}$ is not proper.

Proposition 4.56 (*Characterization of Fields via the Zero Ideal*): Let R be a ring. Then R is a field if and only if the zero ideal $(0) = \{0\}$ is a maximal ideal in R:

$$R \text{ is a field} \iff (0) \text{ is a maximal ideal of } R. \tag{4.53}$$

Why (0) Being Maximal Characterizes Fields:

- **Forward Direction ($\implies$):** If F is a field, its only ideals are (0) and F (Theorem 1.32). Since $(0) \neq F$ (because $1 \neq 0$ in a field, Definition 2.9), (0) is the unique proper ideal, hence maximal: the only proper ideal containing (0) is (0) itself.
- **Backward Direction ($\impliedby$):** If (0) is maximal, then (0) is a proper ideal, so $(0) \neq R \implies R \neq \{0\}$ (ruling out the trivial zero ring). The lecturer stops here and explains in the next lecture why R must then be a field.

Beyond the Lecture: Completing the Backward Direction

Suppose (0) is a maximal ideal of R, so that $R \neq \{0\}$. For any non-zero element $x \in R \setminus \{0\}$, the principal ideal (x) (Definition 4.32) strictly contains (0). By maximality of (0), (x) cannot be a proper ideal, forcing $(x) = R$. Therefore, $1 \in (x)$, meaning there

exists $y \in R$ such that $xy = 1$, so every non-zero element is invertible and R is a field.

Summary of Key Concepts

- **Quotient Rings and Canonical Projection:** For any ideal $I \subset R$, the quotient R/I is a well-defined ring under coset operations $\bar{x} + \bar{y} = \overline{x+y}$ and $\bar{x}\bar{y} = \overline{xy}$. The canonical projection $q\colon R \to R/I$ with $x \mapsto \bar{x}$ is a surjective ring homomorphism with $\ker(q) = I$. By the First Isomorphism Theorem, every homomorphism $f\colon R \to S$ induces an isomorphism $R/\ker(f) \cong \mathrm{Im}(f)$.
- **Pullbacks vs. Pushforwards of Ideals:** Under any ring homomorphism $f\colon R \to S$, the preimage $f^{-1}(J)$ of an ideal $J \subset S$ is always an ideal of R. Conversely, direct images $f(I)$ generally fail to be ideals (e.g., $2\mathbb{Z} \subset \mathbb{Z} \hookrightarrow \mathbb{Q}$), unless the homomorphism is surjective.
- **The Correspondence Theorem for Ideals:** For an ideal $K \subset R$, the lattice of ideals in R/K is in canonical inclusion-preserving bijection with the lattice of ideals in R containing K:
$$\mathrm{Id}_K(R) \longleftrightarrow \mathrm{Id}(R/K), \quad I \mapsto q(I), \quad J \mapsto q^{-1}(J).$$
Surjectivity of q ensures $q(q^{-1}(J)) = J$, while the condition $K \subset I$ ensures $q^{-1}(q(I)) = I$.
- **Maximal Ideals and Fields:** An ideal $\mathfrak{m} \subsetneq R$ is maximal if no proper ideal strictly contains it. In $\mathbb{Z}$, the maximal ideals are precisely the ideals $p\mathbb{Z}$ for primes p. A commutative ring R is a field if and only if the zero ideal (0) is a maximal ideal.